Deriving Standard Model Gauge Structure from Observer Overlap Consistency

B. Müller Mario Poneder Alexander Osika Kai Xue Peter Nguyen
David Matscheko

October 17, 2026

Paper release: r2050 **Released:** October 17, 2026

Author affiliations: Bernhard Mueller, Pragma Research Inc.; Alexander Osika, EtherWorks.

Abstract

Observer-Patch Holography models an observer as a finite self-reading patch with local state, boundary ports, readback records, and repair moves. This paper gives two gauge-reconstruction routes. The categorical route reconstructs a compact group from overlap transport on a declared bosonic sector category. The finite route starts with twelve ports and their icosahedral incidence. Complete reversible port response and endogenous observer transport force the compact Lie-algebra type $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$. Exact finite calculations derive the carrier module and inverse-port response. Compact classification and the fixed-space argument exclude the remaining compact alternative. A conditional matrix realization supplies the familiar current blocks. An exterior-algebra witness contains one fifteen-state generation with exact Standard Model charges and anomaly cancellation. Its common central kernel is $\mathbb{Z}_6$. Among single complete faithful bands in the conditional window $3 \le N_g \le 5$, the exact Laplacian costs $5 - \sqrt{5} < 6 < 5 + \sqrt{5}$ select the rank-three band. The physical matrix current, matter action, and global quotient require same-source response, character-completeness, and loop-to-kernel receipts. The compact locus of the equivariant Jacobi variety is classified into three certified bracket families with exact carrier-integrating admissible global forms. A declared equal-weight oriented-face bracket is exactly $60R_{13}$, has 240 nonzero Jacobi coordinates, and has family G uniquely nearest among the classified compact locus under each of three added coordinate repair norms $\ell_1, \ell_2, \ell_\infty$ and, in the quadratic case, under every invariant carrier metric that is balanced in the two mirror three-dimensional sectors or whose 3^+ scale stays within the exact chirality box $\beta/\delta \in [1/50, 6]$; P is excluded for the complete invariant cone. The rule, the nearest-point law, and the carrier-induced metric class are not source-derived. Over each supplied bracket, ad-invariance leaves one coefficient per simple factor. In particular, the F/G carrier-form spaces drop exactly from dimension three to two, while the P control remains two-dimensional. Constructed two-factor Gibbs kernels bind the complete P family to one color-bearing F family and make the relative coefficient identifiable modulo common scaling. Fixing that coefficient, and attaching the construction to laboratory currents, requires further physical input.

Interpretation and state variables. OPH proposes one structure with two descriptions: a timeless, self-referential complete history and its actual internal computation. Observers are proposed to reverse-engineer their reality's substrate and construct hardware whose execution realizes that same history, including its builders and hardware, in emergent, mutually calibrated local times. The candidate must supply construction, initialization, controls, memory, read/write and record services, clock comparisons, and their resource costs; no external host or additional time is assumed.

Its realization map must specify whether it matches one actual history, a family of preparation and intervention responses, or every independently allowed microstate. These are different requirements. A candidate must specify the history domain and closure operator; neither the schematic equation nor the finite constructions establish this physical realization or select a unique universe. An actual quantum record history also need not assign simultaneous outcomes to incompatible unperformed measurements.

The proposed primitives are patch states, records, interfaces, and admissible operations. Physical space, duration, and matter are intended emergent readouts. A supplied carrier incidence, geometric domain, action, representation, or clock is a premise of the construction using it. Complete-history closure, an instantaneous repair fixed point, a public normal form, and a stationary probability law are different objects. A stationary law can support changing individual histories. An operational observer is defined by its access and record interfaces; consciousness requires an additional psychophysical interpretation.

Basis, premise labels and paper set. OPH has one hypothesis, closure: the universe is a timeless self-referential structure whose observers recover its specification from their records and realize it. Because nothing lies outside that structure, recovered structure needs no a-priori justification, nothing unrecovered may be assumed, and every change of description is implemented by the structure's own operations. Agreement on accepted shared data defines what is public. The executable specification of carrier, repair law, observer algebra and closure inputs is implemented and runs; it is a candidate judged by whether it reproduces what observers measure. The labels A1, A2 and A3 name theorem premises: the carrier, federation and response layer of that specification; the agreement condition with its endogeneity clause; and least-informative state selection. The basis reference states this formally.[1] A flagship synthesis gives the account, an executable observer construction states the specification with its acceptance criteria and premise map, and the theorem library carries conditional implications whose antecedents that premise map tracks.

Time conventions. Host runtime, solver counters, native attempt counts, local event-clock readings, event precedence or logical numbering, model coordinates or lapse, modular parameters, and calibrated observer-clock readings have distinct roles. Their identification requires explicit maps. A modular parameter is dimensionless; its geometric and operational readings require the stated branch and clock premises. Consistent local clock comparisons require no preferred universal clock or identical clock rates. Shared time parameters in restricted benchmarks are declared model choices. Experienced succession and a final instantaneous cosmic endpoint do not follow from a modular-flow or finite-convergence theorem.

Two reconstruction routes

The first route is structural:

$$\begin{aligned}\text{overlap gluing and transport receipts} &\Longrightarrow \text{rigid symmetric sector category}\\ &\Longrightarrow \mathrm{Aut}_{\otimes}(\mathcal{F}).\end{aligned}$$

It reconstructs a compact group on the declared bosonic branch. It does not select the Standard Model without a realized sector witness.

[1] https://github.com/FloatingPragma/observer-patch-holography/blob/r2051/docs/BASIS.md

The second route is finite and carrier-specific:

$$\begin{aligned}&\text{twelve-port incidence and complete reversible response}\\ &\Longrightarrow \text{internal proper-carrier holonomy}\\ &\Longrightarrow \mathfrak{su}(3)\oplus\mathfrak{su}(2)\oplus\mathfrak{u}(1)\\ &\Longrightarrow \text{conditional Standard Model matter image.}\end{aligned}$$

This arrow combines three restrictions. Incidence supplies the twelve-dimensional A_5-module and its single fixed line. Premise A1 supplies the faithful compact commutator-closed response on that module. Premise A2 makes the proper carrier action inner on the same response. Incidence without the response clauses leaves many equivariant laws, while response without endogenous transport leaves the centreless compact branch available. Their intersection removes that branch.

The finite argument constructs its compact current inside the premise A1 unitary response space; an ambient continuous gauge group is absent from its premises. Write $\mathfrak{g} = D(\mathbb{R}^{12}) \subset \mathfrak{u}(H)$. Faithfulness gives $\dim \mathfrak{g} = 12$, and commutator closure makes $\mathfrak{g}$ compact reductive. The transitive action of the alternating group on five letters, A_5, gives $\dim \mathfrak{g}^{A_5} = 1$. Premise A2 implements that action internally on the same response, so it fixes the centre pointwise and $\dim Z(\mathfrak{g}) \leq 1$. A centreless compact algebra of dimension twelve would be $\mathfrak{su}(2)^4$. Its factorwise inner A_5 action has fixed dimension zero or three on each factor and cannot have a one-dimensional fixed space. The centre therefore has dimension one, while the semisimple part has dimension eleven. Compact-simple classification gives the unique split $11 = 3 + 8$, which proves

$$\mathfrak{g} \cong \mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3).$$

The carrier identity $10J = A^3 - 4A^2 - 5A + 10I$ and target-blind readback derive $R = -J$. They do not construct the current derivative. The Lie-type theorem uses the complete-response clauses of A1 and the endogenous-transport clauses of A2. A descent repair law enters only when a branch asks for settled public records. It is not an undeclared premise of the Lie-type theorem. The finite dimension arithmetic and fixed-space exclusion used by the Lie-type classification are machine checked under explicit compact-Lie and classification adapters in the accompanying Lean development [29]; no source current or full Lie-algebra isomorphism is constructed there. The registered recurrence generates the exact commutative algebra $\operatorname{span}\{I, A, A^2, A^3\}$. It contains neither twelve independent current generators nor a nonzero bracket, and its only proper recharting is the identity. This result is bounded to the registered single-recurrence packet. The declared matrix current is an exact conditional witness. The target-free diagonal phase lift has twelve independent first derivatives and an abelian bracket. Adding the registered connected adjacency tangent generates $\mathfrak{u}(12)$, with derived rank 143. This exact control rejects that direct lift and leaves non-diagonal source responses in play. The canonical scalar seam equalizers do not fill that gap. On the eleven-dimensional zero-sum port space, exact closure gives

$$\operatorname{rank}\langle [G_e, G_f] \rangle = 49, \qquad \operatorname{Lie}\langle [G_e, G_f] \rangle = \mathfrak{so}(11), \qquad \operatorname{Alg}\langle G_e \rangle = \operatorname{End}(\mathbb{R}^{11}),$$

of dimensions (49), (55), and (121), respectively. Thus the scalar strict-descent repair maps cannot be relabelled as the desired twelve-dimensional compact reversible current. This exact negative control leaves a separately derived reversible response in play and prevents the repair law from acquiring gauge content by definition.

As a counterfactual, identify the twelve ports in antipodal pairs. The inherited quotient triangulates the real projective plane, with six vertices, fifteen edges and ten triangular surface faces;

its edge graph K_6 has twenty 3-cycles. Its transitive A_5 action retains one fixed line. If a six-dimensional compact current is required to carry that action internally, a one-dimensional centre leaves the impossible semisimple dimension five, whereas a zero-dimensional centre gives $\mathfrak{su}(2)^2$ and fixed dimension zero, three or six. Thus no current satisfies all those premises. The adjacency spectrum $5, -1$, with multiplicities one and five, is exactly the antipode-even part of the icosahedral spectrum; the two $\pm\sqrt{5}$ triplets are antipode-odd. The executable named-menu table at `code/discriminator_strata/carrier_specificity.py` checks these finite facts. Antipodal identification changes topology and port count; forgetting a chosen orientation alone leaves the twelve-port sphere. This comparison supplies neither a calibrated specificity score nor a physical current attachment.

Conditional on the canonical oriented carrier, the complete target-free space $\mathrm{Hom}_{A_5}(\Lambda^2\mathbb{Q}^{12}, \mathbb{Q}^{12})$ has dimension fourteen and an exact rational Reynolds basis. It is a search space only; no bracket is selected. Substitution of the general bracket into Jacobi gives 2,640 coordinate equations in 44 target orbits and an exact coefficient-row rank of 38. A channel change over $\mathbb{Q}(\sqrt{5})$ splits that rowspace into eleven fixed-line derivation-weight products and a rank-27 residual system. The residual equations apply on every weight flat. A three-parameter derivation-only Jacobi family is exact and supplies no compactness or source selection. The compact real locus of that fourteen-parameter equivariant variety is classified, conditional on three named textbook lemmas and exactly over $\mathbb{Q}(\sqrt{5})$, into three certified bracket families, and for each family the coroot lattices, integral carrier weights, and carrier-admitted mathematical global form are computed exactly; the selected forms are adjoint with zero abelian charge lattice, so the abelian sector is undetermined at carrier level and stands exposed rather than repaired.

The source incidence admits a particularly sharp finite discriminator. Applying the declared equal-weight cyclic rule to its twenty oriented faces gives

$$B_{\mathrm{face}} = 60R_{13}.$$

This bracket is A_5-equivariant but fails Jacobi in exactly 240 of the 2640 independent coordinates, split 120 at $+1$ and 120 at -1. Under the explicitly added Euclidean metric on upper-triangular structure constants, orthogonal projection onto the complete compact locus has the unique nearest family G:

$$d_G^2 = \frac{615 - 123\sqrt{5}}{22}, \qquad d_F^2 = \frac{615 + 123\sqrt{5}}{22}, \qquad d_P^2 = 45.$$

Both F/G projections lie in their certified open compact strata. The same conclusion is exact for two endpoint edit norms on the identical 792-coordinate convention. Exact primal–dual certificates give

$$d_1(G) = 30(\sqrt{5} - 1) < \inf d_1(F) = d_1(P) = 60,$$
$$d_\infty(G) = \frac{5 - \sqrt{5}}{10} < d_\infty(F) = \frac{\sqrt{5}}{5} < d_\infty(P) = \frac{1}{2}.$$

The G optimizers lie in its open compact stratum. The displayed ℓ_1 value for F is an infimum, not an attained minimum: the zero point minimizes the ambient linear family and compact F points approach it under scaling. The exact Python replay proves the 792-coordinate primal–dual optimization; the Lean theorem checks the serialized radical values and their strict order. Thus the family verdict is robust across total, quadratic, and worst-coordinate edit scores, but only on the classified compact locus and only after the coordinate norm is supplied. The equal-weight rule and the coordinate norms are declared inputs rather than consequences of A1–A3, so the verdict is a conditional discriminator obtained without reference to the measured value.

The quadratic verdict is additionally metric-robust over the complete invariant class. The carrier splits multiplicity-free as $1+3+3'+5$, and the commutant of the port action is exactly four-dimensional, spanned by the four symmetric spectral projectors; hence every invariant carrier inner product is a positive sector-scale metric $\alpha P_1+\beta P_3+\gamma P_{3'}+\delta P_5$, and every induced Hilbert–Schmidt metric on structure constants is diagonal across the fourteen equivariant channels. The squared distances from B_{face} to the classified compact families are the exact closed forms

$$d_P^2=\frac{15-3\sqrt{5}}{\beta}+\frac{15+3\sqrt{5}}{\gamma}+\frac{15-3\sqrt{5}}{2}\,\frac{\beta}{\delta^2}+\frac{15+3\sqrt{5}}{2}\,\frac{\gamma}{\delta^2},$$

$$d_F^2=\frac{15+3\sqrt{5}}{2}\,\frac{\gamma}{\delta^2}+\frac{15+3\sqrt{5}}{\gamma}+\frac{60+12\sqrt{5}}{11\beta},\qquad d_G^2=\frac{15-3\sqrt{5}}{2}\,\frac{\beta}{\delta^2}+\frac{15-3\sqrt{5}}{\beta}+\frac{60-12\sqrt{5}}{11\gamma},$$

with α absent. Both gaps $d_P^2-d_G^2$ and $d_P^2-d_F^2$ are coefficient-positive Laurent polynomials, so P is strictly excluded for every invariant carrier metric. At $\gamma=\beta$ the gap is $d_F^2-d_G^2=3\sqrt{5}\,\beta/\delta^2+(90\sqrt{5}/11)/\beta>0$, so every sector-balanced metric selects G; and for every metric with $\beta/\delta\in[1/50,6]$ the family G stays uniquely nearest for all γ and δ. The mirror family F genuinely takes over on the $d_F^2<d_G^2$ side of the exact three-scale tie surface, with witness $(\beta,\gamma,\delta)=(8,1,1)$. Conjugating $\sqrt{5}$ and swapping $\beta\leftrightarrow\gamma$ maps d_G to d_F exactly: the F/G asymmetry is the Galois image of the sector swap. Reversing every face orientation negates B_{face} but leaves these distances unchanged, because the compared compact families are centrally symmetric; orientation therefore does not select the branch. A channel-diagonal invariant metric on the bracket space that is not carrier-induced reverses the balanced-point selection, so the carrier-induced restriction is load-bearing. The exact replay derives the closed forms from the pinned tensors and replays six sample metrics through raw normal equations; the Lean companion proves the exclusion, balanced, box, and witness statements as quantified real theorems over the closed forms. The nearest-point law and the carrier-induced class are declared discriminator choices. The committed dual-measure layer adds a measure-conditional point: on the pinned incidence table, equal normalized face weights and the declared barycentric one-third share give port-dual weight exactly $1/12$ at every port, and under the declared diagonal measure-to-metric rule that uniform measure induces the scalar carrier inner product at the balanced cone point $(1/12,1/12,1/12,1/12)$, where G is uniquely nearest with $d_G^2=(3690-738\sqrt{5})/11$, $d_F^2=(3690+738\sqrt{5})/11$, and $d_P^2=540$, twelve times the reference distances by the exact scaling identity $c\,d^2(c,c,c)=d^2(1,1,1)$. The weighting rules and the measure-to-metric identification are declared, so this upgrades the metric-conditional selection to a measure-conditional selection whose measure is pinned by a committed theorem; it is not source selection of the bracket or a physical gauge structure. One composed machine-checked surface (`Lean/Screen/SMStructureAdequacySurface.lean`) bundles the declared weighting, metric, classification-input, and exterior-selection premises and re-exports the forced Lie type, the balanced-point selection, the one-generation exterior selection with anomaly cancellation, and the conditional global-form menu as one package, with the same-source loop-to-kernel identity named as the leading undischarged clause of the physical quotient. A companion composition (`Lean/Screen/SMStructureComposition.lean`) closes the side-by-side gap in that package: one theorem on the one premise bundle proves that the forced type admits a unique type-matched nearest family at the bundle's balanced point and that the fifteen-state count, the anomaly forms, and the $\mathbb{Z}_6$ stabilizer are carried by the bundled selection mask itself. Its capstone is a coarse two-mask statement: among the three classified family labels and the declared 2^{10} exterior table, the admissible pairs are the G label with either of two conjugate masks. Conjugation is proved on the masks only. This does not count points in the continuous G-bracket family and does not classify complete physical Standard Model structures.

The descent bookkeeping itself is exact. On the finite weight lattice of hypercharge, triality, and duality labels in the $q = 6Y$ normalization, a label is fixed by every element of the diagonal $\mathbb{Z}_6$ kernel exactly when $2t + 3d + q \equiv 0 \pmod 6$; the kernel acts on any label by multiples of this center character. Every row of the matter table satisfies the congruence, including both carrier blocks and the scalar, a fractionally charged singlet is blocked by the generator, and exactly one sixth of the thirty-six lattice labels descend, with every character value carried by exactly six labels. The machine-checked receipt composes onto the selected matter mask: every selected component descends through the full kernel (`Lean/Screen/GlobalFormCharacterDescent.lean`). This is the character-lattice half of the global-form question only. The complete character category, the instanton normalization, the laboratory flux attachment, and the selection of one physical quotient among the four admissible ones are additional assumptions, and no physical global form is derived from the congruence.

A three-family bundle types the band-selection rules of the family row as antecedent fields whose triplication arithmetic threads the same selection mask, the chirality row carries one threaded grading-plus-obstruction theorem, and a kernel-checked dictionary identifies the dimension-six baryon census table with the selected generation column by column. The construction supplies no physical-family or Spin/chirality attachment and no source action.

Over the supplied certified brackets, ad-invariance fixes the kinetic shape block-diagonally with exactly zero cross terms. On carrier-projector forms the restriction is stronger and exact: families F and G reduce the three-dimensional carrier-invariant space to two dimensions,

$$F:\ w_{3+} = \sqrt{5}\, w_5, \qquad G:\ w_{3-} = \sqrt{5}\, w_5,$$

with w_{3-} free for F and w_{3+} free for G; family P is the exact two-to-two control. Requiring one form to be invariant for both F and G leaves the ray $(w_{3+}, w_{3-}, w_5) = t(\sqrt{5}, \sqrt{5}, 1)$, but simultaneous mirror invariance is an extra premise, not source selection. The stored minus-one-quarter Killing matrices are exact representatives, not selected couplings. The two-factor history theorem constructs a Gibbs kernel from $aq_1 + bq_2$, proves exact normalizer and kernel factorization, and recovers the additive factor action. If both costs vary, equality of two constructed kernels identifies λa and λb; with nonzero multipliers only common scaling survives, so a/b is not multiplier gauge. Exact instances cover both P factors on 3^6 steps and the full color-bearing F family on 3^{11} steps. These are symbolic finite theorems, not large enumerations. They neither identify an independent source kernel nor select a, b, or their ratio. No theorem supplies an enumeration of noncompact components, a source repair/selection law, action selection, same-current holonomy, relative factor coefficients, or coupling scales [28, 29]. Generation, scalar, global-form, and extra-sector statements carry their own premises.

An assembled structural action

The three committed sector functionals sum as one real function of one field tuple on one typed premise bundle (`Lean/Screen/AssembledActionComposition.lean`). The bundle extends the electroweak-breaking premise data by two declared positive gauge weight parameters in the committed two-parameter G-invariant normal form, and the assembled action is the explicit sum of a gauge kinetic term (one declared magnitude per invariant carrier sector, squared against its weight under a declared diagonal pairing on the committed weight space), the committed quartic potential $\lambda(|h|^2 - v^2)^2$, the committed gauge mass form at the chosen minimum, and the realified committed Yukawa line with coefficient y. Three exact restriction theorems recover each committed sector functional on its field slice over the common base tuple. The assembled gauge coefficients satisfy the committed G ad-invariance constraint row and collapse to the committed one-ray mirror form

under the extra declared F row. The whole sum is exactly invariant under the global abelian phase action at unit scalars with product one, in particular at the committed integer charge balance of the bundled selection mask, with the scalar slot at plus or minus the committed scalar row charge. The imaginary part of the committed complex Yukawa line is recovered as an assembled evaluation, the base tuple evaluates to zero, and the bundle is inhabited with every declared parameter equal to one. This is a finite structural action on the committed coarse grammar: every coefficient is a declared bundle parameter, the covariant kinetic content enters only through the committed constant-direction coefficient at the minimum, and the construction supplies no source-produced spacetime action, measured coefficients, three-physical-family attachment, global-form selection, physical-chirality attachment, or Spin attachment.

The gauge sector at the committed invariant-form level

The gauge kinetic sector of the assembled action extends from the declared diagonal pairing to the committed invariant-form level (`Lean/Screen/GaugeSectorBracketCompletion.lean`). The typed gauge-field slot carries one real coordinate vector per weighted invariant carrier sector of the committed multiplicity-free split $3^+ + 3^- + 5$, and the kinetic functional is the committed weight vector against the sector coordinate squared norms. The declared diagonal pairing is exactly the axis restriction of this functional under the committed basis, the functional factors exactly through the committed pairing at the sector magnitudes, and agreement on the axis slice pins the weights, so the assembled action embeds exactly in the completed one and the completed action factors exactly through it. The invariance is carried at the exact infinitesimal grade: every sector-internal skew-adjoint generator leaves every weight vector invariant, the committed G and F ad-invariance constraint rows are the infinitesimal-invariance rows of two exhibited cross-sector generators at declared coefficient ratio $(1, -\sqrt{5})$, and simultaneous invariance under both leaves exactly the committed one-ray mirror family. The restriction, abelian-invariance, committed-charge, and base receipts persist, the parameter census is unchanged at exactly the two declared normal-form parameters, one exhibited off-axis carrier point evaluates outside the axis slice, and the control weight vector $(1, 1, 1)$ breaks the invariance at an exhibited carrier pair with exact value $1 - \sqrt{5}$. No gauge group action on the carrier is committed or claimed; the generator ratio realizes the committed constraint rows, its derivation from the pinned bracket tensors is certificate content, and no source-produced spacetime action is supplied.

A fermion sector on the assembled action

The assembled action carries one typed fermion sector as a fourth summand, with the committed sum unchanged (`Lean/Screen/FermionSectorAssembly.lean`). The fermion carrier is one complex amplitude per row of the committed ten-component table, and the bilinear row-pair form $\overline{\psi_i}\,\psi_j$ transforms under the abelian phase action through the committed integer charge column $q = 6Y$. On the rows of the bundled selection mask the form is invariant at every unit phase exactly when the committed row charges balance, and the committed charge column separates those rows, so the invariant pairs are exactly the diagonal ones: the charge column forces the kinetic-shaped pairing to be diagonal. The extended action is the committed assembled sum plus the masked diagonal fermion sector with one declared coefficient per row. It restricts exactly to the committed assembled action on the zero fermion slot and at zero fermion coefficients, exactly to the fermion sector on the fermion slice, and the three committed sector restrictions persist. A typed embedding of the fermion amplitudes at the shared doublet and singlet rows into the committed matter slots evaluates the extended action to the committed mass bilinear with coefficient $y\,v$, and the embedding intertwines

the fermion phase action with the committed matter-slot phase action at the committed row charges. The extended sum is exactly invariant under the committed-charge abelian action extended by the fermion charge phase, the parameter census is the committed list plus the fermion coefficients, the bundle is inhabited over the committed inhabitant, and an exhibited charge-imbalanced pair of mask rows breaks the invariance receipt at an exhibited unit phase. The fermion coefficients are declared stand-in weights: the covariant kinetic term, the source-produced spacetime action, and the Spin and physical-chirality attachments are not supplied, and no measured coupling, three-family structure, or Yukawa matrix appears.

Contents

1 Recovered Core: Foundations and Structural Branches **9**
- 1.1 Overview . . . 9
- 1.2 Model and Premises . . . 10
- 1.3 Information-Theoretic Tools . . . 16

2 Finite Screen Spectrum Theorem Package **29**
- 2.1 Overlap Consistency and Gluing . . . 60
- 2.2 Gauge Reconstruction and Standard Model Structure . . . 65
- 2.3 Gauge-chain theorem and falsification boundary . . . 98

A Technical Details for the Specialist Derivations **99**
- A.1 Product-Group Structural Corollaries . . . 99
- A.2 Finite-Quotient Baryogenesis Source Theorem and Gauge-Deck No-Go . . . 100
- A.3 Controlled Worldsheet Effective Description . . . 102
- A.4 Support-Visible Yang–Mills Gap from Repair Dynamics . . . 102
- A.5 A local gauge-covariant action on the declared matter grammar . . . 116
- A.6 A consistent local Cartan–scalar reduction and current response . . . 118
- A.7 A continuous charged-current and separated-detector bound . . . 120
- A.8 Fermionic hypercharge transport on prepared source addresses . . . 121
- A.9 A flux-dressed fermion state satisfying abelian quantum Gauss . . . 123

B A coupled fermion current in a declared hybrid gauge action **125**
- B.1 Spin transport and the information needed to select exchange statistics . . . 128
- B.2 Conditional exclusion from a stable generator and a scalar deformation law . . . 131
- B.3 What finite source operations must supply for exclusion . . . 134

1 Recovered Core: Foundations and Structural Branches

Observer-Patch Holography is a reconstruction program for fundamental physics. On the declared screen-first global-support branch, physical data are charted by a horizon screen S^2, and the effective world is encoded by the mutual consistency of overlapping patch descriptions. This support screen is distinct from a local carrier boundary and a federation overlap nerve. On the producer branch, S^2 enters only after the carrier-to-support map satisfies the spherical incidence, mesh, cross-ratio, normalization, and refinement receipts. The legacy finite reference fixture does not supply that map because it has no composable seam triangle and its higher-overlap condition is vacuous. The source-derived producer instead supplies a twelve-chart, thirty-seam, twenty-face incidence nerve, nonvacuous seam-triangle cocycles, confluent phase repair, and a controlled oriented-S^2 support limit on one declared tower. Event, modular, scale, and laboratory attachments require separate premises. This recovered-core section develops fixed-cutoff overlap repair, transportable sector reconstruction, the axiom-forced local Standard Model gauge Lie algebra, and the conditional Standard Model matter image.

1.1 Overview

Overlap consistency first turns local patch states into a constrained gluing problem. The fixed-cutoff branch gives collar decompositions, edge centers, and finite repair dynamics on the declared physical quotient. The transportable edge-sector branch classifies fixed-stage zero-obstruction sectors and, on a cofinal tail carrying the compact-gauge refinement receipt, reconstructs a compact internal gauge group in the bosonic branch.

The finite-current route is separate. Complete reversible port response in Axiom 1 and endogenous overlap transport in Axiom 2 force the compact Lie type $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$. Icosahedral incidence and inverse-port pairing determine J, with $10J = A^3 - 4A^2 - 5A + 10I$. A target-blind impulse/readback protocol produces $R = -J$ and fixes the relative band signs up to common reversal. It does not reconstruct the current derivative. The charged-double-triplet matrices are an exact conditional realization. Given the declared fermionic Spin category, the matter certificate derives $P_{\mathrm{even}} - P_{\mathrm{vac}}$ and $P_{\mathrm{odd}} - P_{\mathrm{top}}$ as the unique charge-conjugate pair of rank-15 chiral projectors. Anomaly balance and tensor descent give exact hypercharge, $N_c = 3$, the common $\mathbb{Z}_6$ kernel, and the maximal faithful matter image from the declared current and matter premises. The physical current realization, matter action, and global quotient require same-source response, character-completeness, and loop-to-kernel receipts. Under separate complete-band and cost-order premises, an exact theorem selects the rank-three screen response. Its tensor product with the declared fifteen-state table is a conditional rank-45 candidate; chirality and the diagonal $\mathbb{Z}_6$ action are properties of that table. A separate finite local-domain receipt checks the declared operator $D_\sigma \otimes I_{45}$ and conditional gap inheritance. The source does not select this matter action, and no certified bridge carries the twelve-port Spin packet to the 8,662-node operator domain. Laboratory identification of the current and flux sectors, equality of the two reconstruction routes, four-dimensional instanton normalization and theta periodicity, monopole dynamics, scalar attachment and multiplicity, quantum field realization, and physical matter-pole and continuum identification of the finite family carrier are not supplied.

1.2 Model and Premises

1.2.1 Observers and access model

The support-local algebra-state-record reduct of an observer patch is $O_{\mathrm{red}} = (P_O, \mathcal{A}(P_O), \rho_O, R_O)$, where:

- $P_O \subset S^2$ is a connected screen patch (the observer's access region).
- $\mathcal{A}(P_O)$ is the von Neumann algebra associated to P_O.
- ρ_O is the local state, obtained by restricting the global state to $\mathcal{A}(P_O)$.
- R_O is a finite record algebra generated by stable internal record projectors within P_O.

An operational observer additionally carries overlap interface algebras and restriction maps, allowed update and repair instruments, and checkpoint data used for continuation. Observers are internal self-reading patterns in the global state. A patch and its compatible marginal supply a geometric or algebra-state chart. An operational observer requires the interface, record, repair, and checkpoint conditions.

1.2.2 Screen, patches, and algebra net

We work in a single static patch with a horizon screen S^2. Each connected subregion $P \subset S^2$ is assigned a von Neumann algebra $\mathcal{A}(P)$. The net satisfies isotony:

$$P \subset Q \implies \mathcal{A}(P) \subset \mathcal{A}(Q).$$

A global state ω is a positive linear functional on the inductive-limit algebra. Overlap consistency is imposed algebraically: for overlaps $P_1 \cap P_2$, ω restricted to $\mathcal{A}(P_1 \cap P_2)$ is the same from either side.

1.2.3 The premises A1–A3

This paper states the premises A1–A3 because both specialist derivations share their observer-overlap foundation. The flagship synthesis [25] gives the account of the theory, and the executable observer paper [26] records which of these premises the construction supplies. Branch-local hypotheses are written at the theorem that consumes them. Transportability and the fixed-cutoff bosonic sector category are theorem-level results. Refinement/fiber descent and the cofinal compact-gauge witness require the explicit compact-gauge refinement receipt. The finite local Standard Model gauge Lie-algebra implication uses the complete-response and endogenous-transport clauses with compact classification. Its matrix current and matter action are conditional realizations. The $\mathbb{Z}_6$ kernel is exact on the declared matter table, while the physical global quotient requires a source-derived complete character category and loop-to-kernel identity. Laboratory identification of the current and flux sectors, four-dimensional instanton normalization and theta periodicity, monopole dynamics, and scalar multiplicity require separate premises.

The screen premise A1 below defines the screen-first support branch. The producer branch reaches the same premise only after its carrier-to-support receipts establish the global support screen.

A1 (oriented observer-patch federation and spherical support). There exists an observer patch net on an oriented spherical screen: at every finite resolution each local carrier has twelve primitive boundary ports forming the vertices of an oriented triangular boundary with 30 edges and

20 faces, combinatorially the boundary of an icosahedron, and carriers join through typed seams and coherent triple overlaps, refine to an oriented spherical support, and expose local state, readback, records, repair moves, and checkpoints. Their complete infinitesimal port response is represented faithfully on a finite-dimensional unitary response space and is closed under the commutators generated by ordered response composition. Formally, for every regulator r in a directed system there is a typed object $\mathfrak{N}_r = (\mathcal{P}_r, \mathcal{A}_r, \mathcal{R}_r, \mathcal{I}_r, \mathcal{U}_r, \mathcal{C}_r, N_r, S_r, b_r)$: a finite patch/overlap category with an isotone local algebra net and central record algebras; a federation of finite carriers, each with twelve primitive pairwise-orthogonal central port projections summing to one and a boundary packet $K = (P, E, F, o)$ with $|P| = 12$, $|E| = 30$, $|F| = 20$, degree-five ports, five-cycle links, and coherently oriented two-face edges, isomorphic to the icosahedral boundary complex with no preferred labels; seam algebras with unital restrictions and coherent triple-overlap cocycles forming the nerve N_r; a finite edge-midpoint refinement support S_r realized orientation-preservingly in S^2 with mesh tending to zero; and a source-bound degree-one bridge $b_r : N_r \to S_r$, all commuting with refinement. Each carrier also has $V_{r,i} = \mathbb{R}[P_{r,i}]$, a finite-dimensional complex Hilbert space $H_{r,i}$, and an injective real-linear map

$$D_{r,i} : V_{r,i} \longrightarrow \mathfrak{u}(H_{r,i}).$$

The image $\mathfrak{g}_{r,i} = D_{r,i}(V_{r,i})$ is closed under commutators and complete for the declared quotient-visible infinitesimal port response. The primitive port probes span $V_{r,i}$, no additional public response direction is omitted, and $-\operatorname{Tr}(D_{r,i}(v)D_{r,i}(w))$ is positive definite. The response construction is natural under admissible presentation equivalence and refinement. A1 does not select a particular response table, inverse-port law, Lie type, global group, particle interpretation, or metric content.

A2 (observer agreement). Observers operating on the screen agree on the meaning of the data they jointly interpret. Formally, the interpretation map $\mathcal{J}_r : \mathsf{Data}_r \to \mathsf{Meaning}_r$ from observer-accessible data to operational meanings is natural with respect to every visible overlap restriction, recharting, seam translation, higher-overlap map, federation map, and refinement map: every declared data-access diagram for accepted public data commutes after interpretation, and $\mathcal{J}_r \circ c_{s\to r} = C_{s\to r} \circ \mathcal{J}_s$ across resolutions. A2 constrains accepted shared data only, and the domain it quantifies over is supplied by the A1 interfaces rather than by A2 itself.

Accepted reversible overlap transports form a groupoid $\mathcal{O}_r$. For a complete carrier chart o, its closed paths $\mathrm{Hol}_r(o)$ map onto the orientation-preserving incidence automorphisms $\mathrm{Aut}^+(K_{r,i})$. For every proper carrier automorphism a, A2 supplies a closed path γ_a and one projective implementer $[U_a] \in \mathrm{PU}(H_{r,i})$ satisfying

$$\mathrm{Ad}_{[U_a]}\, D_{r,i}(v) = D_{r,i}(a \cdot v).$$

Writing

$$G^0_{D,r,i} = \langle \exp(tD_{r,i}(v)) : t \in \mathbb{R},\ v \in V_{r,i} \rangle^0,$$

the implementer is endogenous:

$$[U_a] = [g_a c_a], \qquad g_a \in G^0_{D,r,i}, \quad c_a \in C_{U(H_{r,i})}(\mathfrak{g}_{r,i}).$$

Closed paths compose projectively and the construction commutes with refinement. A unitary lift need not be unique. Scalar phases and centralizer actions are permitted because they act trivially on the response algebra. An independent spectator carrying a matching projective action cannot replace the response-generated factor or contribute a public response direction. The closed path, port action, and implementer must arise from the same accepted source history in an executable realization.

The two response clauses support an abstract target-free compact Lie-type theorem. They do not certify that an executable producer emitted a particular current fixture. Such a certification must reconstruct the port generators, commutator, and projective implementers from ordered source histories without a named current model upstream. A2 does not imply global state extension, repair termination, confluence, unique normal forms, record durability, Byzantine safety, dissemination, or commutation of state optimization with refinement; raw mismatch and repair precede acceptance.

A3 (conditional maximum randomness). Everything that observer agreement leaves unconstrained is maximally random. Formally, at finite regulator r a state is a compatible family of local normalized states on the accessible algebra net; $\mathcal{K}_r$ is the nonempty convex set of such families satisfying the finite observer-visible constraints supplied by A1 and A2 through an A1-generated constraint grammar with a factorization theorem; and, relative to an exact reference family τ_r, an A1-generated observer cover $\mathcal{G}_r$, and strictly positive exact weights $w_{r,P}$ constructed from quotient-visible A1 data by a rule natural under admissible presentation equivalence, the realized state is the information projection. The cover restriction map is injective on $\mathcal{K}_r$; equivalently, at smooth points no nonzero feasible tangent is invisible to the whole cover. The realized state is

$$\rho_r = \arg\min_{\rho \in \mathcal{K}_r} \sum_{P \in \mathcal{G}_r} w_{r,P}\, D(\rho_{r,P} \| \tau_{r,P}).$$

The same projection rule applies separately to each declared optimizer object type (ontic state, inference state, or transition distribution), and statements about one type do not transfer to another. This is equivalent to weighted local entropy maximization when every local reference density is identity-proportional in its declared trace. "Random" means least informative relative to the declared reference and the complete agreement constraint set. A3 selects one state inside one fixed feasible space. It does not compare unrelated state spaces or field contents, does not imply optimizer compatibility across refinement, and does not imply recovery, alignment, or mixing; each such statement carries its own theorem, interface, or countermodel.

The complete formal statement of A1–A3, including its constraint map and non-implications, is given in the premise reference.

The closure hypothesis. The premises A1–A3 describe the observer screen. The closure hypothesis states that the Universe is a timeless, self-referential structure whose observers recover the specification of their own world from their records and realize it, so that the description and the described system are one object. Every quantity that has both a construction-side reading and a readback-side reading must take the same value once a typed bridge proves that the two readings denote one invariant. The equality is forced by self-identity. Constructing that bridge and the return map is part of the physics. Existence, uniqueness, and stability are separate determinacy tests on the resulting equation. The name given to such an equation carries no mathematical weight: after the typed identification, unequal readings would describe two systems rather than the single self-referential universe.

The closed object is the complete inhabited history, whose internal events and responses to interventions are its proposed intrinsic computation. This reading requires no external computer or playback time. It is distinct from fixedness of a repair state, a public record, or a probability law, and does not itself construct the history or prove a final cosmic endpoint.

The local screen-grain proposal

$$P = \varphi + \sqrt{\pi}/A_T(P),$$

with $A_T(P)$ the inverse coupling returned by the declared source map, has one interval-certified fixed point per declared map on its analytic domain. Interpreting that root as the laboratory coupling additionally requires a target-independent choice of the physical map, a typed bridge identifying the two sides as readings of one quantity, and same-scheme hadronic spectral transport to the Thomson endpoint. The displayed map's certified fixed point sits at $P = 1.6309720959$ and returns $A_T = 136.9948352$; the certified gauge-width variant's fixed point sits at $P = 1.6309682414$ and returns 137.0356601. Through the same outer equation the measured 2022 CODATA coupling $\alpha^{-1} = 137.035999177(21)$ reads back the grain value $P = 1.6309682094$. The comparisons stand at 3.0×10^{-4} relative for the displayed map and at 2.5×10^{-6} for the gauge-width variant, as diagnostics of the declared source maps. The latter absolute residual is 3.3904×10^{-4}, compared with experimental standard uncertainty 2.1×10^{-8}; this is not agreement at measured precision. The gauge width supplies 99.2 percent of the gap between the bare root and the measured value; the residual 3.3904×10^{-4} is addressed to same-scheme hadronic transport to the Thomson endpoint.

The direct global proposal is

$$N = \log M_0(\mathfrak{U}_N).$$

The bounded completion class defined by base agreement, positivity, and the carrier bound does not select a unique cosmic value. Universal all-rung membership in a complete A1–A3 capacity-source contract and the corresponding executable-to-Lean bridge are absent. The incomplete antecedent determines no direct N, and no theorem extends the bounded-class conclusion to the complete source class. A positive result must complete the source antecedent and prove one physical zero, either within the A1–A3 source contract or through a separately named stronger source law. A separate common-load branch starts from

$$N_0 = \pi \exp\!\left(\frac{6\pi}{P\,\alpha_U(P)}\right).$$

On the finite collar branch, the declared total reserve expectation $P/4$ and six-class equidistribution give presence probability $P/24$ for each declared class. If one class is physically selected as the blocked event, its scalar-weighted presence receipt is discharged, and that normalized collar-survival factor is proved to act on the global capacity, the corresponding candidate is

$$N_{\mathrm{pres}} = N_0\left(1 - \frac{P}{24}\right), \qquad \ln\frac{N_{\mathrm{pres}}}{\pi} = \frac{6\pi}{P\,\alpha_U(P)} + \ln\left(1 - \frac{P}{24}\right).$$

The exponential alternative $N_{\mathrm{Pois}} = N_0 e^{-P/24}$ requires a separate mean-count or continuum carrier. Exact neutral, one-class, and six-class-total actions share the declared local datum, obey positive composition and finite cut-count regrouping, and disagree globally. The finite source therefore selects no action or blocked-event semantics. The source class determines no named-law value, and its horizon branch has no capacity object. A stronger source-derived action would require the screen/electroweak bridge, physical common-load bridge, physical $\mathbb{Z}_6$ seam action, scalar-weighted receipt, and horizon-record identification. On the source-forward numerical branch,

$$N_0 = 3.5321 \times 10^{122},$$

so the finite-presence candidate is 3.2921×10^{122} and the exponential candidate is 3.3001×10^{122}. The full weighted Planck base-ΛCDM chain gives the comparison coordinate 3.3129×10^{122}. The respective residuals are -0.63 percent and -0.39 percent. Both comparisons were exposed before the branch choice and carry no predictive weight.

Collar recoverability, generalized-entropy structure, and quantum focusing lie outside A1–A3: they are classified gravity interfaces with their own theorems, countermodels, and required inputs,

consumed field by field where the Einstein composition needs them. Sector content is not selected by A1–A3: the conditional rank-45 tensor candidate does not establish physical families or a local Spin carrier, while scalar multiplicity and extra-sector completeness lack physical attachments. Each theorem below lists the declared completions it consumes.

1.2.4 Closure values and theorem-local technical data

Transportability, the fixed-cutoff bosonic category, regulator bookkeeping, quasi-local propagation, endpoint-Lipschitz control, and collar decomposition are cited from their preceding theorems. Refinement/fiber descent and the cofinal compact-gauge witness additionally require the compact-gauge refinement receipt.

Local information projection at the regulator scale. At the regulator scale ℓ_{UV}, consider the Axiom-3 subbranch carrying a global-extension interface and identity-proportional local references for which the state-determining weighted local objective is represented by one finite density matrix ω. The selected state maximizes the corresponding weighted local entropy subject to:

1. A finite set $\{O_a\}$ of gauge-invariant local operator densities, each supported on a ball of radius $\leq r_0 = O(\ell_{\mathrm{UV}})$.
2. Constraint equations $\langle \sum_x O_a(x) \rangle = C_a$, one homogeneous global sum per density label a, where x runs over the cells of the UV lattice. The constraints are the N_{con} global sums, not one independent constraint per cell, so the number of independent constraints and of Lagrange multipliers is N_{con} (plus the optional global constraints below) on every lattice, independent of the cell count.
3. Optionally, a finite number of global constraints (total energy, charge).

This is the finite-matrix subbranch needed to derive the local Gibbs form of Lemma 2.6 from Axiom 3 with a cutoff-independent multiplier count. The global extension and the objective representation are explicit branch premises.

Clarification (conditional maximum randomness and thermal equilibrium). Axiom 3 is local state selection relative to an exact reference and a complete visible constraint set. It does not state that the universe is in thermal equilibrium. Non-equilibrium physics is modeled in the enlarged *local-multiplier* family obtained by replacing the homogeneous multipliers λ_a with slowly varying fields $\lambda_a(x)$, equivalently by imposing per-cell constraints $\langle O_a(x) \rangle = c_a(x)$. That enlarged family has dimension proportional to the number of UV cells, so it is a function-space approximation regime with gradient error terms. The refinement-stable branch and the induced maps $R_{\ell \to L}$ below concern the homogeneous N_{con}-parameter branch under the separate optimizer-pushforward interface. Controlled violations of exact Markov additivity appear under the explicit mixing hypotheses stated later in Section 2.3. Equilibrium is an approximation regime with explicit error terms.

Rotationally invariant constraint branch. Constraint sets are SO(3)-invariant on S^2.

Gauge as overlap redundancy at fixed cutoff. Overlap identifications are not unique; the freedom that leaves overlap observables invariant forms a local groupoid.

Central-defect subbranch. Assume the overlap centers are identified with one fixed abelian unitary coefficient group Z_Σ on which overlap transport acts trivially. Choose unitary overlap implementers with $U_{ji} = U_{ij}^{-1}$. On triple overlaps, when the failure of strict coherence is central, define

$$\Omega_{ijk} := U_{ij} U_{jk} U_{ki} = z_{ijk}, \qquad z_{ijk} \in Z_\Sigma,$$

equivalently $U_{ij}U_{jk} = z_{ijk}U_{ik}$. The defect is data of the implementer lift; writing only $\mathrm{Ad}(z_{ijk})$ would erase it because conjugation by a central element is the identity.

Collar double-scaling hypothesis. There is one refinement family with $\delta \to 0$, $\ell_{\mathrm{UV}} \to 0$, and uniform finite-range and strong conditional Gibbs mixing constants such that

$$I(A_\delta : D_\delta \mid B_\delta)_\omega \le c|\partial C|_{\mathrm{UV}} e^{-\delta/\xi}, \qquad \frac{\delta}{\xi} - \log|\partial C|_{\mathrm{UV}} \longrightarrow +\infty.$$

The second condition controls the complete boundary-prefactored envelope. The weaker ratio $\delta/\ell_{\mathrm{UV}} \to \infty$ does not imply it. Section 2.3 gives the finite-range theorem, explicit constants, a sufficient logarithmic schedule, and the finite-receipt fields. On the exact central-interface branch the CMI is zero without this mixing premise.

Derived quasi-local propagation and modular-locality control. At scale ℓ_{UV}, the automorphism group generated by the MaxEnt generator of Lemma 2.6, or more generally by any branch generator lying in the same bounded-support algebraic closure, has a finite Lieb–Robinson velocity v_{LR}: for local operators A, B supported on regions separated by distance d,

$$\|[A(t), B]\| \le c\|A\|\|B\| \min(|R_A|, |R_B|) e^{-(d - v_{\mathrm{LR}}|t|)/\xi}$$

for $d > v_{\mathrm{LR}}|t|$, where $\xi = O(\ell_{\mathrm{UV}})$.

This is the branch-internal control statement that turns the quasi-local structure of the local-Gibbs branch into explicit support control for time-evolved operators. It is part of the third-axiom branch stated at regulator scale.

Approximate modular covariance on the fixed-cutoff realized branch. Under the fixed-cutoff realized presentation, Lemma 2.6 (local Gibbs), the derived quasi-local propagation bound above, and the collar double-scaling plus mixing hypotheses of Section 2.3, the modular flow $\sigma_t^{\omega,C}$ maps $\mathcal{A}(R)$ into a slightly thickened region algebra:

$$\sigma_t^{\omega,C}(\mathcal{A}(R)) \subseteq \mathcal{A}(R^{+v_{\mathrm{mod}}|t|}) \quad \text{up to error } \eta(d - v_{\mathrm{mod}}|t|),$$

where R^{+s} denotes the s-neighborhood thickening, v_{mod} is a "modular propagation velocity" controlled by v_{LR} and local norm bounds, and d is the distance from R to ∂C.

Heuristic motivation. On the reduced-modular-locality branch of Lemma 4.1a-b, modular additivity localizes the nonadditive defect of $K_C = -\log \rho_C$ to the collar. The associated propagation receipt then bounds support spreading under $e^{iK_C t}$. In the rate-controlled collar limit of Section 2.3, the thickening vanishes in macroscopic units. This remains heuristic support for the tangent-limit package; a Lieb–Robinson bound for the global Gibbs generator alone does not prove locality of the reduced modular Hamiltonian.

Continuum-limit heuristic. Define the induced region flow

$$f_t^C(R) := \lim_{\ell_{\mathrm{UV}} \to 0} R^{+v_{\mathrm{mod}}|t|}.$$

Then $\sigma_t^{\omega,C}(\mathcal{A}(R)) = \mathcal{A}(f_t^C(R))$ becomes exact in the continuum limit, with error controlled by

$$\eta(\delta) \lesssim 2\sqrt{c|\partial C|_{\mathrm{UV}}}\, e^{-\delta/(2\xi)}$$

when CMI is measured in nats, with the full rate condition imposed on the prefactor.

The discussion above is heuristic support for the tangent-limit package used in Theorem 4.2.

Refinement-stable information-projection branch. Choose a family of coarse-graining channels $\Phi_{\ell\to L}$ between UV scale ℓ and IR scale L satisfying the separately declared optimizer-pushforward and refinement-tail interface. Then on the realized branch one may write

$$\Phi_{\ell\to L}\big(\omega_\ell(\lambda)\big) = \omega_L\big(R_{\ell\to L}(\lambda)\big),$$

for an induced map $R_{\ell\to L}$ on the common finite-dimensional homogeneous multiplier family. The existence of $R_{\ell\to L}$ as a self-map of that same family is the load-bearing cross-scale premise. Reusing the density labels a does not establish it. Coarse-graining a finite-range Gibbs family generically generates interactions outside the retained density list. The closure defect, the moment-matching I-projection realizing $R_{\ell\to L}$, and its trace-norm residual bound are constructed in Definition 2.6a and Lemma 2.6b. Exact vanishing of the defect along the realized branch remains part of the declared optimizer-pushforward interface. Under that interface, the regulator states lie in a shared multiplier family, and the realized branch is a persistent trajectory or invariant subset under the induced refinement maps. This state-side persistence statement does not upgrade fixed-cutoff edge labels to a transportable refinement-persistent sector colimit.

Fixed-cutoff realized presentation. At a UV scale ℓ_{UV}, local patch algebras are type-I with finite-dimensional Hilbert spaces, and gauge-as-gluing is realized as a boundary action on a lifted finite-dimensional presentation. Section 2.3 proves the resulting fixed-cutoff collar package directly from overlap consistency on this realized presentation, while distinguishing the lifted fixed-point algebra from the sector-preserving collar algebra used in the edge-center (EC) theorem.

External mathematical inputs include SSA and recovery theorems (Petz 1986, 1988; Fawzi and Renner 2015). The categorical gauge route also uses Doplicher-Roberts reconstruction once localized zero-obstruction sectors are assembled in the small-region limit. Full citations appear in the References.

1.2.5 Notation

- ρ_C: reduced state on cap C.
- $K_C := -\log \rho_C^{\omega}$: modular Hamiltonian of the reference state.
- B_C: geometric generator of the cap-preserving conformal dilation.
- $S_{\mathrm{gen}}(C)$: generalized entropy on a cap.
- ℓ_{UV}: UV length scale of the refined screen net.
- δ: collar width around a cap boundary.

1.3 Information-Theoretic Tools

1.3.1 Strong subadditivity and Markov states

For any tripartite state ρ_{ABC},

$$I(A:C\mid B) := S(AB) + S(BC) - S(B) - S(ABC) \geq 0.$$

Exact Markov states satisfy $I(A:C\mid B) = 0$ and admit a recovery map:

$$\rho_{ABC} = (\mathrm{id}_A \otimes \mathcal{R}_{B\to BC})(\rho_{AB}).$$

1.3.2 Approximate recovery

If $I(A:C \mid B) \le \varepsilon$ in nats, there exists a CPTP recovery map $\mathcal{R}$ with

$$\|\rho_{ABC} - (\mathrm{id}_A \otimes \mathcal{R})(\rho_{AB})\|_1 \le 2\sqrt{1 - e^{-\varepsilon}} \le 2\sqrt{\varepsilon}.$$

1.3.3 Collar refinement and sufficient mechanisms

Fix a cap $C \subset S^2$ with boundary circle. For collar width δ define

$$B_\delta := \{x \in S^2 : \mathrm{dist}(x, \partial C) \le \delta\}, \qquad A_\delta := C \setminus B_\delta, \qquad D_\delta := (S^2 \setminus C) \setminus B_\delta.$$

Then $S^2 = A_\delta \cup B_\delta \cup D_\delta$ with A_δ and D_δ interacting only through B_δ. The collar double-scaling hypothesis is the requirement that $I(A_\delta : D_\delta \mid B_\delta) \to 0$ in the refinement limit. At fixed regulator scale, the same finite-dimensional realized presentation supplies the patch-net and overlap-gluing data used below. We therefore isolate that realized presentation first, then separate the exact-Markov and quantitative routes.

Finite quotient ensemble theorem. Fix a finite regulator r. The physical presentation space is Σ_r, the presentation redundancy groupoid is Γ_r, and the finite physical quotient is

$$Q_r = \Sigma_r / \Gamma_r, \qquad \pi_r : \Sigma_r \to Q_r.$$

The quotient removes nonphysical presentation data: gauge representatives, port relabelings, mesh labels, shard or worker identifiers, queue order, repair schedule identifiers, retry counters, timestamps unless declared semantic, hidden carrier coordinates, and inert ancillary labels. If only settled configurations carry probability, the probability space is the normal-form subset

$$N_r = n_r(Q_r).$$

The map n_r is a normal-form map, not a probability law. Any physical branch inherits this firewall: it must declare the quotient-intrinsic source law or action before a normal form can be read as a selection or prediction claim.

Observable algebras and reference states. In the finite classical case the quotient observable algebra is

$$\mathcal{O}_r = \ell^\infty(Q_r).$$

In the finite quantum case the physical algebra is a declared quotient algebra $\mathcal{A}_r^{\mathrm{phys}}$ with state

$$\omega_r(A) = \mathrm{Tr}(\rho_r A).$$

When the reference object is obtained from a finite lifted carrier, the load-bearing data are not an abstract groupoid cardinality alone. They are a tracially pointed quotient

$$\left(\mathcal{A}_{r,b}^{\mathrm{phys}}, \tau_{r,b}^0\right), \qquad \mathcal{A}_{r,b}^{\mathrm{phys}} = z_{r,b} B(\widetilde{\mathcal{H}}_r)^{G_r} z_{r,b},$$

where $U_r : G_r \to U(\widetilde{\mathcal{H}}_r)$ is the compact gauge action and $z_{r,b}$ is the central projection for the declared boundary or superselection sector. The reference trace is

$$\tau_{r,b}^0(A) = \frac{\mathrm{Tr}_{\widetilde{\mathcal{H}}_r}(A)}{\mathrm{Tr}_{\widetilde{\mathcal{H}}_r}(z_{r,b})}.$$

If

$$z_{r,b}\widetilde{\mathcal{H}}_r \cong \bigoplus_\alpha V_\alpha \otimes M_\alpha,$$

with $d_\alpha = \dim V_\alpha$ and $m_\alpha = \dim M_\alpha$, then

$$\mathcal{A}_{r,b}^{\mathrm{phys}} \cong \bigoplus_\alpha I_{V_\alpha} \otimes B(M_\alpha), \qquad p_{r,\alpha} = \frac{d_\alpha m_\alpha}{\sum_\beta d_\beta m_\beta}.$$

These are the induced central-sector weights only after the carrier representation and boundary sector have been fixed.

OPH quotient ensemble. An OPH quotient ensemble is specified by a quotient-intrinsic base weight and action

$$m_r : Q_r \to \mathbb{R}_{>0}, \qquad S_r : Q_r \to \mathbb{R} \cup \{+\infty\},$$

and

$$w_r(q) = m_r(q)e^{-S_r(q)}, \qquad Z_r = \sum_{q \in Q_r} w_r(q), \qquad \mu_r(q) = Z_r^{-1} w_r(q).$$

At least one action value must be finite, so that $Z_r > 0$. For a law on settled configurations only, restrict this construction to N_r, or set $S_r = +\infty$ outside N_r. A separately specified prior ν_r on Q_r instead induces the settled law $\mu_r^{\mathrm{nf}} = (n_r)_\# \nu_r$. These constructions agree only if

$$\sum_{x:\, n_r(x)=q} \nu_r(x) = \mu_r(q) \qquad (q \in N_r)$$

and μ_r has no mass outside N_r. For example, a retraction $n(0) = 0, n(1) = n(2) = 1$ sends the uniform law on three states to weights $(1/3, 2/3)$, whereas uniform counting on its two normal forms gives $(1/2, 1/2)$. Uniform quotient counting, uniform representative counting pushed to the quotient, groupoid weights, and tracial central-sector weights are different physical claims. The paper must declare which one is being used.

Normal-form projector non-selection. For any retraction $N : Q \to Q_{\mathrm{nf}}$ onto a subset $Q_{\mathrm{nf}} \subseteq Q$, that is, any map whose restriction to Q_{nf} is the identity, the induced map on laws

$$\mathcal{C}_Q(\mu) = N_\# \mu$$

is idempotent:

$$\mathcal{C}_Q^2 = \mathcal{C}_Q.$$

Every law supported on Q_{nf} is fixed; both statements use the retraction property. Therefore settlement or canonicalization never selects a unique physical probability law by itself.

Selection-gap corollary. Let $X \subseteq Q_{\mathrm{nf}}$ be a finite set of quotient-normal candidates distinguished by visible invariants. Normal-form data determine X and its quotient-visible invariants, but they do not choose a member of X. If two laws μ, ν are supported on X and concentrate on different candidates, both are fixed by $\mathcal{C}_Q$. Unique sector selection therefore requires source data: an intrinsic action with a unique minimizer, a declared physical ensemble, or a refinement-stable gap certificate. A defect or holonomy classification can classify possible sectors without choosing the physical sector, and a contraction or repair generator can certify convergence toward a declared target without creating the target law.

Finite MaxEnt quotient ensemble. For finite Q, positive m, and quotient observables $F_1, \dots, F_k$, maximizing

$$\mathcal{H}_m(\nu) = -\sum_q \nu(q) \log \frac{\nu(q)}{m(q)}$$

subject to

$$\sum_q \nu(q) = 1, \qquad \sum_q \nu(q) F_a(q) = c_a$$

has the full-support solution, when the feasible full-support surface is nonempty,

$$\mu(q) = \frac{m(q) \exp[-\sum_a \theta_a F_a(q)]}{Z(\theta)}.$$

Boundary optima obey the same formula after restricting to their support. On a finite noncommutative quotient algebra with faithful reference state σ_r, self-adjoint constraints and a feasible set containing a faithful state,

$$\rho_r = \frac{\exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}{\operatorname{Tr} \exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}.$$

If the feasible states lie in a proper support projection P, first restrict to their joint support. The same multiplier argument uses $P(\log \sigma_r)P$ and $PF_{r,a}P$, exponentiated and normalized on $P\mathcal{H}$; replacing the reference by $P\sigma_r P$ before taking its logarithm is not valid without a commutation condition. An empty feasible set has no selected state. The finite constraint ledger must name every $F_{r,a}$, its units and support, the target expectation and source, sector or zero-mode treatment, refinement transformation, and proof that no run output or observational output entered the source definition.

Refinement compatibility and RG closure. For $s \succeq r$, let $c_{sr} : Q_s \to Q_r$ be the physical coarse maps, with $c_{tr} = c_{sr} \circ c_{ts}$ for $t \succeq s \succeq r$. Exact compatibility of weighted ensembles is equivalent to the fiber-sum identity

$$\sum_{q' \colon c_{sr}(q') = q} m_s(q') e^{-S_s(q')} = \alpha_{sr} m_r(q) e^{-S_r(q)}$$

for a constant $\alpha_{sr} > 0$ independent of q. Then

$$(c_{sr})_{\#} \mu_s = \mu_r.$$

If the one-step defects are

$$\delta_{k+1,k} = \left\| (c_{k+1,k})_{\#} \mu_{k+1} - \mu_k \right\|_{\mathrm{TV}},$$

then

$$\left\| (c_{nr})_{\#} \mu_n - \mu_r \right\|_{\mathrm{TV}} \le \sum_{k=r}^{n-1} \delta_{k+1,k}.$$

For exponential-family refinement, exact closure requires the fine conditional free energy

$$G_{sr,\theta}(q_r) = -\log \mathbb{E}_{m_s^0} \left[\exp[-\theta \cdot F_s(Q_s)] \mid c_{sr}(Q_s) = q_r \right]$$

to equal $\kappa_{sr}(\theta) + R_{sr}(\theta) \cdot F_r(q_r)$. If the residual is uniformly bounded by ε, the induced total-variation defect is bounded by $\tanh \varepsilon$.

Implementation invariance and representative lifting. If implementations A, B have quotient bijections $h_r : Q_r^A \to Q_r^B$ satisfying

$$m_r^B(h_r q) = m_r^A(q), \qquad S_r^B(h_r q) = S_r^A(q), \qquad h_r \circ c_{sr}^A = c_{sr}^B \circ h_s,$$

then

$$(h_r)_{\#}\mu_r^A = \mu_r^B.$$

For tracially pointed quantum quotients the corresponding equivalence is a trace-preserving quotient equivalence. It is invariant under unitary intertwiners preserving the gauge action and sector, and under inert trivial ancillas $A \mapsto A \otimes I_{\mathrm{anc}}$. It is not invariant under arbitrary changes of gauge-representation multiplicities.

If an implementation stores representatives, a representative-level law must be a conditional lift

$$\widetilde{\mu}_r(x) = \mu_r(\pi_r x)\kappa_r(x \mid \pi_r x), \qquad \sum_{x:\pi_r(x)=q} \kappa_r(x \mid q) = 1.$$

Then $(\pi_r)_{\#}\widetilde{\mu}_r = \mu_r$. Uniform representative sampling yields orbit-size weights and is physical only if representative counting is the declared base measure.

Quotient-lumpable kernels and sampler correctness. A representative kernel $\widetilde{P}(x, y)$ descends to Q_r only when

$$P_Q(q, q') = \sum_{y:\pi(y)=q'} \widetilde{P}(x, y)$$

is independent of the chosen representative $x \in \pi^{-1}(q)$. For $w(q) = m(q)e^{-S(q)}$ and proposal $R(q, q')$ with reciprocal support, the Metropolis–Hastings acceptance rule

$$a(q, q') = \min\left\{1, \frac{w(q')R(q', q)}{w(q)R(q, q')}\right\}$$

gives detailed balance

$$\mu(q)R(q, q')a(q, q') = \mu(q')R(q', q)a(q', q).$$

Repair-informed proposals must include the Hastings asymmetry term; otherwise the stationary law is generically changed.

Repair generators are not selectors. A repair generator of the form

$$L_{\mathrm{rep}} = \sum_C c_C(I - E_C)$$

is a relaxation or sampling object after a law has been selected. Conditional expectations E_C are defined on $L^2(X_r, \pi_r)$, so the reference law π_r is input. On overlapping collars the expectations need not commute. The correct finite gap certificate is the Poincare constant on the orthogonal complement of the common fixed space $\mathcal{F}_r = \bigcap_C \operatorname{Ran} E_C$:

$$\kappa_r = \inf_{0 \neq f \perp \mathcal{F}_r} \frac{\sum_C \|(I - E_C)f\|^2}{\|f\|^2}.$$

If every $c_C \geq \gamma_* > 0$, then

$$L_{\mathrm{rep}} \geq \gamma_* \kappa_r (I - P_0),$$

where P_0 projects onto $\mathcal{F}_r$. In finite dimension, $\kappa_r > 0$ whenever $\mathcal{F}_r^\perp \neq \{0\}$, since zero Dirichlet energy would put f in every fixed range. If all states are fixed, the displayed operator inequality is vacuous. Replacing $f \perp \mathcal{F}_r$ by $f \perp 1$ requires the additional ergodicity condition $\mathcal{F}_r = \mathrm{span}\{1\}$; a conserved nonconstant record otherwise makes that infimum zero. A uniform refinement lower bound $\inf_r \kappa_r > 0$ is a separate theorem or receipt. Multiplying all rates by $s > 0$ rescales the generator and gap by s, while leaving the fixed space, stationarity and detailed balance unchanged. Numerical recognition therefore tests identities after removing a common rate scale and restores that scale to reported gaps. An unresolved positive gap is a precision limitation, not evidence that changing clock units changed the equilibrium.

Finite evidence accuracy. For bounded coarse observables O, if

$$\|\widehat{\mu}_s - \mu_s\|_{\mathrm{TV}} \leq \epsilon_{\mathrm{samp}}$$

and the refinement defects sum to ϵ_{ref}, then

$$\left|\mathbb{E}_{\widehat{\mu}_s}[O \circ c_{sr}] - \mathbb{E}_{\mu_r}[O]\right| \leq 2\|O\|_\infty(\epsilon_{\mathrm{samp}} + \epsilon_{\mathrm{ref}}).$$

Continuum-facing observables require a realization map and correlation Cauchy bound in addition to a finite histogram.

Physical-vacuum requirement. A stationary sampler is not a physical vacuum. For any faithful target law one can build a positive transfer operator with that law as ground state, so positivity alone is not a selector. A physical-vacuum interpretation requires source Euclidean slab data

$$\mathfrak{S}_r^E = (Q_r, m_r^0, J_r, V_r, a_{t,r})$$

whose conductance $J_r(q, q') = J_r(q', q) \geq 0$, local potential V_r, and slab thickness $a_{t,r}$ are derived without using the target law or sampler output. With connected event graph,

$$(H_r^E f)(q) = \frac{1}{m_r^0(q)} \sum_{q'} J_r(q, q')(f(q) - f(q')) + V_r(q) f(q)$$

is self-adjoint and bounded below on $L^2(Q_r, m_r^0)$; its finite Feynman–Kac semigroup is positivity improving. Perron–Frobenius gives a unique positive normalized ground state Ω_r, and the finite vacuum law is

$$\mu_r^{\mathrm{vac}}(q) = |\Omega_r(q)|^2 m_r^0(q).$$

For $T_r = e^{-a_{t,r}(H_r^E - E_{0,r})}$, the Doob kernel is stochastic and detailed-balanced with μ_r^{vac}. A continuum interpretation additionally requires reflection positivity or equivalent reconstruction plus refinement compatibility of the transfer family. On the native source screens the actual density law has such a reflection-positive reconstruction along one common refining subsequence; its one-particle generator is the native continuum operator A rather than $\sqrt{A}$, and it supplies no coherent preparation, relativistic vacuum or Lorentz symmetry.[2]

[2] https://github.com/FloatingPragma/observer-patch-holography/blob/r2051/model/observer_time_fields/REFINEMENT.md

Primordial and cosmological prediction firewall. A screen covariance contains incomplete radial information. The complete one-shell map

$$C_\ell = 4\pi \int_0^\infty \frac{dk}{k} \Delta_\zeta^2(k) j_\ell^2(k\chi_\star)$$

has an infinite-dimensional kernel that persists under positivity. OPH physical primordial interpretation requires the source-only stress, single-clock, entropy-repair, curvature-evolution, adiabatic-mode, phase-coherence, physical mode, radial-null-space, and forward-projection receipts together with a scale-natural physical dilation intertwiner or complete radial cross-covariance tomography. A finite radial prior produces a conditional continuation. Observable CMB comparison also requires declared source, solver, dataset, covariance, nuisance, data-use, and pooled-reducer provenance. On the native source screens the continuum energy is not boundedly congruent to the static boundary action of a smooth compact three-dimensional Dirichlet bulk, whose Steklov eigenvalues grow like $\sqrt{n}$ while the native eigenvalues grow like n, and internal rank-three carrier modes vanish from bounded-energy families; a radial lift needs a radial observation algebra that the native energy does not supply.[3]

Interpretation classes and required receipts. Every ensemble-facing run records its ensemble id, interpretation class, regulator, representative schema, gauge action, canonicalizer, base measure, action coefficients, coarse maps, zero-mode projector, amplitude convention, sampler, smoothing policy, source provenance, and explicit nonclaims. The seed belongs to the run receipt rather than the ensemble definition. The interpretation classes are

seed, proposal, or repair noise,
conventional reference ensemble,
OPH-native quotient ensemble,
OPH physical vacuum,
OPH primordial field,
observable cosmological prediction.

The evidence bundle must keep separate receipts for stationary-law schedule invariance, detailed balance of the aggregate kernel, and pathwise partition invariance. Deterministic replay of semantic random streams or a canonical serial chain is useful, but it is not pathwise partition invariance. Smoothing must preserve raw coefficients, raw spectra, smoothing kernels, smoothed coefficients, smoothed spectra, and hashes of each stage; it is not part of S_r unless explicitly declared.

Derived regulator realization. Choose a finite UV cellulation at scale ℓ_{UV} and let R be a finite union of cells. Hilbertize each cell's finite local data to a finite-dimensional space $\tilde{\mathcal{H}}_i \cong \mathbb{C}^{n_i}$. The extended algebra before quotienting by overlap redundancy is

$$\tilde{\mathcal{A}}(R) = \mathcal{B}(\tilde{\mathcal{H}}_R), \qquad \tilde{\mathcal{H}}_R = \bigotimes_{i \subset R} \tilde{\mathcal{H}}_i.$$

Overlap-preserving changes of local trivialization act only on cut data. On each finite-dimensional chart, their unitary image has compact closure, so one may represent the boundary gluing redundancy by a compact group $G_{\partial R} \subset U(\tilde{\mathcal{H}}_R)$. The lifted boundary-invariant endomorphism algebra is

[3] https://github.com/FloatingPragma/observer-patch-holography/blob/r2051/model/physical_space_time/OCCUPATION_FIELD.md

$$A_{\mathrm{inv}}(R) = \widetilde{\mathcal{A}}(R)^{G_{\partial R}} = \mathcal{B}(\tilde{\mathcal{H}}_R)^{G_{\partial R}}.$$

This is the fixed-cutoff realized presentation used throughout the collar analysis. The EC theorem works on the invariant-state realization and uses the sector-preserving induced collar algebra there; it does not apply to the entire fixed-point algebra on the unreduced tensor product. When the consensus paper speaks of quotient repair, the physical repair law is defined on this fixed-point / overlap-invariant data and any representative-level map is only a lift of that quotient update. Descent to the gauge quotient is therefore built into the physical-algebra formulation as quotient-level repair: the local map is $\mathrm{locRep}_\lambda : Q \to Q$, and the global map is the finite quotient normal-form operator

$$\mathrm{Rep}_\lambda = \overline{\mathrm{nf}}_\lambda : Q \to Q.$$

On the declared fixed-cutoff collar branch, the local repair step itself is read from exact Markov splice or a declared Petz/Fawzi–Renner recovery channel. The declared fixed-cutoff branch adds repair completeness, together with control on the declared Petz domain where that branch is used. The consensus paper proves those clauses on a nontrivial rooted-tree packet-net export: finite packet labels live on a rooted tree, hidden labels are quotient-only, weighted parent-copy repair strictly lowers Φ, normal forms are exactly consistent packet assignments, and the classical full-support Petz channel is CPTP and trace-norm contractive on its positive-support-gap domain. Gauge-invariant observables factor through the quotient normal form on that verified branch. The declared touched-overlap acceptance contract makes Φ the finite-patch Lyapunov functional for primitive repair proposals on the scalar branch, but a proposal becomes a physical step only through the transactional acceptance layer: snapshot-current read sets, unchanged-outside-write validation, preserved boundary/sector/holonomy data, and exact well-founded descent. The read set is validation-complete: it contains every support of every measure term that can change under the write. Overlapping primitive proposals are aggregated by connected conflict component; the restriction-compatible union-collar glue supplies the canonical aggregate payload, while primitive members of that component do not commit independently. On the same fixed-cutoff quantum lift, the terminal expectation functional on each declared physical observable algebra is therefore unique even when microscopic representative lifts differ by gauge labels globally or by sector labels inside one quotient-local glued state. The unqualified coding statement at this stage is only a finite constraint-code statement: the overlap-net codewords are the globally consistent states $C = \Phi^{-1}(0)$. A bare graph does not determine code distance: the same graph can carry constant readouts with distance 1 or repetition constraints with distance $|V|$. Topological-code distance/min-cut, Knill–Laflamme resilience, spectral-gap convergence, BFT wall-clock liveness, and hardware search-work reduction therefore require their own certificates and are not imported by the core overlap graph. Exact descent gives termination. Confluence follows from an independently checked conflict-component diamond on the physical quotient, protected-support and protected-conflict completeness, and repair completeness. The theorem therefore gives one schedule-independent terminal quotient normal form from a fixed initial quotient state. A stronger same-boundary conclusion requires a preserved boundary/sector map and at most one consistent quotient extension in that boundary fiber. The layered functional carrier in the consensus paper proves the finite multi-edge, multi-step witness for $H_B \wedge H_{\mathrm{fib}}$; the functional selected-fiber theorem gives the rooted obstruction-check form by constructing the single candidate from the boundary/root data and then checking all non-tree, sector, and holonomy obstructions. On that branch, multiple same-boundary candidate interiors can be present, inconsistent candidates are eliminated, and all surviving candidates share one quotient normal form. Normal-form hashes are public equality receipts after quotienting, not selectors among physically distinct minimizing endpoints. If accepted schedules terminate at different observer-facing quo-

tient normal forms without a declared holonomy or higher-gauge obstruction, that repair law is outside the OPH consensus theorem. The neutral result of Ref. [16] packages this quantifier split exactly: under observation preservation and completeness, agreement modulo a silent equivalence of normal endpoints reached from all same-boundary sources is equivalent to injectivity of the induced boundary map on the consistent quotient. This cross-source criterion, same-source confluence, weak normalization, and all-schedule liveness are separate obligations. The named layered and functional carriers discharge the cross-source premise only on their stated domains. For separated cofinal refinement systems, the consensus paper adds an inverse-limit bridge: if finite-stage restriction maps commute with the quotient normal-form maps and holonomy maps, and if compatible families are visibly separated on cofinal finite stages, then finite normal forms and holonomy obstructions assemble into unique refinement-limit classes with finite-stage witnesses for nonzero holonomy. It also adds the RG-facing comparison: when a chosen coarse-graining channel shadows the finite-stage normal forms and obstruction maps with declared defects ε^n and ε^h, reconciling first and then coarse-graining gives the same macroscopic law readout as coarse-graining first and then reconciling, up to $\max\{\varepsilon^n, \varepsilon^h\}$; exact naturality is the zero-defect case. Exact normal-form naturality is discharged by a repair-morphism witness, and holonomy naturality by the corresponding cochain map. For distributed implementations, the same paper proves that a worker run presents one finite universe only when it starts from one global carrier and each event projects to a legal monolithic repair path, a physical stutter, or a certified rollback to an earlier committed projection. The required certificate names the global graph, initial state, partition, cut interfaces, observer registry, run/config/code hashes, linearized commits, restart roots, final monolithic normal form, and recomputed readout. Executor order is separate from observer history. Observer-clock naturality additionally requires a history-augmented quotient with semantic event keys independent of worker ID, repair iteration, queue position, timestamp, and retry metadata; a global observer registry descended from local registry groupoids with disjoint patch/cap/future namespaces and explicit lineage arrows; a state-preserving observer-algebra extraction functor from terminal augmented normal forms; a support-visible cap chart map natural with that extraction; and an operational clock instrument whose readout is calibrated up to the declared affine reparameterization and residual bound. The listed objects or their proof-producing finite certificates are prerequisites. In their absence, modular-order/readback naturality is a named premise rather than a consequence of implementation bookkeeping. The same quotient boundary applies to neutral-bulk readout. For each shard s, raw record rows Σ_s must first be quotient by the declared presentation groupoid Γ_s and then sent through the terminal normal-form map n_s, producing

$$X_s = n_s(\Sigma_s/\Gamma_s).$$

Interface maps $\tau_{ts} : U_{st} \to U_{ts}$ between these terminal charts must satisfy the identity, inverse, cocycle, and zero-cycle-holonomy checks before rows from different shards are compared. A global neutral readout is the quotient

$$Q_{\mathrm{vis}} = \left(\bigsqcup_s X_s \right) / \sim_\tau,$$

not the concatenation of shard-local tables. Channel features $F_{s,c}$ descend to Q_{vis} only when their domains and values are transported by the same atlas. For a complete declared channel set $\mathcal{C}_\star$, the neutral distance is

$$d_{\mathrm{neu}}(x, y) = \left(\sum_{c \in \mathcal{C}_\star} w_c \, d_c(F_c(x), F_c(y))^p \right)^{1/p}.$$

Here $p \geq 1$, and $\mathcal{C}_\star$ is finite. An infinite declared channel set is allowed only when the displayed weighted p-sum is finite for every compared pair. Under that finite-distance premise this

is stated first as a quotient-visible pseudometric. It becomes a metric only after quotienting zero-distance feature-collision classes or proving that the channels jointly separate points. Pairwise available-channel comparison is explicitly excluded: missingness is handled only by complete cases, a fixed quotient-visible missing symbol, or a train-only imputation map labelled as an imputed-representation metric. The certificate records a finite channel list or a pairwise weighted-ℓ^p summability witness. Presentation changes (gauge representatives, port relabelings, observer order, schedules, and shard partitions) preserve the metric only when they induce a bijection of Q_{vis} together with channel isometries; refinement claims additionally require a cofinally vanishing distance tail modulus. Finite Euclidean claims require the double-centered Gram certificate

$$B = -\frac{1}{2}H(D^{\circ 2})H \succeq 0$$

and noisy runs must report negative spectral mass, rank/effective-rank data, held-out stress, and planted/shuffled controls. Statistical certificates split independent generative shard batches before preprocessing; chart alignment, scaling, imputation, weights, graph construction, dimension choice, and thresholds are train/validation objects, and a test batch, seed, boundary condition, trajectory family, duplicate, or descendant appearing in more than one split blocks the claim. These theorems establish at most a common finite quotient metric or pseudometric. Identifying it with a physical Riemannian bulk or with Lorentzian/Einstein spacetime (via the source-causal continuum certificate of Theorem 4.3e, under its physical-order, refinement, count–volume, independent-dimension, manifoldlikeness, topology, and curvature receipts) requires the separate modular/geometric branch below. The same consensus surface is classified explicitly as a finite-state exact theorem package plus this controlled inverse-limit bridge: decidable normal-form computation with the Lyapunov step bound, automatic approximate stability only through the collar-local splice and record controls, a conditional fair-block contraction branch for long-run noisy approximate consensus at block indices, with deterministic block endpoints additionally required for the stated all-time expectation bound, and a verified rooted-tree packet-net subdomain. Computational universality for growing patch-net families belongs to the consensus paper's expressive-power boundary and is not a dependency for the recovered-core branches.

Fixed-cutoff packet closure map and invariant simplex. On any declared fixed-cutoff packet quotient Q whose repair relation is terminating and confluent, the normal-form map

$$N : Q \to Q_{\mathrm{nf}}$$

is well-defined and schedule-independent. It induces an affine OPH closure map on the probability simplex over packet states,

$$\mathcal{C}_Q : \Delta(Q) \to \Delta(Q), \qquad \mathcal{C}_Q\left(\sum_{q\in Q} p_q \delta_q\right) = \sum_{q\in Q} p_q \delta_{N(q)}.$$

Since Q is finite, $\Delta(Q)$ is a nonempty compact convex set and $\mathcal{C}_Q$ is a continuous affine self-map. Its image $\Delta(Q_{\mathrm{nf}})$ is invariant, and $\mathcal{C}_Q^2 = \mathcal{C}_Q$. This is an actual closure map on the fixed-cutoff packet-closed quotient branch. It is not the full Appendix/habitat closure-map theorem for arbitrary OPH state-and-law data; the first operation not internalized at that level is the general habitat lift from this finite packet quotient to the full compact-convex observer-supporting sector.

Finite quotient ensemble theorem. The closure map above is an idempotent normal-form projector. It sends any input law to terminal quotient normal forms, and every law supported on

those normal forms is a fixed point. Hence the physical probability law at regulator r must be supplied as an additional quotient-intrinsic object. Let

$$Q_r = \Sigma_r/\Gamma_r$$

be the finite physical quotient, or let $N_r = n_r(Q_r)$ be the quotient-normal-form subset when only settled configurations are intended to carry probability. The declared OPH quotient ensemble is specified by an intrinsic base weight and action

$$m_r : Q_r \to \mathbb{R}_{>0}, \qquad S_r : Q_r \to \mathbb{R} \cup \{+\infty\},$$

and

$$w_r(q) = m_r(q)e^{-S_r(q)}, \qquad Z_r = \sum_{q \in Q_r} w_r(q), \qquad \mu_r(q) = Z_r^{-1} w_r(q).$$

Require $Z_r > 0$, equivalently at least one finite action value. If only normal forms carry probability, restrict the state space to N_r, or put infinite action outside it. A separately supplied prior ν_r induces $\mu_r^{\mathrm{nf}} = (n_r)_{\#}\nu_r$. It equals the displayed Gibbs law only when that law is supported on N_r and the prior's mass in each normal-form fiber equals its target weight. A uniform three-state prior with fibers of sizes one and two yields $(1/3, 2/3)$, not the uniform two-state law. The choice of m_r, S_r, or ν_r is load-bearing: uniform quotient counting, uniform representative counting pushed to the quotient, finite-groupoid weights, and trace weights of a declared quotient algebra are generally different.

Finite MaxEnt form. For a finite Q, positive m, and quotient observables $F_1, \dots, F_k$, maximizing

$$\mathcal{H}_m(\nu) = -\sum_q \nu(q) \log \frac{\nu(q)}{m(q)}$$

over the affine constraint surface

$$\sum_q \nu(q) = 1, \qquad \sum_q \nu(q) F_a(q) = c_a$$

has a unique full-support maximizer of the form below when the feasible set contains a full-support point (equivalently, the prescribed moment vector is in the relative interior of the attainable moment set):

$$\mu(q) = \frac{m(q) \exp[-\sum_a \theta_a F_a(q)]}{Z(\theta)}.$$

This is the finite-dimensional Lagrange-multiplier argument with strict concavity of $\mathcal{H}_m$. If no full-support feasible point exists, the unique optimizer lies on a proper face; the same finite-multiplier formula holds only after restricting the state space and observables to the optimizer's support (or globally as an extended exponential family with infinite penalties). On a finite noncommutative quotient algebra, when the feasible set contains a faithful state, the corresponding density operator with faithful reference state σ_r is

$$\rho_r = \frac{\exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}{\operatorname{Tr} \exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}.$$

Refinement and projective compatibility. For $s \succeq r$, let $c_{sr} : Q_s \to Q_r$ be the physical coarse-graining map. Exact compatibility of the weighted ensembles is equivalent to the fiber-sum identity

$$\sum_{q' :\, c_{sr}(q') = q} m_s(q') e^{-S_s(q')} = \alpha_{sr} m_r(q) e^{-S_r(q)}$$

for a constant $\alpha_{sr} > 0$ independent of q; necessarily $\alpha_{sr} = Z_s/Z_r$. The proof is just the equality

$$(c_{sr})_{\#}\mu_s(q) = Z_s^{-1} \sum_{q' \colon c_{sr}(q')=q} m_s(q')e^{-S_s(q')}$$

compared with $Z_r^{-1} m_r(q)e^{-S_r(q)}$. For the selected-prior route, if

$$c_{sr} \circ n_s = n_r \circ c_{sr}, \qquad (c_{sr})_{\#}\nu_s = \nu_r,$$

then functoriality gives $(c_{sr})_{\#}(n_s)_{\#}\nu_s = (n_r)_{\#}\nu_r$. Along a chain, finite-stage compatibility defines a unique projective-limit probability measure on the cylinder σ-algebra. If the one-step defects are

$$\delta_{k+1,k} = \|(c_{k+1,k})_{\#}\mu_{k+1} - \mu_k\|_{\mathrm{TV}},$$

then total-variation contraction gives

$$\|(c_{nr})_{\#}\mu_n - \mu_r\|_{\mathrm{TV}} \le \sum_{k=r}^{n-1} \delta_{k+1,k}.$$

Implementation invariance and representative lifting. If two implementations A, B carry quotient bijections $h_r : Q_r^A \to Q_r^B$ with

$$m_r^B(h_r q) = m_r^A(q), \qquad S_r^B(h_r q) = S_r^A(q), \qquad h_r \circ c_{sr}^A = c_{sr}^B \circ h_s,$$

then $(h_r)_{\#}\mu_r^A = \mu_r^B$. If an implementation stores representatives $\pi_r : \Sigma_r \to Q_r$, a representative-level law must be a conditional lift

$$\widetilde{\mu}_r(x) = \mu_r(\pi_r x)\kappa_r(x \mid \pi_r x), \qquad \sum_{x:\pi_r(x)=q} \kappa_r(x \mid q) = 1.$$

Then $(\pi_r)_{\#}\widetilde{\mu}_r = \mu_r$. Uniform representative sampling weights quotient states by orbit size and is physical only if representative counting is the declared base measure. A representative Markov kernel must also be quotient-lumpable: the transition probability to an orbit cannot depend on which representative of the source orbit is used.

Sampler correctness. A production sampler is a separate interface from a deterministic settler:

$$\texttt{settle}(q) \to n_r(q), \qquad \texttt{sample}(\mu_r) \to q.$$

For $w(q) = m(q)e^{-S(q)}$ and a proposal kernel $R(q, q')$ with reciprocal support, the Metropolis–Hastings acceptance rule

$$a(q, q') = \min\left\{1, \frac{w(q')R(q', q)}{w(q)R(q, q')}\right\}$$

gives detailed balance

$$\mu(q)R(q, q')a(q, q') = \mu(q')R(q', q)a(q', q),$$

and hence $\mu P = \mu$. Irreducibility and aperiodicity give uniqueness and convergence on the finite state space. If a repair-informed proposal is $q' = R_{\mathrm{repair}}(q) + \eta$ with density g, the Hastings ratio must include

$$\frac{g(q - R_{\mathrm{repair}}(q'))}{g(q' - R_{\mathrm{repair}}(q))};$$

dropping this term generally changes the stationary law. For a lazy reversible finite chain with spectral gap γ,

$$\|P^n(x,\cdot)-\mu\|_{\mathrm{TV}} \le \frac{1}{2}\sqrt{\mu(x)^{-1}-1}\,(1-\gamma)^n,$$

and the usual autocorrelation bound follows from the nontrivial spectral radius. Exact small regulators should therefore enumerate Q_r, w_r, P_r, detailed-balance error, stationarity error, refinement residuals, lumpability/orbit-size checks, and the exact spectral gap.

Gaussian screen continuation. For a real reduced screen field space V_r after background and dipole modes are removed, let K_r be positive definite and impose fixed expected quadratic release energy,

$$\mathbb{E}[q]=0, \qquad \mathbb{E}[q^{\mathsf{T}}K_r q]=d_r A_r, \qquad d_r=\dim V_r.$$

The unique maximum-entropy density relative to the declared Euclidean volume is

$$p_r(q)=\frac{\sqrt{\det K_r}}{(2\pi A_r)^{d_r/2}}\exp\left[-\frac{1}{2A_r}q^{\mathsf{T}}K_r q\right], \qquad \mathrm{Cov}(q)=A_r K_r^{-1}.$$

This is the relative-entropy proof against the displayed Gaussian. If instead every sample obeys $q^{\mathsf{T}}K_r q=d_r A_r$, the MaxEnt law is uniform on the ellipsoid, not Gaussian. Under a linear coarse field map $C_{sr}: V_s \to V_r$, exact Gaussian refinement is the covariance criterion

$$A_r K_r^{-1} = C_{sr}(A_s K_s^{-1})C_{sr}^{\mathsf{T}}.$$

For harmonic truncation, choose real modes $Y_{\ell m}$, $2 \le \ell \le L$, set

$$K_L Y_{\ell m}=\kappa_\ell Y_{\ell m}, \qquad \kappa_\ell=[\ell(\ell+1)]^{1+\theta/2},$$

and sample independent coefficients

$$a_{\ell m}\sim \mathcal{N}(0, A/\kappa_\ell).$$

Projection from L' to L then discards high modes and is exactly refinement-compatible. For irregular meshes, covariance should be primary under coarse-graining:

$$\Sigma_r = C_{sr}\Sigma_s C_{sr}^{\mathsf{T}}, \qquad K_r = A_r\Sigma_r^{-1}.$$

Equivalently, when eliminating fine variables from a precision matrix

$$K_s=\begin{pmatrix} K_{cc} & K_{cf} \\ K_{fc} & K_{ff}\end{pmatrix},$$

the coarse precision is the Schur complement

$$K_r = K_{cc}-K_{cf}K_{ff}^{-1}K_{fc},$$

not the principal block K_{cc}.

Physical vacuum and primordial premises. A stationary sampler is not a physical vacuum: the identity transition kernel leaves every measure stationary. A finite vacuum theorem needs a physical transfer operator. If T_r is real symmetric, positive definite, and strictly positive, Perron–Frobenius gives a simple largest eigenvalue $\lambda_{0,r}$ with positive eigenvector Ω_r. Then

$$H_r=-a_t^{-1}\log(T_r/\lambda_{0,r})$$

is positive semidefinite, Ω_r is the unique zero-energy state, and the Doob kernel

$$P_r(x,y) = \frac{T_r(x,y)\Omega_r(y)}{\lambda_{0,r}\Omega_r(x)}$$

is stochastic and detailed-balanced with

$$\mu_r(x) = \frac{\Omega_r(x)^2}{\sum_z \Omega_r(z)^2}.$$

Continuum identification further requires reflection positivity or an equivalent reconstruction certificate plus refinement compatibility of the transfer family. A screen covariance alone does not determine an unrestricted primordial curvature spectrum. Even the complete noiseless one-shell sequence for the thin-shell map

$$C_\ell = 4\pi \int_0^\infty \frac{dk}{k} \Delta_\zeta^2(k) j_\ell^2(k\chi_\star)$$

determines the three-dimensional correlation only on $0 \le s \le 2\chi_\star$ and has an infinite-dimensional kernel that persists under positivity. Source-derived uniqueness requires a physical dilation-intertwiner theorem or complete radial cross-covariance tomography. A finite prior gives a conditional continuation. Conventional free-scalar and lattice-gauge baselines may therefore be recorded. They instantiate neither an OPH-native physical vacuum nor a primordial source without these premises.

2 Finite Screen Spectrum Theorem Package

The screen-level spectrum theorem supplies the finite source object used by the cosmology branch. It does not supply temperature-temperature, temperature-polarization, polarization-polarization, lensing, likelihoods, or a physical cosmic microwave background interpretation. Those require the Boltzmann and data-contract theorems.

Definition 2.1 (finite screen regulator)**.** *For regulator r, a finite screen regulator is*

$$\mathfrak{S}_r = (\mathcal{T}_r, M_r, \Gamma_r, \mathcal{Q}_r, C_{sr}, J_{rs}),$$

where $\mathcal{T}_r$ is a finite cellulation of the screen, $M_r \succ 0$ is the area/quadrature mass matrix, Γ_r is the hidden-presentation groupoid, $\mathcal{Q}_r = \Sigma_r/\Gamma_r$ is the physical quotient, and C_{sr}, J_{rs} are the coarse-graining and interpolation maps. Shape regularity and quadrature convergence require

$$f^{\mathsf{T}} M_r g \to \int_{S^2} f g \, d\Omega$$

with an explicit band-limited error bound

$$\left| f^{\mathsf{T}} M_r g - \int f g \, d\Omega \right| \le \varepsilon_{M,r}(L) \|f\|_{H^s} \|g\|_{H^s}.$$

Patch count alone is therefore not an angular-resolution certificate.

Definition 2.2 (geometric screen scalar)**.** *Let the quotient-visible collar-volume readout be*

$$J_{X,r}(x) = \lambda_r(x) \sqrt{\det \sigma_{AB,r}(x)} > 0,$$

and let $\bar{J}_{X,r}$ be emitted by an independently defined homogeneous or frozen-background branch, not by a CMB fit. Define

$$q_{0,r} = \frac{1}{3}\log\frac{J_{X,r}}{\bar{J}_{X,r}}.$$

On a certified spherical chart, with $B_r = [1, n_x, n_y, n_z]$,

$$\Pi_{<2,r} = B_r(B_r^{\mathsf{T}} M_r B_r)^{-1} B_r^{\mathsf{T}} M_r, \qquad q_r = (I - \Pi_{<2,r}) q_{0,r}.$$

On an irregular screen, B_r is replaced by the certified generalized eigenprojector onto the finite $\ell = 0, 1$ subspace. Hard-coded feature z-scores or observer-summary weights are diagnostic proxies, not q_r.

Proposition 2.3 (quotient invariance)**.** *If a recharting U preserves the screen inner product and geometric readout,*

$$U^{\mathsf{T}} M_r' U = M_r, \qquad q_{0,r}' = U q_{0,r}, \qquad B_r' = U B_r,$$

then $\Pi_{\geq 2,r}' U = U \Pi_{\geq 2,r}$, and $q_r' = U q_r$.

Proof. Substitution gives

$$\begin{aligned}\Pi_{<2,r}' U &= U B_r (B_r^{\mathsf{T}} U^{\mathsf{T}} M_r' U B_r)^{-1} B_r^{\mathsf{T}} U^{\mathsf{T}} M_r' U \\ &= U B_r (B_r^{\mathsf{T}} M_r B_r)^{-1} B_r^{\mathsf{T}} M_r = U \Pi_{<2,r}.\end{aligned}$$

Subtracting from U gives the high-mode identity. □

Definition 2.4 (physical scalar precision and action)**.** *The scalar precision K_r must have a physical origin on $V_r = \operatorname{im} \Pi_{\geq 2,r}$. Allowed branches are:*

$$K_r^{\mathrm{Hess}} = \nabla^2 \Phi_r \Big|_{q=0},$$

for a quotient-visible mismatch or release free energy; a repair Dirichlet form

$$K_r^{\mathrm{diss}} = H_r - T_r^{\mathsf{T}} H_r T_r, \qquad T_r^{\mathsf{T}} H_r T_r \preceq H_r;$$

or the reversible generator form

$$K_r^{\mathrm{Dir}} = -\frac{1}{2}(L_r + L_r^{\dagger_H}).$$

A raw nonsymmetric repair matrix or caller-supplied κ is not a precision operator. The absolute normalization must be fixed independently, for example by

$$K_0 Y_{\ell m} = \ell(\ell+1) Y_{\ell m}$$

on the $\theta = 0$ branch. Once normalized,

$$S_{\mathrm{scr},r}[q] = \frac{1}{2A_{q,r}} \langle q, K_r q \rangle_r = \frac{1}{2A_{q,r}} q^{\mathsf{T}} M_r K_r q.$$

Theorem 2.5 (finite MaxEnt screen covariance)**.** *Let $K_r e_i = \kappa_i e_i$ with an M_r-orthonormal basis of V_r, $\kappa_i > 0$, and $d_r = \dim V_r$. Among continuous densities in coefficients $q = \sum_i a_i e_i$ with $\mathbb{E}[a_i] = 0$ and*

$$\frac{1}{2}\mathbb{E}\langle q, K_r q \rangle_r = E_{q,r}^{\mathrm{src}},$$

the unique entropy maximizer is

$$d\mu_r(q) = Z_r^{-1} \exp\left[-\frac{1}{2A_{q,r}}\langle q, K_r q\rangle_r\right] dq, \qquad A_{q,r} = \frac{2E^{\mathrm{src}}_{q,r}}{d_r},$$

and

$$\mathbb{E}[a_i a_j] = \delta_{ij}\frac{A_{q,r}}{\kappa_i}.$$

Proof. The Euler–Lagrange equation for entropy with normalization and quadratic-energy constraints gives a Gaussian density with inverse temperature β_r. The expected energy is

$$E^{\mathrm{src}}_{q,r} = \frac{1}{2}\sum_i \kappa_i \frac{1}{\beta_r \kappa_i} = \frac{d_r}{2\beta_r}.$$

Thus $A_{q,r} = \beta_r^{-1} = 2E^{\mathrm{src}}_{q,r}/d_r$. Strict concavity of entropy gives uniqueness. □

Remark 2.6 (discrete quotient branch)**.** *For a discrete finite quotient with intrinsic base weight $m_r(q)$, the exact Gibbs law is*

$$\mu_r(q) = Z_r^{-1} m_r(q) e^{-\beta_r E_r(q)}.$$

A Gaussian statement on that branch requires a separate Laplace or central-limit theorem with controlled higher-order terms.

Definition 2.7 (source release energy)**.** *Let m_r^0 be the tracially pointed base weight on the finite physical collar quotient. Let $F_{r,a}$ be a finite ledger of primitive collar observables: scalar occupancy, protected-center presence, released volume, conserved source charges, and the release clock. The ledger excludes screen energy, sky spectra, CMB residuals, likelihoods, fitted amplitudes, and measurement-calibrated proxies. For source-emitted constraint values $c^{\mathrm{col}}_{r,a}$, the selected release law is*

$$\nu_r^{\mathrm{rel}}(x) = \frac{m_r^0(x)\exp[-\sum_a \lambda_{r,a}F_{r,a}(x)]}{Z_r}, \qquad \mathbb{E}_{\nu_r^{\mathrm{rel}}} F_{r,a} = c^{\mathrm{col}}_{r,a}.$$

Strict concavity of relative entropy gives uniqueness on full support. A boundary optimum uses the same form on its certified support. The constraint ledger, base weight, source DAG, multipliers, support, and refinement maps belong to the receipt.

For this independently selected quotient ensemble on released collar normal forms, define

$$E^{\mathrm{src}}_{q,r} = \frac{1}{2}\int \langle q_r(x), K_r q_r(x)\rangle_r \, d\nu_r^{\mathrm{rel}}(x).$$

The receipt must expose ν_r^{rel}, the base measure, K_r, d_r, $E^{\mathrm{src}}_{q,r}$, $A_{q,r}$, no-observation ancestry, and the same operator normalization used in the action. MaxEnt alone does not determine amplitude.

Proposition 2.8 (microscopic-law necessity)**.** *For fixed positive K_r, the Gaussian family $Z(A)^{-1}\exp[-\langle q, K_r q\rangle/(2A)]$ exists for every $A > 0$. The quadratic MaxEnt form therefore leaves one positive scale free. The source release law in Definition 2.7, or an equivalent source-only vacuum law, supplies the energy that fixes $A_{q,r} = 2E^{\mathrm{src}}_{q,r}/d_r$.*

Theorem 2.9 (rotational screen spectrum)**.** *If the continuum precision K_θ commutes with the $SO(3)$ action on scalar functions, then by Schur's lemma each $\mathcal{H}_\ell$ is an eigenspace. For the exact conformal-shell family,*

$$\Lambda_\ell(\theta) = \frac{\Gamma(\ell + 2 + \theta/2)}{\Gamma(\ell - \theta/2)}, \qquad \ell \geq 2.$$

Then $\Lambda_\ell(0) = \ell(\ell+1)$, and the MaxEnt covariance gives

$$C_\ell^q = A_q \frac{\Gamma(\ell - \theta/2)}{\Gamma(\ell + 2 + \theta/2)}.$$

Consequently $\mathcal{D}_\ell^q = \ell(\ell+1)C_\ell^q/(2\pi) \sim (A_q/2\pi)\ell^{-\theta}$.

Definition 2.10 (source-derived conformal precision)**.** *Let $L_r \succeq 0$ be the normalized detailed-balanced scalar collar Dirichlet operator on V_r, with continuum limit $-\Delta_{S^2}$. Set*

$$B_r = (L_r + \tfrac{1}{4}I)^{1/2}, \qquad K_{\theta,r} = \frac{\Gamma(B_r + \frac{3}{2} + \frac{\theta}{2})}{\Gamma(B_r - \frac{1}{2} - \frac{\theta}{2})}.$$

For $-2 < \theta < 4$ on the retained $\ell \geq 2$ branch, this operator is positive. The gamma recurrence gives $K_{0,r} = L_r$ exactly. The finite receipt carries detailed balance, positivity, the $\ell(\ell+1)$ normalization, anisotropy splitting, and refinement residuals.

Theorem 2.11 (finite refinement error)**.** *Suppose that for $2 \leq \ell \leq L$*

$$\left|\frac{\lambda_{\ell m,r}}{\Lambda_\ell(\theta)} - 1\right| \leq \varepsilon_{K,r}(L), \qquad \left|\frac{A_{q,r}}{A_q} - 1\right| \leq \varepsilon_{A,r}, \qquad \varepsilon_{K,r} < 1.$$

Then $C_{\ell m,r}^q = A_{q,r}/\lambda_{\ell m,r}$ satisfies

$$\left|\frac{C_{\ell m,r}^q}{A_q/\Lambda_\ell(\theta)} - 1\right| \leq \frac{\varepsilon_{A,r} + \varepsilon_{K,r}}{1 - \varepsilon_{K,r}}.$$

Proof. Write $A_{q,r} = A_q(1 + a_r)$ and $\lambda_{\ell m,r} = \Lambda_\ell(1 + b_{\ell m,r})$. Then

$$\frac{C_{\ell m,r}^q}{A_q/\Lambda_\ell} = \frac{1 + a_r}{1 + b_{\ell m,r}},$$

and the displayed bound follows from $|a_r| \leq \varepsilon_A$, $|b_{\ell m,r}| \leq \varepsilon_K$. □

Theorem 2.12 (refinement-semigroup tilt)**.** *Let R_b be the scalar refinement map for scale ratio $b > 1$. If one isolated scalar covariance mode has positive eigenvalue $\lambda(b)$, $\lambda(b_1 b_2) = \lambda(b_1)\lambda(b_2)$, and λ is continuous, then there is a unique real θ with*

$$\lambda(b) = b^{-\theta}, \qquad \theta = -\frac{\log \lambda(b)}{\log b}.$$

Proof. Set $g(t) = -\log \lambda(e^t)$. The semigroup law gives $g(t+s) = g(t) + g(s)$. Continuity makes $g(t) = \theta t$, hence the result. □

Definition 2.13 (edge-center reserve generator receipt)**.** *Write $s = \log b$ for logarithmic refinement thickness. Let $u_{\mathrm{full}}(s)$ be the scalar-conditioned covariance-survival cocycle across a full oriented collar, and let $u_q(s)$ be its source-facing half-collar restriction. The receipt requires*

$$u(s+t) = u(s)u(t), \qquad u(0) = 1,$$

strong continuity at zero, the source-derived full-collar density

$$-u'_{\mathrm{full}}(0) = \frac{P_\star}{24},$$

and the orientation-reversal coarea identity

$$-u_q'(0) = \frac{1}{2}[-u_{\mathrm{full}}'(0)] = \frac{P_\star}{48}.$$

These are infinitesimal generator statements. A one-step survival probability has exponent $-\log u_q(\log b)/\log b$.

Theorem 2.14 (edge-center tilt and repair-clock reconciliation)**.** *Under Definition 2.13,*

$$u_q(s) = e^{-\theta s}, \qquad \theta = \frac{P_\star}{48}, \qquad n_s = 1 - \frac{P_\star}{48}.$$

In the coordinate $\theta = \kappa_{\mathrm{rep}}(P_\star - \varphi)$*, the same branch has*

$$\kappa_{\mathrm{rep}}^{\mathrm{edge}} = \frac{P_\star}{48(P_\star - \varphi)}.$$

The value e *is a separate diagnostic hypothesis unless an additional identity equates it with this source-derived coordinate.*

Proof. For $g(s) = -\log u_q(s)$, the cocycle law gives the continuous Cauchy equation. Hence $g(s) = \theta s$. Differentiation at zero and the half-collar identity give $\theta = P_\star/48$. The formula for $\kappa_{\mathrm{rep}}^{\mathrm{edge}}$ is algebraic. □

Theorem 2.15 (homogeneous radial source family)**.** *Assume source-stress closure, a single clock, freezeout, multicenter consistency, and translation/rotation covariance. Let* $C_\zeta = M_{\Delta_\zeta^2}$ *on the common physical* $d\ln k$ *mode basis. If a scale-natural source embedding transports refinement to physical dilation and its finite operator residual converges to*

$$D_s^{-1} C_\zeta D_s = e^{-\theta s} C_\zeta, \qquad (D_s f)(k, \hat{k}) = f(e^{-s}k, \hat{k}),$$

then every positive measurable representative satisfies

$$\Delta_\zeta^2(k) = A_\zeta (k/k_\star)^{-\theta}$$

almost everywhere. Continuity gives the pointwise identity. A single shell has an infinite-dimensional radial kernel, including positive ambiguities; complete radial cross-covariances give the independent spherical-Hankel tomography route.

Proof. Conjugation of the multiplication operator gives $\Delta_\zeta^2(e^s k) = e^{-\theta s}\Delta_\zeta^2(k)$ almost everywhere. In logarithmic coordinates, rational-translation invariance on a common full-measure set forces $e^{\theta t}\Delta_\zeta^2(e^t)$ to be constant almost everywhere. The one-shell and tomography statements follow from the correlation-restriction and spherical-Hankel theorems in the primordial bridge packet. □

Theorem 2.16 (thin-shell power-law lift)**.** *Assume* $q(\hat{n}) = Z_\star \Pi_{\ell \geq 2} \zeta_\star(R_\star \hat{n})$ *and*

$$\Delta_\zeta^2(k) = A_\zeta (k/k_\star)^{-\theta}.$$

Then

$$C_\ell^q = 4\pi Z_\star^2 \int d\ln k\, \Delta_\zeta^2(k) j_\ell^2(kR_\star),$$

and, for $-2 < \theta < 4$*,*

$$A_q = \pi^{3/2} Z_\star^2 A_\zeta (k_\star R_\star)^\theta \frac{\Gamma(1+\theta/2)}{\Gamma(3/2+\theta/2)}.$$

Thus

$$A_\zeta = \frac{A_q}{\pi^{3/2} Z_\star^2 (k_\star R_\star)^\theta} \frac{\Gamma(3/2+\theta/2)}{\Gamma(1+\theta/2)}.$$

Proposition 2.17 (finite-window bound)**.** *Let* $\Psi_\ell(k) = \int dr\, W(r) j_\ell(kr)$, $W \geq 0$, $\int W dr = 1$, *with mean* $R_\star$ *and variance* σ_R^2. *If*

$$\delta_\ell(k) = \frac{k^2 \sigma_R^2}{2} \sup_{r \in \operatorname{supp} W} |j_\ell''(kr)|,$$

then

$$|C_{\ell,W}^q - C_{\ell,\mathrm{shell}}^q| \leq 4\pi Z_\star^2 \int d\ln k\, \Delta_\zeta^2(k) \left[2\delta_\ell(k) + \delta_\ell(k)^2\right].$$

Proposition 2.18 (radial non-identifiability)**.** *For a finite radial basis,* $C = Ap$. *If* $A \in \mathbb{R}^{N_\ell \times N_k}$ *has rank* r, *then* $\dim \ker A = N_k - r$, *and every* $p + v$ *with* $v \in \ker A$ *gives the same screen spectrum. A physical radial reconstruction therefore requires either a source theorem restricting* p, *or a declared prior with a published null basis, resolution kernels, positivity checks, and prior-sensitivity report.*

Theorem 2.19 (promotion contract for the thin-shell power-law pipeline)**.** *Under the declared thin-shell lift and power-law source family above, one hash-locked source DAG may emit*

$$(q_r, S_{\mathrm{scr},r}, \theta, A_{q,r}, C_{\ell,r}^q, A_{\zeta,r}, \Delta_{\zeta,r}^2)$$

as a conditional primordial packet when the geometric-scalar, collar-law, release-energy, reserve-generator, conformal-precision, refinement, source-stress, single-clock, freezeout, adiabaticity, isocurvature, phase-coherence, thin-shell, radial-null, finite-window, and forward- residual receipts pass on that DAG. Any measurement, likelihood, fitted parameter, or measurement-calibrated ancestor fails the source packet. Physical temperature and polarization spectra require the independent Boltzmann, recombination, nuisance, covariance, and likelihood gates. This contract is specific to the displayed pipeline. A different source-only radial reconstruction or forward projection may use another fully typed contract, provided it certifies its own injectivity or null space, window and convergence control, forward residual, and no-target-leak provenance.

Target 2.20 (requirements for a source-derived screen spectrum)**.** *A concrete source-derived spectrum requires the following receipt families before primordial interpretation:*

1. *geometry, scalar quotient, low-mode projector, scalar precision, and operator normalization;*
2. *quotient ensemble selection, scalar release energy, and MaxEnt screen covariance;*
3. *the infinitesimal reserve generator, scalar refinement tilt, and angular screen spectrum with a finite error budget;*
4. *source-stress, clock, freezeout, radial-window, radial-null, forward-residual, and transfer firewall receipts.*

The theorem and algorithm packet is complete for the declared thin-shell power-law pipeline. No finite source DAG that passes this receipt set is supplied, and the list is not a no-go for a differently typed radial pipeline.

2.0.1 Source-only primordial bridge theorem

The finite screen covariance theorem is a screen theorem. To read it as a source-only primordial curvature spectrum one must add the following bridge objects and receipts. The dependency chain is

$$\boxed{\begin{aligned}&\mathrm{nf}(\mathcal{Q}_r) \to (\chi_r, u_r, h_r) \to \mathsf{BackgroundCurvatureStatus}_r \to (g, T_I) \to q_\zeta = \zeta_\rho \\ &\qquad \to \text{single clock} \to \text{freeze-out} \to \text{coherent growing mode} \to \Delta_\zeta^2(k).\end{aligned}}$$

No successful fit of a screen C_ℓ or a TT spectrum may replace one of these hypotheses.

Definition (background curvature branch label). The primordial bridge carries a finite record

$$\mathsf{BackgroundCurvatureStatus}_r = (\mathsf{BranchType}, \kappa, I_K, I_{\Omega_K}, \mathsf{Basis}, \mathsf{Top})$$

where

$$\mathsf{BranchType} \in \{\textsc{FlatExact}, \textsc{FlatAssumed}, \textsc{OpenCurved}, \textsc{ClosedCurved}, \textsc{Unresolved}\}.$$

FlatExact is allowed only after a direct theorem or conditional CMH receipt. FlatAssumed may run flat kernels but its outputs remain assumption-conditioned. Unresolved blocks physical interpretation. The H^3 observer-facing atlas is not this record; this record comes from the clocked FLRW slice and its spatial Levi–Civita holonomy or declared branch input.

Definition (source stress readout). On a reconstructed branch with relational metric g_{ab}, let

$$S_{\mathrm{src}} = \sum_I S_I[g, \Psi_I], \qquad T^I_{ab} = -\frac{2}{\sqrt{-g}} \frac{\delta S_I}{\delta g^{ab}}, \qquad T^{\mathrm{tot}}_{ab} = \sum_I T^I_{ab},$$

where I ranges over every active sector, including ordinary matter, radiation, neutrinos, OPH anomaly or repair sectors, and any auxiliary source kept in the run. At finite regulator this is a quotient-visible map

$$\mathsf{StressRead}_r : \mathrm{nf}(\mathcal{Q}_r) \longrightarrow \{T^I_{r,ab}\}_I.$$

All source stress components must be read from the same global normal form as the collar scalar.

Source-stress closure theorem. Assume each S_I is relationally diffeomorphism invariant, the source equations hold, sector exchanges satisfy

$$\nabla_a T^{ab}_I = Q^b_I, \qquad \sum_I Q^b_I = 0,$$

and $\mathsf{StressRead}_r$ descends to the physical quotient and converges. Then

$$\nabla_a T^{ab}_{\mathrm{tot}} = 0.$$

On the recovered Einstein branch,

$$G_{ab} + \Lambda g_{ab} = 8\pi G\, T^{\mathrm{tot}}_{ab}.$$

The evidence bundle must also report the non-circular comparison

$$E^{ab}_T = T^{ab}_{\mathrm{src}} - \frac{G^{ab} + \Lambda g^{ab}}{8\pi G};$$

using the geometric right-hand side alone is a geometry diagnostic, not a source-only primordial receipt.

Total-energy frame and curvature covector. When $T^a{}_b$ has a unique future timelike unit eigenvector u^a, define

$$T^a{}_b u^b = -\rho u^a, \qquad h_{ab} = g_{ab} + u_a u_b,$$

and decompose

$$T^{ab} = \rho u^a u^b + p h^{ab} + \pi^{ab}, \qquad u_a \pi^{ab} = 0, \qquad \pi^a{}_a = 0.$$

Let $\Theta = \nabla_a u^a$ and let σ_{ab} be the shear. Total energy conservation gives

$$\dot\rho + \Theta(\rho + p) + \sigma_{ab}\pi^{ab} = 0, \qquad \dot f := u^a \nabla_a f.$$

On an expanding branch, define

$$p_{\mathrm{eff}} = p + \frac{\sigma_{ab}\pi^{ab}}{\Theta};$$

then $\dot{\rho} + \Theta(\rho + p_{\mathrm{eff}}) = 0$.

Theorem (exact total-curvature evolution). Work on an open flow domain on which $\Theta \neq 0$, $\dot{\rho} \neq 0$, and $\rho + p_{\mathrm{eff}} \neq 0$, so every quotient below is defined. Let

$$\dot{\alpha} = \Theta/3, \qquad \zeta_a = \nabla_a \alpha - \frac{\dot{\alpha}}{\dot{\rho}} \nabla_a \rho,$$

and

$$\Gamma_a^{\mathrm{eff}} = \nabla_a p_{\mathrm{eff}} - \frac{\dot{p}_{\mathrm{eff}}}{\dot{\rho}} \nabla_a \rho.$$

Then, exactly and without a long-wavelength approximation,

$$\mathcal{L}_u \zeta_a = -\frac{\Theta}{3(\rho + p_{\mathrm{eff}})} \Gamma_a^{\mathrm{eff}}.$$

Hence a barotropic effective source, $p_{\mathrm{eff}} = p_{\mathrm{eff}}(\rho)$, conserves ζ_a.

Collar scalar corollary. Assume that the quotient record emits a connected uniform-total-density cut and a collar-adapted spatial metric

$$h = \lambda^2 dr^2 + \sigma_{AB}(dx^A + \beta^A dr)(dx^B + \beta^B dr), \qquad \lambda > 0, \qquad \sigma_{AB} \succ 0.$$

The block-determinant identity gives

$$\sqrt{\det h} = \lambda \sqrt{\det \sigma_{AB}}.$$

The native readout premise is that J_X and its frozen reference are these spatial volume densities, with the same reference-density convention. The formula $J_X = \lambda\sqrt{\det \sigma_{AB}}$ without this metric attachment does not establish that premise. For relational rods Lie-dragged by the total-energy flow, the attached volume Jacobian satisfies

$$\frac{d}{d\tau} \frac{1}{3} \log J_X = \frac{\Theta}{3}.$$

Relative to a frozen background,

$$q_\zeta = \frac{1}{3} \log \frac{J_X}{\bar{J}_X}.$$

On the uniform-total-density cut,

$$q_\zeta = \zeta_\rho + \text{constant}.$$

After removing the monopole and dipole,

$$q = \Pi_{\ell \geq 2} q_\zeta = \Pi_{\ell \geq 2} \zeta_\rho, \qquad Z_q = 1.$$

This is the source-only quotient scalar on the OPH screen; by itself it has type ScreenCurvature, not PrimordialCurvature.

Definition (rank-one single-clock branch). Let

$$\mathcal{F}(x) = (\rho, p_{\mathrm{eff}}, \{\rho_I, p_I, Q_I\}_{I=1}^N)$$

be the scalar source map on the source moduli space. A branch is rank-one single-clock when $\operatorname{rank} d\mathcal{F} = 1$, $d\rho \neq 0$, every level set of ρ is connected, all sector velocities agree with u^a, and orthogonal momentum transfer vanishes. An approximate single-clock branch may instead carry an explicit interval bound on that transfer, but the bound must then be propagated into the transfer and curvature-evolution residuals. At finite cutoff this requires a quotient-visible clock χ_r and a factorization

$$\mathcal{F}_r = \widehat{\mathcal{F}}_r \circ \chi_r.$$

Theorem (single-clock adiabaticity). On an exact rank-one single-clock branch, and for the dynamically nondegenerate sectors I, J under discussion with $\dot{\rho}_I \neq 0$ and $\dot{\rho}_J \neq 0$,

$$p_{\text{eff}} = p_{\text{eff}}(\rho), \qquad S_a^{IJ} = 3(\zeta_a^I - \zeta_a^J) = 0,$$

and the intrinsic nonadiabatic pressure and transfer covectors obey

$$\Gamma_a^I = 0, \qquad \Xi_a^I = 0.$$

On the nonvanishing domain of the total-curvature theorem, therefore $\mathcal{L}_u \zeta_a = 0$. The proof is the chain rule through the single clock: every source scalar is a function of the same χ, and connected ρ-fibers remove residual branch labels.

Theorem (entropy repair gap). Let the linearized scalar repair/evolution block be

$$\delta Y_{n+1} = L_r \delta Y_n + \eta_n,$$

and let $P_\perp$ project away gauge directions, constrained zero modes, and the adiabatic growing direction. If there is a positive metric H_r such that

$$\|P_\perp L_r P_\perp\|_{H_r} \leq e^{-\gamma_r} < 1,$$

and the projected sector is not sourced by the complementary block,

$$P_\perp L_r = P_\perp L_r P_\perp,$$

then

$$\|P_\perp \delta Y_n\|_{H_r} \leq e^{-n\gamma_r} \|P_\perp \delta Y_0\|_{H_r} + \sum_{j=0}^{n-1} e^{-(n-1-j)\gamma_r} \|P_\perp \eta_j\|_{H_r}.$$

In the unforced case the orthogonal entropy, decaying, and repair scalar modes decay exponentially and converge to zero; they need not vanish after a finite number of steps. If the block-invariance identity is absent, the term $P_\perp L_r (1 - P_\perp) \delta Y_n$ must instead be included in the forcing bound. Eigenvalues alone do not certify this theorem for nonnormal repair matrices; a norm bound or a discrete Lyapunov inequality

$$L_\perp^\mathsf{T} H L_\perp \preceq e^{-2\gamma} H$$

is required.

Theorem (exact and approximate growing-mode coherence). Let $X(k)$ collect the scalar initial-condition variables for curvature, radiation, baryons, CDM/anomaly, neutrinos, and repair-sector scalars, and let $v_+(k)$ be the normalized adiabatic growing vector. Assume the single-clock and repair-gap theorems hold through readout and the growing/orthogonal splitting is invariant. If the initial orthogonal component is zero and no forcing or independent stochastic source has an orthogonal component, then

$$X(k) = v_+(k)\zeta(k), \qquad \mathcal{P}_X(k) = \mathcal{P}_\zeta(k) v_+(k) v_+(k)^\dagger.$$

Thus rank $\mathcal{P}_X(k) = 1$, source-side isocurvature fractions vanish, and all species share a deterministic growing-mode phase. Without those exact zero-component hypotheses, the repair-gap estimate gives only

$$X_n(k) = v_+(k)\zeta_n(k) + \varepsilon_{\perp,n}(k),$$

with $\varepsilon_{\perp,n}$ bounded by the preceding geometric-series inequality; strict contraction alone does not make it zero at finite readout. Finite runs must report λ_2/λ_1, $\beta_{\mathrm{iso}}(k)$, decaying-mode fraction, and phase-convention defects as source diagnostics.

Theorem (screen-to-primordial bridge). Assume the geometric record and causal-area metric receipts, the collar identity $q_\zeta = \zeta_\rho$, source-stress closure, rank-one normal form, positive repair gap, freeze-out, adiabatic growing-mode, isocurvature, and phase-coherence receipts all pass. Also assume the physical scale-bridge theorem: a source embedding with `source_embedding_hash`, a physical mode basis with `physical_mode_basis_id`, a calibrated comoving coordinate χ, mode-normalization and density-of-states receipts, and a no-post-hoc-calibration receipt, and a $\mathsf{BackgroundCurvatureStatus}_r$ that is not UNRESOLVED. Then the low-mode-removed OPH screen scalar is the source-only primordial curvature field restricted to the declared shell or radial window:

$$q(\hat{n}) = \Pi_{\ell_{\mathrm{src}}\geq 2} \int d\chi\, W(\chi)\zeta_\star(\chi\hat{n}).$$

For a thin shell, $q(\hat{n}) = \Pi_{\ell_{\mathrm{src}}\geq 2}\zeta_\star(R_\star\hat{n})$. The forward spectrum is exactly

$$C_\ell^q = 4\pi \int_0^\infty d\ln k\, \Delta_\zeta^2(k) |\Psi_\ell^{\mathsf{BranchType}}(k)|^2,$$

$$\Psi_\ell^{\mathsf{BranchType}}(k) = \begin{cases} \int dr\, W(r) j_\ell(kr), & \mathsf{BranchType} = \text{FLATEXACT or FLATASSUMED}, \\ \int dr\, W(r)\Phi_{\ell k}^{(-)}(r), & \mathsf{BranchType} = \text{OPENCURVED}, \\ \int dr\, W(r)\Phi_{\ell k}^{(+)}(r), & \mathsf{BranchType} = \text{CLOSEDCURVED}, \end{cases}$$

and for a thin shell

$$C_\ell^q = 4\pi \int_0^\infty d\ln k\, \Delta_\zeta^2(k) \left|\Psi_\ell^{\mathsf{BranchType}}(k; R_\star)\right|^2.$$

At finite regulator the continuum integral is represented by the finite spectral measure

$$d\nu_r(k) = \sum_j w_{rj}\delta(k - k_{rj}),$$

so the actual finite evidence receipt is

$$C_{\ell_{\mathrm{src}},r}^q = 4\pi \sum_j w_{rj}^{\log k}\, \Delta_{\zeta,r}^2(k_{rj}) |\Psi_{\ell_{\mathrm{src}},r}^{\mathsf{BranchType}}(k_{rj})|^2.$$

Every radial node, window, and mode bin is bound to the geometry hash, source-embedding hash, scale-certificate hash, boundary-condition hash, and mode-basis identifier, and the evidence bundle must emit radial kernel geometry compatibility receipt. The observed multipole L_{CMB} is not ℓ_{src}; it is the output index of the line-of-sight transfer solver. Cap eigenmodes, inverse cap-opening angles, and relabeled unit strings remain diagnostic by the dimensional, angular, cap-mode, and unit-string no-go propositions.

Physical scale dependency. The k_{rj} grid in the preceding equation is not a free producer field. It is consumed only after

physical spatial k receipt

recomputes the comoving k-intervals from the physical geometry, scale certificate, scale factor, finite eigenvalues, normalization, and refinement residuals. The radial kernel and source-screen projection additionally require

screen to physical k association receipt.

The source covariance is a finite positive operator on the common physical mode projectors:

$$C_{\zeta,r} \succeq 0.$$

Homogeneity and isotropy on the safe band require

$$P_I C_{\zeta,r} P_I = \mathcal{P}_\zeta(k_I) P_I + E_I$$

with a declared residual bound, together with an off-diagonal leakage report

$$\sum_{I \neq J} |P_I C_{\zeta,r} P_J|^2.$$

The same physical projector family is used by primordial covariance and anomaly response:

common primordial anomaly mode basis receipt.

Mode freezeout and common initial surface are separate gates:

physical mode freezeout map receipt, *physical freezeout surface receipt.*

The latter contains initial data and normal derivatives on one common spacelike surface.

Radial prior, null space, and residual receipt. The map above is a forward projection, not a free inversion. With finite bins p_j for Δ^2_ζ,

$$A_{\ell j} = 4\pi \int_{\text{bin } j} d\ln k\, b_j(k) |\Psi_\ell(k)|^2, \qquad C^q = Ap.$$

The evidence bundle must declare the basis, support, prior p_0, Q, positivity conditions, singular spectrum of $AQ^{-1/2}$, effective rank, null basis, resolution kernels, prior sensitivity, and every forward residual

$$r_\ell = C^q_\ell - \widehat{C}_\ell.$$

A finite prior, regularizer, positivity constraint, or square discretization selects one representative from a nonunique shell equivalence class. Its typed output is ConditionalRadialContinuation. A source-derived primordial packet using either radial route certified below requires the common source, scale, mode, null-space, positivity, ancestry, and non-fitting forward-residual receipts together with that route's exact uniqueness certificate. Another source-restricted injective construction may be used if it supplies its own orbit-separation theorem and the same physical and no-target-leak interfaces. If the geometry packet is imported, the result is *conditional physical*; a *source-native physical* result requires the native $\mathsf{CosmoGeomRead}_r$ theorem.

Complete radial-lift theorem packet. The screen-to-radial map is a forward restriction with a nonunique unrestricted inverse. For a flat source window W and geometric normalization Z_q,

$$C^q_{\ell,W} = 4\pi Z_q^2 \int_0^\infty d\ln k\, \Delta^2_\zeta(k) |\Psi_{\ell,W}(k)|^2, \qquad \Psi_{\ell,W}(k) = \int W(dr) j_\ell(kr).$$

The packet below certifies sufficient uniqueness routes through radial dilation, radial tomography and common-bulk consistency on unbounded radii. They are not asserted to be exhaustive or mutually exclusive. A source-derived parametric or dynamical law, an injective source-restricted basis, analytic or PDE constraints, or another fully typed orbit-separating construction remains admissible. A finite prior continuation has the type ConditionalRadialContinuation.

Theorem (one-shell correlation restriction and exact no-go). For a thin shell of radius R, the complete angular covariance satisfies

$$K_R(\mu) = Z_q^2 \xi_\zeta\Big(R\sqrt{2-2\mu}\Big) = \sum_{\ell\geq 0} \frac{2\ell+1}{4\pi} C_{\ell,R}^q P_\ell(\mu).$$

Hence even the noiseless sequence $\{C_{\ell,R}^q\}_{\ell\geq 0}$ determines the three-dimensional correlation only on $0 \leq s \leq 2R$. The full one-shell operator has an infinite-dimensional kernel, including positivity-preserving ambiguities. Explicitly, for any nonzero $g \in C_c^\infty((2R,\infty))$,

$$h_g(k) = \frac{2k^3}{\pi} \int_0^\infty r^2 g(r) j_0(kr)\, dr$$

has correlation perturbation $\xi_{h_g} = g$ and therefore changes no shell multipole. Choosing $B = |h_g| + w$ with any strictly positive integrable w makes B and $B + h_g$ two distinct positive spectra with the same complete shell covariance.

Proof. The spherical-Bessel addition theorem gives the displayed restriction identity. The map $\mu \mapsto R\sqrt{2-2\mu}$ covers exactly $[0, 2R]$. Radial Fourier inversion gives $\xi_{h_g} = g$ and is injective, so the kernel contains an infinite-dimensional copy of $C_c^\infty((2R,\infty))$. The positive pair differs by the same null element. □

Corollary (exact low-mode-removed kernel). For a continuous signed correlation perturbation ξ_h, invisibility to every shell multipole $\ell \geq 0$ is equivalent to $\xi_h(s) = 0$ on $0 \leq s \leq 2R$. Invisibility to every retained OPH multipole $\ell \geq 2$ is equivalent to

$$\xi_h(s) = a + b\left(1 - \frac{s^2}{2R^2}\right), \qquad 0 \leq s \leq 2R,$$

for constants a, b. Legendre completeness gives the first statement. Removing $\ell = 0, 1$ leaves the span of P_0 and P_1 invisible, which gives the second.

Theorem (one common bulk on unbounded round radii). Assume one homogeneous isotropic generalized field on a common Euclidean three-dimensional time slice, with well-defined spherical restrictions. Its covariance is represented off the diagonal by a locally integrable radial function $\xi(s)$, with no additional contact distributions. Suppose an unbounded sequence R_j of round radii has

$$C_\ell(R_j) = \frac{2\pi A_j}{\ell(\ell+1)}, \qquad \ell \geq 1, \quad A_j > 0,$$

in one common readout normalization. Then all $A_j = A$, and

$$\xi(s) = -A \log s + c \quad \text{for almost every } s > 0, \qquad P_3(k) = \frac{2\pi^2 A}{k^3} \quad (k > 0).$$

The ideal field is understood modulo constants, on smooth mean-zero test observables; it does not have finite point variance. The covariance identity is distributional, with the displayed logarithm as a chosen off-diagonal representative. Gaussianity is not needed for the covariance conclusion. If the hypotheses concern only $\ell \geq 2$, the residual freedom is $c + ds^2$, representing unobserved zero-wave-number or affine data.

Proof. The identity $\int_0^\infty j_\ell(x)^2\,dx/x = 1/[2\ell(\ell+1)]$, or Legendre expansion of the logarithm, identifies each mean-zero spherical kernel. Chord restriction and completeness give $\xi(s) = -A_j \log s + c_j$ almost everywhere on $0 < s < 2R_j$. Every two intervals overlap. Linear independence of $\log s$ and 1 forces their coefficients to agree. Unbounded radii cover all positive separations; radial Fourier uniqueness gives the stated spectrum away from zero. With the dipole removed, use independence of $\log s, 1, s^2$ instead. Contact terms are excluded by hypothesis, so off-diagonal equality does not hide an extra diagonal distribution. □

Separate screen marginals do not establish the common-bulk premise. The metrics, time slice, normalization and cross-screen records must be jointly compatible. For finitely many radii, unobserved separations allow the preceding null-space perturbations; a finite run cannot invoke unbounded coverage.

Example (positive alternatives to the scale-invariant baseline). For a unit-radius screen set

$$h(s) = (s-3)^2(4-s)^2\mathbf{1}_{[3,4]}(s), \qquad H(k) = 4\pi\int_3^4 s^2 h(s)\frac{\sin(ks)}{ks}\,ds.$$

Writing $g(s) = sh(s)$, both g and g' vanish at the support endpoints. Two integrations by parts and $\int_3^4 |g''(s)|\,ds \le 30$ imply $|k^3 H(k)| \le 120\pi$. Hence

$$P_\pm(k) = \frac{2\pi^2 A}{k^3} \pm \frac{\pi A}{120}H(k) \;\ge\; \frac{\pi^2 A}{k^3} > 0.$$

Both positive generalized Gaussian covariances restrict to exactly the same law as the baseline on the unit sphere, since $h = 0$ for all its chord separations. Their bulk covariances differ. This exhibits ambiguity around the scale-invariant baseline itself, without permitting a negative spectrum. The analytic bound proves positivity for all k; numerical quadrature is only a check. The source-pinned radial controls under `evidence/observer_dynamics_20260925/sim-analysis/codex/necessity/boundary_bulk/` check the forward integrals and this perturbation with analytic tail bounds, not formal interval arithmetic.

Producer theorem (source-refinement orbit to physical dilation). Let $\mathcal{H}_r^{\mathrm{src}}$ be the scale-labelled scalar mode space generated by the cofinal refinement orbit of the released screen, including its scale label. A single angular cut is insufficient. Let U_r map the orbit unitarily onto the physical rank-one adiabatic source subspace. If

$$D_{s,r}U_r = U_r R_{s,r}, \qquad C_{\zeta,r} = U_r C_{\mathrm{src},r} U_r^*,$$

then

$$D_{s,r}^{-1}C_{\zeta,r}D_{s,r} - u_r(s)C_{\zeta,r} = U_r\left(R_{s,r}^{-1}C_{\mathrm{src},r}R_{s,r} - u_r(s)C_{\mathrm{src},r}\right)U_r^*.$$

Thus the source and physical operator residuals have equal norm. For a bounded-operator limit on a common Hilbert space, strong covariance convergence with a uniform norm bound, strong convergence of the dilation maps and their inverses, $u_r(s) \to u(s)$, and vanishing norm residual imply $D_s^{-1}C_\zeta D_s = u(s)C_\zeta$ in the continuum. This sufficient criterion does not cover a nonzero tilted power law on all $0 < k < \infty$: unitary conjugation preserves the norm of a bounded operator, so $\|C_\zeta\| = |u(s)|\|C_\zeta\|$ would force $u(s) = 1$ for positive $u(s)$ and $C_\zeta \neq 0$.

For the global power-law branch, use instead a local limit receipt. After embedding on the common physical mode space, let $\mathcal{D}$ be the dense space of L^2 functions supported in compact $\ln k$ bands, and let D_s be the exact physical dilation, preserving $\mathcal{D}$. Require $C_{\zeta,r}f \to C_\zeta f$ for every $f \in \mathcal{D}$ and

$$(D_s^{-1}C_{\zeta,r}D_s - u_r(s)C_{\zeta,r})f \longrightarrow 0 \qquad (f \in \mathcal{D}).$$

If finite dilation maps are used to establish the latter condition, their replacement by D_s must also have vanishing error on this core. The resulting identity holds on $\mathcal{D}$, with a possibly unbounded covariance operator; it does not assert a global uniform covariance norm bound. The commuting square states that source refinement followed by the physical embedding agrees with physical embedding followed by wavelength dilation, and it transfers the source covariance error exactly to the physical covariance.

Proof. Conjugate the source covariance by U_r and use the commuting square; unitary invariance gives the finite residual identity. For the bounded limit, strong convergence and uniform boundedness allow passage through the products. For the local limit, both f and $D_s f$ belong to $\mathcal{D}$, so pass to the limit in each of the two terms using the fixed bounded unitary D_s^{-1}. □

Theorem (source-dilation uniqueness). Let Δ_ζ^2 be a nonnegative measurable radial function, locally essentially bounded in $\ln k$, and nonzero on a set of positive measure. Let $C_\zeta = M_{\Delta_\zeta^2}$ on the common physical $L^2(d\ln k\, d\Omega_k)$ mode basis, with its natural multiplication-operator domain. The operator may be unbounded; the dilation identity below is required on the common compact-band core $\mathcal{D}$. Define

$$(D_s f)(k,\hat{k}) = f(e^{-s}k,\hat{k}).$$

If the source refinement packet proves, for all real s,t and positive $u_q(s)$,

$$D_s^{-1} C_\zeta D_s = u_q(s) C_\zeta, \qquad u_q(s+t) = u_q(s)u_q(t), \qquad -\frac{d}{ds}\ln u_q(s)\Big|_{s=0} = \theta,$$

then $u_q(s) = e^{-\theta s}$ and there is $A > 0$ such that

$$\Delta_\zeta^2(k) = Ak^{-\theta} \quad \text{almost everywhere.}$$

For a continuous source representative and pivot $k_\star$,

$$\boxed{\Delta_\zeta^2(k) = A_\zeta (k/k_\star)^{-\theta}}, \qquad A_\zeta = \Delta_\zeta^2(k_\star).$$

Proof. The stated logarithmic derivative at zero gives continuity there and solves the positive multiplicative Cauchy equation for u_q. Conjugation of the multiplication operator gives $\Delta_\zeta^2(e^s k) = e^{-\theta s}\Delta_\zeta^2(k)$ almost everywhere for each s. In logarithmic coordinates, $e^{\theta t}\Delta_\zeta^2(e^t)$ is invariant under all rational translations on one common full-measure set; bounded-transform convolution makes it constant almost everywhere. Nonnegativity and the nonzero premise make that constant strictly positive. Continuity upgrades the identity to every k. The compact-band functions form a core of the stated locally bounded multiplication operator, so the identity also identifies its natural domains. □

Theorem (exact flat thin-shell lift). On the source-dilation branch, for $-2 < \theta < 4$ and $\ell \geq 2$,

$$\int_0^\infty \frac{dx}{x} x^{-\theta} j_\ell^2(x) = \frac{\sqrt{\pi}}{4}\frac{\Gamma(1+\theta/2)}{\Gamma(3/2+\theta/2)}\frac{\Gamma(\ell-\theta/2)}{\Gamma(\ell+2+\theta/2)},$$

so

$$C_{\ell,R_\star}^q = A_q \frac{\Gamma(\ell-\theta/2)}{\Gamma(\ell+2+\theta/2)},$$

with the complete amplitude conversion

$$\boxed{A_\zeta = \frac{A_q}{\pi^{3/2} Z_q^2 (k_\star R_\star)^\theta}\frac{\Gamma(3/2+\theta/2)}{\Gamma(1+\theta/2)}.}$$

Proof. Insert the source power law into the exact forward map, set $x = kR_\star$, and apply the Weber–Schafheitlin square integral. The convergence strip follows from $j_\ell(x) \sim x^\ell/(2\ell+1)!!$ at zero and $j_\ell(x) = O(x^{-1})$ at infinity. □

At the source-native pivot $k_\star R_\star = 1$ with $Z_q = 1$ and the source-law tilt θ, the conversion factor evaluates to approximately 0.16080676. It is an arithmetic consequence of the symbolic relation and supplies no source amplitude by itself.

Corollary (source-family injectivity and multipole consistency). Fix $\theta, Z_q, k_\star$, and a source window for which the forward coefficient is positive and finite. The forward map is injective in A_ζ on the family $\Delta_\zeta^2 = A_\zeta(k/k_\star)^{-\theta}$. For each retained multipole,

$$A_\zeta^{(\ell,W)} = \frac{C_{\ell,W}^q}{4\pi Z_q^2 k_\star^\theta \int d\ln k\, k^{-\theta}|\Psi_{\ell,W}(k)|^2}.$$

Every $A_\zeta^{(\ell,W)}$ must agree. A disagreement falsifies the declared window/dilation packet and cannot be repaired with an ℓ-dependent amplitude. Once the source law fixes θ, the screen has one amplitude left to determine, and the unused multipoles become consistency checks.

Theorem (finite-window bound). Let $\ell \geq 1$ be an integer, $R_\star, k_\star > 0$, $A_\zeta > 0$, and let W be a nonnegative probability measure on positive radii with finite $\theta/2$ moment. For $0 < \theta < 2\ell$, define

$$I_\ell(\theta) = \int d\ln x\, x^{-\theta} j_\ell^2(x), \quad J_\ell(\theta) = I_\ell(\theta-2) - \left[\ell(\ell+1) - \frac{\theta(\theta+1)}{2}\right] I_\ell(\theta),$$

or, equivalently by gamma recurrence,

$$J_\ell(\theta) = I_\ell(\theta)\left[\frac{\ell(\ell+1)}{\theta} + \frac{(\theta+1)(\theta-2)}{4}\right].$$

$$\eta_{\ell,W} = \frac{2\sqrt{J_\ell(\theta)}}{\theta} \int W(dr)|r^{\theta/2} - R_\star^{\theta/2}|.$$

Then

$$|C_{\ell,W}^q - C_{\ell,R_\star}^q| \leq 4\pi Z_q^2 A_\zeta k_\star^\theta \eta_{\ell,W}\left(2R_\star^{\theta/2}\sqrt{I_\ell(\theta)} + \eta_{\ell,W}\right).$$

Proof. In the Hilbert norm $\|f\|_\theta^2 = \int d\ln k\, k^{-\theta}|f(k)|^2$, $\|j_\ell(kr)\|_\theta = r^{\theta/2}\sqrt{I_\ell}$ and $\|\partial_r j_\ell(kr)\|_\theta = r^{\theta/2-1}\sqrt{J_\ell}$. The fundamental theorem of calculus, Minkowski, and $|\|f\|^2 - \|g\|^2| \leq \|f-g\|(\|f\|+\|g\|)$ give the bound. □

Normalization and positivity of W are premises: for example, $W = 2\delta_{R_\star}$ gives $\eta_{\ell,W} = 0$ while multiplying the shell spectrum by four. A signed window would require its total variation in the norm estimate.

The reference implementation evaluates this analytic bound in binary64; it does not provide an outward-rounded interval certificate. Near $\theta = 0$ it uses the recurrence form of J_ℓ and evaluates the radius difference as $R_\star^{\theta/2}\,\mathrm{expm1}[(\theta/2)\ln(r/R_\star)]$. Direct subtraction of rounded powers can incorrectly return a zero bound for a nontrivial window. Numerical promotion requires a separate rounding-error budget in addition to the exact inequality.

Theorem (finite singular-value and prior continuation). For a declared finite radial basis, write $C = Ap$. If $A \in \mathbb{R}^{N_\ell \times N_k}$ has rank r, then $\dim\ker A = N_k - r$, and every exact solution is $p = p_{\mathrm{part}} + V_0 a$ for a right-null basis V_0. Given a prior center p_0, $Q \succ 0$, and a feasible datum $C \in \mathrm{Ran}\, A$, the unique minimizer over real vectors of $\frac{1}{2}(p-p_0)^{\mathsf{T}} Q (p-p_0)$ subject to $Ap = C$ is

$$\boxed{p_* = p_0 + Q^{-1}A^{\mathsf{T}}(AQ^{-1}A^{\mathsf{T}})^{+}(C - Ap_0).}$$

If $C \notin \operatorname{Ran} A$, no constrained minimizer exists; the displayed pseudoinverse expression instead leaves a nonzero constraint residual. Even when feasible, it need not give a nonnegative spectrum. Positivity therefore requires a separate check or a declared constrained optimization, whose solution is not in general the displayed expression. The run publishes the singular values, rank threshold, right-null basis, resolution operator, unresolved projector, positivity active set, and prior sensitivity. The output type is ConditionalRadialContinuation.

Theorem (unrestricted tomographic uniqueness). For

$$C_\ell(r, r') = \frac{2}{\pi} \int_0^\infty k^2 P_\zeta(k) j_\ell(kr) j_\ell(kr')\, dk,$$

the covariance operator on $L^2(r^2 dr)$ obeys

$$Q_\ell = \mathcal{H}_\ell^{-1} M_{P_\zeta} \mathcal{H}_\ell.$$

Thus a complete radial cross-covariance operator determines P_ζ uniquely almost everywhere on its natural operator domain. Fixing one radius $r_0 > 0$, the family $C_\ell(r, r_0)$ known for all r suffices, provided that this section admits spherical-Hankel inversion in the declared function or distribution class: its spherical-Hankel transform is $\sqrt{2/\pi}\, P_\zeta(k) j_\ell(kr_0)$, and the Bessel zeros are discrete. A finite collection of auto-spectra does not supply this tomography theorem. The positive-radius condition is essential: $j_\ell(kr_0)$ is identically zero at $r_0 = 0$ for $\ell \geq 1$, so that section contains no spectral information.

Proof. Apply spherical-Hankel inversion in the first radial variable. Unitary conjugation exposes the multiplication operator; equal multiplication operators have equal multipliers almost everywhere. □

Theorem (branch-typed radial boundary). The ordinary spherical-Bessel and gamma-function formulas apply to FlatExact and FlatAssumed branches. Open and closed spatial branches use their declared hyperspherical eigenfunctions and spectral measures. An unresolved curvature branch blocks physical radial identification, and the flat thin-shell amplitude formula cannot certify a curved branch.

Physical source requirements. A source-derived primordial packet using either certified route above requires all source-stress, single-clock, freezeout, growing-mode, phase, physical-scale, source-amplitude, null-space, positivity, and non-fitting forward-residual receipts, together with its radial-dilation intertwiner or radial-tomography certificate. An alternative source construction must supply its own injectivity or orbit-separation theorem and the same physical and no-target-leak interfaces. A finite inverse selected before evaluation is a conditional radial continuation. It supplies no physical temperature or polarization spectrum.

Corollary (transfer firewall). Radial identification changes no temperature, polarization, lensing, recombination, foreground, nuisance, covariance, or likelihood conclusion. Those outputs require the independent transfer and likelihood system.

total stress closure receipt	$T_I^{ab}, Q_I^a, E_T, \nabla_a T^{ab}$
single clock normal form receipt	$\chi_r, \operatorname{rank} d\mathcal{F},$ connected fibers
entropy repair gap receipt	$L_\perp, H, \gamma,$ forcing/refinement bounds
curvature evolution receipt	$\zeta_a, \Gamma_a^{\mathrm{eff}},$ evolution residual, freeze bound
adiabatic mode receipt	$S_{IJ}, \Gamma_I, \Xi_I,$ growing-mode projection
isocurvature bound report	$\beta_{\mathrm{iso}}(k)$ and source covariance
primordial phase coherence receipt	$\lambda_2/\lambda_1,$ phase and decaying-mode defects
screen to radial lift receipt	$W(r), R_\star, \Psi_\ell(k),$ source hashes
radial null space report receipt	singular values, null basis, resolution kernels
forward projection residual receipt	$r_\ell,$ aggregate norms, quadrature and solver errors.

Source-provenance and pooled-reducer firewall. Let the source artifact DAG have parent-to-child edges and let M be the set of measurement, residual, likelihood, posterior, fitted-parameter, or measurement-calibrated nodes. Define

$$\tau(v) = \mathbf{1}_{v \in M} \vee \bigvee_{p \in \mathrm{Parents}(v)} \tau(p).$$

Then $\tau(v) = 1$ exactly when v has a forbidden measurement ancestor. The proof is induction over a topological ordering of the DAG. Every node claimed as source-derived for $\eta_R, \Gamma_{\mathrm{rec}}, A_\zeta, q_{\mathrm{IR}}, \ell_{\mathrm{IR}}, B_A(k,a), \rho_A(a)$, and N_{CRC} must have $\tau = 0$. Unknown source metadata fails closed.

For nonlinear estimates, shards may not average nonlinear quantities. Each row must emit additive sufficient statistics in a commutative monoid $(\mathcal{M}, \oplus)$, for example

$$\sum w_i, \qquad \sum w_i x_i, \qquad \sum w_i x_i x_i^{\mathsf{T}}$$

or, for weighted response regression,

$$X^{\mathsf{T}} W X, \qquad X^{\mathsf{T}} W y, \qquad y^{\mathsf{T}} W y.$$

Associativity, commutativity, complete coverage, and duplicate removal make the pooled statistic partition-invariant; the deterministic estimator is then evaluated only after pooling.

Three-dimensional homogeneous lift. A rotationally invariant screen covariance about one observer is not a statistically homogeneous three-dimensional primordial field by itself. A frozen lift

$$\mathcal{L}_r : V_r^{\mathrm{screen}} \to \mathcal{H}_r^{(3)}$$

must define

$$\Sigma_r^{(3)} = \mathcal{L}_r \Sigma_r^{\mathrm{screen}} \mathcal{L}_r^*$$

and certify positivity, refinement convergence, rotation covariance, translation covariance or an equivalent multicenter consistency theorem, and explicit null-space treatment. If the limiting covariance commutes with translations and rotations, Fourier diagonalization gives

$$\langle \zeta_a(\mathbf{k}) \zeta_b^*(\mathbf{k}') \rangle = (2\pi)^3 \delta^{(3)}(\mathbf{k} - \mathbf{k}') P_{ab}(|\mathbf{k}|),$$

with $P_{ab}(k)$ positive semidefinite.

Transfer-ready initial data. The source-to-transfer map must emit a complete initial state

$$Y(k, \eta_i) = \mathcal{M}_{\mathrm{init}}(k; \mathfrak{B}_{\mathrm{src}}) \xi(k),$$

including metric, photon, polarization, baryon, neutrino, anomaly, recipient, and auxiliary response variables. Near a singular early-time point, regular Frobenius modes solve

$$Y' = \left(\frac{M_{-1}}{\eta} + M_0 + O(\eta)\right) Y, \qquad Y = \eta^s(v_0 + v_1\eta + \cdots), \quad M_{-1}v_0 = sv_0.$$

The branch must classify regular adiabatic growing modes, allowed isocurvature modes, decaying modes, gauge modes, and singular interaction modes. On the adiabatic branch,

$$S_{IJ} = 3(\zeta_I - \zeta_J) = 0$$

for all active sectors. The finite evidence receipt must bound the initial Hamiltonian and momentum constraints and the propagated residual

$$\epsilon_{\text{constraint}} = \sup_{k,\eta} \|\mathcal{C}(k,\eta)Y(k,\eta)\|.$$

Receipt contract. Every ensemble run must record its ensemble id, interpretation class, regulator, representative schema, gauge action, quotient canonicalizer, base measure definition, action coefficients, coarse maps, zero-mode projector, amplitude convention, sampler kernel, partition-invariant random-event schema, smoothing policy, source provenance, and explicit nonclaims. The seed belongs to the run receipt rather than the ensemble definition. The interpretation classes are:

seed, proposal, or repair noise,
conventional reference ensemble,
OPH-native quotient ensemble,
OPH physical vacuum,
OPH primordial field,
observable cosmological prediction.

Smoothing must preserve raw coefficients, raw spectra, smoothing kernels, smoothed coefficients, smoothed spectra, and hashes of each stage; it is not part of S_r unless explicitly declared. For bounded coarse observables O, if the implemented law satisfies $\|\widehat{\mu}_s - \mu_s\|_{\text{TV}} \le \epsilon_{\text{samp}}$ and the refinement defects sum to ϵ_{ref}, then

$$\left|\mathbb{E}_{\widehat{\mu}_s}[O \circ c_{sr}] - \mathbb{E}_{\mu_r}[O]\right| \le 2\|O\|_\infty(\epsilon_{\text{samp}} + \epsilon_{\text{ref}}),$$

which is the finite evidence accuracy statement for this theorem.

Theorem 2.3 (EC from regulated overlap gluing). For a collar B_δ around a cap boundary Σ, there is a canonical decomposition

$$H_{B_\delta} = (\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{G_\Sigma} = \bigoplus_\alpha (H_{b_L^\alpha} \otimes H_{b_R^\alpha}),$$

and the sector-preserving collar algebra

$$A_{\text{EC}}(B_\delta) := \bigoplus_\alpha (B(H_{b_L^\alpha}) \otimes B(H_{b_R^\alpha}))$$

with

$$Z(A_{\text{EC}}(B_\delta)) = \bigoplus_\alpha \mathbb{C}\,\mathbf{1}_\alpha,$$

such that $\mathcal{A}(A_\delta B_\delta)$ acts only on $H_{b_L^\alpha}$ and $\mathcal{A}(B_\delta D_\delta)$ acts only on $H_{b_R^\alpha}$ within each block.

Proof. Split the collar into half-collars B_L and B_R meeting on $\Sigma = \partial C$. By the realized regulator presentation above, the physical collar Hilbert space is the diagonal invariant subspace $(\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{G_\Sigma}$. Decompose each side into irreps:

$$\tilde{\mathcal{H}}_{B_L} = \bigoplus_\alpha (V_\alpha \otimes H_{b_L^\alpha}), \qquad \tilde{\mathcal{H}}_{B_R} = \bigoplus_\beta (V_\beta^* \otimes H_{b_R^\beta}).$$

Then

$$\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R} = \bigoplus_{\alpha,\beta} (V_\alpha \otimes V_\beta^*) \otimes (H_{b_L^\alpha} \otimes H_{b_R^\beta}).$$

By Schur's lemma,

$$(V_\alpha \otimes V_\beta^*)^{G_\Sigma} \cong \begin{cases} \mathbb{C}, & \alpha = \beta, \\ 0, & \alpha \neq \beta. \end{cases}$$

Therefore the invariant subspace is

$$H_{B_\delta} = \bigoplus_\alpha (H_{b_L^\alpha} \otimes H_{b_R^\alpha}),$$

as claimed. The induced sector-preserving collar algebra is

$$A_{\mathrm{EC}}(B_\delta) = \bigoplus_\alpha (B(H_{b_L^\alpha}) \otimes B(H_{b_R^\alpha})),$$

so the center is generated by the block projectors. Adjacent region algebras act on the left or right factor only because the gluing action is supported on Σ. QED.

Remark. On the central-defect subbranch, replace G_Σ by its central extension. The sector label α then ranges over irreps of the extension; the decomposition is unchanged.

We refer to the decomposition in Theorem 2.3 as **edge-center completion (EC)**.

Exact Markov route. If $I(A_\delta : D_\delta \mid B_\delta) = 0$, the Hayden–Jozsa–Petz–Winter (HJPW) structure theorem yields a blockwise factorization of the state over *some* direct-sum/tensor decomposition of $\mathcal{H}_{B_\delta}$; that decomposition is state-dependent, and HJPW does not by itself identify it with the preselected EC factors b_L^α, b_R^α of Theorem 2.3. The exact identity used for literal Markov-modular equalities below is the **EC-aligned** normal form

$$\rho_{A_\delta B_\delta D_\delta} = \bigoplus_\alpha p_\alpha (\rho_{A_\delta b_L^\alpha} \otimes \rho_{b_R^\alpha D_\delta}), \qquad I_\omega(A_\delta : D_\delta \mid B_\delta) = 0,$$

and the identification of the HJPW factors with the EC factors is a genuine additional state condition, the **Markov-split alignment hypothesis (MSA)**. It is not implied by exact Markovity: with four qubits A, B_L, B_R, D and $\rho = \Phi_{AB_R} \otimes \Phi_{B_L D}$ (Bell pairs across the cut, one center block), one has $I(A : D \mid B) = 0$ but $I(A : B_R) = 2 \ln 2$, so the state is exactly Markov yet not of the displayed EC-aligned form, which would force $I(A : B_R) = 0$. All literal Markov-modular equalities below therefore assume exact Markovity *plus* MSA, or an explicitly stated idealized limit that supplies both; The spacetime and Einstein paper states MSA formally and records this counterexample. MSA is checkable state by state: The spacetime and Einstein paper proves it equivalent to a splitting of $\log \rho$ into commuting one-sided terms supported on $A b_L^\alpha$ and $b_R^\alpha D$, to the existence of a ρ-preserving conditional expectation onto either one-sided edge algebra (a Takesaki commuting square over the collar center [61]), and to blockwise vanishing one-sided mutual information

$I(Ab_L^\alpha : b_R^\alpha D) = 0$. On the local information-projection branch this reduces the derivation of MSA to a sharp condition on the effective Hamiltonian: every cross-cut coupling must act through the central sector label, as it does in lattice-gauge-type regulators where the interface energy is a function of the conserved flux. On that declared **central-interface branch** The spacetime and Einstein paper derives both MSA and exact collar Markovianity as theorems (descent of one-sided invariants and boundary charges through the edge-center quotient), and it adopts the central-interface normal form as an explicit named branch premise, so the literal Markov-modular identities hold on the declared branch without a further state hypothesis. This premise is independent of A1–A3: The spacetime and Einstein paper exhibits a finite regulator package that satisfies observer agreement, schedule-independent transactional repair, and an exactly closed refinement-channel family while its retained constraint family carries a boundary-invariant noncentral cross-cut coupling. Observer agreement and repair do not force the premise. Off the declared branch MSA is carried as an explicit hypothesis.

On the separately declared rate-controlled family obtained by pushing the optimizer forward, Theorem 2.3 supplies the kinematic edge-center decomposition. Exact Markovity is one idealized route. The following lemma and theorem supply the quantitative decay route when the approximation is kept explicit.

Lemma 2.6 (interior local information projection implies local Gibbs form). Under Axiom 3 on the subbranch with identity-proportional local references, state-determining aggregation equivalent to the declared entropy objective, homogeneous global-sum constraints $\langle \sum_x O_a(x) \rangle = C_a$, and the finite-dimensional regulator realization above, assume also that the constraint vector lies in the relative interior of the attainable moment set, so the selected state is faithful on the retained sector. Then the selected state has the finite-multiplier Gibbs form

$$\omega = \frac{e^{-H_{\mathrm{eff}}}}{\operatorname{Tr} e^{-H_{\mathrm{eff}}}}, \qquad H_{\mathrm{eff}} = \sum_x \sum_a \lambda_a O_a(x) + H_{\mathrm{sec}},$$

where the sum runs over UV cells x and density labels a, with one cell-independent multiplier λ_a per label. For the finite-range collar theorem, H_{sec} is either a sum of uniformly bounded finite-range charge densities or is central and fixed on the chosen superselection sector. A genuinely nonlocal noncentral term falls outside the theorem. Under this condition the effective Hamiltonian is finite-range in UV graph units, and the multiplier count matches the constraint count N_{con} plus the number of retained sector terms on every lattice.

Proof. On a finite-dimensional algebra, an interior entropy maximizer subject to linear constraints $\operatorname{Tr}(\rho O_i) = c_i$ is given by finite Lagrange multipliers:

$$\rho = \frac{e^{-\sum_i \lambda_i O_i}}{\operatorname{Tr} e^{-\sum_i \lambda_i O_i}}.$$

Strict concavity of von Neumann entropy ensures uniqueness. Here the constrained operators are the N_{con} global sums $O_a := \sum_x O_a(x)$ plus the retained sector terms, so the exponent has one shared multiplier per density label. MaxEnt gives the Gibbs form on this faithful/interior branch; the displayed locality restriction on H_{sec} is separate and load-bearing. A boundary constraint can instead force a rank-deficient optimizer (for example $\operatorname{Tr}(\rho|1\rangle\langle 1|) = 0$ on $\mathbb{C}^2$); it has no finite full-algebra multiplier and must be

treated on its support or by infinite penalties. Such a boundary branch does not supply the stagewise faithfulness used below. Had the constraints instead been imposed per cell, $\langle O_a(x)\rangle = c_a(x)$, the multipliers would be cell-dependent fields $\lambda_a(x)$ and the family would grow with the lattice; that enlarged family belongs to the non-equilibrium approximation regime of the clarification above and is not the branch used in the refinement argument. QED.

Definition 2.6a (closure defect and induced refinement map). Fix successive regulator scales ℓ (finer) and L (coarser), let $\{\omega_L(\lambda')\}$ be the homogeneous global-sum exponential family at scale L, and let $\Phi_{\ell\to L}$ be an admissible coarse-graining channel. For fine-scale multipliers λ set $\sigma := \Phi_{\ell\to L}(\omega_\ell(\lambda))$ and define the **closure defect**

$$\varepsilon_{\ell\to L}(\lambda) := \inf_{\lambda'} D(\sigma \,\|\, \omega_L(\lambda')),$$

with D the quantum relative entropy. When the infimum is attained at a unique λ^*, set $R_{\ell\to L}(\lambda) := \lambda^*$.

Lemma 2.6b (I-projection residual bound). If the constrained operators at scale L together with the identity are linearly independent, then $\lambda' \mapsto D(\sigma\|\omega_L(\lambda'))$ is smooth and strictly convex; for faithful σ the minimizer $\lambda^* = R_{\ell\to L}(\lambda)$ exists, is unique, and is characterized by moment matching $\langle S_c\rangle_{\omega_L(\lambda^*)} = \langle S_c\rangle_\sigma$ for every constrained operator S_c; and

$$\|\sigma - \omega_L(R_{\ell\to L}(\lambda))\|_1 \le \sqrt{2\,\varepsilon_{\ell\to L}(\lambda)},$$

so the family reproduces every bounded expectation up to $\|B\|\sqrt{2\varepsilon_{\ell\to L}(\lambda)}$. The separately declared exact optimizer-pushforward interface is the statement $\varepsilon_{\ell\to L}(\lambda) = 0$ along the realized branch. In that case $\sigma = \omega_L(R_{\ell\to L}(\lambda))$, and the induced refinement map is this moment-matching I-projection.

Proof. Writing $\log\omega_L(\lambda') = -\sum_c \lambda'_c S_c - \log Z_L(\lambda')$, the objective is $-S(\sigma) + \sum_c \lambda'_c\langle S_c\rangle_\sigma + \log Z_L(\lambda')$, and the Hessian of $\log Z_L$ is the Duhamel (Kubo–Mori) covariance matrix of the constrained operators, positive definite under the stated independence; this gives strict convexity, uniqueness, and the moment-matching first-order condition, while attainment for faithful σ is standard finite-dimensional exponential-family duality [59]. The displayed residual bound is the quantum Pinsker inequality $D(\rho\|\tau) \ge \frac{1}{2}\|\rho-\tau\|_1^2$ [60] evaluated at λ^*, and $\varepsilon_{\ell\to L}(\lambda) = 0$ forces $\sigma = \omega_L(\lambda^*)$ by strict positivity of relative entropy off the diagonal. QED.

Acceptance check. A finite two-lattice test implementing this package (the constraint/multiplier count against the displayed N_{con}-dimensional family on successive lattices, the moment-matching projection, the residual bound, a generic decimation channel with strictly positive closure defect, and the exactly closed transverse-field subfamily where $R_{\ell\to L}$ acts as the identity) is included with the repository sources.

Numerical projection and closure are distinct. For faithful σ on $\mathbb{C}^d$, choose a Hilbert–Schmidt orthonormal basis of the observable space V modulo identity, and write its traceless Hamiltonian as $H(\theta) = \sum_j \theta_j Q_j$, with $\rho_\theta = e^{-H}/\operatorname{Tr} e^{-H}$. The invariant moment residual is $\eta = \|P_V(\sigma-\rho_\theta)\|_{\mathrm{HS}}$; a raw residual in caller units is not a convergence

criterion. Coercivity and convexity give

$$D(\rho_*\|\rho_\theta) \le B := \eta\left(\|\theta\|_2 + \frac{\log d}{\lambda_{\min}(\sigma)}\sqrt{\frac{d-1}{d}}\right), \qquad \|\rho_* - \rho_\theta\|_1 \le \min\{2, \sqrt{2B}\},$$

where ρ_* is the unique exact projection. This optimizer error is separate from the closure defect $D(\sigma\|\rho_*)$; the Pythagorean identity adds the two. The implementation evaluates these bounds in floating point, without outward-rounded enclosures or an exact-zero certificate. Invertible changes of observable coordinates and identity offsets preserve the state; scalar energy origins shift $\log Z$ and must be retained in partition and free-energy calculations. Compensated scalar accumulation and a stable relative-entropy remainder avoid cancellation of small finite effects; neither supplies a spectral support certificate or recovers input differences lost to rounding.

Strong conditional mixing premise. Let G_ℓ be the UV cell graph. Assume local dimension at most q, interaction degree at most Δ_0, interaction diameter at most r_0 graph layers, and

$$\sup_x \sum_{X \ni x} \|\beta\Phi_\ell(X)\| \le J_0$$

with constants independent of the cut and refinement stage. Let m_ℓ be the resolved collar width in graph layers and set $\delta = m_\ell \ell$. For the faithful Gibbs state, define

$$\mathbf{J}_\rho(A : D \mid B) := \log\rho_{ABD} + \log\rho_B - \log\rho_{AB} - \log\rho_{BD}.$$

The strong mixing premise is a boundary-anchor expansion

$$\mathbf{J}_\rho(A_\delta : D_\delta \mid B_\delta) = \sum_{z \in \partial_{r_0}^{\mathrm{UV}} C} E_{z,\ell,\delta}, \qquad \|E_{z,\ell,\delta}\|_\infty \le \kappa e^{-(m_\ell - r_0)/\zeta},$$

with κ, ζ uniform in the cut, boundary condition, and stage. This conditional or matrix mixing condition is stronger than ordinary two-point exponential clustering. Local Gibbs form, a Lieb–Robinson bound, or a Hamiltonian spectral gap does not imply it for a general noncommuting Gibbs state.

Theorem 2.5 (finite-range conditional Gibbs mixing gives collar recovery). Under Lemma 2.6 and the strong conditional mixing premise,

$$\boxed{I_\omega(A_\delta : D_\delta \mid B_\delta) \le \kappa|\partial C|_{\mathrm{UV}} e^{-(m_\ell - r_0)/\zeta} \le c|\partial C|_{\mathrm{UV}} e^{-\delta/\xi_\ell}}$$

with

$$\xi_\ell = \zeta\ell, \qquad c = \kappa e^{r_0/\zeta}.$$

Proof. Embedding the marginal logarithms in the tripartite algebra gives

$$I(A : D \mid B)_\rho = \mathrm{Tr}[\rho_{ABD}\mathbf{J}_\rho(A : D \mid B)]\,.$$

The boundary expansion and trace-norm duality bound this expectation by the sum of the boundary term norms. The second form follows from $\delta = m_\ell\ell$. QED.

For one uniform family, the sharp sufficient limit condition is

$$\frac{\delta}{\xi_\ell} - \log|\partial C|_{\mathrm{UV}} \longrightarrow +\infty.$$

If

$$|\partial C|_{\mathrm{UV}} \le a(\ell_0/\ell)^p, \qquad \delta(\ell) \ge \zeta(p+\eta)\ell\log(\ell_0/\ell), \quad p \ge 0, \quad \eta > 0,$$

then $\delta/\ell \to \infty$, and

$$I(A_\delta : D_\delta \mid B_\delta) \le ca(\ell/\ell_0)^\eta \longrightarrow 0.$$

This lower bound on δ does not imply $\delta \to 0$. Physical collar shrinking remains the separate double-scaling hypothesis stated above. It is realized, for example, by choosing $\delta(\ell) = \zeta(p+\eta)\ell\log(\ell_0/\ell)$, or by rounding its graph-layer width upward with an additional error at most ℓ. For a counterexample to inferring shrinking from the lower bound alone, take $\ell_n = 1/(4n)$, $\ell_0 = \delta_n = 1/4$, $p = 0$, and $a = \zeta = \eta = 1$. Then $\delta_n \ge \ell_n \log n$ and $\delta_n/\ell_n = n$, but $\delta_n = 1/4$ does not shrink. A zero-Hamiltonian product Gibbs chain has zero conditional matrix defect and CMI and satisfies all the local finite-range and mixing bounds in this example; it fails precisely the separately declared shrinking-collar hypothesis. The ratio $\delta/\ell \to \infty$ alone is insufficient. For $\ell_n = \ell_0 e^{-n^2}$, $\delta_n = \zeta n \ell_n$, and $|\partial C_n|_{\mathrm{UV}} = e^{n^2}$, that ratio diverges while the upper envelope grows as e^{n^2-n}.

A finite receipt records the interaction range and norm bound, treatment of global sector terms, boundary-cell count, graph separation, regional CMI in nats, matrix-defect norm, predeclared κ, ζ, log-envelope slack, Fawzi–Renner error, held-out cuts, and the rate margin $\delta/(\zeta\ell) - \log|\partial C|_{\mathrm{UV}}$. Fitted constants and local random-triplet CMI remain finite proxies. They do not certify a cofinal limit.

This bound is the quantitative hinge for constructive gluing.

2.0.2 Concrete UV realization: quantum link models

Section 2.3 internalizes the regulator package: the fixed-cutoff type-I algebra and boundary fixed-point structure are the realized form of the screen net plus overlap gluing. Quantum link models are included here only as an explicit microscopic example of that structure, not as a separate premise layer.

UV regulator. Triangulate S^2 at scale ℓ_{UV}, giving vertices v, oriented links ℓ, and plaquettes p. Refinement corresponds to $\ell_{\mathrm{UV}} \to 0$ with increasing lattice size.

Degrees of freedom. Attach to every oriented link ℓ a **finite-dimensional** Hilbert space $\mathcal{H}_\ell$. In ordinary Wilson lattice gauge theory, the continuum/refinement-limit edge description is modeled by $L^2(G)$ (infinite-dimensional for continuous G), but the microscopic OPH regulator is instead a **quantum link model** with finite-dimensional link Hilbert spaces that preserve gauge symmetry in operator form [79]. Optionally attach matter Hilbert spaces $\mathcal{H}_v$ at vertices. Then:

$$\tilde{\mathcal{H}}_{\mathrm{total}} = \bigotimes_\ell \mathcal{H}_\ell \otimes \bigotimes_v \mathcal{H}_v,$$

finite-dimensional on any finite lattice. This is a concrete realization of the extended type-I presentation used in Section 2.3.

Boundary gluing as Gauss-law invariants. Define a local gauge transformation group G_v at each vertex v acting on incident links (and matter at v). Physical states satisfy:

$$|\psi\rangle \in \mathcal{H}_{\mathrm{phys}} \quad \Longleftrightarrow \quad U(g_v)|\psi\rangle = |\psi\rangle \ \ \forall\, v,\ g_v \in G_v.$$

Equivalently: $\mathcal{H}_{\text{phys}} = \tilde{\mathcal{H}}_{\text{total}}^{\prod_v G_v}$.

Region algebras. For any region $R \subset S^2$, define an extended Hilbert space $\tilde{\mathcal{H}}_R$ from the links/vertices in R. The **boundary gauge group** $G_{\partial R}$ acts on the cut degrees of freedom (the "half-links" ending on ∂R). Define:

$$\mathcal{A}_{\text{inv}}(R) = \mathcal{B}(\tilde{\mathcal{H}}_R)^{G_{\partial R}}.$$

This is the concrete quantum-link instance of the lifted fixed-point presentation used in Section 2.3. The same gauge constraints then induce the sector-preserving EC algebra on collars.

Why EC follows immediately. Take a cap C and a collar B_δ around ∂C. Because the *only* coupling between inside and outside is through the boundary gauge constraint, the collar Hilbert space decomposes into superselection blocks labeled by boundary irreps:

$$\mathcal{H}_{B_\delta} \cong \bigoplus_\alpha (H_{b_L^\alpha} \otimes H_{b_R^\alpha}),$$

with center generated by the projectors P_α. This is precisely the Schur-lemma mechanism of Theorem 2.3. The labels α are the familiar edge-mode / electric-flux labels appearing whenever one factorizes gauge theories across an entangling cut [80]. Exact Markovity depends on the state and is not forced by the decomposition alone; nor does exact Markovity align the HJPW factors with these edge-mode labels, which is the separate Markov-split alignment hypothesis of Section 2.3.

Dynamics and MaxEnt. The natural Hamiltonian is a 2+1D lattice gauge Hamiltonian on the screen worldvolume: plaquette ("magnetic") terms, electric terms on links, vertex Gauss terms as constraints, plus local matter couplings. In quantum link form this is finite-dimensional per link while behaving like gauge theory in the continuum limit. Then the MaxEnt assumption becomes concrete: the MaxEnt state is a Gibbs state $\rho \propto e^{-\sum_i \lambda_i O_i}$ with quasi-local O_i, precisely the local-Gibbs regime.

Geometry and G. This microphysics naturally supplies the emergent geometric objects:

- **Edge entropy / area operator:** $L_C = \sum_\alpha (\log d_\alpha) P_\alpha$ becomes "log of boundary irrep dimension" in the gauge link model.
- **Newton constant G:** the conversion factor between edge entropy density per boundary UV cell and macroscopic geometric area.

Thus area is an operator living in the center of the boundary algebra, because in gauge systems the center is where the cut labels live.

Scope of this example. The quantum-link realization makes the fixed-cutoff matrix/fixed-point bookkeeping concrete. Modular flow on caps becomes geometric conformal dilation with the 2π KMS normalization through the support-visible BW scaling theorem on the geometric subnet. Viable architectures for this include holographic quantum error-correcting codes [81] and quantum double / string-net Hamiltonians [82], but any use of QECC distance, min-cut resilience, or Knill–Laflamme correction requires the corresponding code subspace, logical-operator, error-family, and recovery certificate.

2.0.3 Conformal-modular fixed-point microphysics

On the faithful/interior local finite-constraint branch of Axiom 3 with identity-proportional reference, the logarithm of the selected state is a quasi-local UV generator. A separately declared

optimizer-pushforward and refinement-tail interface places the branch inside one common finite-dimensional multiplier family. At each regulator stage the patch and cap algebras are type-I matrix algebras. The finite regulator class is not refinement-closed, so the scaling-limit observer algebra may leave that class. The fixed-cutoff analysis proves the local-Gibbs result and propagation. Endpoint-Lipschitz control of *reduced* modular Hamiltonians requires the additional exact-Markov/central-interface and endpoint-control branch stated below; it does not follow from finite-range locality alone. The cross-scale interface supplies refinement stability. The geometric cap pair with geometric modular flow is obtained in the support-visible sense: the target algebra is the extracted geometric subnet, the support-visible modular matrix elements converge under regularization, and weak-$*$/GNS extraction plus support-readable modular covariance and ordered cut-pair rigidity gives the geometric cap automorphism.

Local finite-constraint information-projection branch. The constraint family $\mathcal{C}$ is generated by finitely many gauge-invariant local densities $\{O_a(x)\}$ of UV range $O(\ell_{\mathrm{UV}})$, constrained through their homogeneous global sums $\langle \sum_x O_a(x) \rangle = C_a$. Axiom 3 selects the fixed-stage information projection. Retention of the same finite label set under refinement belongs to the separate optimizer-pushforward and refinement-tail interface.

Theorem 2.6 (Interior local constraints imply a local-Gibbs form). If the MaxEnt constraints are expectations of finitely many quasi-local operators $\{O_a\}$ with bounded support size at scale ℓ_{UV}, and the prescribed moment vector lies in the relative interior of the attainable moment set so that the optimizer is faithful on the retained sector, then the entropy maximizer is

$$\omega \propto \exp\left(- \sum_a \lambda_a O_a \right),$$

so the MaxEnt generator $H_{\mathrm{MaxEnt}} = -\log \omega$ is a UV-range quasi-local sum. This is exactly the local-Gibbs form used later.

Proof. The finite-dimensional interior exponential-family theorem gives finite Lagrange multipliers λ_a. Boundary optima require restriction to their support or extended infinite multipliers and are not included in this faithful branch. QED.

Derived propagation control on the same branch. Because H_{MaxEnt} is a finite-range or quasi-local sum on a finite type-I regulator net, standard Lieb–Robinson estimates [41] apply to the automorphism group it generates, or to any branch generator lying in the same bounded-support algebraic closure. Thus there is a finite propagation velocity v_{LR} and constants C, ξ such that for local observables A_X, B_Y,

$$\left\| [\tau_t(A_X), B_Y] \right\| \leq C \, \|A_X\| \, \|B_Y\| \, \min(|X|, |Y|) \, e^{-(d(X,Y) - v_{\mathrm{LR}}|t|)/\xi}.$$

Additional endpoint-control branch. Locality of the global Gibbs generator does not by itself make reduced interval modular Hamiltonians local; the four-qubit counterexample in Theorem 4.3d rules out that inference. On the exact-Markov/central-interface branch, assume in addition a specified endpoint-to-regulator-cell map and a uniform local norm/site-density bound such that, after subtracting the central endpoint block, the operator-norm sum of terms changed between allowed intervals I, I' is at most $L|I' \Delta I|$. On this explicitly named branch, the local split and that geometric norm bound give bounded-interval endpoint-Lipschitz matrix elements,

$$\left| \langle \psi, (K[I'] - K[I]) \phi \rangle \right| \leq C_{\psi, \phi, I_{\max}} \, |I' \Delta I|,$$

used later in the null-modular bridge. Thus quasi-local propagation follows on the faithful local-Gibbs branch, while reduced-modular endpoint control is a separate exact-Markov/central-interface receipt with the stated endpoint geometry and uniformity.

Refinement-stable multiplier branch. The separately declared optimizer-pushforward and refinement-tail interface preserves the same finite constraint family under coarse-graining and places the regulator states in one common finite-dimensional multiplier family. This is a substantive renormalization condition because coarse-graining a finite-range Gibbs family generically generates interactions outside a fixed density list. The "refinement-stable" language used later means persistence along the stable or fixed branch of this multiplier family. This state-side notion is enough to compare realized states across cutoffs. FixedCutoffBosonicSectorCategory constructs the fixed-stage tensor-generated category and its forgetful fiber. RefinementFunctorAndFiberDescent constructs the refinement pullbacks and compatible fibers only on a cofinal tail carrying the compact-gauge refinement receipt. The cofinal compact-gauge witness uses the same receipt. The separate finite Standard Model route uses complete reversible response and endogenous overlap transport to force the local Standard Model gauge Lie algebra. Incidence and target-blind readback derive the carrier module and $R = -J$. The finite conditional fixture supplies the charge-conjugate rank-15 projector pair, hypercharge lattice, common $\mathbb{Z}_6$ kernel, and maximal faithful matter image. Source derivation of that current, matter action, and global quotient is not supplied. Under the complete-band and operational-cost premises, the exact screen theorem selects the rank-three response band. Physical family attachment, extra-sector completeness, and the charged-lepton and electroweak quantitative branches carry their own premises. Axiom 3 supplies neither the finite current realization nor the sector-colimit data or laboratory attachment.

Scaling limit and algebraic type. The regulator presentation gives a family of finite type-I algebras

$$\mathcal{A}_\ell(C) \cong \mathcal{B}(\mathcal{H}_{C,\ell}).$$

A scaling limit of this family need not be type I. In the continuum-QFT case of interest one expects the local limit algebra $\mathcal{A}_\infty(C)$ to be non-type-I, typically type III. Accordingly the fundamental modular datum in the limit is the automorphism group of the pair $(\mathcal{A}_\infty(C), \omega_\infty^C)$, not a density matrix inside $\mathcal{A}_\infty(C)$.

Corollary 2.6 (Geometric modular action on paired-certified caps). For any support-visible extracted scaling-limit geometric cap pair $(\mathcal{A}_\infty^{\mathrm{geo,sv}}(C), \omega_\infty^{\mathrm{geo},C})$ emitted on a branch carrying both the finite cap-normal certificate and the independently complete same-branch multiresolution algebra-state comparison package, let $\alpha_{\lambda_{\widehat{C}}(s)}$ be the automorphism induced by the independently normalized BW-framed cap flow. Then

$$\sigma_t^{\omega_\infty^{\mathrm{geo},C}} = \alpha_{\lambda_{\widehat{C}}(2\pi t)}.$$

If the limit cap algebra happens to be type I, this may be written as

$$K_C = 2\pi B_C + Z_C, \qquad Z_C \in Z(\mathcal{A}(C))_{\mathrm{sa}}.$$

In the generic continuum case, the automorphism identity is the complete statement.

Proof. This is The spacetime and Einstein paper's finite cap-net BW theorem applied to the extracted geometric cap pair. The finite cap-normal input supplies BW framing, support-order faithfulness, held-out oriented cross-ratio rigidity, the geometric support flow, independently normalized geometric 2π-KMS comparison, and wrong-normalization controls. The separate multiresolution algebra-state comparison input supplies the complete algebra-state reference tower, cap-family-uniform mixed-GNS convergence, comparison maps, and modular support covariance. QED.

Support envelope from net regularity (not an alternative derivation). The outer-regularity / minimal-support condition used later lets one read a support envelope for the modular

image from the scaling-limit net. By itself it does not establish modular covariance; that equality comes from the paired certificate in Corollary 2.6, or must be assumed as an additional two-sided net-automorphism property.

(NR) Outer regularity / minimal support. For any operator O, the intersection of all connected regions P with $O \in \mathcal{A}(P)$ is again a connected region, denoted $\mathrm{supp}(O)$.

Proposition 2.7 (Modular support-envelope inclusion). Under (NR), define for any region $R \subset C$

$$f_t^C(R) := \bigcup_{O \in \mathcal{A}(R)} \mathrm{supp}\Big(\sigma_t^{\omega,C}(O)\Big).$$

Then

$$\sigma_t^{\omega,C}(\mathcal{A}(R)) \subseteq \mathcal{A}(f_t^C(R)).$$

Equality does not follow from (NR). It requires, for example, a two-sided order-isomorphism of the local net compatible with σ_t and σ_{-t}, together with the relevant additivity/local-generation property. Those are substantive covariance data, not consequences of knowing minimal operator supports.

Proof. For every $O \in \mathcal{A}(R)$, (NR) assigns $\sigma_t^{\omega,C}(O)$ a minimal support contained in $f_t^C(R)$; isotony therefore places that operator in $\mathcal{A}(f_t^C(R))$. Taking all such operators gives the stated inclusion. Surjectivity onto the entire envelope algebra is not supplied by the fact that σ_t is an automorphism of the global algebra. QED.

Null-surface modular structure. On the same support-visible scaling branch and extracted geometric-subnet branch, the null-surface modular machinery narrows as follows:

- **Fixed-cutoff null-strip bridge.** The null-strip package proves transferred cut-center data, the theorem-local inherited left/right strip-split condition needed for the spatial-collar-type tensor decomposition, exact-or-controlled four-term strip additivity on one inherited strip model, and renormalized endpoint-Lipschitz control up to the weak tail generator. On the scaling branch the geometric action supplies the half-sided relation; the separately registered common-vector and relative-commutant standardness receipts make it a standard half-sided modular inclusion whose Borchers–Wiesbrock consequence is an explicit positive null-translation generator on its Stone domain with the affine half-line modular relation. The same half-line family fixes the generator's endpoint-derivative identity internally; its local null-stress reading consumes the separately supplied effective-locality and same-source bridge.

- **Derived half-sided modular relation.** Nested null half-line algebras satisfy the half-sided covariance and containment relation on the declared geometric scaling branch by Corollary 5.2e. To obtain a *standard* half-sided modular inclusion, the common reference vector must additionally be cyclic and separating for both half-line algebras and cyclic for their relative commutant. On that separately registered standardness receipt, Borchers–Wiesbrock identifies the positive null-translation generator on its Stone domain together with the affine half-line modular relation [45].

- **Weak continuity and finite variation.** The bounded-interval and half-line endpoint-Lipschitz control follows on the exact-Markov/central-interface branch only after the specified endpoint-to-cell map and uniform norm/site-density bound of the endpoint-control clause. It then defines the weak tail generator at fixed cutoff. The support-visible scaling theorem together with the extracted geometric cap pair supplies the continuum null-generator setting in which that weak-tail data can be matched to the geometric null modular action of the relativistic phase.

Constraint set specification. On the local finite-constraint branch, the "correct fixed-cap constraint set" becomes explicit: constraints are the local conserved charges of the symmetries used in the derivation:

1. **Edge/cap label constraints:** fix the distribution of collar-sector labels, equivalently $\langle L_C \rangle$ for each cap size, giving the area term.

2. **Gauge charges:** fix boundary flux or charge operators.

3. **Geometric (conformal) charges:** fix the expectation of the conformal Killing charges that preserve the cap, i.e. the generator B_C or its microscopic lattice approximation.

MaxEnt therefore selects the unique finite-stage invariant state compatible with those conserved charges. The support-visible BW scaling theorem is the continuum statement: if the limit algebra is non-type-I, the geometric modular action on that branch is outer rather than an inner density-matrix identity.

Focusing boundary. QNEC has rigorous QFT proofs in broad settings [55, 57], with hypotheses (null-surface locality and analyticity, modular-flow/causality/defect-OPE structure) that have not been verified on the reconstructed net, and QFC is strictly stronger than QNEC [56]. Quantum focusing is an optional gravity interface and is not consumed by the Einstein theorem:

- EC plus the declared state, edge-reference, and entropy-split premises identify $S_{\mathrm{gen}} = S_{\mathrm{bulk}} + \langle L_C \rangle$ on the finite branch (Section 5.4).

- The null-stress and Einstein theorem branch supplies the kinematic inputs required by an internal QNEC/QFC proof; that derivation is not supplied.

Summary. The CMFP package therefore separates the dependency structure into two layers:

- **Fixed-stage and cross-scale consequences:** Axiom 3 with the declared local constraint and reference package gives the local-Gibbs form and quasi-local propagation on the faithful/interior branch. Bounded-interval reduced-modular endpoint-Lipschitz control additionally consumes the exact-Markov/central-interface split and the specified endpoint-to-cell uniform bound. The separate optimizer-pushforward interface supplies the realized state-side refinement branch along which later transport questions are asked.

- **Support-visible scaling statement:** the scaling limit emits the support-visible geometric cap pair and ordered cut-pair rigidity holds on it. On that branch the limit algebra may be non-type-I, the modular action may be outer, and the 2π normalization follows. The later null half-sided-inclusion bridge additionally consumes its registered common-vector and relative-commutant standardness receipts.

The theorem route works on the support-visible quotient. The realized transported cap-local system packages the geometric cap-local test family, the projectively compatible transported marginal family, and the asymptotic transport-equivalence certificate. The regularized transport theorem below replaces the unavailable full-algebra lower spectral floor. Local weak-$*$ extraction and GNS gluing then emit the support-visible scaling-limit cap pair, support-readable modular covariance reads the modular group as a cap-local support map, and ordered cut-pair rigidity collapses the residual cap-preserving conformal freedom to the unique hyperbolic subgroup.

Proposition 2.6a (Recoverability is not modular geometry: common-floor collapse countermodel). Exact or asymptotically exact Markov recovery at finite cutoff does not by itself

imply a full-algebra lower spectral floor along refinement. In a two-dimensional matrix algebra, let

$$\rho_n = \begin{pmatrix} e^{-n} & 0 \\ 0 & 1 - e^{-n} \end{pmatrix}.$$

Each ρ_n is faithful at finite n, and tensoring it with any fixed finite exact-Markov collar factor gives a full-rank exact-Markov collar family. Nevertheless $\lambda_{\min}(\rho_n) = e^{-n} \to 0$, so there is no positive lower bound along the refinement family. On the off-diagonal matrix unit E_{12}, the modular generator carries the logarithmic gap

$$[-\log \rho_n, E_{12}] = n\, E_{12} + O(1) E_{12},$$

and the unregularized modular transport has no finite common-floor limit on that direction. Thus collar Markovity and finite-stage faithfulness are not enough for the false stronger full-algebra BW lift. The observer-facing theorem uses the support-visible regularized replacement, with the exact-Markov comparison family checked only on fixed collars and made replacement-independent after weak-$*$/GNS extraction.

Theorem 2.6b (regularized support-visible modular transport). Fix a local collar model of finite dimension $d_{m,\delta}$. Let ρ_n be the transported physical collar marginal, $\widehat{\rho}_n$ the exact-Markov comparison marginal on the same collar model, and $\Delta_n = \|\rho_n - \widehat{\rho}_n\|_1$. For $a > 0$, set $K_a(\rho) = -\log(\rho + a\mathbf{1})$. For every bounded collar observable O in the support-visible algebra,

$$\left|\operatorname{Tr} \rho_n O\big(K_a(\rho_n) - K_a(\widehat{\rho}_n)\big)\right| \le \|O\| \left(\frac{4\Delta_n}{a} + d_{m,\delta} a + 4\Delta_n |\log a| \right).$$

Consequently, if $a_n \downarrow 0$ is chosen with

$$\Delta_n / a_n \to 0, \qquad d_{m,\delta} a_n \to 0, \qquad \Delta_n |\log a_n| \to 0,$$

then the regularized support-visible modular matrix elements converge on that fixed collar model.

Proof sketch. On the spectral interval $[a, \infty)$, the logarithm is operator-Lipschitz at the scale used above by its integral representation. Splitting the comparison into the support above a, the a-tail, and the trace-distance error gives the displayed bound. The three displayed conditions make the three error terms vanish. QED.

Definition 2.6c (explicit multiresolution reference branch). The finite-to-continuum bridge is evaluated on a declared reference branch, not on an unstructured sequence of finite plots. A regulator is

$$r = (m, L, b),$$

where m is the refinement depth, L is the finite support or volume scale, and b is a boundary/phase label. The screen branch uses nested icosahedral subdivisions of S^2; Euclidean branches use dyadic mesh $a_m = a_0 2^{-m}$, with an additional temporal spacing a_t when a transfer matrix is present. Each added cell, collar, edge-center, or support-visible channel j carries

$$D_j = \bigoplus_{\alpha \in S_j} M_{d_{j,\alpha}}(\mathbb{C})$$

and a faithful detail state τ_j. At regulator r,

$$\widetilde{M}_r = \bigotimes_{j \in \mathcal{J}_r} D_j, \qquad \widetilde{\omega}_r = \bigotimes_{j \in \mathcal{J}_r} \tau_j.$$

A finite-depth local presentation circuit $\Gamma_r : \widetilde{M}_r \to M_r$ changes from multiresolution coordinates to the OPH patch, port, collar, and gauge coordinates. For $r \preceq s$, write

$$\widetilde{M}_s = \widetilde{M}_r \otimes D_{s\setminus r}, \qquad \tau_{s\setminus r} = \bigotimes_{j\in\mathcal{J}_s\setminus\mathcal{J}_r} \tau_j.$$

The canonical refinement and coarse-graining maps are

$$\iota_{rs}(A) = \Gamma_s\Big[\Gamma_r^{-1}(A)\otimes \mathbf{1}\Big],$$

and

$$Q_{sr}(X) = \Gamma_r\Big[(\mathrm{id}\otimes\tau_{s\setminus r})(\Gamma_s^{-1}(X))\Big].$$

The embedded conditional expectation is $E_{sr} = \iota_{rs}Q_{sr}$. The reference state is $\widehat{\omega}_r = \widetilde{\omega}_r \circ \Gamma_r^{-1}$.

Theorem 2.6d (exact modular-compatible reference tower). The data

$$(M_r, \widehat{\omega}_r, \iota_{rs}, E_{sr})_{r\preceq s}$$

form a directed tower. The maps ι_{rs} are unital injective $*$-homomorphisms with $\iota_{st}\iota_{rs} = \iota_{rt}$. The Q_{sr} are faithful state-preserving completely positive coarse-grainings with $Q_{tr} = Q_{sr}Q_{ts}$. The maps E_{sr} are faithful conditional expectations onto $\iota_{rs}(M_r)$, and

$$\widehat{\omega}_s \circ \iota_{rs} = \widehat{\omega}_r, \qquad \widehat{\omega}_r \circ Q_{sr} = \widehat{\omega}_s, \qquad \widehat{\omega}_s \circ E_{sr} = \widehat{\omega}_s.$$

The finite reference modular groups are also exactly compatible:

$$\sigma_t^{\widehat{\omega}_s}(\iota_{rs}A) = \iota_{rs}(\sigma_t^{\widehat{\omega}_r}(A)).$$

This theorem uses no full-algebra spectral floor uniform in r; in bare coordinates it is only the identity $A \mapsto A\otimes\mathbf{1}$, partial expectation against faithful detail states, and conjugation by Γ_r, Γ_s.

Theorem 2.6e (canonical renormalization and correlation Cauchy bound). Let $\mathfrak{A}$ be the C^*-inductive limit with embeddings $j_r : M_r \to \mathfrak{A}$. There is a unique state $\widehat{\omega}$ with $\widehat{\omega}\circ j_r = \widehat{\omega}_r$, and the maps Q_{sr} induce state-preserving conditional expectations

$$\mathbb{E}_r : \mathfrak{A} \to j_r(M_r)$$

such that $\|\mathbb{E}_r(A) - A\| \to 0$ for every $A \in \mathfrak{A}$. The finite-regulator realization of a continuum test observable is

$$\mathcal{R}_r(A) = j_r^{-1}(\mathbb{E}_r A).$$

For $\|A_k\| \le M$,

$$\left|\widehat{\omega}_r\left(\prod_{k=1}^n \mathcal{R}_r(A_k)\right) - \widehat{\omega}\left(\prod_{k=1}^n A_k\right)\right| \le M^{n-1}\sum_{k=1}^n \|A_k - \mathbb{E}_r A_k\|.$$

Thus the operator mixing matrix and additive subtraction at finite cutoff are outputs of $\mathcal{R}_r$, not free fits to the target continuum answer. Independent lattice-spacing, volume, and boundary sweeps must publish the corresponding martingale-tail or physical-error envelopes before finite regulator data are read as continuum evidence.

Theorem 2.6f (compact-time modular convergence on fixed local algebras). Let $\rho_{s\downarrow r}^{\mathrm{phys}}$ be the density matrix of the physical state at fine stage s, restricted to the embedded fixed local algebra M_r, and let

$$\epsilon_{s,r} := \|\rho_{s\downarrow r}^{\mathrm{phys}} - \widehat{\rho}_r\|_1.$$

For a regularizer $\eta_s > 0$, define

$$K_{s\downarrow r}^{(\eta_s)} = -\log(\rho_{s\downarrow r}^{\mathrm{phys}} + \eta_s \mathbf{1}).$$

If $\lambda_r = \lambda_{\min}(\widehat{\rho}_r) > 0$, then for every $A \in M_r$ and every $T < \infty$,

$$\sup_{|t|\le T} \left\| e^{-itK_{s\downarrow r}^{(\eta_s)}} A e^{itK_{s\downarrow r}^{(\eta_s)}} - \sigma_t^{\widehat{\omega}_r}(A) \right\| \le 2T\|A\| \left[\frac{\epsilon_{s,r}}{\eta_s} + \log\left(1 + \frac{\eta_s}{\lambda_r}\right) \right].$$

Hence $\eta_s \to 0$ and $\epsilon_{s,r}/\eta_s \to 0$ give uniform-on-compact-time convergence on each fixed support-visible local algebra. For a countable cofinal local test tower, pointwise $\epsilon_{s,r} \to 0$ admits a diagonal subsequence and one cutoff schedule. The phrase "finite-stage modular action" below means this transported, regularized local action; it does not mean the unregularized full finite-cap modular group.

Cap-family uniformity clause. For application to the finite cap-net BW theorem, the compact-time estimate is required uniformly over the declared finite nondegenerate cap family and the fixed separating support-visible test tower. The operator-norm estimate then implies the quadratic mixed-GNS Cauchy residual because $\|XD\|_{2,\omega} \le \|X\|\|D\|_{2,\omega}$. The reference tower supplies the analytic engine; the finite cap-normal certificate adds the support-order, BW-frame, oriented cross-ratio, geometric 2π-KMS, and wrong-scale controls.

Theorem 2.6g (finite positive transfer and Lorentzian handoff condition). On Euclidean repair branches, let $P_{C,r}$ be weighted resampling inside the fibers of the complete repaired visible datum for active collar C. Ref. [16] proves that this finite operator is the orthogonal conditional expectation and gives the exact support, equal-fiber-row, and weighted detailed-balance matrix receipt that recognizes it. A concrete active collar repair kernel is identified with $P_{C,r}$ only when its independently derived transition matrix passes that receipt; constructing the matrix from the target formula is not a verification. Choose $c_{C,r} \ge 0$, and set

$$L_r^{\mathrm{rep}} = \sum_C c_{C,r}(I - P_{C,r}), \qquad T_r(a_t) = e^{-a_t L_r^{\mathrm{rep}}}.$$

Then $L_r^{\mathrm{rep}} = L_r^{\mathrm{rep}\,*} \ge 0$, $0 < T_r(a_t) \le I$, $T_r(a_t)\mathbf{1} = \mathbf{1}$, and the stationary Euclidean path measure is reflection positive:

$$\mathbb{E}_r[(\Theta F)F] \ge 0$$

for positive-time cylinder functions F. A Lorentzian unitary statement requires this finite positivity plus either exact shell-additive direct-limit semigroup identities or a Mosco / strong-resolvent substitute, including vacuum-projection and intertwiner convergence. Weak-$*$ state convergence alone is not a transfer theorem.

BW-side closure boundary. Proposition 2.6a blocks the stronger unregularized full-algebra conclusion from the proved finite-cutoff package. Theorems 2.6c–2.6g supply the explicit reference-tower certificate needed for the finite-to-continuum reading: continuum correlations come from the conditional-expectation renormalization map, compact-time modular group convergence is only for transported regularized physical marginals on fixed support-visible local algebras, and Lorentzian transfer needs reflection positivity or an equivalent positive-transfer certificate. Theorem 4.2 then uses exactly the observer-facing support-visible content to close the BW/geometric cap-pair automorphism statement. Without the multiresolution regulator certificate, finite regulator data remain regulator evidence. A black-box AQFT/BW reconstruction route would require separately verifying conformal-net properties such as locality, additivity, duality, standardness, positive energy, and suitable modular inclusions.

2.1 Overlap Consistency and Gluing

The constructive part of overlap consistency is the tree-gluing theorem below. The structural part is the origin of the gluing redundancy itself. On the ordinary or central-defect branch, gauge-as-gluing is the finite-regulator overlap redundancy of local chart presentations: once the finite-dimensional overlap algebras and their allowed chart-change groupoid are put in the block-complete spatial presentation specified below, overlap-consistent rechartings of a cut form a compact unitary transition system. The collar theorem of Section 2.3 should therefore be read with its boundary group G_Σ understood as shorthand for this spatialized compact boundary action, not as an automatic consequence of an arbitrary algebra representation. On the genuinely noncentral branch, the corresponding declared weak gluing data are encoded by a compact crossed-module change system. Block permutations and other groupoid data are retained rather than silently treated as inner automorphisms. A second structural point is UV underdetermination. If each patch Hilbert space is tensored with an inert finite ancillary factor and the observable patch algebras are embedded as $\mathcal{A}(P) \otimes \mathbf{1}$, then the physical observables, collar conditional mutual information, carried Markov errors, and quotient normal forms are unchanged. So the physical UV branch is fixed only modulo gauge or implementation hiding together with such ancillary stabilization, not a unique microscopic presentation; the unique theorem-grade object is the induced family of terminal expectation functionals on the declared physical observable algebras, not a preferred microscopic representative.

2.1.1 Constructive gluing on tree covers

Theorem 3.1 (tree gluing). Let a rooted tree of patches satisfy a tree-ordered overlap structure and a tripartite factorization (A_k, B_k, C_k) at step k. If a target state ρ^* obeys $I(A_k : C_k \mid B_k) \leq \varepsilon_k$, then there exist recovery maps $\mathcal{R}_k$ such that

$$\|\rho^*_{A_k B_k C_k} - (\mathrm{id}_{A_k} \otimes \mathcal{R}_k)(\rho^*_{A_k B_k})\|_1 \leq \delta_k,$$

with

$$\delta_k = 2\sqrt{1 - e^{-\varepsilon_k}} \leq 2\sqrt{\varepsilon_k},$$

The iteratively glued state $\hat{\rho}$ satisfies

$$\|\hat{\rho} - \rho^*\|_1 \leq \min\left(2, \sum_{k=2}^{n} \delta_k\right).$$

Proof. Induct on k. The recovery error contracts under CPTP maps, so the errors add. QED.

2.1.2 Gauge-as-gluing and loops

At finite regulator scale, the fixed-cutoff gauge-as-gluing package identifies the overlap-consistency redundancy of local chart presentations. Choose block-complete faithful finite-dimensional spatial presentations of the overlap algebras on a connected cut Σ, and require the allowed overlap-chart isomorphisms to be represented by their unitary normalizers. A finite-dimensional C^*-algebra is a direct sum of full matrix blocks: its star automorphisms permute isomorphic central blocks and act inner within each block. They are spatial in the canonical block-complete presentation, but need not be inner in the algebra itself (an abelian two-block swap is the elementary counterexample). Fixing a reference chart and retaining any allowed block permutations, the compact closure of the subgroup generated by the spatial recharting unitaries is the boundary gluing group K_Σ; before fixing the reference chart, the same data form a compact unitary groupoid.

Proposition 3.2a (Spatialized gauge-as-gluing at finite regulator). Let a finite regulator chart be chosen for the patches meeting along a connected interface Σ, together with the block-complete spatial presentations and unitary-normalizer realization of the allowed chart-change groupoid just stated. Then overlap consistency determines a compact unitary transition system on the cut data. On the ordinary or central-defect branch, after a reference chart is fixed, this transition system reduces to a compact boundary group K_Σ, and when the triple-overlap defect is central its projective composition law lifts to a genuine action of a compact central extension $\widehat{K}_\Sigma$. On this realized presentation, gauge names the overlap redundancy; the spatialization data are explicit hypotheses rather than consequences of finite dimensionality alone.

Proof. The structure theorem for finite-dimensional C^*-algebras decomposes each chart algebra into full matrix blocks. An allowed star isomorphism permutes isomorphic central blocks and is inner on each matched block; the chosen block-complete presentation implements the combined operation by a unitary normalizer. The subgroup generated by those unitaries has compact closure inside a finite-dimensional unitary group. If triple-overlap defects are central, the resulting projective composition law lifts to a central extension. QED.

Lemma 3.2b (trees vs loops). If the patch adjacency graph is a tree, one can choose local charts h_i so that the overlap labels satisfy $g_{ij} = h_i^{-1} h_j$ on all edges. If loops exist, the loop holonomy

$$H(\gamma) = g_{i_1 i_2} g_{i_2 i_3} \cdots g_{i_n i_1}$$

transforms by conjugation at the chosen basepoint under local frame changes. Its conjugacy class, and therefore every class-function value (including every one-dimensional character value), is invariant. In particular, nontriviality is invariant and obstructs global trivialization. QED.

Proposition 3.2ba (Finite path ratio, character phase, and cyclic quantization). For reversal-compatible group-valued edge labels on a typed finite path system, the ordered transport ratio of two paths with common endpoints is exactly the holonomy of the loop obtained by following the first and reversing the second. Every one-dimensional unitary character sends this identity to an exact relative phase. A vertex rechart conjugates based holonomy, so holonomy itself is invariant for an abelian structure group and the character phase is invariant for an arbitrary group. There is an explicit four-vertex path/face control whose two declared triangular faces have trivial holonomy while a separate undeclared, unfilled loop and its two-arm relative character phase are nontrivial. Finally, if the loop holonomy factors through a supplied $\mathbb{Z}_n$ sector, every character phase is an nth root of unity.

Scope. The result is machine checked for supplied finite paths, edge labels, character, faces, and cyclic-sector factorization. The path/face control declares exactly two triangular faces; it neither claims flatness on undeclared triangles nor proves a puncture or noncontractibility statement. No theorem here constructs the connection from observer records or identifies the phase with a physical gauge flux, spacetime loop, charge, laboratory arm, detector, or interference fringe.

Corollary 3.2c (Collar consequence on the ordinary or central-defect branch). Let $B_\delta = B_L \cup B_R$ be a collar around a cap boundary Σ, and set $\widehat{K}_\Sigma = K_\Sigma$ on the ordinary branch. Then the EC theorem of Section 2.3 has the derived interpretation

$$H_{B_\delta} = (\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{\widehat{K}_\Sigma} \cong \bigoplus_\alpha (H_{b_L^\alpha} \otimes H_{b_R^\alpha}),$$

with

$$Z(A_{\mathrm{EC}}(B_\delta)) = \bigoplus_\alpha \mathbb{C}\,\mathbf{1}_\alpha.$$

The right half-collar carries the contragredient action because it sees the inverse transport across the same cut. Exact Markov normal forms used later require the additional state hypothesis $I_\omega(A_\delta : D_\delta \mid B_\delta) = 0$ *together with* the Markov-split alignment hypothesis of Section 2.3, or the explicitly stated idealized recoverability limit supplying both; the block decomposition itself is kinematic and follows from the derived boundary action.

Proof sketch. Decompose the left boundary data into irreps ($V_\alpha \otimes H_{b_L^\alpha}$) of $\widehat{K_\Sigma}$ and the right boundary data into the dual modules ($V_\beta^* \otimes H_{b_R^\beta}$). Then Schur's lemma leaves a singlet only when $\alpha = \beta$, producing the displayed direct sum. QED.

2.1.3 Loop obstruction class (central defect)

On the central-defect subbranch, identify the overlap centers with one fixed abelian unitary coefficient group Z_Σ and assume overlap transport acts trivially on it. Choose unitary overlap implementers normalized by $U_{ji} = U_{ij}^{-1}$ and define central defects z_{ijk} at the implementer level by

$$\Omega_{ijk} := U_{ij}U_{jk}U_{ki} = z_{ijk} \in Z_\Sigma,$$

or equivalently $U_{ij}U_{jk} = z_{ijk}U_{ik}$. This implementer-level relation records the projective multiplier; the automorphism-level expression $\mathrm{Ad}(z_{ijk})$ alone would be the identity and would not determine z_{ijk}.

Then $\{z_{ijk}\}$ is an untwisted Čech 2-cocycle with coefficients in Z_Σ, and its cohomology class $[z]$ is gauge invariant. Central defects do not obstruct the collar block decomposition: they only replace K_Σ by its central extension $\widehat{K_\Sigma}$. The condition $[z] = 0$ is exactly the strictifiability of the projective triangle defect. It is not by itself global path independence: at least one resulting strict edge 1-cocycle must also have trivial represented holonomy around every closed overlap loop. With a nontrivial action on varying centers, the corresponding statement uses the twisted/local-system Čech differential instead. (A full proof for the fixed-coefficient subbranch appears in Section 6.4 below, in the algebra-net language.)

2.1.4 Non-central obstruction (2-group cocycle)

When defects are not central, the natural coefficient data is a crossed module $(H \to G)$ with an action of G on H by conjugation. Here G is the reconstructed gauge group, and H is the unitary group acting on edge multiplicity spaces, with boundary map $\partial : H \to G$.

A crossed module is a homomorphism $\partial : H \to G$ together with an action of G on H such that

$$\partial(g \triangleright h) = g\,\partial(h)\,g^{-1}, \qquad \partial(h) \triangleright h' = hh'h^{-1}.$$

On a good cover $\{P_i\}$, a weakly coherent gluing is encoded by:

$$g_{ij} : P_{ij} \to G, \qquad h_{ijk} : P_{ijk} \to H,$$

obeying the 2-cocycle conditions

$$g_{ij}g_{jk} = \partial(h_{ijk})g_{ik},$$

and on quadruple overlaps,

$$h_{jkl}h_{ijl} = (g_{ij} \triangleright h_{ikl})h_{ijk}.$$

Gauge changes act by 1- and 2-cochains in the standard way for crossed-module cohomology.

Write

$$q_\Sigma = [(g,h)] \in \check{H}^2(N_\Sigma, H_\Sigma \to G_\Sigma)$$

for the full crossed-module gluing orbit. It classifies the equivalence orbit of the pair (g,h) but does not in general determine a unique ordinary G_Σ-valued 1-cocycle class after strictification, because the map below need not be injective. Let

$$\begin{aligned} \iota_\Sigma : \check{H}^1(N_\Sigma, G_\Sigma) &\longrightarrow \check{H}^2(N_\Sigma, H_\Sigma \to G_\Sigma), \\ [g] &\longmapsto [(g,1)] \end{aligned}$$

be the natural strict-locus map. Define the separate $\{0,1\}$-valued strictification obstruction by

$$\begin{aligned} o_\Sigma^{(2)}(g,h) = 0 \quad &\Longleftrightarrow \quad q_\Sigma \in \mathrm{im}(\iota_\Sigma) \\ &\Longleftrightarrow \quad [(g,h)] \text{ has a representative } (g^{\mathrm{str}}, 1) \\ &\qquad\qquad \text{with } g^{\mathrm{str}}_{ij} g^{\mathrm{str}}_{jk} = g^{\mathrm{str}}_{ik}. \end{aligned}$$

and set $o_\Sigma^{(2)}(g,h) = 1$ otherwise.

Theorem 3.4 (non-central higher-defect strictification). The higher associator defect of (g_{ij}, h_{ijk}) is removable iff $o_\Sigma^{(2)}(g,h) = 0$. In that case a higher-gauge change gives $h_{ijk} = 1$ and a genuine G-valued edge 1-cocycle. Strict endpoint-only transport additionally requires that at least one allowed strict representative have trivial represented holonomy. The full orbit q_Σ need not determine a unique ordinary H^1 class.

Proof sketch. Removing the higher associator corresponds to $h_{ijk} = 1$ and $g_{ij}g_{jk} = g_{ik}$. Crossed-module coboundaries preserve the full orbit, so this strictification exists exactly when that orbit meets the image of the ordinary G-valued 1-cocycle locus, which is the definition of $o_\Sigma^{(2)} = 0$. The map ι_Σ need not be injective: H-valued edge changes can relate strict representatives and alter their ordinary holonomy through ∂. Thus the invariant residual test is whether at least one allowed strict representative has trivial represented holonomy, as in Theorem 3.4b. QED.

The central-defect case is the abelian truncation with H central and trivial action, which reduces to Section 3.3. A genuinely noncentral class is the point where the fixed-cutoff collar theorem upgrades from the ordinary-group package to the higher-gauge one below.

Corollary 3.4a (Fixed-cutoff higher-gauge EC and transportability). For a finite regulator chart on a connected cut Σ, the genuinely noncentral branch admits a compact crossed-module change system

$$\mathcal{T}_\Sigma = C^1(N_\Sigma, H_\Sigma) \rtimes C^0(N_\Sigma, G_\Sigma).$$

The physical higher-gauge collar is

$$\mathcal{H}_B^{2g} = (\widetilde{\mathcal{H}}_{B_L} \otimes \widetilde{\mathcal{H}}_{B_R})^{\mathcal{T}_\Sigma} \cong \bigoplus_\lambda (\mathcal{H}_{b_L^\lambda} \otimes \mathcal{H}_{b_R^\lambda}),$$

with

$$Z(\mathcal{A}_{2g}(B)) = \bigoplus_\lambda \mathbb{C}\,\mathbf{1}_\lambda,$$

and the full orbit q_Σ is invariant under local rechartings and classifies the fixed-cutoff genuinely noncentral gluing datum. Its separate invariant property $o_\Sigma^{(2)} = 0$ holds iff the higher associator is removable. The resulting strict representatives can have different ordinary G_Σ-valued 1-holonomies, so transportability is the separate existential test of Theorem 3.4b rather than a uniquely defined holonomy attached to q_Σ.

Proof sketch. Pair-overlap rechartings are inner, while triple-overlap associators strictify to compact crossed-module data. Finite-dimensional unitary $\mathcal{T}_\Sigma$-modules decompose semisimply, and Schur matching leaves only diagonal left/right blocks. Crossed-module cohomology classifies the higher defect; Theorem 3.4b separately tests the ordinary loop holonomy that remains after strictification. QED.

Theorem 3.4b (TransportabilityFromOverlapGluing). Fix a connected cut Σ and a finite regulator charting nerve N_Σ. Transport of a collar charge is defined by the path composite in the overlap recharting groupoid: along $p = (i_0 \to \cdots \to i_m)$, the charge block is moved by $U_p = U_{i_{m-1}i_m} \cdots U_{i_0 i_1}$. This construction uses the overlap unitary transition system itself, not an added DHR transportability assumption.

After a central or crossed-module strictification, let U^{str} denote a resulting genuine edge 1-cocycle. Write $\mathcal{C}_\alpha$ for the full transported collar-charge block, including its boundary carrier and multiplicity/intertwiner data, and define

$$\mathrm{Hol}_{\alpha,U^{\mathrm{str}}} : \pi_1(|N_\Sigma|, i_*) \longrightarrow U(\mathcal{C}_\alpha), \qquad [\gamma] \longmapsto U^{\mathrm{str}}_\gamma\big|_{\mathcal{C}_\alpha}.$$

On the ordinary branch, transport is strictly path-independent iff every closed overlap loop has trivial holonomy on the collar-sector block. On the central branch, elementary triangle moves accumulate the central cocycle z_{ijk}. Strict path-independent transport exists iff $[z]_\Sigma = 0$ and there is a central 1-cochain strictification for which $\mathrm{Hol}_{\alpha,U^{\mathrm{str}}} = 1$. The first condition removes the projective triangle defect; the second removes the residual ordinary loop action. If $[z]_\Sigma = 0$ but the second condition fails, the lifts form a genuine 1-cocycle and transport is path dependent.

On the genuinely noncentral branch, path comparisons are H_Σ-valued 2-morphisms in the crossed module $H_\Sigma \to G_\Sigma$. Strict ordinary transport exists iff $o^{(2)}_\Sigma = 0$ and at least one crossed-module strictification gives a genuine G_Σ-valued 1-cocycle with trivial represented holonomy on $\mathcal{C}_\alpha$. If $o^{(2)}_\Sigma \neq 0$, the sector has higher transport rather than ordinary compact-group DR transport. If $o^{(2)}_\Sigma = 0$ but every allowed strict representative has nontrivial represented G_Σ-holonomy, the higher associator has strictified but ordinary path dependence remains. The full orbit $q_\Sigma = [(g, h)]$ need not determine a unique ordinary H^1 class and is not itself the strictification test. Thus "zero obstruction" below means the combined associator-strictification and trivial-holonomy criterion, not vanishing of triangle or higher-associator data alone. QED.

Triangle-free cycle check. Let $N_\Sigma = C_n$, $n \geq 4$, have no filled 2-simplices. On $\mathcal{C}_\alpha = \mathbb{C}^2$, put identity transport on every oriented cycle edge except one, where the sector action is the non-scalar unitary $V = \mathrm{diag}(-1, 1)$, and use inverse transports on reversed edges. All central 2-data and higher associator data are vacuously trivial, so $[z]_\Sigma = 0$ and $o^{(2)}_\Sigma = 0$. Its ordered cycle holonomy is $V \neq 1$, which no central rephasing removes. For a genuinely noncentral instance, take the crossed module

$$\mathrm{SU}(2) \hookrightarrow \mathrm{U}(2)$$

with conjugation action and the standard action on $\mathcal{C}_\alpha$. The band $\mathrm{U}(2)/\mathrm{SU}(2) \cong \mathrm{U}(1)$ is detected by the determinant, and the cycle has band holonomy $\det V = -1$. An SU(2)-valued edge change has determinant one and a vertex-frame change only conjugates the based loop, so no allowed strict representative has trivial holonomy. The two routes between the endpoints of the distinguished edge differ, and the combined criterion rejects both the central and noncentral assignments.

2.2 Gauge Reconstruction and Standard Model Structure

2.2.1 Edge sector category and gauge group reconstruction

At any fixed UV cutoff, edge-center completion provides finitely many visible seed charges and finite-dimensional intertwiners. Their rigid tensor closure is defined below and may have infinitely many simple classes. The local MaxEnt / collar-mixing package established above controls only fixed-cutoff recoverability, modular-support localization, and carried error terms on the realized branch. It is logically separate from the question whether zero-obstruction edge sectors survive refinement. The fixed-stage category is theorem-produced; the refinement/fiber ladder and cofinal compact-gauge witness require the explicit compact-gauge refinement receipt defined below.

On the ordinary or central-defect branch, path-independent movement of collar charges is supplied by overlap gluing. TransportabilityFromOverlapGluing constructs transport from overlap paths and proves the exact combined criterion: the central triangle class $[z]_\Sigma$ must vanish and at least one allowed strictification must have trivial represented sector holonomy. On the genuinely noncentral branch, $o_\Sigma^{(2)} = 0$ strictifies the higher associator, after which at least one allowed strict G_Σ-valued representative must have trivial represented holonomy. A nonzero $o_\Sigma^{(2)}$ remains a higher-gauge sector; an orbit for which every strict representative has residual loop action is also outside the ordinary path-independent DR sector. The full orbit q_Σ need not determine a unique ordinary H^1 class and is not itself the strictification test. Thus the overlap obstruction calculus classifies and routes sectors; it does not by itself select the Standard Model.

For the tensor construction at cutoff r, choose one common stagewise strict edge cocycle U_r^{str} from the allowed representatives (the ordinary branch uses its given strict system). Retain only seeds on which this same representative has trivial action around every closed overlap loop. Sectors requiring incompatible strictifications define alternative fixed-stage categories; they are not unioned before tensor closure.

Classification is not realization.

Proposition (obstruction neutrality of the finite Standard Model gauge implication). The ordinary trivial-holonomy condition, the combined central condition $[z]_\Sigma = 0$ plus a trivial-holonomy strictification, and the combined noncentral condition $o_\Sigma^{(2)} = 0$ plus an allowed strict representative with trivial represented G_Σ-holonomy are transportability conditions. After one common stagewise strict representative is chosen, they permit an ordinary transportable bosonic sector category generated only by sectors trivial under that choice; they do not select

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}.$$

On a cofinal tail carrying the compact-gauge refinement receipt, DR/Tannaka reconstruction returns $G = \mathrm{Aut}_\otimes(\mathcal{F})$ for the constructed sector category and fiber functor. If that category is trivial, G is trivial; in general G is whatever compact group the tensor-fiber data reconstructs. The separate finite response producer and the conditional charge-conjugate pair of rank-15 chiral projectors give determinant balance, the Standard Model charge lattice, and a common $\mathbb{Z}_6$ tensor kernel. Quotienting by the full kernel gives the maximal faithful matter image displayed above. The cover and its intermediate quotients carry the same local tensors. The six-axis artifact computes the full order-six menu after diagonal and zero-sum axis relations are declared. The carrier does not select those relations, so this menu cannot independently select the physical quotient. A complete source character category, a same-source loop-to-kernel identity, laboratory identification, and four-dimensional bundle data are not supplied. This implication uses only the stated premises and does not identify the finite current with the independently reconstructed Tannaka group. QED.

Definition (fixed-cutoff tensor-generated sector category). Let $S_r = \{W_{\alpha,r} : P_{\alpha,r}$ is visible and has trivial holonomy under the common $U_r^{\mathrm{str}}\}$. Define

$$\mathsf{Sect}_r^{\mathrm{bos}} = \langle \mathbf{1}, S_r, S_r^* \rangle_{\oplus,\otimes,\text{subobjects,isomorphism}} \subset \mathrm{Rep}_{\mathrm{fd}}(\widehat{K}_{\Sigma,r}).$$

This is the replete full additive Karoubi rigid tensor subcategory generated by the finitely many visible carriers. The original projectors $P_{\alpha,r}$ identify only one-collar seeds. The simple objects are all irreducible summands of finite tensor words in seeds and conjugates; they need not have projectors in the original one-collar algebra, and there may be infinitely many of them even though every object and Hom space is finite-dimensional. A subobject $X \subset T$ is supported by its invariant idempotent in $\mathrm{End}_{\widehat{K}_{\Sigma,r}}(T)$, or by the corresponding block in a supplied finite concatenated-collar realization of that tensor word.

Theorem (FixedCutoffBosonicSectorCategory). At every fixed regulator cutoff, on the ordinary or central-defect zero-obstruction branch and in the bosonic internal-gauge sector of the $3+1$-dimensional EFT regime, the preceding construction is a semisimple rigid symmetric C^*-tensor category. Complete reducibility of finite-dimensional representations of the compact group gives semisimplicity and subobjects; the operator adjoint and norm give the C^*-structure; conjugate representations give duals; and the standard flip gives the bosonic symmetry. For a closed loop, U_r^{str} acts as the identity on every retained seed, hence as the identity on direct sums, tensor products, conjugates, and invariant subobjects. Thus the whole generated category has one monoidally coherent path-independent transport, rather than an objectwise choice of strictification. Collar concatenation realizes tensor words whenever the corresponding finite tensor-realization receipt is supplied. The canonical carrier-forgetful functor

$$F_r : \mathsf{Sect}_r^{\mathrm{bos}} \longrightarrow \mathsf{Hilb}_{\mathrm{fd}}$$

is faithful, *-preserving, and symmetric strong monoidal. Fermionic signs, spinorial matter, and chirality are not part of this bosonic internal-gauge category; they belong to the later super-Tannakian or matter-sector lift.

Definition (compact-gauge refinement receipt). On a cofinal tail, first require a surjectivity/finite-extendability certificate for each finite-state restriction $\rho_{sr} : Q_s \to Q_r$. Only then is the commutative pullback $C(Q_r) \to C(Q_s)$, $f \mapsto f \circ \rho_{sr}$, injective. This does not produce a map of noncommutative collar algebras. For

$$\mathcal{A}_{\mathrm{EC},r} = \bigoplus_\alpha M_{d_{\alpha,r}}(\mathbb{C}), \qquad \mathcal{A}_{\mathrm{EC},s} = \bigoplus_\beta M_{d_{\beta,s}}(\mathbb{C}),$$

the receipt separately supplies multiplicities $m_{\beta\alpha}^{rs}$ and unitaries V_β^{rs}, with $d_{\beta,s} = \sum_\alpha m_{\beta\alpha}^{rs} d_{\alpha,r}$, and defines

$$\big(\jmath_{rs}(a)\big)_\beta = V_\beta^{rs} \left[\bigoplus_\alpha (a_\alpha \otimes \mathbf{1}_{m_{\beta\alpha}^{rs}}) \right] (V_\beta^{rs})^*.$$

Injectivity requires every coarse α to occur. The whole center maps into the fine center exactly when each β has one coarse shadow; on that branch $\jmath_{rs}(P_{\alpha,r}) = \sum_{\beta:\mathrm{sh}_{sr}(\beta)=\alpha} P_{\beta,s}$. Composition requires $m_{\gamma\alpha}^{rt} = \sum_\beta m_{\gamma\beta}^{st} m_{\beta\alpha}^{rs}$ and coherent V's. The same receipt supplies continuous surjections

$$\pi_{sr}^\Sigma : \widehat{K}_{\Sigma,s} \twoheadrightarrow \widehat{K}_{\Sigma,r}, \qquad \pi_{tr}^\Sigma = \pi_{sr}^\Sigma \circ \pi_{ts}^\Sigma,$$

and one common strict representative U_r^{str} at each stage. The pullbacks carry coarse generators into $\mathsf{Sect}_s^{\mathrm{bos}}$, intertwine U_r^{str} with U_s^{str}, preserve trivial action of every fine overlap loop without

changing representative object by object, and agree with the displayed collar supports. The receipt also supplies compatible finite concatenated-collar realizations for every finite tensor word and invariant subobject projector. The finite-state consensus maps alone imply none of these matrix, center, compact-group, or tensor-realization data.

Theorem (RefinementFunctorAndFiberDescent). On a cofinal tail carrying that receipt, define

$$U_{rs} := (\pi_{sr}^{\Sigma})^* : \mathsf{Sect}_r^{\mathrm{bos}} \longrightarrow \mathsf{Sect}_s^{\mathrm{bos}}.$$

Then U_{rs} is fully faithful, symmetric strong monoidal, and *-preserving; it preserves the tensor unit, duals, irreducibles, and the full zero-obstruction transport criterion, with $U_{st}U_{rs} = U_{rt}$. For the canonical forgetful fibers, $F_sU_{rs} = F_r$ literally on carrier Hilbert spaces and linear maps.

Proof. For $X = (H_X, \varrho_X)$, set $U_{rs}X = (H_X, \varrho_X \circ \pi_{sr}^{\Sigma})$ and $U_{rs}(f) = f$. Surjectivity gives the exact identity

$$\mathrm{Hom}_{\widehat{K}_{\Sigma,s}}(U_{rs}X, U_{rs}Y) = \mathrm{Hom}_{\widehat{K}_{\Sigma,r}}(X, Y),$$

so pullback is fully faithful and a coarse simple cannot split. Pullback commutes with tensor products, unit, conjugates, duality maps, and the bosonic flip, while composition of the π's gives composition of the U's. The receipt intertwines the common stagewise strict representatives, so zero holonomy persists for the whole tensor-generated category without a sector-specific gauge change. Since pullback changes only the action, the comparison $F_sU_{rs}X = H_X = F_rX$ is the identity; naturality, monoidality, and refinement coherence are identity diagrams. The displayed block formula, not finite-set duality, separately proves the noncommutative algebra embedding and its center behavior. QED.

The U(1) **charge-one test.** If $W_1(z) = z$ is visible and retained by the common stagewise representative U_r^{str}, the chosen category contains $W_n = W_1^{\otimes n}$ for $n \geq 0$ and $W_{-n} = (W_1^*)^{\otimes n}$ for $n > 0$. Charge two is $W_2 = W_1 \otimes W_1$; its subobject idempotent, equivalently the unique nonzero central projection in its intertwiner algebra, is $\mathbf{1}_{W_2} \in \mathrm{End}_{\mathrm{U}(1)}(W_1^{\otimes 2}) \cong \mathbb{C}$, represented physically in the certified two-collar algebra, not necessarily in the original one-collar algebra. At the next certified refinement, $U_{rs}W_2 = ((\pi_{sr}^{\Sigma})^*W_1)^{\otimes 2}$ remains simple and $F_s(U_{rs}W_2) = \mathbb{C} = F_r(W_2)$ with identity comparison. Thus the fixed collar remains finite while the tensor-generated simple skeleton may be infinite. A finite fusion/truncation alternative would require a separately stated associative fusion quotient.

Consequently, on a cofinal tail carrying the compact-gauge refinement receipt, the EFT branch has a directed ladder $(\mathsf{Sect}_r^{\mathrm{bos}}, U_{rs}, F_r)$ of tensor-generated bosonic edge-sector categories satisfying that zero-obstruction criterion, with fully faithful monoidal pullback functors and compatible forgetful fibers. Write

$$\mathsf{Sect}_\infty := \varinjlim_r \mathsf{Sect}_r^{\mathrm{bos}},$$

for the directed colimit retaining the sectors and intertwiners that persist in that certified system. Without the receipt, only the fixed-stage categories and forgetful functors are constructed; Sect_∞ and $\mathrm{Aut}_\otimes(\mathcal{F})$ are conditional objects. The theorem below is neutral about whether Sect_∞ is trivial or nontrivial: if the realized branch furnishes only the tensor unit, the reconstructed compact group is the trivial group.

Persistence lemma. Write $U_{rs} : \mathsf{Sect}_r^{\mathrm{bos}} \to \mathsf{Sect}_s^{\mathrm{bos}}$ for the pullback functor on a receipt-certified refinement tail. If a realized zero-obstruction edge-sector class α_{r_0} appears on a sufficiently fine collar, has overlap-visible support, and has representatives $\alpha_s \in \mathsf{Sect}_s^{\mathrm{bos}}$ with $U_{st}\alpha_s \simeq \alpha_t$ for all later $t \succeq s$, then it determines a unique object $[\alpha] \in \mathsf{Sect}_\infty$. Fully faithful pullback along a surjective compact-group map cannot erase or split a coarse simple, and the receipt preserves its

full zero-obstruction data. This is a persistence result conditional on the certified ladder, not a consequence of the finite-state refinement maps and not a proof that a nontrivial colimit exists.

This gives a refinement-limit bosonic tensor category Sect_∞ of edge charges:

- objects: zero-obstruction sector labels that persist in the assumed directed system,
- morphisms: intertwiners between sectors,
- tensor product: fusion by collar concatenation, $\alpha \otimes \beta = \bigoplus_\gamma N_{\alpha\beta}{}^{\gamma}\, \gamma$,
- duals: orientation reversal / charge conjugation $\alpha \leftrightarrow \bar{\alpha}$,
- symmetric braiding in the EFT regime (no anyonic statistics in 3+1D).

The monoidal refinement maps descend the tensor unit, associators, duals, and bosonic symmetry to the colimit, while the compatible stagewise forgetful carriers descend to a faithful bosonic fiber functor $\mathcal{F} : \mathsf{Sect}_\infty \to \mathsf{Hilb}_{\mathrm{fd}}$. The fibers are finite-dimensional objectwise even though Sect_∞ can have infinitely many simple objects.

Theorem 6.1 (Receipt-conditional bosonic sector category and Tannaka/DR reconstruction). On the ordinary or central zero-obstruction bosonic EFT branch, suppose a cofinal tail carries the compact-gauge refinement receipt used by RefinementFunctorAndFiberDescent. Then the colimit Sect_∞ is a rigid symmetric C^* tensor category with faithful bosonic fiber functor $\mathcal{F}$. Let $\mathrm{Aut}_\otimes(\mathcal{F})$ denote the group of unitary symmetric monoidal natural automorphisms of $\mathcal{F}$. Then

$$G := \mathrm{Aut}_\otimes(\mathcal{F})$$

is a compact group and $\mathsf{Sect}_\infty \simeq \mathrm{Rep}(G)$. In particular G is unique up to isomorphism, and this group is a compact subgroup of a product of unitary groups.

Proof. The MaxEnt/local-Gibbs/collar-mixing package is not used here to prove nontriviality or to supply the refinement receipt. The fixed-cutoff categories are constructed by FixedCutoffBosonicSectorCategory, and on the certified tail the monoidal pullback functors and compatible forgetful fibers are constructed by RefinementFunctorAndFiberDescent. Since those functors preserve tensor products, unit, duals, and *-structure, the directed colimit inherits a rigid symmetric C^*-tensor structure, and the compatible stagewise carrier spaces descend to a faithful bosonic fiber functor $\mathcal{F}$. Fix a small skeleton of Sect_∞. For every object X, a unitary symmetric monoidal natural automorphism $\eta \in \mathrm{Aut}_\otimes(\mathcal{F})$ has a unitary component $\eta_X \in U(\mathcal{F}(X))$, so

$$\mathrm{Aut}_\otimes(\mathcal{F}) \hookrightarrow \prod_X U(\mathcal{F}(X)), \qquad \eta \mapsto (\eta_X)_X.$$

For each intertwiner $f : X \to Y$, naturality gives the closed relation $\mathcal{F}(f)\eta_X = \eta_Y \mathcal{F}(f)$. For each pair X, Y, the monoidal structure isomorphism $J_{X,Y}$ of $\mathcal{F}$ gives the closed relation

$$\eta_{X\otimes Y} = J_{X,Y}\,(\eta_X \otimes \eta_Y)\, J_{X,Y}^{-1}, \qquad \eta_{\mathbf{1}} = \mathrm{id}_{\mathbb{C}}.$$

Hence $G = \mathrm{Aut}_\otimes(\mathcal{F})$ is a closed subgroup of the compact product $\prod_X U(\mathcal{F}(X))$, so G is compact. The DR/Tannaka hypotheses are therefore satisfied on the constructed pair $(\mathsf{Sect}_\infty, \mathcal{F})$, and the Doplicher–Roberts/Tannaka reconstruction theorem gives a symmetric C^*-tensor equivalence $\mathsf{Sect}_\infty \simeq \mathrm{Rep}(G)$ [66, 67]. If another compact group G' gave the same category through a faithful bosonic fiber functor, the induced symmetric tensor equivalence $\mathrm{Rep}(G) \simeq \mathrm{Rep}(G')$ compatible with the forgetful functors would identify both groups as the automorphism group of the same fiber functor. Thus $G \cong G'$, so the reconstruction is unique up to isomorphism. QED.

Definition 6.1r (DHR/DR realization hypotheses). The field-algebra step below uses the following net-realization package, stated explicitly because it is logically additional to the Tannakian data of Theorem 6.1:

1. **(R1) Observable net.** An isotonous net $O \mapsto \mathcal{A}(O)$ of C^*-algebras over the directed set of small-region collar localizations of the EFT regime, with quasi-local algebra $\mathcal{A}$ and a faithful vacuum representation.
2. **(R2) Locality, duality, Property B.** Spacelike commutativity; Haag duality $\mathcal{A}(O')' = \mathcal{A}(O)''$ in the vacuum representation, or the explicitly weaker essential-duality variant if that is the version invoked; and Property B.
3. **(R3) Localized endomorphisms.** Each persistent class $[\alpha] \in \mathsf{Sect}_\infty$ is realized by an endomorphism ρ_O^α of $\mathcal{A}$ acting trivially on $\mathcal{A}(O_1)$ for every localization region O_1 disjoint from O.
4. **(R4) Transportability.** For every admissible pair of regions $O, \widetilde{O}$ there exists a unitary $u \in \mathcal{A}$ with $u\,\rho_O^\alpha(\cdot)\,u^* = \rho_{\widetilde{O}}^\alpha(\cdot)$. This is a net-level statement about unitary intertwiners. Vanishing collar/represented holonomy supplies path-independent collar transport at the categorical level and is a necessary input to this hypothesis, but it is not by itself equivalent to DHR transportability.
5. **(R5) Statistics and conjugates.** Each ρ^α has finite statistics with permutation-symmetric (bosonic) statistics operator matching the symmetry of Sect_∞, and a conjugate endomorphism realizing the categorical dual $\bar{\alpha}$.
6. **(R6) Completeness.** The assignment $[\alpha] \mapsto [\rho^\alpha]$ is an equivalence of symmetric tensor C^*-categories between Sect_∞ and the category of localized transportable finite-statistics endomorphisms of $\mathcal{A}$.

This paper constructs the categorical side: the fixed-cutoff categories, the refinement functors and fibers, and the colimit pair $(\mathsf{Sect}_\infty, \mathcal{F})$. It does not construct the net realization (R1)–(R6); in particular no observable net is specified here on which the persistent sectors act as localized transportable endomorphisms. (R1)–(R6) therefore enter as explicit hypotheses, and every fixed-point identity $\mathcal{A} = \mathcal{F}_{\mathrm{net}}^G$ in this paper is conditional on them.

Corollary 6.1 (two-level gauge reconstruction on zero-obstruction sectors). The reconstruction conclusion splits into two levels with strictly different hypotheses.

(i) Tannakian level (proved). If in the small-region limit the edge sectors satisfy the zero-obstruction transport criterion of Theorem 3.4b and lie on a cofinal tail carrying the compact-gauge refinement receipt, then Theorem 6.1 yields a compact group $G = \mathrm{Aut}_\otimes(\mathcal{F})$ and a symmetric tensor equivalence $\mathsf{Sect}_\infty \simeq \mathrm{Rep}(G)$. This level uses only the constructed category and fiber functor and involves no observable net.

(ii) DHR/DR field-net level (conditional). If in addition the realization hypotheses (R1)–(R6) of Definition 6.1r hold, then the Doplicher–Roberts reconstruction theorem [66, 67] supplies a field algebra $\mathcal{F}_{\mathrm{net}}$ with normal (here bosonic) commutation relations, carrying an action of G with field operators implementing every persistent sector, such that

$$\mathcal{A} = \mathcal{F}_{\mathrm{net}}^G.$$

Proof. Level (i) is Theorem 6.1. For level (ii): under (R1)–(R6) the localized transportable finite-statistics endomorphisms of $\mathcal{A}$ form a rigid symmetric C^*-tensor category equivalent to Sect_∞ by (R6), hence to $\mathrm{Rep}(G)$ by level (i). The Doplicher–Roberts theorem [66, 67], whose locality, duality, Property B, transportability, statistics, conjugate, and completeness inputs are exactly (R2)–(R6) on the net (R1), then produces the crossed-product field net $\mathcal{F}_{\mathrm{net}}$, its field operators, and the identification $\mathcal{A} = \mathcal{F}_{\mathrm{net}}^G$ with G as gauge group. Without (R1)–(R6) the displayed fixed-point identity is not asserted; the unconditional content of this corollary is level (i). QED.

Corollary 6.1a (The compact-gauge setup and realized witness). On the central branch, the categorical collar-transport input to Corollary 6.1 is the combined theorem-level condition

$[z]_\Sigma = 0$ plus trivial represented holonomy for an allowed strictification; this condition feeds the sector-category construction and hypothesis (R4) of Definition 6.1r, but it is not itself the net-level DHR-transportability statement, which additionally requires localized endomorphisms and unitary transporters on an observable net. On the genuinely noncentral branch, strict ordinary transport requires both $o_\Sigma^{(2)} = 0$ and at least one allowed strict representative with trivial represented G_Σ-holonomy; a nonzero $o_\Sigma^{(2)}$ remains a higher-gauge fixed-cutoff sector, while an orbit for which every strict representative has residual loop action remains nontransportable for the ordinary DR corollary. The full orbit q_Σ classifies the crossed-module representatives but need not determine a unique ordinary H^1 class. FixedCutoffBosonicSectorCategory constructs the fixed-stage category. RefinementFunctorAndFiberDescent constructs the fully faithful pullback functors and compatible forgetful fibers only from the explicit compact-gauge refinement receipt. Thus DR/Tannaka reconstruction yields a compact group on the receipt-certified branch at the Tannakian level; the field-algebra identity $\mathcal{A} = \mathcal{F}_{\mathrm{net}}^G$ additionally requires the DHR/DR realization hypotheses (R1)–(R6). The local Standard Model gauge Lie algebra follows from the complete-response and endogenous-transport clauses. Its declared matrix current is a conditional realization. Equality with the Tannaka group requires a same-source current construction and an additional commuting-square identification.

Theorem (categorical gauge-sector classification). On the OPH compact-gauge branch, conditional on a cofinal tail carrying the compact-gauge refinement receipt, the categorical reconstruction factors as

$$\begin{aligned}\text{overlap/gluing data} &\longrightarrow \text{associator strictification test} \longrightarrow \text{strict-representative holonomy test}\\ &\longrightarrow \mathsf{Sect}_\infty \longrightarrow G_{\mathrm{Tan}} = \mathrm{Aut}_\otimes(\mathcal{F}).\end{aligned}$$

The first arrow computes and, where possible, removes the central or crossed-module associator defect. The second computes represented holonomy across the allowed strict representatives. The third keeps only sectors for which at least one such representative has trivial loop action. The fourth reconstructs the compact group from the persistent tensor category and fiber functor. QED.

Theorem (finite-current fixed-space theorem). Assume the complete reversible twelve-port response of premise A1 and endogenous proper-carrier transport on that same response as required by premise A2. The finite-current implication is independent of the categorical refinement receipt and assumes no ambient continuous gauge group. It factors as

$$\begin{aligned}\text{complete reversible port response} &\longrightarrow \text{compact closed twelve-dimensional current}\\ \text{endogenous proper-carrier transport} &\longrightarrow \text{inner } A_5 \text{ action}\\ &\longrightarrow \mathfrak{su}(3)\oplus\mathfrak{su}(2)\oplus\mathfrak{u}(1).\end{aligned}$$

The twelve-port module has dimension twelve and one fixed line. Inner automorphisms fix the centre pointwise, so the centre has dimension at most one. Compact reductivity and compact-simple classification [77] leave the desired $1+3+8$ split and a centreless $3+3+3+3$ alternative. In the centreless branch, the inner action preserves each simple factor. Simplicity of A_5 makes each restricted action trivial or faithful, giving fixed dimension three or zero. Their sum cannot equal one, which excludes that branch. The centre consequently has dimension one. Its semisimple complement has dimension eleven, and compact-simple classification gives the unique split $11 = 3+8$. Hence

$$\mathfrak{g} \cong \mathfrak{u}(1)\oplus\mathfrak{su}(2)\oplus\mathfrak{su}(3).$$

QED.

Conditional matrix and matter continuation. The finite source producer independently derives $10J = A^3 - 4A^2 - 5A + 10I$, $R = -J$, and the relative band signs. It does not reconstruct the current derivative or same-current holonomy. Twelve diagonal port phases have an abelian bracket. Adjoining the connected adjacency tangent generates $\mathfrak{u}(12)$, with derived rank 143. Conditional on the canonical oriented carrier, the complete target-free A_5-equivariant alternating-bracket space over $\mathbb{Q}$ has dimension fourteen and an exact rational Reynolds basis. This search-space result does not select a Jacobi bracket, compact current, ordered source history, or holonomy [28, 29]. The exact charged-double-triplet matrices are a conditional realization of the forced Lie type. Conditional on this realization, the finite matter certificate derives $P_{\mathrm{even}} - P_{\mathrm{vac}}$ and $P_{\mathrm{odd}} - P_{\mathrm{top}}$ as the unique charge-conjugate pair of rank-15 chiral projectors. Determinant balance and tensor descent give the exact hypercharge lattice, $N_c = 3$, and the conjugation-insensitive $\mathbb{Z}_6$ kernel. Quotienting by it gives the maximal faithful matter image for the declared tensor representation. The cover and intermediate quotients carry the same local tensors. The axis coefficient system gives a conditional order-six match after its relations are declared. Source selection of the matter action, complete character category, and loop-to-kernel identity is not supplied. Laboratory identification of the current and flux sectors and the equality $G_{\mathrm{Tan}} \cong G_{\mathrm{packet}}$ are not supplied. Generation count, family multiplicity, extra-sector completeness, and the charged-lepton and electroweak quantitative branches carry their own declared premises. Scalar attachment and multiplicity remain separate premises.

2.2.2 Finite Standard Model gauge packet and the diagnostic complexity order

Theorem 6.1 yields *some* compact G_{Tan}. Independently, the certified twelve-port incidence data and inverse-port pairing determine J, with $10J = A^3 - 4A^2 - 5A + 10I$. The equivariant linear-response commutant is four-dimensional. A target-blind impulse/readback protocol solves the common farthest-shell filter and produces $R = -J$. Its restrictions to $\mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}$ have fixed relative signs, with simultaneous reversal an algebraic response convention. The response theorem does not identify that sign with physical or matter charge conjugation. These facts constrain the response without selecting a Lie bracket. The complete-response and endogenous-transport clauses force the local Standard Model gauge Lie algebra by the compact fixed-space theorem. The charged-double-triplet matrix current is a conditional realization. On that realization, the finite matter certificate derives the two rank-15 projectors $P_{\mathrm{even}} - P_{\mathrm{vac}}$ and $P_{\mathrm{odd}} - P_{\mathrm{top}}$ as a unique charge-conjugate pair, and the finite hypercharge and common-kernel results do not depend on which representative is chosen. No target-dependent selector enters the Lie-type implication. The generation, extra-sector, charged-lepton, and electroweak quantitative branches each carry their own declared premises, stated at the results that consume them. Source derivation of the matrix current, matter action, complete character category, and physical $\mathbb{Z}_6$ quotient is not supplied. Laboratory identification of the current and flux sectors is also not supplied.

Definition 6.1a (admissible sector packages). A sector package $\mathfrak{S}$ consists of the connected Lie gauge-sector image relevant in the low-energy EFT, its light chiral matter content, and one Higgs doublet. Call $\mathfrak{S}$ admissible when it is (i) loop-coherent / transportable (the central triangle class vanishes, or the noncentral higher associator is strictifiable, and at least one allowed strict 1-cocycle representative has trivial represented sector holonomy), (ii) anomaly-free, (iii) refinement-stable with light chiral matter, (iv) single-Higgs Yukawa-completable with one connected abelian charge factor acting nontrivially on the coupled carrier, (v) intrinsically quark-sector CP-capable, and (vi) weak-sector UV-completable on that same one-Higgs branch. Admissibility is a property,

and the class of admissible packages is a boundary object: nothing below ranks its members, and no minimality argument enters any conclusion.

Definition 6.1b (realization space and physical equivalence). Let $\mathfrak{A}_{\mathrm{adm}}$ be the set of isomorphism classes of finite low-energy sector packages

$$\mathfrak{S} = (G^0, \mathcal{R}_{\mathrm{light}}, H, \mathcal{Y}, \mathcal{F})$$

on the ordinary or central zero-obstruction bosonic branch of Theorem 6.1, together with the explicit realized one-Higgs chiral matter package used below. Here G^0 is the connected Lie gauge-sector image, $\mathcal{R}_{\mathrm{light}}$ is the light chiral matter representation family, H is one Higgs doublet, $\mathcal{Y}$ denotes the Yukawa-completable charge data, and $\mathcal{F}$ is the descended finite bosonic fiber functor on the retained transportable sector package. A package belongs to $\mathfrak{A}_{\mathrm{adm}}$ exactly when it satisfies the six admissibility clauses of Definition 6.1a.

Two packages are physically equivalent when they are related by a compact-group isomorphism and a fiber-compatible symmetric monoidal equivalence that preserve the observer-visible representations, Yukawa invariants, anomaly polynomial, hypercharge lattice up to the allowed U(1) normalization, and the one-Higgs branch, modulo generation relabeling, charge-conjugation convention, gauge-center quotienting, implementation hiding, and inert ancillary stabilization.

Proposition 6.1e (scope of the admissible class). Membership in $\mathfrak{A}_{\mathrm{adm}}$ is a property modulo the equivalence relation of Definition 6.1b; it is not uniqueness of a microscopic regulator representative, and it selects nothing. The complete-response and endogenous-transport clauses fix the connected local Standard Model gauge Lie algebra without reference to this class. The declared matrix current and matter packet give the exact hypercharge lattice, $N_c = 3$, and the common $\mathbb{Z}_6$ kernel as conditional algebraic results. They do not provide a source-selected current, matter action, or global quotient. The scalar scan fixes compatible scalar charges and Yukawa channels, not scalar attachment or multiplicity. The CP and weak-sector clauses leave the generation window $3 \le N_g \le 5$. A separate exact screen theorem conditionally selects a rank-three response under additional realization and cost premises. Tensoring it with the declared generation table gives a conditional rank-45 candidate whose chirality and diagonal $\mathbb{Z}_6$ action come from that table. A separate local-domain receipt checks the declared operator $D_\sigma \otimes I_{45}$ and conditional gap inheritance. No certified source, domain, or transport bridge joins the twelve-port Spin packet to the local operator domain. The construction supplies no physical matter-pole or continuum family identification, scalar attachment or multiplicity, laboratory current or flux identification, global-form selection, or completeness theorem for extra light sectors.

Note on transport obstruction. The gluing obstruction is explicit in the gauge-branch conditions. TransportabilityFromOverlapGluing identifies the central criterion as $[z]_\Sigma = 0$ together with an allowed trivial-holonomy strictification, and the genuinely noncentral criterion as $o_\Sigma^{(2)} = 0$ together with at least one allowed strict representative having trivial represented G_Σ-holonomy. When $o_\Sigma^{(2)} \neq 0$, the sector is handled by crossed-module data rather than by the ordinary DR field-algebra corollary; when $o_\Sigma^{(2)} = 0$ but every strict representative has residual holonomy, it also does not enter that corollary. The full orbit q_Σ classifies the crossed-module representatives but need not determine one ordinary H^1 class. FixedCutoffBosonicSectorCategory constructs the fixed-cutoff category. On a cofinal tail carrying the compact-gauge refinement receipt, RefinementFunctorAndFiberDescent constructs the refinement/fiber descent and the realized witness theorem supplies nonempty ordinary/central compact-gauge witness data. These categorical transport statements are independent of the conditional matrix current. No source construction identifies that current with the Tannaka group.

What the finite packet derives. Incidence determines the inverse-port pairing J. The

target-blind impulse/readback producer derives $R = -J$, and the finite certificate derives the unique charge-conjugate pair of rank-15 chiral projectors. On that contract-conditional branch, the finite packet gives:

- **Product structure**: the rank-15 packet contains the coupled carrier $\mathbb{C}^3 \otimes \mathbb{C}^2$, which enforces commuting color and weak actions.
- **Finite sector content**: the response algebra supplies the weak and color Lie types, while either representative of the derived conjugate projector pair supplies the doublet and triplet matter roles.
- **Charge and global form**: anomaly-forced determinant balance fixes the normalized charge lattice, and tensor descent fixes the conjugation-insensitive $\mathbb{Z}_6$ kernel on the realized tensors.
- **Scalar compatibility**: the finite scan fixes the compatible scalar charges and Yukawa channels; it does not fix scalar attachment, multiplicity, or a one-Higgs completion.

The Lie-type implication uses the complete-response and endogenous-transport clauses with the carrier module and compact classification. The matrix current, rank-15 matter witness, and $\mathbb{Z}_6$ kernel are exact on their declared conditional branch. Under the complete-band and operational-cost premises, the exact screen theorem selects the rank-three response band. Source selection of the current, matter action, global quotient, and laboratory flux sectors is not supplied. Physical family attachment, extra-sector completeness, the charged-lepton branch, and the electroweak quantitative branch carry their own premises.

The conditional matrix and matter realization uses the following standard lemmas. None enters the finite-current fixed-space theorem above. In particular, their compact connected groups, doublet and triplet representations, and connected abelian factor are premises of the conditional realization rather than premises of the forced local Lie-algebra result.

Lemma 6.2 (Fiber-functor-compatible product factorization). Let each symmetric tensor factor Sect_i carry a faithful symmetric tensor fiber functor $\mathcal{F}_i : \mathsf{Sect}_i \to \mathsf{Hilb}_{\mathrm{fd}}$, and set

$$G_i := \mathrm{Aut}^{\otimes}(\mathcal{F}_i).$$

Suppose the factorization equivalence $\mathsf{Sect} \simeq \mathsf{Sect}_1 \boxtimes \mathsf{Sect}_2$ identifies the reconstructed fiber functor $\mathcal{F}$ on Sect with $\mathcal{F}_1 \boxtimes \mathcal{F}_2$. Then

$$\mathrm{Aut}^{\otimes}(\mathcal{F}) \cong G_1 \times G_2, \qquad \mathsf{Sect}_i \simeq \mathrm{Rep}(G_i).$$

Indeed, a tensor-natural automorphism of the external-product fiber functor restricts to one on each factor by evaluating on $(X, \mathbf{1})$ and $(\mathbf{1}, Y)$, while the two restrictions determine its action on every (X, Y) by tensor naturality. Tannaka reconstruction applied to each $(\mathsf{Sect}_i, \mathcal{F}_i)$ gives the displayed representation equivalences. A bare categorical factorization without these compatible fiber functors does not establish the product-group statement. QED.

Lemma 6.3 (SU(2) from a pseudoreal doublet). Let H be a positive-dimensional compact connected Lie group with a faithful *irreducible* 2D pseudoreal unitary representation V. Then the connected derived subgroup of the image is $SU(2)$ acting as the fundamental doublet. Irreducibility is load-bearing: $U(1)$ acting by $\mathrm{diag}(z, z^{-1})$ is faithful and pseudoreal but reducible, and contains no $SU(2)$. Proof: a compact connected abelian group has only one-dimensional irreducibles, so the image is nonabelian; by Schur the center acts by scalars, so the derived subgroup acts irreducibly

on $\mathbb{C}^2$; the only connected compact groups acting irreducibly on $\mathbb{C}^2$ have derived subgroup $SU(2)$. Finite or disconnected counterexamples are not part of the theorem package, which only applies the lemma to the identity component on the relevant nonabelian image. QED.

Lemma 6.4 (SU(3) from an intrinsically complex triplet). Let H be a positive-dimensional compact connected Lie group with a faithful irreducible 3D unitary representation W that is *intrinsically complex on its nonabelian factor*: the connected derived subgroup of the image acts by an irreducible representation not equivalent to its conjugate. Abelian twisting does not qualify: a character twist can make the full representation inequivalent to its conjugate while the nonabelian factor stays real or pseudoreal, and such twists are excluded by the hypothesis. Then the connected derived subgroup of the image is conjugate to $SU(3)$ acting as the fundamental triplet, up to finite central kernel. The hypothesis rejects the defining representation of $SO(3)$ on $\mathbb{C}^3$ (real on the nonabelian factor) and its twists $(z, R) \mapsto zR$ under $U(1) \times SO(3)$ (real on the nonabelian factor), and retains the fundamental of $SU(3)$. Proof: by Schur the center acts by scalars, so the derived subgroup acts irreducibly; two nontrivial simple factors would force a tensor decomposition of dimension at least 4, so exactly one simple factor acts; among compact simple Lie algebras only $\mathfrak{su}(2)$ and $\mathfrak{su}(3)$ carry nontrivial irreducibles of dimension at most 3; the 3D representation of $\mathfrak{su}(2)$ is real, so the intrinsically complex hypothesis leaves the fundamental of $\mathfrak{su}(3)$. This restates Lemma `lem:su3class` of The Standard Model gauge paper. The intrinsically-complex-nonabelian hypothesis is load-bearing: without it, $SO(3)$ on $\mathbb{C}^3$ satisfies every remaining hypothesis and refutes the conclusion. Finite or disconnected counterexamples are not part of the theorem package, which only applies the lemma to the identity component on the relevant nonabelian image. QED.

Lemma 6.5 (Connected abelian factor criterion). If the admissible sector package contains a connected abelian charge factor acting nontrivially on the coupled carrier, then the identity component of the abelianized reconstructed group contains a one-torus. Under the single connected abelian-factor admissibility condition, this factor is $U(1)$. QED.

Proposition 6.6 (conditional maximal faithful matter image). Under the declared charged-double-triplet current and matter representation, the finite matter certificate derives $P_{\mathrm{even}} - P_{\mathrm{vac}}$ and $P_{\mathrm{odd}} - P_{\mathrm{top}}$ as the unique charge-conjugate pair of rank-15 chiral projectors. For either representative, the kernel acting trivially on all realized sectors is Z_6. The maximal faithful matter image is therefore

$$G_{\mathrm{packet}} = \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}.$$

Charge conjugation changes the representative and leaves this kernel unchanged. The cover and its Z_2 and Z_3 quotients also carry the local tensors, so local data alone do not select among them. The axis-center-descent certificate computes an order-six menu inside a declared coefficient system. Its relation lattice is not source-selected, so it does not promote this maximal faithful image to the physical global form. The proposition does not attach laboratory currents or identify G_{packet} with the separately reconstructed Tannaka group. QED.

Proposition 6.6a (finite Standard Model matter-image implication from the named response and matter premises). Assume the certified twelve-port carrier, the complete reversible response clause of Axiom 1, and the endogenous transport clause of Axiom 2. The compact fixed-space theorem forces the current Lie algebra

$$\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1).$$

Incidence and target-blind readback independently produce $R = -J$ and its relative band signs. The declared charged-double-triplet matrix current is an exact conditional realization of the forced

Lie type. Under that realization and the declared matter premises, the finite matter certificate derives $P_{\text{even}} - P_{\text{vac}}$ and $P_{\text{odd}} - P_{\text{top}}$ as the unique charge-conjugate pair of rank-15 chiral projectors. Anomaly-forced determinant balance and tensor descent on every realized matter tensor then give:

- either projector representative supplies the one-generation chiral matter module with the weak-doublet and color-triplet roles on a common coupled carrier;
- determinant balance and anomaly freedom fix the exact Standard Model hypercharge lattice, up to the conventional simultaneous charge reversal;
- the color multiplicity is $N_c = 3$;
- exhaustive center action on the realized tensors fixes the conjugation-insensitive kernel to $\mathbb{Z}_6$;
- the scalar scan fixes the compatible scalar charges and Yukawa channels, without fixing scalar attachment or multiplicity.

Consequently, the maximal faithful matter image is

$$G_{\text{packet}} = \frac{SU(3) \times SU(2) \times U(1)}{Z_6}.$$

The Lie-type implication uses the carrier, complete response, endogenous transport, and compact classification. The remaining conclusions use the declared matrix and matter premises. The response-sign convention and the choice between the conjugate matter projectors are separate; only the latter is the declared matter charge-conjugation choice, and no theorem here identifies the two signs. The registered adjacency recurrence generates a four-dimensional commutative word algebra. It cannot supply twelve independent generators, a nonzero bracket, or nonidentity proper rechartings. That obstruction is bounded to the registered single-recurrence packet. The diagonal-phase and adjacency controls land respectively at derived ranks zero and 143. The exact canonical-carrier alternating search space has dimension fourteen, with no selected Jacobi bracket. Its complete Jacobi system has quadratic coefficient-row rank 38 and an exact 11+27 rowspace decomposition. The noncompact or residual components outside the certified compact locus remain unclassified; the compact real locus is classified conditionally on the named inputs. Source derivation of an order-sensitive matrix current, matter action, complete character category, and physical Z_6 quotient is not supplied. No laboratory identification of the current and flux sectors, physical attachment of the conditional rank-45 family candidate, scalar attachment or multiplicity, quantum field realization, or identification of G_{packet} with the independently reconstructed Tannaka group is supplied. QED.

Theorem 6.6aa (witness landing under separate completion declarations). On the ordinary or central zero-obstruction bosonic branch, suppose a cofinal tail carries the compact-gauge refinement receipt. Take the conditional matrix current of Proposition 6.6a and declare the fermionic Spin category used by its matter certificate. The certificate then supplies the rank-15 chiral matter module through its derived conjugate projector pair. Under separate formal three-generation and one-Higgs completion declarations, the conditional compact-gauge construction admits a formal cofinal heat-kernel edge-sector label witness

$$\mathcal{W}_{\text{SM}} = \{Q_i, u_i^c, d_i^c, L_i, e_i^c, H\}_{i=1}^3$$

with

$$Q_i = (3,2)_{1/6}, \quad u_i^c = (\bar{3},1)_{-2/3}, \quad d_i^c = (\bar{3},1)_{1/3}, \quad L_i = (1,2)_{-1/2}, \quad e_i^c = (1,1)_1, \quad H = (1,2)_{1/2}.$$

The fixed-cutoff edge heat-kernel law gives $p_R(t) \propto d_R e^{-tC_2(R)}$, hence positive support for every finite-dimensional witness sector at finite t. This positivity concerns the bosonic internal-gauge

sector labels only; positive heat-kernel weight on a boundary label does not supply a light chiral fermion multiplet or scalar carrying that label. The conditional matter certificate supplies the one-generation chiral matter content. The three-family copy and one-Higgs scalar label are separate formal completion declarations, not physical attachments. The pullback functors constructed from the receipt carry the zero-obstruction labels cofinally. Proposition 6.6a gives the maximal faithful matter image

$$\frac{SU(3)\times SU(2)\times U(1)}{Z_6}, \qquad N_c = 3,$$

with the Standard Model hypercharge lattice, from the stated premises alone. On the declared one-Higgs branch the CP-capability and weak-sector UV clauses give $3 \le N_g \le 5$; the generation count inside that window is not selected by those clauses. A separate exact screen theorem selects the rank-three band from the complete faithful bands in the declared operational window, under its single-band realization and cost-order premises. Its finite simulator receipt places that band at the lowest positive generator frequency of the declared unitary response channel. The channel conserves mode norms, so this is a frequency statement rather than a relaxation statement. No identification of the selected screen band with physical matter families or exclusion theorem for extra light sectors is supplied. Source selection of the matrix current, matter action, and physical global quotient is not supplied. Laboratory current and flux identification, scalar attachment and multiplicity, physical matter-pole and continuum attachment of the conditional rank-45 candidate, selection of the local matter action, cross-domain Spin transport, equality with the Tannaka group, and quantum field realization are not supplied. QED.

Theorem (Conditional Standard Model packet under a declared minimal completion). Assume the certified incidence branch, the complete-response and endogenous-transport clauses, the conditional matrix current, the declared fermionic Spin category, the resulting conjugate pair of rank-15 chiral projectors, anomaly-forced determinant balance, and tensor descent on every realized tensor. Assume separately the ordinary or central zero-obstruction bosonic branch; a cofinal tail carrying the compact-gauge refinement receipt; and the fixed-cutoff bosonic sector category, refinement/fiber descent, and compact-group reconstruction above. Finally assume a nonempty admissible class in the sense of Definition 6.1a, a one-Higgs scalar attachment witnessed by $\mathcal{W}_{\mathrm{SM}}$, the CP-capable and weak-sector UV clauses of Definition 6.1a, and a separately declared physical attachment of the exact rank-three screen band to the matter packet. Then the completed matter and declared scalar packet has maximal faithful image data

$$\left(\frac{\mathrm{SU}(3)\times \mathrm{SU}(2)\times \mathrm{U}(1)}{\mathbb{Z}_6}, \mathcal{R}_{\mathrm{SM}}, H, \mathcal{Y}_{\mathrm{SM}}\right), \qquad N_c = 3, \qquad N_g = 3,$$

with

$$\mathcal{W}_{\mathrm{SM}} = \{Q_i, u_i^c, d_i^c, L_i, e_i^c, H\}_{i=1}^3,$$

$$Q_i = (3,2)_{1/6}, \quad u_i^c = (\bar{3},1)_{-2/3}, \quad d_i^c = (\bar{3},1)_{1/3}, \quad L_i = (1,2)_{-1/2}, \quad e_i^c = (1,1)_1, \quad H = (1,2)_{1/2}.$$

The abstract gauge Lie type uses no clause of the declared completion. The displayed matter type, charge lattice, and $N_c = 3$ are exact inside the declared matrix-current and matter fixture. Within the declared class, the CP-capability and weak-sector UV conditions give $3 \le N_g \le 5$. The exact screen-band theorem discussed after Proposition 6.9 singles out rank three under its additional realization and cost premises. The displayed physical $N_g = 3$ value enters through the completion declaration; no matter-family identification or source-completeness theorem for extra light sectors is supplied. The one-Higgs scalar is an assumption in this completed packet; the finite scan fixes only compatible scalar charges and Yukawa channels. The displayed quotient is

the maximal faithful image of the declared matter tensors. The axis-center-descent certificate obtains its order-six result after the coefficient relations are supplied. It does not source-select the complete character lattice or physical quotient. The theorem does not attach a laboratory current. The rank-three response band and the declared fifteen-state table give a conditional rank-45 tensor candidate. Proposition 4.3ea checks a separately declared local tensor-identity operator; it neither source-selects that action nor transports the twelve-port Spin packet to the local domain. Physical identification requires the source-derived matter-pole map, continuum Spin/locality, physical seam selection, residue, excluded-band, and complement-complete refinement receipts. The gauge proof is Proposition 6.6a, the hypercharge theorem, three-color corollary, and $\mathbb{Z}_6$ kernel computation. Proposition 6.9 provides the conditional generation window; the exact screen-band theorem provides the conditional finite rank-three candidate; the displayed physical completion is a declaration. Equality with the independently reconstructed Tannaka group requires a separate commuting-square receipt. QED.

Corollary 6.6b (Three colors from the finite matter packet). On the one-generation matter package supplied by the derived conjugate projector pair in Proposition 6.6a, the color factor is the fundamental triplet of SU(3), so

$$N_c = 3.$$

This exact finite implication uses the stated premises alone. The measured non-split deck cover and exhaustive matter scan supply the odd-Weyl Spin typing at finite source-model scope. QED.

Corollary 6.6c (structural electroweak force and charges). Under the finite response of Proposition 6.6a, assume separately that one compatible scalar slot is physically attached as a single Higgs doublet. On that declared one-Higgs branch, the electroweak gauge factor has Lie algebra

$$\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_Y.$$

The Higgs doublet $H = (1,2)_{1/2}$ selects a nonzero neutral vacuum vector

$$\phi_0 = \frac{v}{\sqrt{2}}\begin{pmatrix}0\\1\end{pmatrix}, \qquad v \neq 0,$$

with

$$Q\phi_0 = (T_3 + Y)\phi_0 = 0,$$

so its vector stabilizer has unbroken generator

$$Q = T_3 + Y,$$

and the unbroken gauge factor is $\mathrm{U}(1)_Q$. The charged weak generators give

$$W^{\pm} = \frac{1}{\sqrt{2}}(W^1 \mp iW^2),$$

while the neutral $\mathrm{SU}(2)_L$ and $\mathrm{U}(1)_Y$ gauge fields span the Z/A basis on the electroweak quantitative branch. Explicitly,

$$e^{i\alpha T_3}e^{i\beta Y}\phi_0 = e^{i(\beta-\alpha)/2}\phi_0,$$

so exact vector invariance ties the phases by $\beta = \alpha$, locally, and leaves $\mathrm{U}(1)_Q$. The projective ray $[\phi_0]$, however, is unchanged for independent α and β: its stabilizer is the two-torus $\mathrm{U}(1)_{T_3} \times \mathrm{U}(1)_Y$, modulo the inherited finite center. Projectivization therefore classifies the carrier ray but cannot by

itself select the electromagnetic diagonal. Thus, under the finite current theorem and separate one-Higgs attachment premise, the structural package contains the weak charged-current carriers $W^{\pm}$, the neutral weak carrier Z, the electromagnetic carrier A, and the exact charge operator $Q = T_3 + Y$. The mixing angle, v, and numerical W/Z masses belong to the running and matching surface rather than to this finite structural corollary. A machine-checked composition on the committed doublet carrier (`Lean/Screen/ElectroweakBreakingComposition.lean`) proves the corresponding exact breaking package for declared constants $\lambda, v, g, g' > 0$ and Yukawa coefficient y: minimizer shell $|h|^2 = v^2$, one unbroken line with three Goldstone directions, normalized broken-direction mass values $g^2v^2/4$ and $(g^2 + g'^2)v^2/4$ with tree ratio $g^2/(g^2 + g'^2)$, and a Yukawa bilinear of exact coefficient yv; the five constants are declared bundle parameters and no source action or physical vacuum is supplied. QED.

Corollary 6.6d (conditional Maxwell sector on the realized electromagnetic branch). On the conditional, separately declared one-Higgs branch of Corollary 6.6c, assume the finite current is realized by a support-visible compact-gauge connection, and let A_Q be its component on the unbroken electromagnetic factor $\mathrm{U}(1)_Q$. The abelian restriction of the curvature gives

$$F_Q = dA_Q.$$

Assume in addition the Lorentzian low-energy Maxwell action

$$S_{\mathrm{EM}}[A_Q, J_Q] = -\frac{1}{2g_Q^2}\int F_Q \wedge *F_Q + \int A_Q \wedge *J_Q,$$

the Euler-Lagrange equations are

$$dF_Q = 0, \qquad d * F_Q = g_Q^2 * J_Q.$$

After canonical electromagnetic normalization, these are Maxwell's equations. The numerical value of g_Q, equivalently α_{em}, belongs to the Ward-projected electromagnetic readout. The compact-group reconstruction and connection label do not themselves supply this action or its nonzero kinetic coefficient. QED.

2.2.3 Refinement stability and relevant transverse perturbations

The state-side refinement stability used later in the gauge branch is a separate optimizer-pushforward and refinement-tail interface. It preserves the same finite local constraint family across cutoffs, so the realized UV states lie in one common finite-dimensional information-projection family. This is a substantive renormalization condition: coarse-graining a finite-range Gibbs family generically generates interactions outside any fixed finite density list. The closure defect, the moment-matching I-projection realizing $R_{\ell\to L}$, and its trace-norm residual bound are quantified in Definition 2.6a and Lemma 2.6b. Exact vanishing of the defect along the realized branch remains assumed in this interface. Axiom 3 supplies the fixed-stage optimizer and does not supply this cross-scale statement or the compact-gauge refinement receipt.

Concretely, if

$$\omega_\ell(\lambda) = Z_\ell(\lambda)^{-1}\exp\left(-\sum_x\sum_a \lambda_a O_a(x) - \sum_b \mu_b Q_b\right),$$

then any refinement channel $\Phi_{\ell\to L}$ satisfying the optimizer-pushforward and refinement-tail interface acts on the realized branch by an induced finite-dimensional map

$$\Phi_{\ell\to L}(\omega_\ell(\lambda)) = \omega_L(R_{\ell\to L}(\lambda)).$$

The resulting refinement-stable state branch is simply a trajectory or invariant subset of this finite-dimensional multiplier map. Accordingly, when sections speak of a refinement-stable directed colimit of sectors, persistence along this MaxEnt branch is only the state-side input. The additional compact-gauge receipt must supply finite extendability, center-compatible block embeddings, coherent surjective compact-group maps, preservation of the full transport obstruction, and finite tensor realizations. The MaxEnt/mixing package supplies none of those data and does not decide whether the resulting certified colimit is trivial or nontrivial.

The consensus paper makes the state-side coarse-graining connection explicit. Given quotient normal-form maps n_r, obstruction maps h_r, and coarse-graining maps (ρ_{sr}, χ_{sr}), reconciliation commutes with coarse-graining at the macroscopic readout scale whenever the two square defects

$$d_r^{\mathcal{Q}}(\rho_{sr} n_s(x), n_r \rho_{sr}(x)), \qquad d_r^{\mathcal{H}}(\chi_{sr} h_s(x), h_r \rho_{sr}(x))$$

are controlled. Exact refinement naturality gives zero defect; approximate RG matching is theorem-grade only to the extent that these defects are bounded on the selected branch. This keeps the MaxEnt refinement map $R_{\ell \to L}$ connected to the reconciliation law. Arbitrary coarse-graining channels do not thereby become OPH laws.

A relevant operator absent from the retained constraint family is a potentially unstable direction transverse to that family. Relevance controls the growth of a nonzero transverse component; by itself it neither inserts the operator nor proves that its coupling becomes nonzero.

Diagnostic 6.7 (conditional growth of a relevant transverse component). Assume the local finite-constraint branch of Axiom 3 and the separate optimizer-pushforward and refinement-tail interface. Let $\mathcal{O}$ be a gauge-invariant Lorentz-scalar deformation, and assume in addition that the induced refinement maps admit a differentiable extension to a neighborhood in an enlarged EFT coupling space carrying a transverse coordinate $g_{\mathcal{O}}$. Along the selected trajectory, suppose the linearized transverse update over the scales being compared is

$$\delta g_{n+1} = \Lambda_n \, \delta g_n, \qquad |\Lambda_n| \geq q > 1.$$

Then every nonzero infinitesimal transverse component grows by at least q^m after m such linearized steps. An additive source or operator-mixing term creates that component only when its transverse projection is nonzero. If the zero-coupling surface is invariant and the trajectory starts on it, relevance alone does not generate $\mathcal{O}$.

Argument. Iterating the displayed linearization gives $|\delta g_{n+m}| = \prod_{j=0}^{m-1} |\Lambda_{n+j}| \, |\delta g_n| \geq q^m |\delta g_n|$. Generation from zero is a separate statement about the transverse part of the full extended map, which is not a coordinate of $R_{\ell \to L}$ on the retained multiplier family and is not supplied by Axiom 3. Exact tuning, an invariant or stable manifold, vanishing operator mixing, cancellations, an additional symmetry, fixed-point dynamics, or another source-selected mechanism can therefore keep the transverse component zero. The diagnostic establishes linear instability only after a nonzero component and the stated growth bound are supplied; it is not an exhaustive protection theorem.

Conditional consequence 6.8 (chirality as one protection mechanism). A gauge-invariant Dirac mass term is a relevant scalar when the matter representations allow it. If an explicit extended refinement map has nonzero mixing into that mass coordinate and satisfies Diagnostic 6.7, then a nonzero mass perturbation is linearly amplified unless a cancellation or protecting mechanism intervenes. Chiral matter content is one robust way to forbid the Dirac mass by symmetry. The diagnostic does not exclude a vector-like branch with vanishing mixing, exact tuning, an invariant manifold, fixed-point protection, or another dynamical mechanism, and therefore does not select chirality by itself.

2.2.4 The conditional generation window from CKM CP capability and weak-sector completability

Anomaly cancellation is generation-by-generation, so it does not fix the number of generations. On the separately declared one-Higgs branch, the lower and upper bounds come from the declared CP-capability and weak-sector UV clauses. No selection inside the resulting window follows from those clauses or from the diagnostic order.

Proposition 6.9 (conditional generation window). On a separately declared one-Higgs quark branch built over the conditional matter packet of Proposition 6.6a, with the derived $N_c = 3$ from Corollary 6.6b and with the declared CP-capability and weak-sector UV clauses, the admissible generation window is

$$3 \leq N_g \leq 5.$$

The finite theorem does not select a physical generation count inside this window.

Inputs.

1. **Intrinsic CKM CP capability** is part of the declared quark branch through clause (v) of Definition 6.1a.
2. **Weak-sector UV completability** is part of the same declared one-Higgs branch through clause (vi) of Definition 6.1a.
3. Use the derived $N_c = 3$ from Corollary 6.6b.

Step 1: CKM CP-capability lower bound. The number of physical CP-violating phases in an N_g × N_g CKM matrix is:

$$\#(\text{CP phases}) = \frac{(N_g - 1)(N_g - 2)}{2}.$$

- For N_g = 1, 2: this is 0 → **no intrinsic CKM CP capability**.
- For N_g = 3: this is 1 → **intrinsic CKM CP capability is available**.

So intrinsic CKM CP capability requires:

$$N_g \geq 3.$$

Step 2: SU(2) asymptotic freedom upper bound. The one-loop coefficient is:

$$b_{\mathrm{SU}(2)} = \frac{22}{3} - \frac{1}{3} N_g (N_c + 1) - \frac{1}{6},$$

where the final $-1/6$ is the contribution of one complex Higgs doublet. Asymptotic freedom means $b_{\mathrm{SU}(2)} > 0$, i.e.,

$$N_g (N_c + 1) < \frac{43}{2}.$$

With N_c = 3, we have N_c + 1 = 4, so:

$$4 N_g < \frac{43}{2} \quad \Rightarrow \quad N_g \leq 5.$$

Combining: 3 ≤ N_g ≤ 5.

Step 3: the window is the complete conclusion. The two bounds are the numerical content of clauses (v) and (vi) of Definition 6.1a on the declared one-Higgs branch. They leave the window $\{3, 4, 5\}$ and select nothing inside it. Selection of a generation count requires a source-derived family attachment. QED.

Why this is convincing.

- The clause set fixes the exact three-member window $\{3, 4, 5\}$; clause (v) (CKM CP capability) is a declared input that sets the lower edge.
- It uses two explicit declared clauses (intrinsic CKM CP capability and weak-sector UV completability) and nothing else; the window is their complete numerical content.
- It is not a fit to a continuous number.
- Under the stated finite gauge-packet and declared-branch hypotheses, this is a conditional window result. It is not forced by anomaly cancellation, the A_5 graph, or the target-free source reduct.

Proposition (conditional screen-band selection). The 12-port coefficient module of the finite screen decomposes into bands of ranks $1, 3, 3', 5$. The exact operational costs of the nontrivial bands obey

$$5 - \sqrt{5} < 6 < 5 + \sqrt{5}.$$

Assume that one complete faithful band in the declared rank window $3 \le r \le 5$ realizes the family carrier and that the lower operational cost selects between admissible complete bands. The rank-three band is then unique. Tensoring it with the separately certified rank-15 one-generation packet gives a conditional rank-45 candidate. Chirality and the diagonal $\mathbb{Z}_6$ action remain properties of the generation table. These implications are kernel-decided in the Lean screen certificate.

The response receipt uses the declared combinatorial graph Laplacian as a unitary generator. Its pole-residue reconstruction recovers the same exact cost bands and places the rank-three residue at the lowest positive generator frequency. Mode norms are conserved by this channel; no relaxation rate is inferred. The result is an exact conditional screen selection and a finite simulator realization. Its physical interpretation as three matter families requires matter-pole identification, the continuum Spin/locality limit, physical seam selection, refinement compatibility, and laboratory attachment. It does not establish the absence of other light sectors.

Corollary 6.9a (finite gauge packet with a declared three-family completion). Assume the incidence branch, the conditional matrix current, and the fermionic Spin category of Proposition 6.6a. On the ordinary or central zero-obstruction bosonic branch, assume the admissible class of Definition 6.1a is nonempty and contains the separately declared one-Higgs chiral matter packet used through Proposition 6.9. Then the finite packet fixes

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}, \qquad N_c = 3,$$

with the hypercharge lattice and finite quotient of Proposition 6.6, from those premises alone. The conditional screen-band proposition supplies an exact rank-three finite candidate under its additional premises. A separately declared three-family completion attaches that candidate to the physical matter packet and displays $N_g = 3$ inside the conditional window. No source-derived physical attachment or source-completeness theorem for extra light sectors is supplied. The displayed quotient is the maximal faithful matter image. The finite source does not select the physical $\mathbb{Z}_6$ quotient. Laboratory current and flux identification, scalar and family attachment, equality with the Tannaka group, quantum field realization, and source-law uniqueness are not supplied.

Proof. Proposition 6.6a forces the finite current Lie type and gives the conditional matter, charge, and faithful-image implications. Corollary 6.6b fixes $N_c = 3$, and the hypercharge and quotient propositions fix the visible charge lattice and $\mathbb{Z}_6$ kernel. Proposition 6.9 supplies the conditional window, while the conditional screen-band proposition supplies the exact finite rank-three candidate. The displayed physical $N_g = 3$ is a declaration attaching that candidate to the matter packet. The equivalence relation removes relabelings, the conventional simultaneous charge reversal, gauge-center conventions, and inert implementation data. QED.

2.2.5 Hilbert-space formulation of gluing data

Let {P_i} be a good cover of the screen. For each patch, fix a representation

$$\pi_i : \mathcal{A}_i \to \mathcal{B}(\mathcal{H}_i).$$

For each overlap, choose a unitary intertwiner

$$U_{ij} : \mathcal{H}_j \to \mathcal{H}_i$$

such that for all $O \in \mathcal{A}_{ij}$,

$$\pi_i(O) = U_{ij}\pi_j(O)U_{ij}^\dagger.$$

Normalize U_ii = 1 and U_ji = U_ij†.

Lemma 6.10 (centrality on triple overlaps). On a triple overlap define

$$\Omega_{ijk} := U_{ij}U_{jk}U_{ki}.$$

For all $O \in \mathcal{A}_{ijk}$,

$$\Omega_{ijk}\pi_i(O) = \pi_i(O)\Omega_{ijk}.$$

Proof. Conjugation by U_ki sends π_i(O) to π_k(O), by U_jk to π_j(O), by U_ij back to π_i(O). Thus conjugation by Ω_ijk fixes π_i(O), so Ω_ijk commutes with π_i(O). QED.

Lemma 6.11 (gauge behavior). If $\tilde{U}$_ij = V_i U_ij V_j† with V_i acting trivially on overlap observables, then

$$\tilde{\Omega}_{ijk} = V_i\Omega_{ijk}V_i^\dagger.$$

In particular, if Ω_ijk is central, its class is gauge invariant. QED.

2.2.6 Loop obstruction class (central defect)

Assume the defect is central, the overlap centers are identified with one fixed abelian unitary coefficient group Z_Σ, and overlap transport acts trivially on Z_Σ. With $U_{ji} = U_{ij}^{-1}$, define the implementer-level multiplier by

$$\Omega_{ijk} = U_{ij}U_{jk}U_{ki} = z_{ijk} \in Z_\Sigma,$$

equivalently $U_{ij}U_{jk} = z_{ijk}U_{ik}$. This is the abelian truncation of the full 2-group obstruction in Section 3.4. The multiplier must be defined at the implementer level: $\mathrm{Ad}(z_{ijk})$ is the identity for central z_{ijk} and does not by itself record the defect.

Theorem 6.12 (central defect strictification and residual holonomy). The family $\{z_{ijk}\}$ is an untwisted Čech 2-cocycle with coefficients in Z_Σ, and its class $[z] \in \check{H}^2(N_\Sigma, Z_\Sigma)$ is gauge invariant. On any quadruple overlap P_{ijkl},

$$z_{jkl} z_{ikl}^{-1} z_{ijl} z_{ijk}^{-1} = 1.$$

A central edge rephasing by a cochain in $C^1(N_\Sigma, Z_\Sigma)$ removes every triangle defect iff $[z] = 0$. After such a rephasing, write U_{ij}^{str} for the resulting genuine edge 1-cocycle. Let $\mathcal{C}_\alpha$ be the full transported charge block, including its carrier and multiplicity/intertwiner data. Loop-coherent, endpoint-only transport exists on that block iff, in addition,

$$\mathrm{Hol}_{\alpha, U^{\mathrm{str}}}([\gamma]) = U_\gamma^{\mathrm{str}}\big|_{\mathcal{C}_\alpha} = \mathbf{1}_{\mathcal{C}_\alpha} \qquad \text{for every } [\gamma] \in \pi_1(|N_\Sigma|, i_*).$$

Thus strict path independence is equivalent to the combined conditions $[z] = 0$ and trivial represented holonomy for one allowed strictification.

Proof. The fixed-coefficient and trivial-action hypotheses place every multiplier in the same group Z_Σ. Associativity compares $(U_{ij}U_{jk})U_{kl}$ with $U_{ij}(U_{jk}U_{kl})$ on a quadruple overlap. Substituting $U_{ij}U_{jk} = z_{ijk}U_{ik}$ gives the displayed Čech cocycle identity. A vertex-frame change $U_{ij} \mapsto V_i U_{ij} V_j^\dagger$ conjugates z_{ijk}, as in Lemma 6.11, and therefore leaves a central multiplier unchanged. Separately, a central edge rephasing $U_{ij} \mapsto a_{ij}^{-1} U_{ij}$, with $a \in C^1(N_\Sigma, Z_\Sigma)$, replaces z by the corresponding Čech coboundary. Hence $[z] = 0$ is equivalent to removing all triangle multipliers by a central 1-cochain. This yields a strict 1-cocycle, not automatically endpoint-only transport. For two paths p, p' with the same endpoints, the remaining discrepancy is the holonomy of the closed loop $p'^{-1}p$; it vanishes for every pair exactly when the displayed holonomy representation on $\mathcal{C}_\alpha$ is trivial. Conversely, strict path independence makes every triangle comparison and every represented closed-loop action trivial. QED.

If overlap transport acts nontrivially on a local system of centers, the same comparison contains the transported factor $\varphi_{ij}(z_{jkl})$ and must be written with the corresponding twisted Čech differential; that branch is not the untwisted theorem stated here.

2.2.7 EFT reduction to anomaly cancellation

Assume ExtEFT: a low-energy 3+1D chiral gauge theory exists with group G. Then the obstruction class $[z]$ is structurally analogous to the EFT 't Hooft anomaly class and is expected to map to it after a separate anomaly-descent construction. That map lies outside this manuscript, so $[z] = 0$ is used here as the internal triangle-defect strictifiability condition rather than a proved equivalence to EFT anomaly cancellation; the separate trivial-holonomy condition of Theorem 6.12 is required for transportability.

2.2.8 Hypercharge from anomaly freedom and Yukawas

Theorem 6.13 (Hypercharge from anomaly freedom and Yukawas). Assume gauge group SU(N_c) × SU(2) × U(1)_Y and one generation of left-handed Weyl fermions (Q, u^c, d^c, L, e^c), with a compatible scalar charge and Yukawa channels

$$QHu^c, \qquad QH^\dagger d^c, \qquad LH^\dagger e^c.$$

Then anomaly freedom and Yukawa invariance fix the hypercharges up to an overall normalization, yielding the Standard Model pattern for N_c = 3. Simultaneous reversal of all charges is the conventional charge-conjugation choice. In the OPH application, the conditional finite matter

certificate derives the conjugate rank-15 projector pair and scans the compatible scalar charges and Yukawa channels. It does not fix scalar attachment or multiplicity.

Proof. Yukawa invariance gives

$$Y_u = -(Y_Q + Y_H), \quad Y_d = -Y_Q + Y_H, \quad Y_e = -Y_L + Y_H.$$

Anomaly cancellation yields

$$\begin{aligned} SU(2)^2U(1): &\quad N_c Y_Q + Y_L = 0, \\ \text{grav}^2U(1): &\quad 2N_c Y_Q + N_c Y_u + N_c Y_d + 2Y_L + Y_e = 0. \end{aligned}$$

Solving gives

$$Y_L = -N_c Y_Q, \quad Y_H = N_c Y_Q, \quad Y_u = -(N_c+1)Y_Q, \quad Y_d = (N_c-1)Y_Q, \quad Y_e = 2N_c Y_Q.$$

With these relations, SU(N_c)^{2}U(1) and U(1)^3 anomalies vanish automatically. Fixing the normalization by Q = T_3 + Y and Q(ν_L)=0 gives

$$Y_Q = \frac{1}{2N_c}.$$

For N_c = 3,

$$Y_Q = \frac{1}{6}, \quad Y_L = -\frac{1}{2}, \quad Y_e = 1, \quad Y_u = -\frac{2}{3}, \quad Y_d = \frac{1}{3}, \quad Y_H = \frac{1}{2}.$$

Without the compatible Yukawa channels, the cubic anomaly leaves two discrete branches (Y_u, Y_d exchange). The channel scan selects the compatible scalar charge branch. It does not select scalar attachment, multiplicity, or a one-Higgs completion. QED.

Corollary 6.13a (Exact rational hypercharges). With the conditional $N_c = 3$ of Corollary 6.6b and one fixed overall charge convention, the hypercharge assignments are uniquely fixed to exact rational values:

$$Y_Q = \tfrac{1}{6}, \quad Y_L = -\tfrac{1}{2}, \quad Y_u = -\tfrac{2}{3}, \quad Y_d = \tfrac{1}{3}, \quad Y_e = 1, \quad Y_H = \tfrac{1}{2}.$$

Why this is convincing.

- These are **exact rationals**, not approximate numbers.
- Their ratios are fixed by anomaly freedom + Yukawa invariance, and the absolute lattice is fixed by the standard normalization, with no continuous parameters to adjust.
- The resulting lattice coincides exactly with the Standard Model assignments on the declared matter packet.

2.2.9 Witten anomaly on the realized color branch

Theorem 6.14 (Witten anomaly consistency on the realized color branch). On the conditional one-generation package derived in Proposition 6.6a and Corollary 6.6b, the global SU(2) anomaly is absent generation by generation.

Inputs.

1. The realized color factor is the SU(3) triplet from Corollary 6.6b.
2. The matter content per generation includes:

- one left-handed quark doublet Q which is an SU(2) doublet and carries color,
- one left-handed lepton doublet L which is an SU(2) doublet and color singlet.

3. **Witten's global SU(2) anomaly constraint** (Witten, 1982): the number of left-handed SU(2) doublets must be even.

Proof. Count SU(2) doublets per generation:

- Quark doublets: N_c copies (one per color),
- Lepton doublets: 1 copy.

Total doublets per generation:

$$N_c + 1.$$

Substituting the realized value $N_c = 3$ gives

$$N_c + 1 = 4.$$

Witten anomaly cancellation therefore requires an even number of doublets, and the realized branch satisfies it exactly:

$$4 \equiv 0 \pmod 2.$$

So the realized triplet-doublet package is globally consistent generation by generation. QED.

Role in the derivation. Under the conditional matrix current and matter typing, the finite matter packet fixes $N_c = 3$ and carries it into the anomaly and tensor-descent theorem. Witten's anomaly is a consistency check on that derived triplet-doublet package, not a residual selector that upgrades oddness to 3.

Why this is convincing.

- It checks a **global anomaly constraint** on the conditional matter branch without introducing an additional selector.
- The parity constraint is independent of continuous parameters, RG running, masses, or Yukawa values.
- It confirms that the realized SU(3) triplet plus lepton doublet package is globally consistent generation by generation.

2.2.10 Bond-dimension constraint

In tensor-network or code realizations, gauge actions act on edge factors of size χ, so emergent compact gauge groups embed in $\mathrm{U}(\chi)$. For a faithful representation of $SU(3) \times SU(2)$, the general minimal construction is the block sum $\mathbf{3} \oplus \mathbf{2}$, giving only $\chi \geq 5$. The stronger bound $\chi \geq 6$ holds only when the edge factor is required to be a single irreducible tensor-product carrier nontrivial and faithful on both factors, whose minimal choice is $\mathbf{3} \otimes \mathbf{2}$. That extra carrier hypothesis must be declared; it does not follow from compact-group embedding alone.

2.2.11 Classical carrier modes and the quantum-particle criterion

An abstract compact group identifies a symmetry type, and a classical field equation identifies a classical dynamical branch. Neither datum alone constructs a propagating quantum particle. The following receipt separates those levels.

The upstream structural results remain separate receipts. Conditional on the compact-gauge refinement receipt, Theorem 6.1 constructs the refinement-limit bosonic edge-sector category and its compact group; a field-algebra realization with that group as a local gauge symmetry requires the DHR/DR hypotheses (R1)–(R6) of Definition 6.1r. On the realized electroweak branch, a chosen nonzero neutral Higgs vacuum *vector* has stabilizer $\mathrm{U}(1)_Q$, whereas its projective ray alone has the larger two-torus stabilizer recorded above. On the gravitational side, Theorems 4.3c–4.3e and 5.1 conditionally supply the observer-facing geometry, effective Lorentzian spacetime, and classical Einstein branch. None of these structural outputs supplies the action Hessian or quantum data required below.

Definition 6.18 (carrier-mode and quantum-particle receipts). Fix a structural invariant speed $c_\star$. This is the common invariant null-cone conversion speed. Its representation as $299\,792\,458\ \mathrm{m\,s^{-1}}$ is exact by the SI definition of the metre and is not a prediction of that decimal magnitude. A *classical massless carrier-mode receipt* for a field q_X consists of: (C1) a stated background and phase; (C2) an explicit quadratic action with positive nonzero physical kinetic coefficient; (C3) a gauge fixing or constraint reduction with a finite physical projector Π_X; and (C4) a positive reduced Hamiltonian. Separately, the action Hessian restricted to the physical subspace must have the form

$$K_X^{\mathrm{phys}}(\omega,\mathbf{k}) = Z_X(\omega^2 - c_\star^2|\mathbf{k}|^2)\Pi_X, \qquad Z_X > 0.$$

Equivalently, the free reduced Green function exhibits the action-level pole

$$D_X^{\mathrm{phys}}(\omega,\mathbf{k}) = \frac{i\,\Pi_X/Z_X}{\omega^2 - c_\star^2|\mathbf{k}|^2 + i0}.$$

This receipt proves a classical massless carrier mode and the absence of an algebraic mass term in that declared quadratic action. It does not prove a particle.

A *quantum-particle receipt* additionally requires: (Q1) a positive-energy vacuum quantization and a physical Hilbert space obtained by constraint reduction or BRST cohomology; (Q2) a physical two-point function whose Källén–Lehmann measure contains a positive-residue massless contribution

$$d\rho_X(\mu^2) = Z_X^{\mathrm{pole}}\,\delta(\mu^2)d\mu^2 + d\rho_X^{\mathrm{cont}}(\mu^2), \qquad Z_X^{\mathrm{pole}} > 0;$$

or an equivalent joint energy–momentum spectrum containing a positive-residue $p^0 > 0,\ p^2 = 0$ mass shell (at fixed $\mathbf{k}$, $\omega = c_\star|\mathbf{k}|$); and (Q3) the stability/asymptotic-state or LSZ hypotheses appropriate to the phase, including a deconfined asymptotic sector for a colored carrier. Only a receipt passing (C1)–(C4) and (Q1)–(Q3) licenses a quantum-particle claim. A reconstructed group, a connection label, or a classical field equation by itself fails this criterion.

Theorem 6.18 (action-level modes on the Maxwell, pure-Yang–Mills, and Einstein branches). The following statements hold on the stated additional action and phase branches.

i. **Electromagnetic branch.** Suppose the source-free ordinary electromagnetic vacuum $J_Q = 0$ has an unbroken $\mathrm{U}(1)_Q$ connection with the Lorentzian Maxwell action

$$S_Q = -\frac{1}{4g_Q^2}\int F_{Q,\mu\nu}F_Q^{\mu\nu}\,d^4x, \qquad 0 < g_Q^2 < \infty,$$

about the ordinary vacuum, with no Higgs, Stueckelberg, medium, or nonlocal quadratic mass operator. In radiation gauge the reduced variables are the two transverse components and

$$H_Q^{(2)} = \frac{1}{2g_Q^2}\int (|\mathbf{E}_T|^2 + |\mathbf{B}|^2) d^3x, \qquad K_{Q,T} = g_Q^{-2}(\omega^2 - c_\star^2|\mathbf{k}|^2)\Pi_T, \qquad \operatorname{rank}\Pi_T = 2.$$

Equivalently, covariant ξ-gauge gives

$$D^Q_{\mu\nu}(k) = \frac{-ig_Q^2}{k^2+i0}\left(\eta_{\mu\nu} - (1-\xi)\frac{k_\mu k_\nu}{k^2+i0}\right),$$

whose longitudinal part decouples from a conserved current. Thus this branch has two classical massless electromagnetic carrier modes. Calling them photons requires the quantum-particle receipt.

ii. **Pure Yang–Mills branch.** Suppose a compact group G is supplied with the Lorentzian two-derivative pure Yang–Mills action with $0 < g^2 < \infty$, expanded about the topologically trivial background in a gauge $\bar{A}_\mu = 0$, in a perturbative or deconfined phase with no Higgs condensate. Writing the perturbation as a^a_μ, its quadratic action is

$$S^{(2)}_{\mathrm{YM}} = -\frac{1}{4g^2}\sum_a \int (\partial_\mu a^a_\nu - \partial_\nu a^a_\mu)(\partial^\mu a^{a,\nu} - \partial^\nu a^{a,\mu}) d^4x.$$

Lorenz gauge gives the standard covariant inverse, while Gauss' law and radiation gauge leave two transverse components per generator, with

$$H^{(2)}_{\mathrm{YM}} = \frac{1}{2g^2}\sum_a \int (|\mathbf{E}^a_T|^2 + |\mathbf{B}^a|^2) d^3x \geq 0,$$

with strict positivity on nonzero reduced finite-energy modes of $\mathbf{k} \neq 0$ under the stated boundary conditions. Their quadratic transverse kernel and free Green function are

$$K^{ab}_{\mathrm{YM},T} = \delta^{ab} g^{-2}(\omega^2 - c_\star^2|\mathbf{k}|^2)\Pi_T, \qquad D^{ab}_{\mathrm{YM},T} = \frac{ig^2\delta^{ab}\Pi_T}{\omega^2 - c_\star^2|\mathbf{k}|^2 + i0}, \qquad \operatorname{rank}\Pi_T = 2$$

so there are $2\dim G$ perturbative classical transverse modes and no hard mass in that quadratic expansion. It does not produce a gauge-invariant asymptotic gluon. In the confining QCD phase, and on a gauge-invariant Yang–Mills branch with a positive physical gap, the colored free pole need not occur in the physical spectrum.

iii. **Pure Einstein branch.** Suppose, in addition to the conditional smooth Einstein equation on the physical effective-spacetime and curvature/stress identification branch, that the field-content branch is the two-derivative Einstein–Hilbert action with $G > 0$ about a Minkowski vacuum with $\Lambda = 0$, without higher-curvature kinetic terms, bimetric mixing, or extra scalar/vector fields. After de Donder gauge and residual-gauge reduction, the two transverse-traceless components obey

$$S^{(2)}_{\mathrm{TT}} = \frac{1}{64\pi G}\int \left[(\partial_t h^{\mathrm{TT}}_{ij})^2 - c_\star^2(\nabla h^{\mathrm{TT}}_{ij})^2\right] d^4x,$$

$$H^{(2)}_{\mathrm{TT}} = \frac{1}{64\pi G}\int \left[(\partial_t h^{\mathrm{TT}}_{ij})^2 + c_\star^2(\nabla h^{\mathrm{TT}}_{ij})^2\right] d^3x, \qquad D^{\mathrm{TT}}_{ij,kl}(k) \propto \frac{i\Pi^{\mathrm{TT}}_{ij,kl}}{k^2+i0}, \qquad \operatorname{rank}\Pi^{\mathrm{TT}} = 2.$$

Thus the pure Einstein linearization has two classical massless spin-two wave modes on this background. On a curved Einstein background the relevant statement is instead about the Lichnerowicz operator and its boundary conditions; a Poincaré particle pole is not automatic. The present OPH derivation does not construct the metric-field quantization, physical graviton Hilbert space, or positive-residue two-point pole required by (Q1)–(Q3).

Proof. For (i), expand the Maxwell action to second order. Gauss' law removes the longitudinal canonical pair, leaving two transverse polarizations with the displayed positive Hamiltonian and dispersion $\omega^2 = c_\star^2|\mathbf{k}|^2$. Inverting the gauge-fixed Hessian gives the displayed covariant Green function. For (ii), write $F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + O(A^2)$; the quadratic Hessian is one Maxwell Hessian per Lie-algebra generator, while the nonabelian interactions begin at cubic order. For (iii), the second variation of the Einstein–Hilbert action is the Fierz–Pauli kinetic form. De Donder gauge and its residual gauge freedom reduce it to the transverse-traceless action shown, with the + and × polarizations. These Hessian calculations establish (C1)–(C4). None constructs the quantum data in (Q1)–(Q3), so no unconditional particle conclusion follows. □

Corollary 6.18a (conditional particle interpretation). If a branch in Theorem 6.18 is also supplied with its quantum-particle receipt, its positive-residue $\delta(\mu^2)$ term defines a massless particle pole with the displayed physical polarizations. Without that receipt, this paper claims only the corresponding classical or perturbative carrier mode. In particular, the Tannaka group alone does not imply a Maxwell kinetic term or deconfined phase, and the classical Einstein equation alone does not imply a graviton state.

Hard-term versus physical-spectrum boundary. The vanishing quadratic parameters $\mu^2_{Q,\mathrm{quad}}$, $\mu^2_{\mathrm{YM,quad}}$, and $\mu^2_{\mathrm{EH,TT}}$ in the displayed actions are action-level statements, not universal exact particle masses. Gauge or diffeomorphism redundancy can coexist with Higgs/Stueckelberg completions, plasma or medium self-energies, confinement, higher-derivative poles, bimetric sectors, or additional massive fields. Excluding those possibilities requires the declared field-content and phase hypotheses rather than the symmetry label alone.

2.2.12 Quotient-protected charge quantization

Theorem 6.19 (Conditional charge quantization for packet representations). If the maximal faithful matter image is

$$G_{\mathrm{packet}} = \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{Z_6},$$

as fixed in Proposition 6.6 by tensor descent on the derived projector pair, and if one separately assumes the electroweak branch with $Q = T_3 + Y$, then

Every color-singlet packet representation has integer Q-charge.

Equivalently, the conditional representation packet contains no color-singlet representation with charge such as $\pm 1/3$. This statement does not establish QFT realization or a physical particle spectrum.

Proof. The Z_6 quotient identifies the center elements (e^{2πi/3}, -1, e^{iπ/3}) ∈ SU(3) × SU(2) × U(1) with the identity. For a color-singlet state ($\tau = 0$), the SU(3) factor acts trivially. The resulting identification requires the SU(2) × U(1) quantum numbers to satisfy

$$(-1)^{2j} \cdot e^{i\pi n/3} = 1,$$

where j is the SU(2) spin and n = 6Y is the integer hypercharge label. This gives n ≡ -6j (mod 6), i.e., n ≡ 0 (mod 6) for integer j and n ≡ 3 (mod 6) for half-integer j. Equivalently: Y is integer when j is integer, and Y is half-integer when j is half-integer.

After electroweak breaking, Q = T_3 + Y. For integer j, $T_3 \in \mathbb{Z}$ and $Y \in \mathbb{Z}$, so $Q \in \mathbb{Z}$. For half-integer j, $T_3 \in \mathbb{Z} + 1/2$ and $Y \in \mathbb{Z} + 1/2$, so Q = (half-integer) + (half-integer) $\in \mathbb{Z}$. In both cases, $Q \in \mathbb{Z}$. QED.

Experimental bound: No fractionally charged color-singlet particles have been observed. Three independent high-precision bounds confirm this:

1. **Neutrality of matter** (PDG 2024): The proton-electron charge sum satisfies

$$|q_p + q_e|/e < 1 \times 10^{-21},$$

   confirming charge quantization to 21 decimal places.

2. **Fractional charge searches in bulk matter**: Silicone oil drop experiments limit fractionally charged particle abundance to

$$\text{(fractionally charged particles)/nucleon} \lesssim 10^{-22}.$$

3. **Collider searches** (CMS, PRL 134, 2025): Exclusions for stable particles with $q \in [e/3, 0.9e]$ up to masses ~ 640 GeV (95% CL).

Existing searches are consistent with this conditional structural consequence.

The representation-theoretic implication is exact under its stated response and electroweak premises. The searches do not discharge those premises or establish a QFT realization.

2.2.13 Coupling extraction from edge-sector probabilities

The edge-center completion (Theorem 2.3) yields sector probabilities p_α on collar boundaries. A conditional heat-kernel/Laplacian weighting ansatz relates these probabilities to the renormalized gauge coupling.

Abelian case (Z_n). For a Z_n gauge theory, the edge sectors are labeled by charge q ∈ {0, 1, ..., n-1}. The correct "Casimir" eigenvalue is the Laplacian eigenvalue of the boundary random walk:

$$\lambda_q = 4 \sin^2 \left(\frac{\pi q}{n} \right).$$

Note: only in the limit n → ∞ and q ≪ n does $\lambda_q \approx (2\pi q/n)^2 \propto q^2$. For finite n, the exact form is essential.

The heat-kernel ansatz assigns the sector probabilities

$$p_q \propto e^{-t(\mu)\lambda_q},$$

where t(μ) is the dimensionless heat-kernel parameter encoding the renormalization scale. It is distinct from the BW modular parameter and from operational time. The extraction formula is:

$$t(\mu) = -\frac{\log(p_q/p_0)}{\lambda_q}, \qquad g^2_{\text{ent}}(\mu) = \frac{t(\mu)}{2\pi}.$$

Consistency requires that t extracted from different charges q agrees; this is probed by finite diagnostics (see Section 6.14), without a limit or general validation theorem.

Electric-center measurement. The edge sectors are measured using the *electric-center* prescription. For a region A and boundary vertex v ∈ ∂A, define the restricted star operator:

$$Q_v^{(A)} = \prod_{\ell \in \mathrm{star}(v) \cap A} X_\ell^{\pm 1},$$

where X_ℓ is the shift operator on link ℓ. The sector projectors are:

$$P_{v,q} = \frac{1}{n} \sum_{m=0}^{n-1} \omega^{-mq} \left(Q_v^{(A)}\right)^m, \qquad \omega = e^{2\pi i/n},$$

and the probabilities are p_{v,q} = ⟨P_{v,q}⟩. This electric-center operator, built from X's rather than Z's, correctly captures the boundary gauge charge/flux that labels entanglement edge sectors.

Non-abelian generalization. For SU(N) gauge theories, the edge sectors are labeled by irreducible representations, and the conditional ansatz assigns the probabilities

$$p_j \propto d_j \, e^{-t(\mu) C_2(j)},$$

where d_j is the dimension and C_2(j) the quadratic Casimir. Extraction:

$$t(\mu) = \frac{\log(p_0/d_0) - \log(p_j/d_j)}{C_2(j)}, \qquad g_{\mathrm{ent}}^2(\mu) = \frac{t(\mu)}{2\pi}.$$

Theoretical derivation. The fixed-cutoff same-overlap edge law is carried by the finite microphysics model once the declared overlap-sector projectors, reversible thermal branch, and quasi-local local-Gibbs generator of Theorem 2.6 are in hand. On this synthesis surface, the point is to summarize that fixed-cutoff model package and to state the refinement/Peter–Weyl lift boundary explicitly rather than to relocate the leaf-level finite calculation. The finite calculation is not a physical federation, source-instrument, or support-S^2 receipt. The Laplacian spectrum selection and the one-sided sector rank do not follow from MaxEnt, gauge invariance, and locality alone; both are named hypotheses of the theorem.

Theorem 6.20 (Fixed-cutoff edge-sector law and refinement lift boundary). Under the premises A1–A3, the fixed-cutoff overlap-gauge realization of Sections 2.3 and 3.2, the quasi-local local-Gibbs form supplied by Theorem 2.6, and hypotheses (EH-1) and (EH-2) below, the regulated same-overlap edge-sector probability distribution satisfies:

$$p_R = \frac{d_R \, e^{-t\lambda_R}}{\sum_{R'} d_{R'} \, e^{-t\lambda_{R'}}}$$

where λ_R is the Laplacian eigenvalue on the R-isotypic component and t is determined by the collar Gibbs parameter.

Hypothesis (EH-1, edge Hamiltonian). The MaxEnt edge generator is the factorwise group Laplacian, $H_{\mathrm{edge}} = \Delta_G$ (for a product group $G = \prod_i G_i$, a positive combination $\sum_i c_i \Delta_{G_i}$); the separate Gibbs parameter is t. For finite groups, the scalar formula below assumes that the declared Cayley generating multiset is conjugacy-invariant and inverse-symmetric (with real weights), so its convolution Laplacian is central and self-adjoint: $\lambda_R = |S| - (1/d_R) \sum_{s \in S} \chi_R(s)$. Without centrality the Fourier transform is a matrix on each irreducible block and the character average is only its normalized trace, not a sector eigenvalue. This is a declared microphysics input, not a MaxEnt

consequence: gauge invariance and the Gibbs form force only $H_{\text{edge}} = \sum_R h_R P_R$, and any function of the sector label (for instance $h_R = C_2(R)^2$, or a finite projector energy) is gauge invariant and realizable by a spatially local lattice Hamiltonian at fixed cutoff. Differential order on group space is not spatial finite-range locality, so no locality argument excludes those alternatives. Within second-order bi-invariant differential operators the Laplacian is unique up to the stated scale freedom, and the appendix supplies a model-specific mechanism that derives (EH-1) for one declared Markov chain: a proposal kernel with weighted symmetry and a Casimir acceptance rule has the stated law as its stationary distribution. That derivation covers that chain, not every OPH/MaxEnt edge branch. Universality of (EH-1) across branches is not proved; Section 6.14 provides numerical tests.

Hypothesis (EH-2, one-sided edge algebra). The probability space is the one-sided electric-center edge algebra $\mathcal{A}_{\text{edge}} = \bigoplus_R B(V_R)$, the Gauss-law constrained algebra visible from one side of the cut, on which the sector projector has rank d_R. The convention is fixed here once: on the full two-sided space $L^2(G) \cong \bigoplus_R V_R \otimes V_R^*$ the R-isotypic projector has rank d_R^2, and a Gibbs trace taken there gives sector weights $\propto d_R^2 e^{-t\lambda_R}$ (the closed-surface partition sum $\sum_R d_R^2 e^{-tC_2(R)}$ is that two-sided object). The d_R law of the theorem is a statement about the one-sided algebra and is not obtained by tracing the two-sided Gibbs state, since the two-sided reduction keeps weight $d_R^2 e^{-t\lambda_R}$. (EH-2) declares the one-sided state assignment directly, consistent with the Markov normal form in which the edge factor on one side contributes $\log d_R$ to the entropy.

Proof.

Step 1 (Edge Hilbert space). From the fixed-cutoff overlap-gauge realization, the edge degrees of freedom at a boundary circle $\Sigma = \partial$C live in a Hilbert space transforming under the gauge group G. Microscopically this is a finite-dimensional quantum-link edge space on the declared overlap interface; the effective representation-theoretic description used only for the refinement lift is modeled by $L^2(G)$ with its Peter–Weyl decomposition [76], and the physical one-sided carrier is the (EH-2) algebra.

Step 2 (Gauge invariance). The Gauss law constrains physical states. For an entanglement cut at Σ, the physical edge observables from one side decompose sectorwise as $\mathcal{A}_{\text{edge}} = \bigoplus_R B(V_R)$ by (EH-2).

Step 3 (Edge Hamiltonian). By (EH-1) the MaxEnt generator restricted to edge modes is $H_{\text{edge}} = \sum_R \lambda_R P_R$ with P_R the (EH-2) sector projector of rank d_R.

Step 4 (MaxEnt selection). MaxEnt with the (EH-1) generator selects the Gibbs state on the (EH-2) algebra:

$$\rho_{\text{edge}} = \frac{1}{Z} e^{-tH_{\text{edge}}} = \frac{1}{Z} \sum_R e^{-t\lambda_R} P_R.$$

Step 5 (Sector probabilities). The probability of sector R is $p_R = \text{Tr}(\rho_{\text{edge}} P_R)$, and on the (EH-2) algebra $\text{Tr}\, P_R = d_R$, so

$$p_R = \frac{d_R\, e^{-t\lambda_R}}{Z}.$$

QED.

Scope. The fixed-cutoff same-overlap derivation is the theorem-bearing part of the edge-law package, conditional on (EH-1) and (EH-2). The quasi-local local-Gibbs generator used here is derived from Theorem 2.6 on its faithful/interior branch: if MaxEnt constraints are expectations of finitely many quasi-local operators and the prescribed moment vector is in the relative interior, the entropy maximizer is a finite-multiplier Gibbs state with a quasi-local generator. Boundary

optima require support restriction and do not enter this branch. The additional representation-theoretic step summarized here is the compact-group / Peter–Weyl refinement lift. The Casimir-ratio predictions, coupling extraction, α_U closure, and the 2D-YM/worldsheet bridge downstream of this law are valid on models satisfying (EH-1)/(EH-2), not as consequences of the listed premises alone; the numerical tests of Section 6.14 probe (EH-1) on explicit models. Model-family uniqueness and large-edge continuations have their own claim boundaries.

Normalization anchor: 2D Yang-Mills. The parameter t can be exactly matched to a conventional coupling in 2D Yang-Mills, where the physical Hamiltonian is literally the group Laplacian:

$$H = -\frac{g^2}{2}\Delta_G, \qquad \Delta_G \chi_R = -C_2(R)\chi_R \quad \Rightarrow \quad E_R = \frac{g^2}{2}C_2(R).$$

Euclidean evolution for "time" A (the area of a cylinder in 2D YM) gives weight(R) $\propto$ exp(-A E_R) = exp(-g^2 A C_2(R)/2). Comparing with the heat-kernel expansion K_t(U) = Σ_R d_R χ_R(U) e^{-t C_2(R)} yields the exact identification:

$$t_{\text{phys}} = \frac{g^2 A}{2} \quad \text{(in 2D YM, no ambiguity).}$$

This shows that the Laplacian + MaxEnt $\rightarrow$ heat-kernel structure is exact in a solvable two-dimensional Yang-Mills case. The coefficient in front of C_2 is fixed. This normalization anchor is a two-dimensional edge-theory check, separate from The Standard Model gauge paper's conditional support-visible compact-gauge repair-gap theorem. In any regime where the edge theory reduces to an effective 2D YM with known "Euclidean thickness" A_eff:

$$g^2(\mu) = \frac{2}{A_{\text{eff}}(\mu)} \cdot \frac{\Delta_R(\mu)}{C_2(R)},$$

and the RHS must be R-independent. This R-independence is an internal precision consistency test; the formula itself is the normalization map that connects t to the conventional gauge coupling.

2.2.14 Numerical diagnostics of the heat-kernel law

The heat-kernel/Laplacian weighting of edge sectors is probed in explicit 2D Z_n gauge models on closed geometries.

Model. A 2×2 periodic lattice gauge theory (8 links) with Z_n link Hilbert spaces and Hamiltonian:

$$H = -K\sum_p \mathrm{Re}(B_p) - h\sum_\ell \mathrm{Re}(X_\ell) - \Gamma\sum_v \mathrm{Re}(A_v),$$

where X_ℓ is the Z_n shift on link ℓ, B_p is the oriented plaquette operator (product of Z's around plaquette p), and A_v is the oriented star/Gauss operator (outgoing X, incoming X†). With K = 1 and Γ = 5, the ground state satisfies $\langle$A_v$\rangle$ = 1 at all vertices to numerical precision.

Region and edge operator. Region A consists of links whose tail has x = 0 ("half-lattice" cut). At each boundary vertex v, the electric-center edge charge is the restricted star Q_v^{(A)} = $\prod$_{$\ell \in$ star(v) $\cap$ A} X_ℓ^{$\pm$1}.

Results for $\mathbf{Z_2}$. With $\lambda_1 = 4\sin^2(\pi/2) = 4$:

h	p_0	p_1	t	g_ent
0.5	0.8266	0.1734	0.391	0.249
1.0	0.9612	0.0388	0.803	0.357
2.0	0.9917	0.0083	1.194	0.436

Z_2 has a single nontrivial Laplacian eigenvalue, so a one-parameter fit of t can always be made exactly for any admissible nontrivial weight. The Z_2 table is therefore an implementation and convention check, not a test of the heat-kernel form.

Results for Z_3 (symmetry-forced consistency check). With $\lambda_1 = \lambda_2 = 4\sin^2(\pi/3) = 3$:

h	p_0	p_1	p_2	t(q=1)	t(q=2)	g_ent	m_plaq
0.2	0.4395	0.2803	0.2803	0.1500	0.1500	0.154	2.22
0.5	0.7509	0.1245	0.1245	0.5989	0.5989	0.309	1.75
1.0	0.9606	0.0197	0.0197	1.2956	1.2956	0.454	4.07
1.5	0.9851	0.0074	0.0074	1.6288	1.6288	0.509	7.06
2.0	0.9921	0.0039	0.0039	1.8440	1.8440	0.542	10.10

Both equalities in this table are forced by symmetry before any heat-kernel ansatz is imposed. The equality $p_1 = p_2$ is exact charge-conjugation symmetry in Z_3, and the two nontrivial Fourier modes share the degenerate eigenvalue $\lambda_1 = \lambda_2 = 3$, so extracting t from q = 1 and q = 2 must give the same value for *any* conjugation-invariant distribution. The observed agreement (t_{q=1} ≈ 1.2956389318579 versus t_{q=2} ≈ 1.2956389318521 at h = 1.0, i.e. ~10^{-14}) therefore validates the implementation and the symmetry conventions only; it cannot distinguish the heat-kernel form from an arbitrary conjugation-symmetric distribution on Z_3, and it is not counted as an independent confirmation of the heat-kernel law. The substantive overconstrained tests are the Z_5 and S_3 calculations below, which involve two distinct nonzero eigenvalues.

Region-choice stability. At h = 1, the extracted g_ent is nearly independent of region size:

- 2 links (one vertex's outgoing links): g_ent ≈ 0.453
- 4 links (half-lattice): g_ent ≈ 0.454
- 6 links (three vertices): g_ent ≈ 0.453

The closeness of these three values is consistent with near-cut dominance on this finite torus. Three region sizes on one 2×2 geometry do not isolate locality or control complement, symmetry, or finite-size effects.

Results for Z_5 (golden ratio test). The Z_5 case provides a stringent test because the Laplacian eigenvalues have a distinctive ratio involving the golden ratio $\phi = (1+\sqrt{5})/2$:

$$\lambda_q = 4\sin^2\left(\frac{\pi q}{5}\right), \qquad \frac{\lambda_2}{\lambda_1} = \frac{\sin^2(72^\circ)}{\sin^2(36^\circ)} = \phi^2 \approx 2.618.$$

This ratio distinguishes the Laplacian law from naive alternatives: a linear model ($\lambda_q \propto q$) would predict ratio 2, while a quadratic model ($\lambda_q \propto q^2$) would predict ratio 4.

Direct diagonalization of the displayed Hamiltonian on the full 2×2-torus link basis (5^8 states, K = 1, Γ = 5), measuring the restricted-star edge charge at a boundary vertex of the half-lattice region, gives:

h	Measured ratio $\ln(p_2/p_0)/\ln(p_1/p_0)$	Deviation from ϕ^2
0.50	2.876	9.9%
0.20	2.750	5.1%
0.10	2.671	2.0%
0.05	2.643	1.0%

Across the four displayed weak-field samples, the measured ratio moves toward the golden ratio squared and the deviation decreases monotonically as h decreases. No limit proof, asymptotic error bound, or interval enclosure is provided, so these points do not establish convergence as h $\to 0$. The retained link-basis values are also reproduced by the public script's symmetry-reduced electric calculation, with the same displayed coupling h. For $h > 0$, the unique strictly positive link-basis ground vector is fixed by the commuting star and noncontractible electric-cut permutations. Its electric fields therefore have zero divergence and winding: they are plaquette boundaries, parametrized by four potentials modulo one common constant. This gives n^3 states instead of n^8, including 125 instead of 390,625 for Z_5, and the charge probabilities are positive sums of squared amplitudes. The reduction does not reverse or refit the electric coupling; original-link and separate electric-incidence controls check it. Its numerical solve is subject to explicit precision refusals, and at $h = 0$ the full model has degenerate topological ground sectors. The test is overconstrained at each sampled point: Z_5 has two distinct nonzero eigenvalues, t is fitted on the q = 1 sector alone, the q = 2 weight is a held-out comparison, and the deviation column reports that residual. Its decreasing values are finite-model numerical evidence consistent with the Laplacian structure, not a convergence certificate.

Significance. This supplies finite-model numerical evidence for the law (sector probabilities weighted by Laplacian eigenvalues) on the declared 2D gauge-invariant fixtures with non-flat sector distributions. Because the nontrivial Z_3 eigenvalues are degenerate, the Z_5 test (and the S_3 test below) is the more informative overconstrained comparison. The Z_5 samples do not prove a weak-field limit. The S_3 identity below establishes a large-electric-coupling asymptotic only for its declared one-plaquette model; neither calculation validates the law generally or transfers it to SU(2)/SU(3).

Results for S_3 (first nonabelian test). The abelian tests above use charge-sector projectors that reduce to Fourier modes. For nonabelian groups, the edge-sector projector must be generalized to character projectors:

$$P_{v,R} = \frac{d_R}{|G|} \sum_{h \in G} \chi_R(h^{-1}) Q_v^{(A)}(h),$$

where d_R is the dimension of irrep R, χ_R is its character, and Q_v^{(A)}(h) is the restricted gauge action at boundary vertex v acting only on links in region A.

For S_3 (the smallest nonabelian group, order 6), there are three irreps: trivial (d=1), sign (d=1), and standard (d=2). The Cayley-graph Laplacian eigenvalues for the transposition generating set are:

$$\lambda_{\text{triv}} = 0, \qquad \lambda_{\text{sign}} = 6, \qquad \lambda_{\text{std}} = 3.$$

Exact reduction on one plaquette. For the single-plaquette model (4 links), imposing Gauss's law at all vertices means the physical wavefunction depends only on the plaquette holonomy's conjugacy class. Since S_3 has exactly 3 conjugacy classes, the gauge-invariant Hilbert space

is 3-dimensional, spanned by the character states $\{|\chi_R\rangle\}$. In this basis, the edge-sector probabilities are exactly p_R = |c_R|^2 where $|\psi_0\rangle = \Sigma_R$ c_R $|\chi_R\rangle$. This is an exact identity for the one-plaquette gauge-invariant sector.

The heat-kernel ansatz predicts p_R $\propto$ d_R exp(-t λ_R). Extracting t independently from the sign and standard irreps provides an overconstrained test: the ratio λ_sign/λ_std = 6/3 = 2 is a parameter-free prediction. With $K = 1$ and four electric links, the declared reduced Hamiltonian is

$$H(h) = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 24h & -1 \\ -1 & -1 & 12h-1 \end{pmatrix}, \qquad h \geq 0.$$

The Gauss penalty with $\Gamma = 5$ contributes only a common energy shift on this physical subspace. The following values are recomputed from this matrix; the earlier table did not reproduce the declared Hamiltonian.

h	p_triv	p_sign	p_std	t (sign)	t (std)	Δt/t	log-ratio
0.5	0.962927	2.476×10^{-4}	0.036825	1.37766	1.31898	4.3518%	2.088972
1.0	0.991861	1.400×10^{-5}	0.008125	1.86139	1.83261	1.5580%	2.031406
2.0	0.998116	8.158×10^{-7}	0.001883	2.33619	2.32203	0.6080%	2.012197
5.0	0.999713	1.993×10^{-8}	2.871×10^{-4}	2.95511	2.94950	0.1897%	2.003798
12	0.999951	5.895×10^{-10}	4.890×10^{-5}	3.54195	3.53963	0.0656%	2.001312
100	0.999999	1.208×10^{-13}	6.956×10^{-7}	4.95749	4.95721	0.0056%	2.000112

The "Δt/t" column shows the fractional difference (t_sign - t_std) / $\bar{\text{t}}$. The "log-ratio" column shows log(p_sign/p_0) / log(p_std/(2 p_0)), which should equal λ_sign/λ_std = 2 if the heat-kernel form holds exactly. Equivalently, t is fitted from the standard sector alone and the sign-sector weight is a held-out prediction; log-ratio minus 2 and Δt/t are the held-out residuals. Unlike Z_3, the two nonzero eigenvalues are distinct (6 versus 3), so no symmetry forces the held-out comparison to close.

Exact finite mismatch and its model-specific asymptotic. Write the ground energy as $-x$, where $x > 0$. Its first two eigenvector equations give relative amplitudes $(1, x/(24h+x), x)$; the last gives

$$F(x,h) := x + 12h - 1 - \frac{1}{x} - \frac{1}{24h+x} = 0.$$

For each $h \geq 0$, F increases strictly from $-\infty$ to $+\infty$ as x runs from zero to infinity. Its unique positive root has a strictly positive eigenvector, which selects the ground state of this irreducible matrix with nonpositive off-diagonal entries. The root decreases from $x(0) = 2$ to zero. Put $z = x+12h$ and $y = x(x+24h)$. Multiplying the secular equation by $x(x+24h)$ gives $(z-1)y = 2z$, hence

$$\frac{p_{\text{sign}}}{p_{\text{sign}}^{\text{pred}}} = \frac{4}{y^2} = \left(\frac{z-1}{z}\right)^2 < 1, \qquad \log\frac{p_{\text{sign}}}{p_{\text{sign}}^{\text{pred}}} = -2\log\left(1+\frac{1}{z-1}\right), \quad p_{\text{sign}}^{\text{pred}} = \frac{p_{\text{std}}^2}{4p_{\text{triv}}}.$$

Thus the standard-sector fit overpredicts the sign-sector weight at every finite h, even though the discrepancy becomes small. With $a = 12h$, implicit differentiation gives

$$\frac{dz}{da} = \frac{x^{-2} - (2a+x)^{-2}}{1 + x^{-2} + (2a+x)^{-2}} > 0 \quad (a > 0).$$

Since $z(0) = 2$ and $z \to \infty$, the held-out probability ratio increases strictly from 1/4 toward one as $h \to \infty$. This is a limit within one fixed three-dimensional model, not a refinement or continuous-group limit.

The fitted diffusion time is $t_{\mathrm{std}} = \log(2/x^2)/3$. It is positive precisely for

$$h > h_* := \frac{1 + \sqrt{3} - \sqrt{2}}{24} \simeq 0.0549098852.$$

At $h = h_*$, the relative log residual and log-ratio divide by zero; below it the fitted time is negative and is outside the positive-diffusion ansatz. The unnormalized log discrepancy above remains finite throughout. On $h > h_*$, t_{std} increases and the magnitude of that log discrepancy decreases strictly. Writing the discrepancy as $D < 0$ gives $t_{\mathrm{sign}} - t_{\mathrm{std}} = -D/6$. Consequently the magnitude of the relative log residual, the tabulated $\Delta t/\bar{t}$, and log-ratio minus two all decrease strictly to zero. These identities establish the asymptotic comparison for this Hamiltonian without assuming exact heat-kernel weights at any finite coupling; they are not interval error bounds for a numerical eigensolver.

This is a nonabelian finite-model diagnostic of the edge-sector extraction mechanism. Character projectors, Cayley-graph Laplacian eigenvalues, and an overconstrained t extraction provide a template for SU(2) and SU(3). The S_3 identity proves a strict finite mismatch and transfers nothing to either continuous group.

Parameter-free predictions for SU(2) and SU(3). The heat-kernel law yields exact, parameter-free ratio predictions that require no scheme matching. Define the "Casimir log-gap":

$$\Delta_R \equiv \ln\left(\frac{p_0}{d_0}\right) - \ln\left(\frac{p_R}{d_R}\right) = t\, C_2(R).$$

Ratios of Δ_R cancel all unknowns (t, partition function):

$$\frac{\Delta_{R_1}}{\Delta_{R_2}} = \frac{C_2(R_1)}{C_2(R_2)} \quad \text{(exact, parameter-free)}.$$

SU(2) predictions. Irreps labeled by spin j have d_j = 2j+1 and $C_2(j) = j(j+1)$. The framework predicts:

- $\Delta_1/\Delta_{1/2} = 2/(3/4) = \mathbf{8/3 \approx 2.667}$
- $\Delta_{3/2}/\Delta_{1/2} = (15/4)/(3/4) = \mathbf{5}$
- $\Delta_{3/2}/\Delta_1 = (15/4)/2 = \mathbf{15/8 = 1.875}$

SU(3) predictions. Irreps labeled by Dynkin indices (p,q) have $C_2(p,q) = (p^2 + q^2 + pq + 3p + 3q)/3$. Using the fundamental $\mathbf{3} = (1,0)$ with $C_2 = 4/3$ as the reference:

- $\Delta_8/\Delta_3 = 3/(4/3) = \mathbf{9/4 = 2.25}$
- $\Delta_6/\Delta_3 = (10/3)/(4/3) = \mathbf{5/2 = 2.5}$
- $\Delta_{10}/\Delta_3 = 6/(4/3) = \mathbf{9/2 = 4.5}$
- $\Delta_{15}/\Delta_3 = (16/3)/(4/3) = \mathbf{4}$
- $\Delta_{27}/\Delta_3 = 8/(4/3) = \mathbf{6}$

These are the SU(2)/SU(3) analogs of the Z_5 golden-ratio test: exact rational numbers fixed entirely by group theory, with no adjustable parameters.

SU(3) custody boundary. No committed producer, input fixture, receipt, or regression test supports the earlier truncated one-plaquette SU(3) numerical table or its weak-coupling extrapolation. Those numbers are therefore withheld from the scientific result surface. The exact

SU(2)/SU(3) Casimir ratios above are conditional algebraic consequences of the heat-kernel ansatz, not source-selected coupling predictions. A quantitative SU(3) diagnostic requires a committed producer, truncation and convergence controls, a replayable receipt, and an independently checked error model.

Finite diagnostic summary. The finite-group calculations separate into symmetry-forced implementation checks and overconstrained finite comparisons:

- **Z_2, Z_3**: implementation and convention checks. Z_2 has one nontrivial eigenvalue (one parameter, one datum); in Z_3 the equalities $p_1 = p_2$ and t_{q=1} = t_{q=2} are forced by charge conjugation plus the degenerate eigenvalue $\lambda_1 = \lambda_2$, so the ~10^{-14} agreement is expected from symmetry alone
- **Z_5**: overconstrained golden-ratio test ($\lambda_2/\lambda_1 = \phi^2$, two distinct nonzero eigenvalues), held-out residual 1.0% at h = 0.05, with residuals decreasing across the four displayed samples as h decreases from 0.50 to 0.05
- **S_3**: overconstrained nonabelian Casimir log-ratio test (λ_sign/λ_std = 2), held-out relative log residual of magnitude approximately 0.0056% at $h = 100$, with the fit overpredicting the sign sector

The Z_3 equality is forced by symmetry and is retained only as a regression check of the implementation; it is not an independent confirmation of the heat-kernel law. The Z_5 and S_3 calculations are more informative finite diagnostics: each fits t using fewer spectral sectors than it compares, involves two distinct nonzero eigenvalues, and reports a held-out residual. The Z_5 samples supply no limit theorem or monotonicity proof. For the declared S_3 one-plaquette Hamiltonian, the exact scalar identity proves a strict held-out mismatch at every finite coupling and monotone asymptotic agreement on the positive-diffusion branch. Neither result supplies a refinement limit, interval enclosure, controlled truncation error, or continuous-group transfer. These calculations test the finite implementation of the proposed "MaxEnt + Laplacian => heat-kernel sector weights" mechanism; they do not validate it generally or for physical SU(2)/SU(3). A self-contained held-out refit script for the finite-group models is provided with the public code bundle; it fits t on a declared subset of sectors and prints held-out residuals for the remaining sectors.

2.2.15 IBM Quantum Cloud engineering checks

The IBM Quantum Cloud runs form an engineering and reproducibility data set. They test preparation and readout of OPH-motivated reduced-sector structures on superconducting-qubit devices, together with a programmed record-gated controller against restricted controller nulls. Standard quantum mechanics predicts every prepared state and dynamic circuit in the data set. The runs therefore have no power to distinguish OPH from quantum mechanics, regardless of shot count or blinding quality. Evidentiary use requires a source-closed observable for which OPH and quantum mechanics give different numerical predictions with an identifiable effect size.

Detailed circuits, representative derived summaries, the complete blinded-run receipts, and rerun instructions are public in the canonical code and data archive, beside the executable producers and frozen run records.

- **Stage 1 (Markov/recoverability benchmark).** On `ibm_marrakesh` and `ibm_fez`, the structured state was reported below both controls in conditional mutual information and above both controls in squared Uhlmann fidelity after Petz recovery. On `ibm_marrakesh`, the archived ordering is $0.2309 < 0.3890 < 0.9474$ for CMI in bits and $0.9297 > 0.8649 > 0.6066$ for squared fidelity; on `ibm_fez`, it is $0.1498 < 0.4992 < 0.9166$ and $0.9479 > 0.8117 > 0.6449$.

These summaries lack complete tomography counts and reconstructed states, so their measured rows cannot be revalidated after the numerical recovery repair. The historical values are retained without asserting a fresh reproduction. The current pipeline retains acquired counts before analysis, pools unequal shot counts, rejects incomplete tomography, and preserves valid CMI when Petz support is numerically unresolved. In that case it reports unavailable recovery metrics explicitly, without deleting small positive support. Complete-count local controls replay these behaviors; they are software checks, not new hardware runs.
The Fawzi–Renner bound is 2^{-I} for squared fidelity ($2^{-I/2}$ for root fidelity) and concerns a theorem-guaranteed recovery channel, not necessarily the tested unrotated Petz map. The Petz calculation is offline. The benchmark does not distinguish OPH from standard quantum mechanics.

- **$\mathbf{Z_3}$ hardware sanity check.** The two-sector Z_3 extraction (whose two-sector agreement is expected a priori from the degenerate eigenvalues and charge-conjugation symmetry, so this is an implementation check, not an overconstrained test) passes cleanly on hardware: across prepared $t = 0.30, 0.60, 0.90$, the mean extracted t is within about 0.02 of the target, the two independent extractions agree to within 0.0142, 0.0034, and 0.0043, and leakage stays below 0.1%. This shows that the reduced-sector preparation and readout path is internally coherent on-device before sharper ratio claims are interpreted.
- **$\mathbf{Z_5}$ exact-ratio preparation.** The programmed target is $\Delta_2/\Delta_1 = \varphi^2 \approx 2.618$. The measured values span approximately 2.407 to 2.878, and the focused high-shot $t = 0.90$ interval lies below the exact target. Because the target amplitudes are directly prepared, QM predicts the same ideal ratio.
- **$\mathbf{S_3}$ nonabelian layout diagnostic.** The first hardware run revealed a real layout-dependent bias. Reversing the qubit layout moved the mitigated ratio from 1.8724 to 2.0299, and a repeat returned 2.0657. Because the better mapping was selected after observing the bias, this is a post-hoc device diagnostic rather than a blinded confirmation.

These runs document preparation, readout, layout, and feedback engineering on real devices. They add no evidence for OPH over standard quantum mechanics because the two descriptions make identical predictions for the implemented interventions. Their role is reproducibility and controller validation.

2.3 Gauge-chain theorem and falsification boundary

The finite carrier calculation derives the incidence polynomial for J, the target-blind response $R = -J$, and its relative sector signs. Complete reversible port response and endogenous overlap transport force the local Standard Model gauge Lie algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$. The declared matrix current is an exact conditional realization. Given that realization, the declared fermionic Spin category, and rank-15 matter projector pair, the matter certificate derives anomaly balance, exact hypercharge up to conjugation, three colors, the common $\mathbb{Z}_6$ tensor kernel, and the maximal faithful matter image. The Lie-type theorem and the conditional matter theorem have separate premise sets.

The registered adjacency recurrence and inverse-port readback generate the four-dimensional commutative word algebra $\mathrm{span}\{I, A, A^2, A^3\}$. They cannot reconstruct twelve independent generators, a nonzero bracket, or nonidentity proper rechartings. This is a bounded statement about that response packet. A source-positive current requires order-sensitive port perturbations and

closed overlap words on the same carrier. Twelve diagonal phases have an abelian bracket, while adjoining connected adjacency generates $\mathfrak{u}(12)$ with derived rank 143. Conditional on the canonical oriented carrier, the complete target-free equivariant alternating-bracket space over $\mathbb{Q}$ has dimension fourteen and an exact rational basis. It selects no Jacobi bracket, compact current, source reconstruction, or same-current holonomy [28, 29].

Four central global forms act consistently on the same local tensors: the cover and its $\mathbb{Z}_2$, $\mathbb{Z}_3$, and $\mathbb{Z}_6$ quotients. The kernel theorem identifies the maximal faithful local matter image. The six-axis artifact computes an order-six flux menu after its coefficient relations are declared. This conditional menu matches the $\mathbb{Z}_6$ quotient and carries the declared matter single-valuedly. It does not select the physical global form. Source selection of the relation lattice, character-category completeness, and a same-source loop-to-kernel identity are not supplied. Laboratory identification of the current and flux sectors, four-dimensional instanton normalization, theta periodicity, monopole dynamics, equality between the finite current and the categorically reconstructed group, scalar attachment and multiplicity, physical matter-pole and continuum identification of the conditional rank-45 candidate, selection of the local matter action, cross-domain Spin transport, and quantum field realization are not supplied.

The order-six seam classification keeps two actions separate. Every pure-hypercharge character is enumerated: the fixed-space dimension menus are $\{15, 7\}$ at order two, $\{15, 3\}$ at order three, and $\{15, 7, 3, 1\}$ at order six. Particular nontrivial characters give the dimensions 7, 3, and 1. The canonical diagonal $\mathbb{Z}_6$ quotient kernel acts trivially on all fifteen realized matter states. These are different representations of the same abstract cyclic order. Refinement preserves the classified actions. The finite calculation selects no physical seam character, flux action, or 2-representation, and it does not exhaust every possible seam grammar.

The gauge result fails if the incidence identity, target-blind response, equivariant lift, anomaly scan, tensor-character enumeration, or refinement persistence receipt fails its registered negative controls. The categorical route fails outside its combined strictification, trivial-holonomy, localization, and refinement/fiber receipts.

A Technical Details for the Specialist Derivations

This appendix carries the structural gauge details that support the Standard Model derivation. It sharpens the product-group, baryogenesis-boundary, effective-worldsheet, and compact-gauge Yang–Mills surfaces without widening the recovered core beyond its stated branch conditions.

A.1 Product-Group Structural Corollaries

Theorem A.1 (Conditional factorization from product data)**.** *Assume a declared compact product group $G = G_1 \times G_2 \times G_3$, the corresponding one-sided edge algebra with factorizing trace, and a factorwise quadratic-Casimir generator*

$$H_\partial = H_\partial^{(1)} + H_\partial^{(2)} + H_\partial^{(3)} \qquad \textit{with} \qquad [H_\partial^{(i)}, H_\partial^{(j)}] = 0,$$

then

$$p(R_1, R_2, R_3) \propto \prod_{i=1}^{3} d_{R_i} e^{-t_i C_2(R_i)}.$$

This is a forward implication. Factorized weights on one represented state do not reconstruct a product group, and a product group may carry coupled invariant constraints that do not factorize. Conversely, a product-group reconstruction is licensed when the symmetric tensor category has a factorization $\mathsf{Sect} \simeq \boxtimes_i \mathsf{Sect}_i$*, each factor carries a faithful symmetric tensor fiber functor* $\mathcal{F}_i$*, and the reconstructed functor is identified with* $\boxtimes_i \mathcal{F}_i$*. Under those hypotheses,*

$$\mathrm{Aut}^{\otimes}(\mathcal{F}) \cong \prod_i \mathrm{Aut}^{\otimes}(\mathcal{F}_i).$$

Applied to the assumed transportable refinement-directed edge-sector colimit on the realized branch, together with the rigid symmetric C^**-tensor and faithful bosonic fiber-functor conditions of the compact-gauge theorem, Tannaka–Krein reconstruction first yields some compact group* G*. The displayed hypotheses alone do not select its Lie type. If, independently, the finite local-Lie-type theorem and its category/current identification certify the two nonabelian factors as* $SU(3)$ *and* $SU(2)$*, and the charge theorem certifies one connected* $U(1)$ *factor, then the connected realization is* $SU(3) \times SU(2) \times U(1)$ *up to the separately computed finite kernel.*

Corollary A.2 (No simple-GUT X/Y gauge channel)**.** *On the independently certified connected* $SU(3) \times SU(2) \times U(1)$ *realization (up to its finite kernel) and the declared twelve-dimensional current algebra, the adjoint representation is*

$$(8,1,0) \oplus (1,3,0) \oplus (1,1,0),$$

equivalently the derived nonabelian adjoint is

$$(8,1,0) \oplus (1,3,0).$$

There are therefore no gauge generators in mixed representations $(3,2,\pm 5/6)$*, i.e. no simple-GUT* X, Y *bosons on the declared product branch. The repository instance is the declared charged-double-triplet current fixture on the certified twelve-port carrier; its physical source gate is false, so the corollary is a representation-theoretic identity of the declared type, with no identification with the physical Standard Model gauge group. It excludes the ordinary* X/Y *exchange channel inside the twelve-dimensional current algebra. Heavier* X/Y*-type modes outside that algebra, scalar mediators, and every other gauge-mediated ultraviolet mechanism stay open. General proton stability and a proton lifetime require separate baryon-violating operator and ultraviolet data.*

Conditional on the declared one-generation matter table and its baryon and lepton labels, an exhaustive dimension-six census makes that boundary exact. Four Hermitian-conjugacy classes survive the gauge, Lorentz, and diagonal $\mathbb{Z}_6$ tests: $QQQL$, $QQ\,u_R e_R$, $QLu_R d_R$, and $u_R u_R d_R e_R$. Exact exterior-algebra contractions remain nonzero after the Schouten, Fierz, and identical-field Pauli relations. The gauge structure and the diagonal quotient therefore do not prove general proton stability. The census supplies no source coefficient, decay amplitude, QCD matrix element, or lifetime.

A.2 Finite-Quotient Baryogenesis Source Theorem and Gauge-Deck No-Go

At regulator r, let

$$\mathfrak{B}_r = (Q_r, C_r, E_r, \omega_{R,r}, L_{r,T}, p_{r,i}, \tau_r, T_r, \rho_{R,r})$$

be a finite baryogenesis source packet. Here Q_r is the physical quotient, C_r is its CP involution, $\omega_{R,r} : E_r \to \mathbb{Z}$ is a CP-odd integral winding cocycle, $L_{r,T}$ is a quotient-intrinsic continuous-time Markov generator, $p_{r,i}$ is the source initial law, τ_r and $T_r(\tau)$ supply the physical clock and temperature map, and $\rho_{R,r}$ assigns record charges r_ψ to left-handed Weyl multiplets. Let $\theta_0 = 2\pi/m_R$, where m_R is the primitive winding period.

Theorem A.3 (Finite-quotient anomaly-and-current theorem)**.** *If the record phase acts as* $\psi \mapsto e^{ir_\psi \Theta_R}\psi$*, then its electroweak topological coefficient and expected phase velocity are*

$$k_R = \sum_{\psi\,\mathrm{LH}} r_\psi\, 2T_2(R_\psi), \qquad \dot{\Theta}_R(T) = \theta_0 \sum_{q,q'\in Q_r} p_r(q,T) L_{r,T}(q,q')\omega_{R,r}(q,q').$$

The probability law satisfies $dp_r/d\tau = p_r L_{r,T(\tau)}$*. If* $p_{r,i}(C_r q) = p_{r,i}(q)$ *and* $L_{r,T}(C_r q, C_r q') = L_{r,T}(q,q')$*, then*

$$\dot{\Theta}_R(T) = Y_B = 0.$$

For the determinant/deck direction of the realized Standard Model branch, the natural central gauge attachment is hypercharge and

$$k_R^{YWW} = N_g(3Y_Q + Y_L) = 3\left(3\cdot\frac{1}{6} - \frac{1}{2}\right) = 0.$$

The OPH $\mathbb{Z}_6$ *gauge/deck phase therefore supplies no direct* $\Theta_R W\widetilde{W}$ *baryogenesis coupling. A conventional global* $B{+}L$ *attachment would instead give* $k_R = 2N_g = 6$*, conditional on a quotient-visible gauge-singlet record phase carrying that attachment.*

Proof. The chiral measure gives the mixed $SU(2)_L^2 U(1)_R$ anomaly index, with an $SU(2)$ doublet contributing $r_\psi 2T_2(\mathbf{2}) = r_\psi$, including spectator multiplicities. The expected number of transitions $q \to q'$ in proper time $d\tau$ is $p_r(q,T)L_{r,T}(q,q')d\tau$; weighting by the integral winding increment gives the displayed current. Pairing every transition with its CP image cancels the current under a CP-symmetric law. The gauge/deck value is the Standard Model mixed hypercharge anomaly, which vanishes generation by generation. For $B+L$, the quark and lepton weak doublets contribute two units per generation. □

Source boundary. The determinant/deck cocycle supplies a candidate phase coordinate and its CP inversion. Normal-form settlement and the oriented 24-slot register supply no transition probabilities and select no baryon sign. A nonzero branch requires a distinct anomalous record attachment, a source-only $L_{r,T}$, a physical clock and initial or boundary law, a quotient-intrinsic CP-odd affinity, and cosmological domain coherence. On the conditional $B{+}L$ branch, the transport functional requires

$$\left\langle \frac{\dot{\Theta}_R}{T} \right\rangle_{\mathrm{fo}} = (4.463 \pm 0.028)\times 10^{-9}.$$

This number is an evaluation target for the source generator. It is not an input from which the generator may be selected.

Proton-spin continuation benchmark. On the realized finite matter branch the color factor is SU(3), so the structural OPH input for proton-spin bookkeeping is

$$C_F = \frac{4}{3}, \qquad C_A = 3.$$

If one adds the QCD-dependent continuation ansatz that quark/gluon spin sharing equilibrates according to these Casimirs, then

$$\Delta\Sigma \approx \frac{C_F}{C_F + C_A} = \frac{4}{13} \approx 0.308.$$

Against a representative lattice benchmark $\Delta\Sigma \approx 0.286$, this lands within about 8%, but it is only a deferred continuation benchmark. A first-principles OPH derivation of proton spin would require the nonperturbative light-quark/hadron completion plus an explicit map from OPH data to renormalized proton matrix elements.

A.3 Controlled Worldsheet Effective Description

Exact OPH-to-2D-Yang–Mills bridge. On the compact-group heat-kernel branch, the edge-partition theorem proves the exact identity

$$Z_{\mathrm{edge}}(t) = K_t(1)$$

by gluing the open-edge weights $p_R(t) \propto d_R e^{-tC_2(R)}$ into the closed partition sum

$$Z_{\mathrm{edge}}(t) = \sum_R d_R^2 e^{-tC_2(R)}.$$

This is the precise theorem-level sense in which the OPH edge-sector partition reorganizes into the two-dimensional Yang–Mills heat-kernel surface, and the Chapman–Kolmogorov law supplies the matching collar-sewing rule. This bridge is a two-dimensional heat-kernel partition identity on the stated compact-group branch. The four-dimensional compact-gauge repair-gap theorem is the separate support-visible result in Theorem A.29.

Large-N_{edge} boundary. The large-N_{edge} worldsheet interpretation is carried only when one fixes a distinct large-N_{edge} sequence, with $N_{\mathrm{edge}} \neq N_c = 3$, a fixed-$\tau$ window for

$$\tau = tN_{\mathrm{edge}},$$

and the uniform genus-remainder control stated for that expansion. On that branch the edge free energy satisfies the Standard Model gauge paper's theorem-level criterion for a controlled genus expansion, so the standard Gross–Taylor rewriting becomes a controlled worldsheet effective description of edge dynamics.

External items. The exact bridge theorem uses the compact-group heat-kernel branch and the quadratic-Casimir normalization declared in the Standard Model gauge paper. The conditional worldsheet theorem uses the declared large-N_{edge} regime and the imported Gross–Taylor large-N worldsheet dictionary for two-dimensional Yang–Mills. It does not identify a critical worldsheet CFT. Worldsheet supersymmetry, critical dimension, modular invariance, anomaly cancellation, GSO projection, and full massless-spectrum matching use separate physical inputs beyond the compact recovered-core chain.

A.4 Support-Visible Yang–Mills Gap from Repair Dynamics

Assumption A.4 (Support-visible compact-gauge Yang–Mills branch)**.** *The compact-gauge branch used in this subsection is the ordinary or central zero-obstruction compact-gauge sector with compact simple structure group G, a four-dimensional Euclidean scaling chart, and a cofinal tail carrying the compact-gauge refinement receipt of the compact-gauge refinement receipt stated in the main derivation. It also has a reflection-positive ordinary vacuum, topological angle $\theta = 0$, gauge-invariant local finite-constraint MaxEnt/Gibbs refinement, no additional relevant dimension-four pure-gauge operator on the branch besides the positive quadratic curvature invariant, active exact-Markov repair collars, bounded-color collar covers, and repair completeness. It is carried on the separated cofinal regulator system stated in the main derivation. No continuum cylinder state, GNS space, transfer generator, or vacuum projection is included in this assumption; those objects are constructed, or separately certified, below.*

Definition A.5 (Finite support-visible compact-gauge cylinder system)**.** *At regulator r, make the following construction twice: for the finite four-dimensional Euclidean slab $\Lambda_r^{(4)} = (V_r^{(4)}, E_r^{(4)})$ cut out by the scaling chart, and for its time-zero spatial slice $\Lambda_r^{(0)} = (V_r^{(0)}, E_r^{(0)})$. In the formulas below, $\Lambda_r = (V_r, E_r)$ denotes either choice and the superscript is restored whenever the two must be distinguished. Let D_r be the finite collection of repaired collar records and zero-obstruction sector labels retained by the compact-gauge patch carrier. In the gauge-register presentation, let*

$$\widetilde{X}_r \subseteq G^{E_r} \times D_r$$

be the closed set satisfying the finite Gauss, overlap, and repaired-record constraints, let $\mathcal{G}_r = G^{V_r}$ act by endpoint gauge transformations, and let $\sim_{\mathrm{ov}}$ be the closed equivalence relation of support-visible overlap/presentation indistinguishability. The finite-stage support-visible configuration space is

$$X_r := (\widetilde{X}_r/\mathcal{G}_r)/\!\sim_{\mathrm{ov}}.$$

Thus X_r is compact Hausdorff. "Finite stage" means that Λ_r and D_r are finite; X_r need not be a finite set when G is a compact Lie group. A finite quantum-link truncation gives the same commuting Euclidean readout algebra after restriction to its joint cylinder spectrum; no noncommutative support-quotient claim is made here.

For each active collar C, let

$$\rho_{C,r} : X_r \longrightarrow Y_{C,r}$$

be the continuous complete repaired readback. On the classical compact-holonomy branch used by the repair lemma, the complementary conditional fiber contains every record or holonomy coordinate actually resampled by collar repair. It is either finite with its uniform conditional law or a standard atomless probability space with its conditioned MaxEnt/Haar law. The complete readback retains exactly the fixed repaired datum, not every pre-repair holonomy; otherwise nonconstant Wilson cylinders would incorrectly lie in the repair kernel.

For a bounded cell region $O \subset \Lambda_r$, let $\mathfrak{C}_r^{G,\mathrm{sv}}(O)$ be the unital *-algebra generated by

(a) *contracted Peter–Weyl spin-network functions, Wilson-loop characters, and any explicitly retained gauge-invariant boundary-carrier contractions supported in O;*

(b) *the overlap-sector projectors whose collars lie in O; and*

(c) *continuous functions of the complete repaired readbacks $\rho_{C,r}$ for those collars.*

Its uniform closure

$$\mathcal{A}_r^{G,\mathrm{sv}}(O) := \overline{\mathfrak{C}_r^{G,\mathrm{sv}}(O)}^{\|\cdot\|_\infty} \subseteq C(X_r)$$

is the finite-stage local cylinder algebra. Equivalently, start from the freely presented readout-generator *-algebra and quotient by the kernel of its evaluation homomorphism on X_r, then complete. That kernel is a closed two-sided overlap-trivial ideal fixed before a state is chosen. The relation $\sim_{\mathrm{ov}}$ identifies exactly the configurations on which all declared spin-network, sector, and repaired-readback generators agree. These generators are self-adjoint, contain the constants, and therefore separate points of X_r. Stone–Weierstrass gives

$$\mathcal{A}_r^{G,\mathrm{sv}}(\Lambda_r) = C(X_r).$$

Consequently, for any selected stage measure π_r, their GNS cylinder closure is the full $L^2(X_r, \pi_r)$ after the finite π_r-null support reduction.

Write the resulting spaces and algebras as

$$(X_r^{(4)}, \mathcal{A}_r^{G,\mathrm{sv},(4)}(O)) \quad \textit{and} \quad (X_r^{(0)}, \mathcal{A}_{r,0}^{G,\mathrm{sv}}(B)).$$

Restriction of a Euclidean history to the time-zero slice gives a continuous map

$$\tau_{0,r} : X_r^{(4)} \longrightarrow X_r^{(0)}$$

and the corresponding pullback embeds time-zero cylinders into four-dimensional cylinders.

For $r \preceq s$, the deterministic coarse shadow multiplies the ordered fine-edge holonomies lying over each coarse edge, restricts collar readbacks, and sends a fine sector label to its coarse shadow. Gauge covariance makes this a map

$$p_{sr} : X_s \longrightarrow X_r, \qquad p_{tr} = p_{sr} \circ p_{ts} \quad (r \preceq s \preceq t).$$

The four-dimensional and time-zero coarse shadows satisfy the finite commuting square

$$\tau_{0,r} p_{sr}^{(4)} = p_{sr}^{(0)} \tau_{0,s}.$$

On local cylinders it gives the unital *-map

$$\iota_{rs}^{O} : \mathcal{A}_r^{G,\mathrm{sv}}(O) \longrightarrow \mathcal{A}_s^{G,\mathrm{sv}}(O_s), \qquad \iota_{rs}^{O}(a) = a \circ p_{sr}.$$

On the compact-gauge refinement branch, the sector-projector part must agree with the separately supplied block-multiplicity injection $\jmath_{rs}^{B}$ of the compact-gauge refinement receipt; it is not inferred from the deterministic coarse-shadow map. The holonomy and readback parts are its declared compatible extension. The maps preserve isotony and obey

$$\iota_{rt}^{O} = \iota_{st}^{O_s} \iota_{rs}^{O}.$$

If every coarse configuration on the selected branch has a fine extension, p_{sr} is surjective and ι_{rs}^{O} is injective. Without that finite extendability receipt, the C^*-inductive limit quotients the common refinement-null kernel; the extraction proof below does not assume prelimit injectivity of the full cylinder map. That broader quotient extraction is not, by itself, the receipt-certified sector ladder of the refinement/fiber descent theorem in the main derivation. Write $\mathcal{A}_r^{G,\mathrm{sv}} := \mathcal{A}_r^{G,\mathrm{sv}}(\Lambda_r)$. Choose a countable cofinal regulator sequence and a countable bounded-region exhaustion of the four-dimensional chart. This is the cylinder family used below. For a non-countable regulator presentation, the same construction uses the full directed family and a subnet instead of this cofinal sequence.

Proposition A.6 (Projective compact-gauge cylinder extraction)**.** *For every regulator s, let $\mu_s^{(4)}$ be the declared finite quotient-Gibbs probability measure on $X_s^{(4)}$, let*

$$\pi_s := (\tau_{0,s})_{\#} \mu_s^{(4)}$$

be its stationary time-zero marginal on $X_s^{(0)}$, and put $\mu_s^{(0)} := \pi_s$. For $d \in \{4, 0\}$, write

$$\omega_s^{(d)}(a) := \int_{X_s^{(d)}} a \, d\mu_s^{(d)}, \qquad \omega_{s\downarrow r}^{(d)} := \omega_s^{(d)} \circ \iota_{rs}^{(d)}.$$

The statements below hold for both $d = 4$ and $d = 0$; the superscript is suppressed in the displayed proof. Then the following statements hold.

(i) *The marginals of any one fine state are exactly projective: for $q \preceq r \preceq s$,*

$$\omega_{s\downarrow r} \circ \iota_{qr} = \omega_{s\downarrow q}.$$

(ii) *There is a cofinal subsequence, or a cofinal subnet in the general directed case, and states $\overline{\omega}_r \in S(\mathcal{A}_r^{G,\mathrm{sv}})$ such that*

$$\omega_{s_\alpha\downarrow r} \underset{\alpha}{\to} \overline{\omega}_r \quad \textit{weak-}^*$$

on every fixed local cylinder algebra. The limit family is projective:

$$\overline{\omega}_r = \overline{\omega}_t \circ \iota_{rt} \qquad (r \preceq t).$$

(iii) *The compatible local states define a state ω_∞ on the algebraic inductive limit and extend uniquely to its local C^*-completion*

$$\mathcal{A}_\infty^{G,\mathrm{cyl}}(O) := \overline{\varinjlim_r \mathcal{A}_r^{G,\mathrm{sv}}(O_r)}.$$

The resulting local algebras are isotone and their union is dense in the global cylinder algebra.

(iv) *Let $\mathcal{N}_\omega(O)$ be the null ideal of the limiting cylinder measure,*

$$\mathcal{N}_\omega(O) := \{a \in \mathcal{A}_\infty^{G,\mathrm{cyl}}(O) : \omega_\infty(a^*a) = 0\}.$$

Because the Euclidean cylinder algebra is commutative, this is a closed two-sided ideal. On

$$\mathcal{A}_\infty^{G,\mathrm{supp}}(O) := \mathcal{A}_\infty^{G,\mathrm{cyl}}(O)/\mathcal{N}_\omega(O)$$

the induced state is faithful. The local GNS spaces glue isometrically, their local cylinder images have dense union, and their Hilbert direct limit is the GNS space $(K^{(d)}, \Pi^{(d)}, \mathbf{1}^{(d)})$ of $\omega_\infty^{(d)}$. Denote the four-dimensional Euclidean cylinder GNS space by $K^E := K^{(4)}$ and the time-zero repair GNS space by $K := K^{(0)}$. Each is a faithful cyclic GNS pair on its support quotient; the time-zero vector is identified as the physical vacuum only by the transfer/OS receipt below.

If, in addition, the finite four-dimensional quotient presentation satisfies the fiber-sum identity stated after the Local MaxEnt and Refinement Stability axiom, then $(p_{sr}^{(4)})_\# \mu_s^{(4)} = \mu_r^{(4)}$. The time-zero commuting square gives $(p_{sr}^{(0)})_\# \pi_s = \pi_r$, so both original state families are projective. Without that finite receipt, the bars on $\overline{\omega}_r$ are essential: compactness extracts a projective cluster family but does not turn the originally chosen coarse Gibbs measures into an exactly projective family. In the continuous compact-holonomy presentation, the corresponding receipt is the pushforward identity $(p_{sr}^{(4)})_\# \mu_s^{(4)} = \mu_r^{(4)}$, expressed through disintegration rather than a finite sum.

Proof. Functoriality of the coarse shadows gives

$$(\omega_s \circ \iota_{rs}) \circ \iota_{qr} = \omega_s \circ \iota_{qs},$$

which proves (i) without a continuum assumption.

For each fixed cylinder algebra, its state space is weak-* compact by Banach–Alaoglu. The algebras in the chosen cylinder exhaustion are separable: compact metrizable G has a countable Peter–Weyl test algebra, and only finitely many readbacks and sector projectors occur at one stage. Their state spaces are therefore weak-* metrizable. Starting with the first cylinder, extract a subsequence on which its marginals converge; from that subsequence extract one for the second

cylinder, and continue. Choose the k-th diagonal index beyond the k-th regulator stage. The resulting diagonal subsequence is cofinal and converges on every cylinder. In the general directed presentation, compactness of the product of the local state spaces gives the same conclusion with a cofinal subnet: unresolved early coordinates may be filled by arbitrary states because every fixed coordinate is genuinely resolved on a cofinal tail. For $q \preceq r$, weak-* continuity of precomposition by ι_{qr}, together with (i), gives

$$\overline{\omega}_r \circ \iota_{qr} = \lim_{\alpha} \omega_{s_\alpha \downarrow r} \circ \iota_{qr} = \lim_{\alpha} \omega_{s_\alpha \downarrow q} = \overline{\omega}_q.$$

This proves (ii).

If $a \in \mathcal{A}_r^{G,\mathrm{sv}}$ represents an element $[a]_r$ of the algebraic inductive limit, set

$$\omega_\infty([a]_r) := \overline{\omega}_r(a).$$

Projectivity makes this independent of the representative. Positivity and normalization hold at the finite stage containing any given algebraic element. For every $t \succeq r$,

$$|\overline{\omega}_r(a)| = |\overline{\omega}_t(\iota_{rt}(a))| \le \|\iota_{rt}(a)\|.$$

Taking the infimum along the directed tail gives the C^*-inductive-limit norm bound, including when the maps are noninjective. Hence the state extends uniquely to the completion. The same construction region by region proves isotony and density, giving (iii).

By the Riesz representation theorem, a commutative local limiting state is integration against a Radon probability measure μ_O. Its null ideal consists precisely of the continuous functions vanishing on the measure support, so the quotient is canonically $C(\operatorname{supp}\mu_O)$ and the induced state is faithful. If $O \subseteq O'$ and $\iota_{OO'}$ is the local inclusion, state compatibility gives

$$a \in \mathcal{N}_\omega(O) \quad \Longleftrightarrow \quad \iota_{OO'}(a) \in \mathcal{N}_\omega(O').$$

Thus the local inclusions descend to injective maps of the support quotients, which therefore form an isotone faithful local net. For $r \preceq t$, projectivity makes

$$V_{rt} : K_{\overline{\omega}_r} \longrightarrow K_{\overline{\omega}_t}, \qquad V_{rt}[a]_r = [\iota_{rt}(a)]_t$$

well defined and isometric, because

$$\|V_{rt}[a]_r\|^2 = \overline{\omega}_t(\iota_{rt}(a^*a)) = \overline{\omega}_r(a^*a).$$

The maps compose, preserve $\mathbf{1}$, and intertwine left multiplication. Their Hilbert direct limit is therefore the GNS completion of the algebraic cylinder union and has dense local images. This proves (iv). Finally, suppressing the $d = 4$ superscript, the finite fiber-sum identity gives

$$Z_s = \sum_{x \in X_r} \sum_{x' : p_{sr}(x') = x} m_s(x') e^{-S_s(x')} = \alpha_{sr} Z_r,$$

and hence

$$(p_{sr})_{\#}\mu_s(x) = \frac{\alpha_{sr} m_r(x) e^{-S_r(x)}}{\alpha_{sr} Z_r} = \mu_r(x).$$

This is equivalent to $\omega_s^{(4)} \circ \iota_{rs}^{(4)} = \omega_r^{(4)}$; the commuting time-zero square gives the corresponding identity for $d = 0$. □

Remark A.7 (Two different quotients)**.** *The overlap-trivial presentation ideal in Definition A.5 is not the GNS null ideal $\mathcal{N}_\omega(O)$. The overlap-trivial ideal removes data that no support-visible readout can distinguish before a state is selected. The GNS null ideal removes directions that have zero norm in the extracted state. Faithful finite-stage states can have a nonfaithful weak-* limit, so quotienting only the overlap-trivial ideal would not prove faithfulness of the limiting pair.*

Assumption A.8 (Renormalized four-dimensional Yang–Mills identification receipt)**.** *On the selected four-dimensional cluster family, compatible renormalized holonomy cylinders obey uniform local Cauchy and regularity bounds. Their infinitesimal rectangle tests converge, in the declared distributional sense, to a connection/curvature pair*

$$U_{\mu\nu}(\varepsilon, x) = \mathbf{1} + \varepsilon^2 F_{\mu\nu}(x) + O(\varepsilon^3), \qquad F = dA + A \wedge A.$$

The renormalized local actions converge to the unique reflection-even positive dimension-four pure-gauge density on this branch, while the declared irrelevant remainders vanish. This is the regularity/universality receipt that identifies the projective generalized-holonomy state with the four-dimensional Yang–Mills cylinder state. The notation $e^{-S_E[A]}DA/G$ below abbreviates that cylinder family; it is not a literal infinite-dimensional Lebesgue measure.

Theorem A.9 (Conditional OPH four-dimensional Euclidean Yang–Mills form)**.** *Under Assumptions A.4 and A.8, on an extracted cylinder family from Proposition A.6 that lies on the declared local four-dimensional scaling branch, the continuum gauge-sector Euclidean action is*

$$S_E[A] = \frac{1}{4g^2} \int_{\mathbb{R}^4} \langle F_{\mu\nu}, F_{\mu\nu} \rangle \, d^4x, \qquad F = dA + A \wedge A,$$

with compact simple structure group G. The extracted gauge-quotient cylinder family is denoted

$$d\mu_{\mathrm{YM}}(A) = Z^{-1} e^{-S_E[A]} \, DA/G$$

in the OPH support-visible GNS representation. When Assumption A.22 also holds, its support-visible continuum transfer semigroup is the corresponding Euclidean Yang–Mills semigroup.

Proof. The proof spine is included here to make the branch target explicit. The present Yang–Mills branch takes compact simple G as branch data. When G is imported from OPH reconstruction, that import is conditional on the compact-gauge refinement receipt; on that tail the zero-obstruction transportable bosonic sector category and its faithful forgetful fiber functor reconstruct G. At fixed cutoff, the finite gauge-register / quantum-link presentation gives support-visible link holonomies and plaquette holonomies. In the refinement limit, the zero-obstruction gluing law makes infinitesimal rectangle holonomies multiplicative and path-local. The regularity and Cauchy bounds in Assumption A.8 upgrade this generalized-holonomy limit to a local connection A on the four-dimensional scaling chart, with infinitesimal plaquette defect

$$U_{\mu\nu}(\varepsilon, x) = \mathbf{1} + \varepsilon^2 F_{\mu\nu}(x) + O(\varepsilon^3), \qquad F = dA + A \wedge A.$$

The Euclideanized MaxEnt/local-Gibbs branch supplies a local finite-range action density built from support-visible gauge-invariant collar data. Gauge quotienting permits only class functions of the curvature and its covariant derivatives. Four-dimensional scaling, Euclidean rotation invariance, locality, and reflection positivity leave one relevant dimension-four positive quadratic invariant in the pure gauge sector:

$$\langle F_{\mu\nu}, F_{\mu\nu} \rangle.$$

The possible topological density $\langle F\wedge F\rangle$ is reflection odd and belongs to a separate topological-angle sector; it is absent on the ordinary reflection-positive zero-obstruction vacuum branch used here. Higher curvature powers and covariant-derivative terms are irrelevant under the declared continuum scaling; their disappearance in the extracted state is the remainder bound in Assumption A.8. Normalizing the unique positive quadratic invariant defines the coupling g. Definition A.5 identifies the finite-stage sources of the gauge cylinders, and Proposition A.6 proves their projective weak-$*$ / GNS extraction. Identification of its transfer generator is the separate dynamic statement in Theorem A.23; it is not a consequence of weak-* compactness alone. □

Prize-facing proof separation. The proof has two separate claims. Theorem A.9 conditionally identifies the four-dimensional Euclidean Yang–Mills form on the certified support-visible compact-gauge branch. The repair-dynamics theorem proves a spectral gap for that Hamiltonian.

Standing compact-gauge setup. Fix a compact simple gauge group G carried by an OPH compact-gauge zero-obstruction vacuum branch on a cofinal tail satisfying the compact-gauge refinement receipt, realized at fixed cutoff by the declared compact-gauge patch-carrier architecture. For each regulator r, let

$$(\mathcal{H}_r,\Omega_r,H_r),\qquad T_r(t)=e^{-tH_r},$$

be the physical Euclidean Hilbert space, vacuum, Hamiltonian, and transfer semigroup. Let $X_r:=X_r^{(0)}$ be the support-visible time-zero compact-gauge quotient configuration space, and let π_r be the stationary time-zero measure from Proposition A.6, and set

$$K_r:=L^2(X_r,\pi_r).$$

Write $\omega_r(a):=\int_{X_r}a\,d\pi_r$ for its state functional. All statements in K_r are automatically made after the finite π_r-null support reduction. Let $\mathcal{C}_r$ be the finite family of active repair collars. For each $C\in\mathcal{C}_r$, use the complete repaired visible datum $\rho_C:=\rho_{C,r}:X_r\to Y_{C,r}$, and let

$$E_C:K_r\to K_r$$

be conditional expectation onto the ρ_C-measurable functions. This subsection stays on the ordinary or central zero-obstruction vacuum branch where the compact-gauge reconstruction ladder is carried by the explicit refinement receipt used in the Standard Model gauge paper.

Proposition A.10 (Local exact repair equals conditional expectation)**.** *For each active collar C, the exact-Markov repair map on the support-visible quotient is the π_r-preserving conditional expectation E_C.*

Proof. On the exact-Markov branch, repair preserves exactly the repaired visible datum ρ_C, changes only complementary invisible fiber data, and acts on the quotient-first physical algebra rather than on representatives. Let Φ_C be the Heisenberg repair map and let $\mathcal{N}_C$ be the repaired local fixed algebra. Exact repair semantics require $\mathcal{N}_C$ to be exactly the ρ_C-measurable subalgebra. Then

$$\Phi_C(a)=a\quad(a\in\mathcal{N}_C),\qquad\Phi_C(\mathcal{A}_r^{G,\mathrm{sv}})\subseteq\mathcal{N}_C,\qquad\omega_r\circ\Phi_C=\omega_r,$$

and Φ_C is $\mathcal{N}_C$-bimodular. Hence, for $a\in\mathcal{N}_C$ and $x\in\mathcal{A}_r^{G,\mathrm{sv}}$,

$$\omega_r(a^*\Phi_C(x))=\omega_r(a^*x).$$

Since $\Phi_C(x)\in\mathcal{N}_C$, this characterizes its class in K_r as the orthogonal projection of x onto the ρ_C-measurable subspace, uniquely modulo π_r-null functions. That projection is the π_r-preserving conditional expectation E_C. □

Lemma A.11 (Fiber-homogeneous orbit condition)**.** *Fix an active collar C and a repaired value $y \in Y_{C,r}$. Let $F_C(y) = \rho_{C,r}^{-1}(y)$, with conditional law $\nu_{C,y}$. Assume this is either a finite uniform probability space or a standard atomless probability space, and that the primitive collar relaxation is invariant under every $\nu_{C,y}$-preserving automorphism. Then the complete conditional fiber, including every holonomy coordinate actually resampled by repair, is homogeneous for the local repair receipt.*

Proof. This is the complete-fiber homogeneity receipt. Quotienting removes declared implementation labels, but the proof must also check that no remaining source constraint or relaxation rate distinguishes points in the conditional fiber. Under that check, the conditioned state and primitive relaxation carry the stated full measure-preserving symmetry. □

Lemma A.12 (Scalar relaxation on a homogeneous conditional fiber)**.** *Let (F, ν) be either a finite uniform probability space or a standard atomless probability space. Let E_F be expectation onto constants, and let D_F be a bounded positive self-adjoint Markov relaxation generator with*

$$\ker D_F = \mathrm{Ran}(E_F)$$

that commutes with every measure-preserving automorphism of (F, ν). Then $D_F = c_F(I - E_F)$ for a scalar $c_F > 0$.

Proof. For a finite uniform fiber this is the symmetric-group argument. For a standard atomless fiber, restrict first to step functions on a partition into n equal-measure pieces. Automorphisms within pieces and permutations of pieces force every bounded operator in the commutant to be scalar on the mean-zero step functions. Compatibility under equal-measure refinements gives one scalar, and such step functions are dense in $L_0^2(F, \nu)$. Thus the commutant on L_0^2 is scalar. Positivity and the stated kernel make that scalar strictly positive. □

Assumption A.13 (Finite ground-state-transform and cross-fiber receipt)**.** *For each regulator, the reflection-positive finite transfer matrix supplies a unitary $U_r : \mathcal{H}_r \to K_r$, with $U_r\Omega_r = \mathbf{1}_r$, whose ground-state-transformed generator decomposes as*

$$U_r H_r U_r^{-1} = \sum_{C \in \mathcal{C}_r} D_C.$$

Each D_C is the bounded positive self-adjoint detailed-balance Markov generator acting on the complete complementary conditional fiber of collar C, including every record or holonomy coordinate changed by repair, with fixed algebra exactly the $\rho_{C,r}$-measurable algebra. If the fiberwise scalar supplied by Lemma A.12 over repaired value y is $c_C(y)$, the finite receipt also verifies the cross-fiber equality $c_C(y) = c_C$ on the support of π_r. A Gibbs state alone does not imply this transfer-matrix/decomposition receipt.

Theorem A.14 (Exact Euclidean-consensus law under homogeneous fibers)**.** *Under the fiber-homogeneous orbit condition for every active collar and Assumption A.13, there are positive constants $c_C > 0$ such that the ground-state transformed physical Euclidean generator is exactly*

$$L_r^{\mathrm{EC}} = \sum_{C \in \mathcal{C}_r} c_C(I - E_C),$$

and therefore

$$U_r e^{-tH_r} U_r^{-1} = e^{-tL_r^{\mathrm{EC}}} \qquad (t \geq 0),$$

for a unitary $U_r : \mathcal{H}_r \to K_r$ with $U_r\Omega_r = \mathbf{1}_r$.

Proof. The finite receipt supplies the ground-state transform and a decomposition into primitive local pieces D_C. Each such piece is supported on collar C, preserves the repaired visible datum $\rho_{C,r}$, and relaxes the complete complementary conditional fiber. Thus

$$\ker D_C = \mathrm{Ran}(E_C).$$

Lemma A.11 gives full hidden-fiber permutation symmetry. Applying Lemma A.12 fiberwise gives $D_{C,y} = c_C(y)(I - E_{C,y})$. The cross-fiber part of the finite receipt makes $c_C(y) = c_C$, so

$$D_C = c_C(I - E_C)$$

for a collar-type constant $c_C > 0$. Summing over the active collars gives the displayed generator. Positivity and self-adjointness give the transfer-semigroup identity. □

Under a common clock rescaling $c_C \mapsto sc_C$, $s > 0$, the generator and gap scale by s, while stationarity, detailed balance and the fixed space remain unchanged. Numerical identity checks use the generator divided by a common rate scale; their acceptance cannot depend on the clock unit. This supplies no physical time calibration.

Definition A.15 (Source-defined admissible atomic collar tower)**.** *For every regulator and allowed boundary condition b, suppose the finite source presentation gives an atomic register set V_r with finite local alphabets, a finite-range Gibbs law $\pi_{r,b}$, and, for each $v \in V_r$, a rooted active collar $C(v)$ and the heat-bath projection*

$$P_{v,r,b} f := \mathbb{E}_{\pi_{r,b}}[f \mid x_{V_r \setminus \{v\}}].$$

The source type *of $C(v)$ is the isomorphism class of the finite tuple consisting of its rooted interaction-radius neighbourhood, boundary/sector flags, local register alphabets, nonzero Gibbs-potential templates, complete repaired readback, conditional kernel, and normalized rate. The tower is admissible when the following data are printed by the source rather than inferred from the target gap:*

(G1) *one finite set $\mathfrak{T}_{\mathrm{act}}$ contains every active source type at every location, size, allowed boundary condition, and stage of the selected cofinal refinement tail;*

(G2) *the active family is exactly $\mathcal{C}_r = \{C(v) : v \in V_r\}$, with $E_{C(v)} = P_{v,r,b}$; type-equivalent collars have the same exact heat-bath kernel and rate, every atomic register has one active rooted collar, and $c_{v,r,b} = c_{\tau(v)} \geq c_* > 0$;*

(G3) *for the one-site conditional kernels, let $a^{r,b}_{vu}$ be the supremum total-variation change at root v when two exterior configurations differ only at u. Outward-rounded rational interval bounds printed in the finite type table prove*

$$\sup_{r,b,v} \sum_{u \neq v} a^{r,b}_{vu} \leq \eta_* < 1. \tag{INF}$$

The Dobrushin Poincaré comparison then derives, rather than assumes,

$$\mathrm{Var}_{\pi_{r,b}}(f) \leq \frac{1}{1 - \eta_*} \sum_{v \in V_r} \|(I - P_{v,r,b}) f\|^2_{2,\pi_{r,b}} \tag{AT}$$

uniformly over the family [86];

(G4) *coarse shadow sends each rooted fine collar to a rooted collar of the same declared type or to an explicitly listed type transition, preserves the conditional kernels on pulled-back coarse functions, and introduces no type outside* $\mathfrak{T}_{\mathrm{act}}$.

Collar CMI decay and qualitative finite-range Gibbs mixing are not item (G3).

Proposition A.16 (Finite classification and uniform local-rate floor)**.** *For an admissible atomic collar tower, two active collars have the same source type exactly when their tuples in Definition A.15 are isomorphic. Thus the admissible active collar classes are precisely the nonempty fibers of* $\tau : \bigsqcup_r V_r \to \mathfrak{T}_{\mathrm{act}}$*; in particular there are at most* $|\mathfrak{T}_{\mathrm{act}}|$ *classes. Their rates obey*

$$c_{v,r,b} \geq c_* := \min_{t \in \mathfrak{T}_{\mathrm{act}}} c_t > 0$$

uniformly in location, boundary condition, system size, and cofinal refinement.

Proof. The tuple is a complete source signature, so equality of types is exactly rooted signature-isomorphism. Item (G1) makes its image finite. Item (G2) makes the rate a positive function on that finite image, and item (G4) prevents refinement from creating an unlisted or rate-degenerate signature. The displayed finite minimum is therefore positive and has all four stated uniformities. □

Theorem A.17 (Uniform collar-projection and transfer gap)**.** *On an admissible atomic collar tower satisfying the finite ground-state-transform receipt, define*

$$L^{\mathrm{col}}_{r,b} := \sum_{v \in V_r} c_{v,r,b}(I - P_{v,r,b}), \qquad P_{0,r,b} f := \pi_{r,b}(f)\mathbf{1}.$$

Then, with the single explicit modulus

$$\delta_* := c_*(1 - \eta_*) > 0, \tag{GAP}$$

one has, simultaneously for all locations, allowed boundary conditions, system sizes, and stages of the cofinal refinement tower,

$$L^{\mathrm{col}}_{r,b} \geq \delta_*(I - P_{0,r,b}), \qquad \|e^{-tL^{\mathrm{col}}_{r,b}} - P_{0,r,b}\|_{2\to 2} \leq e^{-t\delta_*}.$$

Consequently the collar transfer operator has a positive projection/transfer gap at least δ_**. Under item* (G2)*,* $L^{\mathrm{col}}_{r,b} = L^{\mathrm{EC}}_r$.

Proof. Conditional expectation is an orthogonal projection, hence

$$\langle f, (I - P_{v,r,b})f \rangle = \|(I - P_{v,r,b})f\|_2^2.$$

For $f \perp \mathbf{1}$, item (G3) gives (AT), and item (G2) gives

$$\langle f, L^{\mathrm{col}}_{r,b} f \rangle \geq c_* \sum_v \|(I - P_{v,r,b})f\|_2^2 \geq c_*(1 - \eta_*)\|f\|_2^2.$$

This is the operator inequality. The spectral theorem gives the semigroup estimate. All constants are source-type constants and item (G4) carries the same table and receipt up the cofinal tower, so none depends on $r, b, |V_r|$, or the collar location. □

Proposition A.18 (Sharp hypothesis countermodels)**.** *Neither locality nor mixing may be deleted from the preceding theorem.*

(i) Without uniform mixing, take the faithful two-site law

$$\pi_\varepsilon(00) = \pi_\varepsilon(11) = \frac{1-\varepsilon}{2}, \qquad \pi_\varepsilon(01) = \pi_\varepsilon(10) = \frac{\varepsilon}{2}.$$

The exact spectrum of the two single-site heat-bath generator is $\{0, 2\varepsilon, 2(1-\varepsilon), 2\}$*, so its gap is* $2\varepsilon \to 0$*. The limiting zero-gap example is therefore not doing any work hidden by a loss of faithfulness.*

(ii) Without source locality, put the product fair-bit law on $\{0,1\}^m$*, list its* $N = 2^m$ *states in cyclic Gray-code order, and let* E_0, E_1 *average over the two alternating perfect matchings of that cycle. The law has exact product mixing and zero separated CMI, but selecting which bit to change uses the entire configuration. For* $L_m = (I - E_0) + (I - E_1)$*, exact Fourier diagonalization of the cycle gives*

$$\mathrm{gap}(L_m) = 1 - \cos(2\pi/2^m) \longrightarrow 0.$$

Both examples are finite at every displayed m*; the second is a finite family with an explicit vanishing gap.*

Remark A.19 (Claim boundary)**.** *Finite-range Gibbs form and collar-CMI decay alone do not prove* (AT)*, do not classify the kernel/rate signatures, and do not bound the Friedrichs angles between collar projections. Accordingly they do not imply a uniform repair gap. A numerical receipt is admissible only when outward-rounded intervals certify* $c_*^{\mathrm{lo}} > 0$ *and* $\eta_*^{\mathrm{hi}} < 1$*, in which case* $\delta_*^{\mathrm{lo}} = c_*^{\mathrm{lo}}(1 - \eta_*^{\mathrm{hi}})$*; a point estimate is not a certificate.*

Proposition A.20 (Executable finite calibration, not a physical receipt)**.** *The exact-rational four-dimensional Ising calibration table has* 244 *active collar types on the declared four-dimensional cofinal family* $N_r = 3 \cdot 2^r$*, for periodic, free, fixed-plus, and fixed-minus boundary conditions. Its complete deterministic table has*

$$c_* = 1, \qquad \eta_* \le \frac{1}{2}, \qquad A_* \le 2, \qquad \delta_* \ge \frac{1}{2}.$$

Thus its declared finite generator obeys

$$L_{r,b} \ge \frac{1}{2}(I - P_{0,r,b}), \qquad \|e^{-tL_{r,b}} - P_{0,r,b}\|_{2\to 2} \le e^{-t/2}.$$

The bundled verifier recomputes every rational conditional-row variation and the receipt hash, and rejects binary floating-point inputs, nonpositive rate floors, and noncontractive influence rows.

Remark A.21 (Physical interpretation boundary)**.** *Proposition A.20 validates the finite data model and the stated implication for a declared calibration family. It does not identify that Ising family with the compact-simple-gauge OPH tower. The intentionally uninstantiated physical manifest fails closed until quotient-first gauge collar types, source conditional kernels, rate floors, refinement closure, gauge and topological zero-mode handling, an independent ground-state-transform receipt, and the separate continuum/transfer/OS-noncollapse receipts are supplied.*

Assumption A.22 (Finite transfer and vacuum compatibility receipt)**.** *On the cofinal family selected in Proposition A.6, suppose the following finite-stage identities are certified.*

(T1) *The quotient ensembles obey the finite fiber-sum identity, or its continuous push-forward/disintegration analogue, so* $(p_{sr}^{(4)})_{\#}\mu_s^{(4)} = \mu_r^{(4)}$ *and, by time-zero restriction,* $(p_{sr}^{(0)})_{\#}\pi_s = \pi_r$*. Hence pullback gives coherent isometries*

$$J_{rs}^K : K_r \longrightarrow K_s, \qquad J_{rs}^K f = f \circ p_{sr}^{(0)}.$$

(T2) *There are coherent physical refinement isometries* $J_{rs}^{\mathcal{H}} : \mathcal{H}_r \to \mathcal{H}_s$ *satisfying*

$$J_{rs}^K \mathbf{1}_r = \mathbf{1}_s, \qquad J_{rs}^{\mathcal{H}} \Omega_r = \Omega_s, \qquad U_s J_{rs}^{\mathcal{H}} = J_{rs}^K U_r.$$

(T3) *With one refinement-independent physical Euclidean-time normalization, the finite semigroups are coherent:*

$$J_{rs}^K e^{-tL_r^{\mathrm{EC}}} = e^{-tL_s^{\mathrm{EC}}} J_{rs}^K, \qquad J_{rs}^{\mathcal{H}} e^{-tH_r} = e^{-tH_s} J_{rs}^{\mathcal{H}} \quad (t \geq 0).$$

(T4) *Let* $\iota_{0,r}$ *pull a time-zero observable into the four-dimensional cylinder algebra and let* $\tau_t^{(r)}$ *be finite Euclidean time translation. The finite time-zero Markov/transfer identity holds on the cylinder core:*

$$\omega_r^{(4)}\Big((\iota_{0,r} f)^* \tau_t^{(r)} (\iota_{0,r} g)\Big) = \langle f, e^{-tL_r^{\mathrm{EC}}} g \rangle_{K_r} = \langle U_r^{-1} f, e^{-tH_r} U_r^{-1} g \rangle_{\mathcal{H}_r}.$$

The reflection-positive finite OS time-zero quotient is therefore the same finite $(\mathcal{H}_r, \Omega_r, H_r)$*, through* U_r*, rather than an unrelated Hilbert space.*

These are finite commuting-diagram receipts. They are not consequences of the sector-category refinement functors or of weak- state compactness.*

Theorem A.23 (Continuum exact transfer identification)**.** *Under Assumption A.22, the finite repair and physical Hilbert spaces have Hilbert direct limits* K *and* $\mathcal{H}$*. Their semigroups induce strongly continuous self-adjoint contraction semigroups with nonnegative generators* L^{rep} *and* H*, and the finite unitaries induce a unitary* $U : \mathcal{H} \to K$ *such that*

$$U e^{-tH} U^{-1} = e^{-tL^{\mathrm{rep}}} \qquad (t \geq 0),$$

and hence

$$U H U^{-1} = L^{\mathrm{rep}}.$$

The constant vectors and physical vacua define limit vectors $\mathbf{1} \in K$ *and* $\Omega \in \mathcal{H}$*, with*

$$U\Omega = \mathbf{1}, \qquad P_0^K = |\mathbf{1}\rangle\langle\mathbf{1}|, \qquad P_0^{\mathcal{H}} = |\Omega\rangle\langle\Omega|, \qquad U P_0^{\mathcal{H}} U^{-1} = P_0^K.$$

If $L_r^{\mathrm{EC}} \geq \delta_*(I - P_{0,r})$ *for one* $\delta_* > 0$*, then*

$$L^{\mathrm{rep}} \geq \delta_*(I - P_0^K), \qquad H \geq \delta_*(I - P_0^{\mathcal{H}}).$$

Proof. The fiber-sum identity makes J_{rs}^K isometric, since

$$\|f \circ p_{sr}^{(0)}\|_{K_s}^2 = \int_{X_r} |f|^2 \, d(p_{sr}^{(0)})_{\#}\pi_s = \|f\|_{K_r}^2.$$

It also makes $\overline{\omega}_r^{(0)} = \omega_r^{(0)}$, so these J_{rs}^K are exactly the time-zero GNS isometries $V_{rs}^{(0)}$ in Proposition A.6. Thus their Hilbert direct limit is canonically the support-visible GNS space K constructed

there, not a second continuum repair space. Take the Hilbert direct limits under J^K_{rs} and $J^{\mathcal{H}}_{rs}$. On the dense finite-stage images define

$$S^K(t)J^K_r f := J^K_r e^{-tL^{\mathrm{EC}}_r} f, \qquad S^{\mathcal{H}}(t)J^{\mathcal{H}}_r \psi := J^{\mathcal{H}}_r e^{-tH_r}\psi.$$

The finite coherence identities make these definitions independent of the representative. They are contraction semigroups. Strong continuity holds on every finite-stage image and therefore, by density and uniform contractivity, on the whole direct limits. For each fixed t, the finite-stage inner-product identity makes the bounded extension symmetric on a dense union and hence symmetric everywhere; a bounded everywhere-defined symmetric operator is self-adjoint. Positivity also passes from the finite semigroups. Thus the spectral theorem gives nonnegative self-adjoint generators L^{rep} and H.

Define $UJ^{\mathcal{H}}_r\psi := J^K_r U_r\psi$. The finite unitary square makes this well defined and isometric; the same construction with U_r^{-1} shows that its range is dense and closed, so it is unitary. The finite transfer identity of Theorem A.14 gives $US^{\mathcal{H}}(t)U^{-1} = S^K(t)$ on a dense union and hence everywhere. Uniqueness of self-adjoint semigroup generators gives $UHU^{-1} = L^{\mathrm{rep}}$.

Unitality and the physical refinement identity make the constant and vacuum vectors coherent. For $f \in K_r$,

$$P_{0,s}J^K_{rs} f = \langle \mathbf{1}_s, J^K_{rs} f\rangle \mathbf{1}_s = J^K_{rs} P_{0,r} f,$$

so the rank-one vacuum projections induce the displayed limit projections; the same holds on the physical side and the unitary identifies them.

The finite operator inequality implies

$$\|e^{-tL^{\mathrm{EC}}_r}(I - P_{0,r})\| \le e^{-\delta_* t}.$$

Passing this norm bound through the direct-limit definition gives $\|e^{-tL^{\mathrm{rep}}}(I - P^K_0)\| \le e^{-\delta_* t}$. The spectral theorem is equivalent to $L^{\mathrm{rep}} \ge \delta_*(I - P^K_0)$, and conjugation by U^{-1} gives the physical bound. □

Remark A.24 (Asymptotic alternative)**.** *If the exact semigroup identities in Assumption A.22 are not available, the valid replacement is generalized Mosco convergence of both finite transfer forms, strong cylinder-core convergence of U_r and U_r^{-1}, and strong convergence of the vacuum projections. The Mosco theorem then supplies the two limit semigroups and the same intertwining conclusion. Weak-* convergence of cylinder states by itself supplies none of those dynamic statements.*

Assumption A.25 (OS regularity and noncollapse receipt)**.** *The selected finite cylinder family carries reflection-compatible refinement maps and exact finite reflection positivity. Its renormalized local Schwinger functions have uniform bounds sufficient to pass Euclidean covariance, locality, OS regularity, and clustering to the extracted family. There are gauge-invariant local cylinders A_r, transported from one fixed cylinder test, such that*

$$\inf_r \mathrm{Var}_{\pi_r}(A_r) > 0, \qquad \sup_r \langle A_r\mathbf{1}_r, L^{\mathrm{EC}}_r A_r \mathbf{1}_r\rangle < \infty.$$

The centered vectors $A_r\mathbf{1}_r - \omega_r(A_r)\mathbf{1}_r$ converge under the same cylinder/Hilbert identifications used in the extraction and transfer receipts. This is the separate OS/nontriviality receipt. It is not part of the weak- extraction theorem.*

Theorem A.26 (Osterwalder–Schrader reconstruction on the compact-gauge branch)**.** *Under Assumptions A.4, A.8, A.22, and A.25, with the extraction of Proposition A.6, the continuum support-visible compact-gauge cylinder family is Euclidean invariant, reflection positive, regular on gauge-invariant local cylinder observables, and cyclic for the vacuum sector. Thus Osterwalder–Schrader*

reconstruction gives a four-dimensional quantum Yang–Mills theory $(\mathcal{H}, \Omega, H, \mathcal{A}^G_{\mathrm{loc}})$ *on the support-visible gauge-invariant local algebra, with* $H \geq 0$ *and* e^{-tH} *equal to the Euclidean transfer semigroup of Theorem A.9.*

Proof. For every finite list of positive-time cylinders, finite reflection positivity says that the associated reflected Gram matrix is positive semidefinite. Its entries converge along the extraction subnet, and the cone of positive semidefinite matrices is closed, so reflection positivity passes to the limit. Euclidean covariance, locality, regularity, and clustering pass by the compatible transformation maps and uniform bounds in Assumption A.25; weak-* compactness alone would not supply those bounds. The vacuum vector is cyclic for the GNS closure of the gauge-invariant local cylinder algebra. The finite time-zero identity (T4) passes to the limit on the dense cylinder core: the left side converges by four-dimensional cylinder extraction and the right side by the coherent semigroup limit. Consequently the OS time-zero inner product and translation semigroup are unitarily identical to the direct-limit $(\mathcal{H}, \Omega, e^{-tH})$ of Theorem A.23; the notation does not identify two unrelated Hamiltonians. The Osterwalder–Schrader reconstruction theorem then produces the Hilbert space, vacuum, local algebra, positive Hamiltonian, and Euclidean transfer semigroup [87, 88]. □

Remark A.27 (Scope of the compact-gauge OS theorem)**.** *The preceding reconstruction is restricted to the support-visible bosonic compact-gauge cylinder algebra on its declared branch. It falls short of the full chiral* G_6 *exact finite quantization, does not by itself instantiate the Standard-Model continuum observable sector, and supplies no resonance-sheet continuation. Those are separate typed inputs and implications in Appendix A.*

Proposition A.28 (Nontriviality of the support-visible compact-gauge theory)**.** *Under Assumptions A.22 and A.25, The support-visible compact-gauge local algebra on the zero-obstruction vacuum branch strictly contains the vacuum scalars and admits a non-vacuum finite-energy local excitation.*

Proof. The compact-gauge witness and physical-UV landing theorem supplies finite support-visible gauge-invariant local observables, for example nonconstant Wilson/plaquette cylinders. The uniform positive variance in Assumption A.25 says that the chosen centered cylinder does not enter the GNS null ideal in the limit, so it gives a nonzero vector orthogonal to the vacuum. The uniform form bound and lower-semicontinuity of the certified transfer-form limit give this vector finite continuum energy. Thus the support quotient has not erased the excitation that proves nontriviality. □

Theorem A.29 (Conditional positive compact-gauge Yang–Mills mass gap from OPH repair dynamics)**.** *Let* G *be a compact simple gauge group carried by a support-visible compact-gauge OPH vacuum branch satisfying the standing setup, Assumptions A.22 and A.25, the renormalized Yang–Mills identification receipt of Assumption A.8, the finite ground-state-transform and cross-fiber receipt of Assumption A.13, and all hypotheses of Theorem A.14, Lemma A.16, and Proposition A.17. The theory reconstructed in Theorem A.26 is nontrivial by Proposition A.28, and its continuum support-visible Hamiltonian* H *satisfies*

$$H \geq \delta_*(I - P_0^{\mathcal{H}}),$$

where $P_0^{\mathcal{H}}$ *projects onto the physical vacuum. Therefore*

$$\mathrm{Spec}(H) \cap (0, \delta_*) = \varnothing, \qquad \Delta_{\mathrm{YM}} \geq \delta_* > 0.$$

The Yang–Mills gap is the repair gap:

$$\Delta_{\mathrm{YM}} = \Delta_{\mathrm{rep}}.$$

Proof. Proposition A.17 gives

$$L_r^{\mathrm{EC}} \geq \delta_*(I - P_{0,r})$$

at every finite stage. Theorem A.23, using the explicit finite transfer/vacuum receipt, transports the bound to the support-visible continuum:

$$L^{\mathrm{rep}} \geq \delta_*(I - P_0^K).$$

Using $UHU^{-1} = L^{\mathrm{rep}}$, conjugation by U^{-1} gives the Hamiltonian lower bound. The spectral-gap statement follows immediately. Since U is unitary and maps the vacuum to the constant sector, it preserves the nonzero spectrum, so the Yang–Mills gap and repair gap are equal. □

Exact gap accounting. The theorem gives an identity stronger than a phenomenological estimate:

$$\mathrm{Spec}(H) \setminus \{0\} = \mathrm{Spec}(L^{\mathrm{rep}}) \setminus \{0\}.$$

Thus

$$\Delta_{\mathrm{YM}} = \inf\big(\mathrm{Spec}(H) \setminus \{0\}\big) = \inf\big(\mathrm{Spec}(L^{\mathrm{rep}}) \setminus \{0\}\big) = \Delta_{\mathrm{rep}}.$$

The finite-stage projection argument proves positivity of that same quantity.

Relation to the Clay/Jaffe–Witten statement. The Clay problem asks for a nontrivial quantum Yang–Mills theory on $\mathbb{R}^4$, for each compact simple G, satisfying axiomatic properties at least as strong as the stated Wightman or Osterwalder–Schrader references and possessing a positive mass gap [85, 84]. Theorem A.9 supplies the four-dimensional Euclidean Yang–Mills form on the OPH support-visible compact-gauge branch, and Theorem A.26 supplies the support-visible OS reconstruction on the gauge-invariant local algebra. Proposition A.28 supplies nontriviality. Theorem A.29 supplies the exact spectral gap accounting on that same branch. Proposition A.6 proves the algebra/state/GNS extraction only. Full Clay admissibility additionally requires the finite transfer/vacuum and OS/noncollapse receipts in Assumptions A.22 and A.25, together with the renormalized Yang–Mills identification, finite ground-state-transform/cross-fiber, and uniform-gap receipts; weak-* compactness is not a substitute for those inputs. A focused Clay presentation uses the same premises and conclusions.

A.5 A local gauge-covariant action on the declared matter grammar

Fix an oriented flat Lorentzian chart with metric $\eta = \mathrm{diag}(1, -1, -1, -1)$, its two-component Spin convention, and smooth compactly supported variations. These geometric and Spin data are inputs. Take the declared algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ on its fundamental, dual and trivial carriers. The exterior grammar supplies

$$Q : (3,2)_{1/6}, \quad u^c : (\bar{3},1)_{-2/3}, \quad d^c : (\bar{3},1)_{1/3}, \quad L : (1,2)_{-1/2}, \quad e^c : (1,1)_1.$$

The Higgs choice $H : (1,2)_{1/2}$, family multiplicity, minimal first-derivative kinetic law, scalar polynomial truncation and Yukawa intertwiners are declared. No coupling, mass or Yukawa entry is selected.

Use anti-Hermitian connections, with coupling normalization absorbed into their coordinates, and set

$$D_\mu = \partial_\mu + \rho(A_\mu), \qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu]. \tag{1}$$

Fermion values and first derivatives generate a complex exterior algebra. Conjugation is the antilinear anti-automorphism exchanging fermions with independent conjugate generators and reversing products. Scalar and connection coefficients are even. Write $\bar\sigma^\mu = (1, -\sigma^1, -\sigma^2, -\sigma^3)$ and $\epsilon_{01} = 1$. With color index i, weak indices a, b, spin indices α, β and family indices r, s, define

$$T^u_{rs} = \epsilon_{\alpha\beta}\epsilon_{ab} Q^{ia\alpha}_r H^b (u^c)^\beta_{s,i}, \qquad T^d_{rs} = \epsilon_{\alpha\beta} Q^{ia\alpha}_r \overline{H}_a (d^c)^\beta_{s,i}, \qquad T^e_{rs} = \epsilon_{\alpha\beta} L^{a\alpha}_r \overline{H}_a (e^c)^\beta_s. \tag{2}$$

The superscript c denotes charge conjugation. These are all-left-Weyl expressions, not four-component barred expressions with unchanged charges.

Theorem A.30 (Conditional local action composition)**.** *Under the stated representation, Spin and kinetic assumptions, the density*

$$\begin{aligned}\mathcal{L} = {} & \frac{1}{2}\sum_{s=3,2,1} \kappa_s \operatorname{Tr}(F^s_{\mu\nu} F^{\mu\nu}_s) + \sum_{r,\psi} \frac{i}{2}\left[\psi^\dagger_r \bar\sigma^\mu D_\mu \psi_r - (D_\mu \psi_r)^\dagger \bar\sigma^\mu \psi_r\right] \\ & + (D_\mu H)^\dagger D^\mu H - m^2 H^\dagger H - \lambda (H^\dagger H)^2 - \sum_{r,s,\chi=u,d,e} \left(Y^\chi_{rs} T^\chi_{rs} + \overline{Y^\chi_{rs}} (T^\chi_{rs})^\dagger\right)\end{aligned} \tag{3}$$

is Grassmann even, Hermitian and invariant under arbitrary infinitesimal local gauge transformations. The real m^2, λ, positive κ_s and complex Y^χ_{rs} remain free coefficients. The construction composes varying-field gauge, scalar and fermion kinetic terms and all three Yukawa channels.

Proof. For an anti-Hermitian local parameter θ, set $\delta\psi = \rho(\theta)\psi$, $\delta H = \rho_H(\theta) H$ and $\delta A_\mu = -\partial_\mu \theta + [\theta, A_\mu]$. The product rule gives

$$\delta(D_\mu \psi) = \rho(\theta) D_\mu \psi, \qquad \delta F_{\mu\nu} = [\theta, F_{\mu\nu}].$$

The first identity cancels the two opposite $(\partial_\mu\theta)\psi$ terms. The second uses symmetry of the second parameter jet and cancels the first-derivative terms against the connection commutators. No equation of motion is used. Cyclic trace proves gauge-kinetic invariance; anti-Hermiticity and commutation of internal generators with $\bar\sigma^\mu$ prove kinetic invariance. The trace forms are ad-invariant, with the free normalizations permitted by the invariant-form classification.

The color fundamental–dual pairing and weak determinant tensor are invariant. The hypercharge sums are $1/6 + 1/2 - 2/3 = 0$, $1/6 - 1/2 + 1/3 = 0$ and $-1/2 - 1/2 + 1 = 0$. The supplied Spin epsilon contracts anticommuting Weyl components. Each T^χ_{rs} and its adjoint are gauge neutral, separately for every family pair. Every term has even Grassmann degree and the displayed conjugate combinations are Hermitian. The symmetrized kinetic expression differs from $i\psi^\dagger\bar\sigma^\mu D_\mu\psi$ by a total derivative. □

Proposition A.31 (Matter current of the same local action)**.** *For an anti-Hermitian basis element T_a, define $\delta\mathcal{L}_{\text{matter}} = j^\mu_a \delta A^a_\mu$ with matter jets fixed. Then*

$$j^\mu_a = i\sum_{r,\psi} \psi^\dagger_r \bar\sigma^\mu \rho_\psi(T_a)\psi_r + (\rho_H(T_a)H)^\dagger D^\mu H + (D^\mu H)^\dagger \rho_H(T_a) H. \tag{4}$$

Its covariant divergence vanishes on the matter Euler–Lagrange equations, using the dual adjoint connection on the current index.

Proof. Differentiate the two covariant kinetic terms. Anti-Hermiticity doubles the fermion contribution in the symmetrized expression; the potential and Yukawa tensors have no connection dependence. For a compactly supported gauge variation, the matter equations remove matter variations from the first variation of the action. Substituting $\delta A_\mu = -\partial_\mu\theta + [\theta, A_\mu]$ and integrating by parts gives the dual covariant-divergence identity. □

Exact local checks and scope. The executable fixture retains three families, 900 independent Grassmann value, derivative and conjugate generators, and all 27 complex Yukawa slots. Connection and scalar jets are exact Gaussian-rational test values, not physical coupling constants or measurements. Independent arithmetic checks reconstruct the density, retain first and symmetric second jets of the gauge parameter, and reject omitted inhomogeneous connection terms, omitted curvature commutators, wrong dual color action and mixed Yukawa charges. Grassmann anticommutation is checked separately from neutrality. Lean proves the noncommutative first- and second-jet identities, the cyclic-trace Ward identity and the spin-sign control. The full action and on-shell current argument are analytic, not full Lean formalizations.

The flat chart, Spin interpretation, kinetic choice, Higgs/Yukawa content, relative normalizations, family multiplicity and local connection remain declared. Identifying the connection with one reconstructed from source response histories is separate. There is no source-produced Standard Model action or current, broken-vacuum selection, gauge fixing or ghost construction, quantum anomaly cancellation from a fermionic measure, renormalization, continuum quantum-field existence theorem or numerical particle prediction. Global gauge quotient and physical line-spectrum attachments remain separate. The local action admits a consistent classical charged-scalar slice, with both Cartan gauge equations and their variational current retained. Its finite source-address execution uses a declared logarithmic plaquette action and model clock, as follows.

A.6 A consistent local Cartan–scalar reduction and current response

Work on the flat Spin chart and local classical action of Theorem A.30. All kinetic coefficients are positive constants; the mass, quartic and Yukawa coefficients remain declared. Use the standard weak Cartan generator $i\sigma_3/2$ and central generator i. Writing the two real Cartan connection coordinates as $a = (a_2, a_1)$ gives

$$q = (1/2, 1/2)^T, \qquad K = \mathrm{diag}(\kappa_2, 2\kappa_1), \qquad d = q^T K^{-1} q, \qquad w = K^{-1} q/d.$$

Here the gauge density is $-\frac{1}{4} f^T K f$: the fundamental traces are $\mathrm{Tr}((i\sigma_3/2)^2) = -1/2$ and $\mathrm{Tr}(i^2) = -1$. The executable weak basis uses $i\sigma_3$, in which $q = (1, 1/2)^T$ and $K = \mathrm{diag}(4\kappa_2, 2\kappa_1)$; d agrees.

Theorem A.32 (Local classical solution embedding)**.** *Set all fermions and their independent conjugates, color connections and off-diagonal weak connections to zero. Let $H = (\varphi, 0)$ and $a = wA$. Every smooth solution, including the temporal gauge equation, of*

$$\mathcal{L}_{\mathrm{red}} = -\frac{F_{\mu\nu} F^{\mu\nu}}{4d} + |(\partial_\mu + iA_\mu)\varphi|^2 - m^2|\varphi|^2 - \lambda|\varphi|^4 \tag{5}$$

lifts to a solution of the full local classical equations, with ansatz-compatible initial and boundary data. All complex Yukawa entries remain unrestricted.

Proof. The identities $q^T w = 1$, $Kw = q/d$ and $w^T K w = 1/d$ give the density and both Cartan equations. In the current convention $\delta\mathcal{L}_{\mathrm{matter}} = j^\mu \delta A_\mu$, these equations are $q_a[d^{-1}\partial_\nu F^{\nu\mu} + j^\mu] = 0$. If $q^T n = 0$, the decomposition $a = wA + nB$ has no scalar coupling to B and no kinetic cross term, since $w^T K n = 0$; $B = 0$ solves its free equation. The Higgs and its covariant derivative lie in the upper weak line, so both off-diagonal weak currents vanish. The gauge covariant divergence of a Cartan curvature is Cartan. Color curvature and sources vanish. The diagonal connection preserves the lower Higgs line, whose potential derivative vanishes at zero. Every fermion equation contains a fermion factor; every Yukawa contribution to a Higgs equation contains two. Thus all discarded variations vanish without setting any Yukawa coefficient to zero. The temporal Cartan equation is included, so Gauss's law is retained. □

This is a local real-connection statement. Generic kinetic coefficients give a direction incommensurate with the full charge lattice. No compact subgroup, global quotient, large-gauge identification, electromagnetic vacuum, perturbative stability or quantum truncation follows. Retaining a real unwrapped connection is essential: its exponential alone need not determine the embedded connections in every representation.

A compatible spatial discretization. The same reduction holds for a declared semidiscrete gauge–scalar action with covariant link differences, invariant electric metric, and the local logarithmic plaquette energy. For the weak holonomy use its principal matrix logarithm on the smooth identity chart. On the Cartan slice, $\log U_f = i\sigma_3 w_{\mathrm{w}}(\mathrm{d}A)_f/2$ when $|w_{\mathrm{w}}(\mathrm{d}A)_f| < 2\pi$, where w_{w} is the weak component of w. The hypercharge curl is real and unwrapped. Consequently the plaquette energy restricts to the quadratic Maxwell curl with coefficient $1/d$. The first variation of $\mathrm{Tr}(\log U)^2$ is $2\,\mathrm{Tr}(\log U\, U^{-1}\delta U)$; discarded weak variations are off-diagonal and have zero trace pairing. Scalar link currents also have zero off-diagonal component. Electric transport along the Cartan line is geodesic for the invariant metric. These facts give the same omitted equations and weighted Cartan square. This statement requires the logarithm chart throughout the history. A Wilson cosine plaquette does not supply this quadratic reduction for arbitrary w.

On the prepared golden $q = 5$ tensor family use dimensionless positions x/L, $L = 2/\sqrt{\phi+2}$. The ordered one-dimensional nodes are $0, 2\phi-3, 4\phi-6, \phi-1, 3\phi-4, 1$. The 4^3 interior nodes have lumped masses v_i; an internal edge has conductance c_e equal to its transverse dual area divided by its gap. The fixed zero scalar boundary contributes the usual nonnegative onsite stiffness b_i. Declare $\kappa_1 = \kappa_2 = m^2 = 1$, $\lambda = 1/4$, so $d = 3/8$. For each internal plaquette set its Maxwell weight γ_f to $1/d$ times normal dual length divided by area, and set $\epsilon_e = c_e/d$. These are action quadratures, not causal counting weights.

With $\pi_i = v_i D_0 \psi_i$ and canonical real link momentum Π_e, the Hamiltonian is

$$\begin{aligned}\mathcal{H} = &\sum_i \frac{|\pi_i|^2}{v_i} + \sum_e \frac{\Pi_e^2}{2\epsilon_e} + \sum_{e=i\to j} c_e |e^{ia_e}\psi_j - \psi_i|^2 \\ &+ \sum_i [(v_i + b_i)|\psi_i|^2 + v_i|\psi_i|^4/4] + \tfrac{1}{2}\sum_f \gamma_f (\mathrm{d}a)_f^2. \end{aligned} \tag{6}$$

The complex scalar symplectic form is $2\,\mathrm{Re}(\mathrm{d}\overline{\pi}_i \wedge \mathrm{d}\psi_i)$. Differentiating this same energy gives

$$J_{i\to j} = -2c_e\,\mathrm{Im}(\overline{\psi}_i e^{ia_e}\psi_j), \qquad \rho_i = 2\,\mathrm{Im}(\overline{\psi}_i \pi_i), \qquad G_i = \rho_i + (\mathrm{div}\,\Pi)_i.$$

Here divergence is outward; $\dot\rho + \mathrm{div}\, J = 0$ and $\dot\Pi = J - \mathrm{d}^T\gamma \mathrm{d}a$ imply $\dot G = 0$.

Proposition A.33 (Local split evolution preserves Gauss constraints)**.** *Scalar kinetic drift, electric kinetic drift and every individual edge, onsite and plaquette potential kick preserve each G_i in ideal arithmetic. Their symmetric composition is symplectic and preserves the initial Gauss constraints. Disjoint local subflows commute.*

Proof. Scalar drift adds a real multiple of π_i to ψ_i, leaving ρ_i unchanged; electric drift changes only a. An edge kick changes the two charges by $-hJ_e, +hJ_e$ and Π_e by hJ_e. Onsite kicks add a real multiple of ψ_i to π_i. A plaquette kick adds a curl transpose to Π, whose divergence vanishes. Each map is its Hamiltonian subflow; disjoint subflows have disjoint dynamical variables. The potential kicks also commute because their coordinates are fixed during the kicks. □

Finite current and feedback control. The executable packet uses 64 scalar sites, 144 internal links and 108 plaquettes, with two steps of declared model duration 1/200. It compares the real preparation $\psi_i = 1/5$, $\pi_i = a_e = \Pi_e = 0$ with a local phase intervention $(3 + 4i)/5$ and a genuine gauge copy, which also changes the incident unwrapped connections. The intervention leaves charge and Gauss constraints unchanged but changes the edge current. At the selected edge $c_e = 1/4$ exactly, so $J_e = 2/125$ and its first electric half-kick is 1/25000. The gauge copy preserves invariant readouts. Complete operation records bind every dynamic read to its version's actual writer; decoded checkpoints contain scalar intensity, charge, edge current and electric feedback. Independent high-precision replay checks these binary64 records, including a nonempty off-influence control. This numerical agreement is separate from the exact local identities and gives no continuous-history error enclosure. The prepared addresses, local execution law, action, boundary and model tick remain supplied. No causal-count clock attachment, inherited fine-grid detector error, selected physical current or full Standard Model quantum evolution is asserted.

A.7 A continuous charged-current and separated-detector bound

The same finite Hamiltonian admits a useful continuous-time comparison without increasing the charged execution. Work with its exact golden weights and exact initial scalar magnitude $s_0 = 1/5$, zero scalar and electric momenta, and zero initial connections. Compare the baseline with the phase intervention $(3 + 4i)/5$ at site 21. The distinct detector site 37 is joined to it by the oriented edge of conductance $c = 1/4$. The reference gauge copy is the exact gauge transform of the baseline. Times are the declared dimensionless action parameter.

Theorem A.34 (Resolved nonlinear charged readout)**.** *The exact classical solutions exist on* $0 \le t \le T = 1/100$ *within the principal-log chart. On the selected edge their intervention current satisfies*

$$J(t) \ge 0.015110268, \qquad \Pi(T) > 0.0001543.$$

Writing $I_i = |\psi_i|^2$*, the spatially distinct detector obeys*

$$I_{37}^{\text{intervention}}(t) - I_{37}^{\text{baseline}}(t) \le -\frac{9}{20}t^2 \qquad (0 < t \le T).$$

Thus the response magnitude is at least 0.000045 *at* T *and at least* 0.00001125 *throughout* $[1/200, 1/100]$*. For the nine initial/completed decoded intensity checkpoints across all three preparations, the exact continuous-reference error is less than* 0.000006 *per checkpoint.*

Proof. Set $S = 1/4$, $A_* = 10^{-5}$, $B = 88$, and $D = 13/250$. On the box $|\psi_i| \le S$, $|a_e| \le A_*$, differentiation of the Hamiltonian gives

$$|\ddot{\psi}_i| \le \frac{2S\sum_{e\ni i} c_e + (v_i + b_i + v_i S^2/2)S}{v_i} < B,$$
$$|\ddot{a}_e| \le \frac{2c_e S^2 + 4A_*\sum_{f\ni e}\gamma_f}{\epsilon_e} < D.$$

Exact rational enclosures of the golden weights verify every inequality and $\sum_{f\ni e}\gamma_f < 78$. Since the initial velocities vanish, $|\psi_i(t) - \psi_i(0)| \le Bt^2/2$ and $|a_e(t)| \le Dt^2/2$. Both bounds lie strictly inside the box through T; a first-exit argument and local ODE continuation prove existence there. Also $|(4/3)(\mathrm{d}a)_f| \le (16/3)A_* < 1 < 2\pi$, preserving the chart. Gauge covariance transfers the invariant bounds to the ideal gauge copy.

The current product estimate gives

$$|J(t) - 2/125| \le C_2 t^2 + C_4 t^4, \quad C_2 = \tfrac{1}{4}(2s_0 B + s_0^2 D) = 8.80052, \quad C_4 = B^2/8 = 968.$$

The magnetic force is at most $2D(78)t^2$. Integrating $\dot{\Pi} = J - \mathrm{d}^T \gamma \mathrm{d}a$ therefore bounds $|\Pi(t) - (2/125)t|$ by $(C_2 + 156D)t^3/3 + C_4 t^5/5$, yielding the stated current and feedback.

For the detector, let $u = \sum_{e \ni 37} c_e / v_{37}$; its boundary stiffness vanishes. The cubic map $z \mapsto |z|^2 z$ is $3S^2$-Lipschitz on the complex disk. The phase estimate $|e^{ia} - 1| \le |a|$ consequently gives

$$|\ddot{\psi}_{37}(t) - \ddot{\psi}_{37}(0)| \le E t^2/2, \quad E = u(2B + s_0 D) + (1 + 3S^2/2)B < 30759.$$

The quadratic Taylor remainder is at most $r = 30759 t^4/24$. Put $v = v_{37}$ and

$$a_0 = -51/250, \qquad a_1 = a_0 - 1/(50v) + i/(25v).$$

These are the exact initial accelerations of the two preparations. Each intensity differs from $|s_0 + a_j t^2/2|^2$ by at most $2(s_0 + Bt^2/2)r + r^2$. Subtraction gives a quadratic coefficient $s_0 \operatorname{Re}(a_1 - a_0)$ and quartic coefficient $(|a_1|^2 - |a_0|^2)/4$. Exact rational bounds on v show that their sum with both remainders is at most $-(9/20)t^2$ for $0 < t \le T$.

Finally, decode ψ_{37} from the actual versioned reads at each initial or completed checkpoint. Interpret every stored binary64 component as its exact dyadic value. Its intensity discrepancy from the appropriate quadratic Taylor expression, plus the preceding analytic remainder, is an exact bound against the continuous IVP. The independent consumer checks all nine values, each below 6×10^{-6}, and excludes the intermediate split kick. It replays the parent custody and operations; copied public frames cannot replace the read events. □

This is a nonlinear classical current and readout result on the supplied family. The checkpoint bound does not enclose every state component or every intermediate observer operation. The exact reference preparation is distinct from its binary64 encoding. Quantum evolution, full fermion generation, a physical clock and native selection of the action or population are not supplied by this estimate.

A.8 Fermionic hypercharge transport on prepared source addresses

Fix the same 64 prepared golden addresses and 144 oriented nearest-neighbour links as the scalar execution. Give each address two Spin components of each of the five registered matter multiplets. Their internal multiplicities and integer charges are

$$(d_s) = (6, 3, 3, 2, 1), \qquad (q_s) = 6(Y_s) = (1, -4, 2, -3, 6).$$

This finite model supplies its spatial hopping action, canonical anticommutation relations, state preparation and model step. The source addresses do not select those inputs. The colour and weak indices are retained as multiplicities of an abelian transport law; the model does not execute the complete nonabelian or Yukawa dynamics.

Let $c_{s,a,v,\alpha}$ be the annihilation operators on the finite fermion Fock space, where $a = 1, \dots, d_s$ and $\alpha = 1, 2$. They satisfy $\{c_i, c_j^\dagger\} = \delta_{ij}$ and $\{c_i, c_j\} = 0$. For an oriented link $e = (u, v)$ in direction j, a unit complex link U_e and a supplied real hopping weight κ_e, put

$$K_{s,e} = i\sigma_j U_e^{q_s}, \tag{7}$$

$$\widehat{H}_e = \kappa_e \sum_{s,a} (c_{s,a,u}^\dagger K_{s,e} c_{s,a,v} + c_{s,a,v}^\dagger K_{s,e}^\dagger c_{s,a,u}). \tag{8}$$

The Pauli matrices act on Spin, not on the internal multiplicity index. The sign and link orientation in this definition fix the current convention. If $U_e \mapsto g_u U_e \overline{g}_v$ and $c_{s,a,w} \mapsto g_w^{q_s} c_{s,a,w}$, the quadratic expression is unchanged. The link is held fixed during this matter update.

Occupy one normalized spatial–Spin orbital ψ_s for every internal index a. Distinct internal indices and multiplets are orthogonal, so their exterior product is a normalized fifteen-fermion Slater state. Its one-body density is the orthogonal projector

$$P = \bigoplus_s I_{d_s} \otimes |\psi_s\rangle\langle\psi_s|.$$

The complex orbital coefficients describe this fermionic state. They are not commuting replacements for the field operators. In particular $0 \le P \le 1$ and $\operatorname{Tr} P = 15$.

For one occupied orbital write $z = (\psi_{s,u}, \psi_{s,v})$ and

$$h_{s,e} = \kappa_e \begin{pmatrix} 0 & K_{s,e} \\ K_{s,e}^\dagger & 0 \end{pmatrix}.$$

For step $\delta > 0$, take the implicit midpoint equation

$$i\frac{z' - z}{\delta} = h_{s,e}\bar{z}, \qquad \bar{z} = \frac{z' + z}{2}. \tag{9}$$

Since $K_{s,e}$ is unitary, $h_{s,e}^2 = \kappa_e^2 I$. The update is the unitary Cayley transform $(I + i\delta h_{s,e}/2)^{-1}(I - i\delta h_{s,e}/2)$. It preserves the projector property and particle number, with $P \mapsto VPV^\dagger$, and is exact over Gaussian rationals when the link, step, weight and preparation lie in that field.

Proposition A.35 (Variational current and mean Gauss preservation)**.** *Define the mean site charge and midpoint current by*

$$\rho_w = \sum_s d_s q_s \|\psi_{s,w}\|^2, \tag{10}$$

$$J_e = -2\kappa_e \sum_s d_s q_s \operatorname{Im}(\bar{\psi}_{s,u}^\dagger K_{s,e} \bar{\psi}_{s,v}). \tag{11}$$

Then J_e is the derivative of the quadratic midpoint functional of (8) under $U_e \mapsto e^{i\theta} U_e$, with the midpoint orbitals fixed. These averages need not be normalized occupied orbitals; only the step endpoints represent the normalized fifteen-fermion Slater states. The charge changes satisfy

$$\rho_u' - \rho_u = -\delta J_e, \qquad \rho_v' - \rho_v = \delta J_e.$$

Let E_e be a classical link momentum and update $E_e' = E_e - \delta J_e$. With div *positive at the initial endpoint of each oriented link, every site residual $G_w = (\operatorname{div} E)_w - \rho_w$ is unchanged. This remains true for any finite composition of these edge factors.*

Proof. For the finite canonical anticommutation relations, $[c_i^\dagger c_j, c_k^\dagger c_l] = \delta_{jk} c_i^\dagger c_l - \delta_{li} c_k^\dagger c_j$; the degree-four terms cancel with their fermionic signs. Thus the quadratic Hamiltonian acts through its one-body matrix. Differentiating its link phase at fixed state gives the displayed J_e. The identity

$$\|x'\|^2 - \|x\|^2 = 2\operatorname{Re}\left(\frac{x' + x}{2}\right)^\dagger (x' - x)$$

and (9) give opposite endpoint transfers. The electric update cancels those changes in G_u, G_v; all other site charges and incident momenta are unchanged. Induction proves the finite composition statement. Gauge covariance follows by conjugating the one-body matrices and their Cayley transforms. □

Write $U_e = e^{i\theta_e}$. The same first-order action is

$$S_e = \int \left(i \sum_s d_s \psi_s^\dagger \dot{\psi}_s + E_e \dot{\theta}_e - \langle \widehat{H}_e \rangle \right) dt.$$

Its variations give the orbital equation, $\dot{E}_e = -\partial_{\theta_e} \langle \widehat{H}_e \rangle$ and $\dot{\theta}_e = 0$, since this factor has no E_e dependence in its Hamiltonian. Thus frozen links and electric feedback belong to one declared action factor. The midpoint scheme discretizes these equations. Exact discrete conservation does not identify its step with the continuous Hamiltonian exponential or prove a trajectory-error estimate. The electric momentum is driven by a fermionic expectation: this is a semiclassical update, not a quantized gauge field coupled to fermion Fock space. Electric drift, plaquette and Yukawa factors are absent.

A sharp quantum-constraint boundary. At every address choose $\psi_s = (3/40, 1/10)$ before the local phase intervention. Each orbital has squared site norm $1/64$ and unit global norm. Since $\sum_s d_s q_s = 0$, the initial mean charge vanishes at every site, and $E = 0$ solves the mean Gauss constraint. Nevertheless the local charge operator has variance

$$\mathrm{Var}(\widehat{Q}_w) = \sum_s d_s q_s^2 \frac{1}{64} \left(1 - \frac{1}{64} \right) = \frac{945}{512} > 0. \tag{12}$$

Each occupied internal orbital contributes an independent Bernoulli site occupation; this proves the formula. A site phase preserves it. Consequently the product preparation with sharp classical electric data is not annihilated by the quantum Gauss operator. In ordinary hypercharge units the variance is divided by 36. Vanishing mean residuals cannot certify a physical gauge-constrained quantum state.

Execution and interpretation. The executable construction in `code/sm_fermion_current/` retains the local orbital state, incident link ports, current readback and electric feedback as versioned software records. Its independent verifier rebuilds the charge dictionary, source addresses, Cayley updates, variational currents and consumed writer/value versions. Baseline, local phase intervention and gauge-related execution are separate controls. The generation's linear, cubic and mixed anomaly sums are representation identities; a finite canonical-anticommutation implementation does not construct a chiral continuum fermion measure.

This packet has one declared generation and supplied quantum preparation. It proves neither native source selection, all-matter Standard Model evolution, a controlled spatial or temporal continuum limit, laboratory current, a physical clock, nor compatibility with a sourced curved family. The preceding scalar reduction and its continuous-readout enclosure retain their own action and preparation; their error bounds do not transfer to this fermionic transport.

A.9 A flux-dressed fermion state satisfying abelian quantum Gauss

The mean-field obstruction has a precise remedy within a separately supplied quantum link model. It changes both the gauge-field Hilbert space and the preparation. Choose one oriented link (u, v) of the same golden carrier. Its electric space is $\ell^2(\mathbb{Z})$, with

$$\widehat{E}|n\rangle = n|n\rangle, \qquad \widehat{U}|n\rangle = |n+1\rangle, \qquad [\widehat{E}, \widehat{U}] = \widehat{U}.$$

The ambient flux space is infinite. A finite reachable support is computed from the preparation; no cyclic wrap or truncated shift replaces this unitary operator.

Index the fifteen internal channels by a, retaining their charges q_a. In each channel occupy the normalized left Spin orbital $\chi = (3/5, 4/5)$ and leave its orthogonal orbital empty. The active right orbital is $-i\sigma_j\chi$, so the Pauli link maps it back to χ. The two active modes have annihilators l_a, r_a. Fix the canonical fermion order $(l_1, r_1, l_2, r_2, \ldots)$ and set

$$\widehat{h}_a = \kappa_a \left(l_a^\dagger \widehat{U}^{q_a} r_a + r_a^\dagger \widehat{U}^{-q_a} l_a \right). \tag{13}$$

Each channel retains exactly one fermion. Distinct channel Hamiltonians commute: their even fermion bilinears have disjoint mode support and the flux shifts commute. Initially all fifteen fermions are at u, the right modes are empty, and the link is in $|0\rangle$. Since $\sum_a q_a = 0$, this preparation is locally neutral.

Proposition A.36 (Exact finite-support abelian quantum Gauss state)**.** *Let $\widehat{Q}_u = \sum_a q_a l_a^\dagger l_a$ and $\widehat{Q}_v = \sum_a q_a r_a^\dagger r_a$. Every $\widehat{h}_a$ commutes with*

$$\widehat{G}_u = \widehat{E} - \widehat{Q}_u, \qquad \widehat{G}_v = -\widehat{E} - \widehat{Q}_v.$$

Its Cayley transform preserves their joint kernel. Starting from the neutral preparation, the product of these transforms is the normalized state

$$|\Psi\rangle = \sum_{x \in \{0,1\}^{15}} \left[\prod_a A_a^{1-x_a} (-iB_a)^{x_a} \right] |x\rangle_F \otimes \left| -\sum_a q_a x_a \right\rangle_E, \tag{14}$$

where $x_a = 1$ means that channel a occupies its right mode, and

$$\eta_a = \delta\kappa_a/2, \qquad A_a = \frac{1-\eta_a^2}{1+\eta_a^2}, \qquad B_a = \frac{2\eta_a}{1+\eta_a^2}.$$

Every branch, and hence the complete state, satisfies both quantum Gauss constraints. All other vertices retain zero charge and incident flux.

Proof. In $l_a^\dagger \widehat{U}^{q_a} r_a$, both left charge and electric flux increase by q_a, while right charge decreases by q_a. The adjoint reverses all three changes. This proves the two commutators on the invariant finite-support core of the electric operator. The Hamiltonians are bounded; their Cayley transforms preserve the joint Gauss sectors and the electric domain. On the one-fermion channel sector, $\widehat{h}_a^2 = \kappa_a^2 I$, which gives the displayed two coefficients. The interleaved fermion convention fixes the even-hop signs. Channel commutation gives the product expansion. Its orthogonal Fock branches have squared norms summing to $\prod_a (A_a^2 + B_a^2) = 1$. On branch x, $Q_v = \sum_a q_a x_a$, $Q_u = -Q_v$ and $E = -Q_v$. Both Gauss residuals therefore vanish individually; no cancellation of opposite residuals in an expectation is used. □

The span of these 2^{15} dressed branches is invariant under every $\widehat{h}_a$: each admissible hop changes the occupation and flux by the same signed charge. Thus the finite electric support is exact for the coupled evolution of this sector, although the standalone shift $\widehat{U}$ does not preserve it. Channel commutation also gives the continuous solution of the supplied Hamiltonian $\sum_a \widehat{h}_a$: replace A_a, B_a in (14) by $\cos(\kappa_a t), \sin(\kappa_a t)$. This is an exact finite-sector quantum history, not a time-step approximation.

For $\delta = 1/2$, choose $\kappa_a = 1$ except on the e^c channel, where $\kappa_a = 2$. These species weights are declared; they do not establish a universal physical propagation speed. The coefficients are

$(A,B) = (15/17, 8/17)$ on the first fourteen channels and $(3/5, 4/5)$ on the last. All 2^{15} Fock branches occur, with electric support $\{-18,\dots,18\}$. The link-phase current operator is

$$\widehat{J}_a = iq_a\kappa_a\left(l_a^\dagger \widehat{U}^{q_a} r_a - r_a^\dagger \widehat{U}^{-q_a} l_a\right), \qquad \dot{\widehat{E}} = -\sum_a \widehat{J}_a.$$

Exact evaluation in (14) gives

$$\langle \widehat{E}\rangle = -\frac{18144}{7225}, \qquad \sum_a \langle \widehat{J}_a\rangle = \frac{47232}{7225}.$$

The sum of the factor-midpoint currents is instead $36288/7225$; multiplied by $-\delta$, it equals the electric change. A final-state current and an integrated discrete transfer are different readouts.

For the continuous solution with the same species weights,

$$\langle \widehat{E}\rangle_t = 6(\sin^2 t - \sin^2 2t), \qquad \langle \widehat{J}\rangle_t = -6\sin 2t + 12\sin 4t.$$

The electric derivative equals minus the current. At the supplied model time $t = \pi/6$ they give -3 and $3\sqrt{3}$, respectively, with electric variance $45/2$. The flux fluctuations are correlated with charge, so quantum Gauss holds throughout this history. The time parameter, action and preparation are inputs; this solution calibrates no physical clock.

The executable producer and independent verifier in `code/sm_fermion_current/quantum_link.py` and `code/sm_fermion_current/verify_quantum_link.py` bind the channel and Spin maps, CAR signs, complete branch population, flux distribution and current matrix elements. This is a supplied finite quantum state and action factor on one prepared link. It is not the uniform semiclassical Slater preparation, a measured quantum history, full gauge-field evolution, nonabelian Gauss constraints or Yukawa evolution, or a chiral continuum regulator. Native state and operation selection, source-causal attachment and laboratory realization remain additional requirements.

B A coupled fermion current in a declared hybrid gauge action

The fermionic current can be coupled to electric drift and magnetic plaquette forces on the same prepared carrier. The construction below declares a regulated Hamiltonian and a finite splitting word. Its gauge fields are classical, while its matter state belongs to finite fermion Fock space. This hybrid interpretation differs from quantizing the gauge field in the preceding one-link construction.

Take the direct-product gauge cover $SU(3)\times SU(2)\times U(1)$, with the five registered representations and integer hypercharges $q_s = 6Y_s$ in each of three declared families. The compact abelian link is $U_e = e^{i\theta_e}$ with momentum E_e. For fixed $M, \lambda > 0$, choose

$$T_\lambda(E) = \frac{2E}{\lambda}\arctan\frac{\lambda E}{2M} - \frac{2M}{\lambda^2}\log\left(1+\frac{\lambda^2E^2}{4M^2}\right), \tag{15}$$

$$W_f = \prod_{e\in\partial f} U_e^{\epsilon_{fe}}, \qquad V_f = \beta_f(1-\operatorname{Re} W_f). \tag{16}$$

Here ϵ_{fe} is the oriented face boundary. The reduced Hamiltonian is

$$\mathcal{H}_{\rm red} = \sum_e T_\lambda(E_e) + \sum_f V_f + \sum_e \langle \Psi, \widehat{H}_{{\rm CAR},e}(U_e)\Psi\rangle,$$

where the hopping operators and Spin matrices are those defined above. In temporal gauge, its first-order action contains $\sum_e E_e\dot\theta_e + i\langle\Psi,\dot\Psi\rangle - \mathcal{H}_{\rm red}$. The link-phase variation of the same matter Hamiltonian defines its current.

This is a specified kinetic law. In particular,

$$T'_\lambda(E) = \frac{2}{\lambda}\arctan\frac{\lambda E}{2M}, \qquad T''_\lambda(E) = \frac{1}{M(1+(\lambda E/(2M))^2)} > 0.$$

Integrating $x - \arctan x = \int_0^x t^2/(1+t^2)\,dt$ gives

$$0 \le \frac{E^2}{2M} - T_\lambda(E) \le \frac{\lambda^2 E^4}{48M^3}.$$

The quadratic electric energy is therefore a controlled small-field comparison of functions. It is not the exact kinetic energy of this model, and the displayed bound is not a trajectory-error estimate. The parameter λ remains fixed when describing this action; varying it changes the law. The separate charge-one abelian Wilson term is specified on the direct-product cover. No descent of that term to the global central quotient is assumed.

Proposition B.1 (Invariant expectation-level matter sector)**.** *Extend the reduced Hamiltonian by classical nonabelian links and momenta, quadratic nonabelian electric energies, Wilson magnetic energies, the covariant Higgs kinetic and spatial terms, and a potential with vanishing first derivative at zero Higgs field. Couple the finite CAR matter through its registered representation matrices and all twenty-seven complex Yukawa coefficients of three families. Interpret every matter force as its Fock-state expectation in the canonical classical-field/quantum-matter action.*

Set the nonabelian links to identity, their electric momenta to zero, and the Higgs field and conjugate momentum to zero. In each species of one family take normal covariance $P_s = I_{d_s} \otimes |\psi_s\rangle\langle\psi_s|$, with no covariance between distinct species or families. Let the other two families be Fock vacuum, and take zero anomalous covariance $\langle c_i c_j\rangle = 0$. These conditions define an invariant hybrid subsystem with Hamiltonian $\mathcal{H}_{\mathrm{red}}$.

Proof. Every nonabelian matter charge or link-current variation inserts a traceless internal generator T_s. The remaining Spin and spatial factor is scalar in that internal index, so the expectation factors through $\operatorname{tr} T_s = 0$. This checks the omitted nonabelian matter forces, including each color and weak generator. The nonabelian electric velocities vanish at zero momenta, and their magnetic first variations vanish at identity plaquette holonomy. Scalar kinetic, spatial and gauge-current variations vanish at zero Higgs field and momentum.

In all-left-Weyl notation, the Yukawa terms have the gauge-neutral forms QHu^c, $Q\overline{H}d^c$ and $L\overline{H}e^c$, with the required Spin, color and weak contractions and their adjoints. Their Higgs first variation is a linear combination of anomalous expectations $\langle cc\rangle$ and $\langle c^\dagger c^\dagger\rangle$. Both vanish by fermion-number selection, for every value of every Yukawa coefficient. This checks the Higgs force rather than only the value of the Yukawa energy at $H = 0$.

The surviving hopping Hamiltonian is number conserving, family diagonal and scalar on each internal multiplicity space. It preserves the stated covariance form, the vanishing anomalous covariance and the two vacuum families. The field equations hence preserve every imposed condition. No stability of the zero-Higgs equilibrium is needed for this invariance statement. □

The expectation hypothesis is essential. It does not identify independent Grassmann components with one commuting orbital, and it does not annihilate the nonabelian Gauss operators. For example, in the initial uniform preparation $p = 1/64$, the weak generator $\sigma_z/2$ on the occupied quark and lepton doublets has local charge variance

$$\operatorname{Var}(\widehat{Q}_T(v)) = \sum_s \operatorname{tr}(T_s^2)p(1-p) = \frac{63}{2048} > 0.$$

The absent nonabelian mean source therefore cannot certify a full gauge-constrained quantum state with sharp zero nonabelian electric data.

Proposition B.2 (Coupled transport and conserved readout)**.** *On a finite oriented lattice with closed face boundaries, compose the following factors of the declared hybrid action:*

1. *a fermionic implicit-midpoint hopping factor, with its variational current and the same classical electric kick;*

2. *an electric factor of duration λ,*

$$U_e' = \frac{1 + i\lambda E_e/(2M)}{1 - i\lambda E_e/(2M)} U_e, \qquad E_e' = E_e;$$

3. *a magnetic factor of duration h on face f,*

$$E_e' = E_e - h\beta_f \epsilon_{fe} \operatorname{Im} W_f \quad (e \in \partial f), \qquad U_e' = U_e.$$

For Gaussian-rational initial data and rational parameters these maps are exact over Gaussian rationals. Each preserves the expectation Gauss residual at every vertex. Their recorded current and charge transfers satisfy exact discrete continuity under finite composition, and the construction is covariant under residual time-independent vertex gauge transformations.

Proof. The midpoint matter identity proved above gives opposite endpoint charge transfers and precisely the electric kick that cancels them in Gauss. The midpoint equations are applied to the orbital, link-angle and momentum variables of that same factor action; its angle is constant during the factor because its Hamiltonian has no electric-momentum dependence.

The electric factor is the exact flow of T_λ for the stated duration: exponentiating $i\lambda T_\lambda'(E) = 2i \arctan(\lambda E/(2M))$ gives the rational phase. It leaves charge and electric divergence unchanged. For the magnetic factor, $\partial_{\theta_e} V_f = \beta_f \epsilon_{fe} \operatorname{Im} W_f$. Its momentum kick is the oriented face boundary, whose vertex divergence vanishes. These facts prove Gauss preservation for every factor and hence for their composition. The unit-modulus rational phase, Hermitian Cayley matter transform and rational Wilson product also prove the asserted arithmetic closure. Abelian gauge transformations conjugate the matter updates, leave E and W_f invariant and commute with the electric drift. □

This factor word is a declared discrete transport. It is not the exact exponential of the summed Hamiltonian; exact Gauss conservation does not imply exact total-energy conservation or a continuous-trajectory error bound.

Finite execution and returned current. Use the same sixty-four prepared addresses, 144 oriented links and 108 plaquettes, with $\lambda = h = 1/2$ and $M = \beta = \kappa = 1$. A full matter sweep is followed by all electric drifts and all magnetic kicks. One subsequent matter factor reads the evolved link and momentum on edge 14. That final factor is a readout in a partial second word, not a second completed full-Hamiltonian time step.

Across baseline, local intervention, gauge copy and a disabled-drift control, the retained prefixes contain 2,444 parent events. Each of the four continuations has 255 events: 144 electric factors, 108 magnetic factors, one returning matter factor and two public readouts. The control leaves the link unchanged instead of applying electric drift; it is a counterfactual operation word, not another execution of the full declared action. In the intervention and gauge copy the return current lies in $[-0.005355160745, -0.005355160744]$, and the final edge momentum lies in

$[0.002253305815, 0.002253305816]$. The difference from the disabled-drift return current obeys the exact outward enclosure

$$0 < \frac{242747271}{2^{40}} \le J_{\mathrm{return}} - J_{\mathrm{disabled}} \le \frac{30343409}{2^{37}}.$$

Thus the electric record changes the link that the later fermionic current actually consumes. Gauge-related executions agree on the invariant readouts.

The package `code/sm_fermion_current/` contains the producer `coupled.py` and independent verifier `verify_coupled.py`. They reconstruct the closed face boundaries, exact factor updates, current, Gauss residuals and consumed writer/value versions. The observer-like software patches retain local orbital state, incident ports, public records and electric feedback; declared parent labels cannot substitute for actual consumption. Local magnetic energies are exact rationals, with their large sum bounded by outward dyadic intervals.

The action, regulator, family population, prepared addresses, routing and time labels are supplied. The invariant reduction does not execute an excited Higgs or nonabelian sector, select a chiral continuum regulator, identify a laboratory current or physical clock, or establish compatibility with a sourced curved geometry. Those statements require their own common-action and physical realization maps.

B.1 Spin transport and the information needed to select exchange statistics

A one-particle spin representation and a many-particle exchange rule are different data. Likewise, a current obtained from a supplied fermionic action need not distinguish that action from a bosonic alternative on the preparations actually used. The following finite statements separate these two questions without assuming a relativistic field reconstruction.

Proposition B.3 (A spin lift does not select a two-particle exchange sector)**.** *Let $\mathcal{H}$ be a finite-dimensional complex Hilbert space with $\dim \mathcal{H} \ge 2$, and let $D : G \to U(\mathcal{H})$ be a supplied unitary representation. Suppose a central element $z \in G$ acts as $D(z) = -I$. On $\mathcal{H}\otimes\mathcal{H}$, let $\tau(v\otimes w) = w\otimes v$. Both eigenspaces*

$$\mathcal{H}_+ = \ker(\tau - I) = \mathrm{Sym}^2 \mathcal{H}, \qquad \mathcal{H}_- = \ker(\tau + I) = \bigwedge^2 \mathcal{H}$$

are nonzero invariant subspaces of $D \otimes D$. The central element acts as $+I$ on both. For every nonzero $v \in \mathcal{H}$, the vector $v \otimes v$ is a nonzero member of $\mathcal{H}_+$, whereas its antisymmetric projection is zero. Thus the one-particle spin action, including its central minus sign, does not by itself determine the exchange sign or exclude two particles in the same one-particle mode.

Proof. For any linear operator A on $\mathcal{H}$,

$$\tau(A \otimes A) = (A \otimes A)\tau,$$

as is checked on simple tensors, which span the tensor product. Hence the orthogonal projections $\Pi_\pm = (I \pm \tau)/2$ commute with $D(g) \otimes D(g)$ for every g. The restrictions are therefore unitary representations. Since $(-I) \otimes (-I) = I$, they have the stated central action. If $v \neq 0$, then $\|v \otimes v\| = \|v\|^2 > 0$ and $\tau(v \otimes v) = v \otimes v$, so $\Pi_-(v \otimes v) = 0$. For orthonormal u, v, the unit vector $(u \otimes v - v \otimes u)/\sqrt{2}$ belongs to $\mathcal{H}_-$. Both sectors therefore exist for the same supplied spin representation, but they disagree on the doubled-mode state. □

This statement applies, in particular, to a supplied two-dimensional unitary realization of a binary rotation group whose central element is $-I$. A nonsplit double cover or a unique spin structure does not remove the displayed choice of two-particle sector. The statement does not construct a relativistic bosonic spinor field satisfying locality and a positive-energy spectrum condition, and it is not a counterexample to a relativistic spin–statistics theorem. It establishes only the insufficiency of the finite one-particle representation data just specified.

Proposition B.4 (Conserved single-occupancy channels cannot select statistics)**.** *Let*

$$\mathcal{K} = \bigoplus_{\lambda\in\Lambda} \mathcal{K}_\lambda$$

be a finite orthogonal decomposition into one-particle channel spaces. In both the bosonic and fermionic Fock spaces over $\mathcal{K}$, retain the sector with exactly one particle in every channel. Use the same initial normalized channel vectors ψ_λ, the same classical registers, and the same ordered finite sequence of the following operations:

1. *evolution by a self-adjoint channel-preserving quadratic Hamiltonian, or second quantization of a specified channel-preserving one-particle unitary;*
2. *readout of expectations of channel-preserving one-body operators;*
3. *classical register updates whose values are determined by those readouts and the retained classical registers.*

The two statistics give identical one-body densities, all such readouts, and every resulting classical register value. In particular, an expectation current, mean Gauss residual, and classical feedback history formed from these data cannot determine the exchange statistics.

Proof. Order the finite channel set once. On the algebraic tensor product define

$$J_\pm\Big(\bigotimes_\lambda v_\lambda\Big) = \prod_\lambda a_\pm^\dagger(v_\lambda)\,|0\rangle,$$

where $+$ denotes the bosonic and $-$ the fermionic creation operators, and the product follows the fixed order. Orthogonality of the channel spaces and the canonical relations show that the inner product of two such vectors is $\prod_\lambda \langle v_\lambda, w_\lambda\rangle$ in both cases. The maps therefore extend to unitary identifications between $\bigotimes_\lambda \mathcal{K}_\lambda$ and the respective fixed-channel-occupation sectors.

For a one-body operator B_λ supported in channel λ, its number-conserving second quantization obeys

$$J_\pm^*\, d\Gamma_\pm(B_\lambda)\, J_\pm = I\otimes\cdots\otimes B_\lambda\otimes\cdots\otimes I.$$

In the fermionic calculation, the signs from moving annihilation and creation through the preceding occupied channels cancel. In the bosonic calculation there is no sign, and occupation one gives the same matrix coefficient. Consequently the restriction of a channel-diagonal quadratic Hamiltonian $d\Gamma_\pm(\bigoplus_\lambda h_\lambda)$ is the same sum of tensor-slot Hamiltonians. The restrictions of second-quantized channel-preserving unitaries are likewise the same tensor products. Starting from the same tensor-product vector therefore gives the same channel vectors after every such operation. Their common one-body density is

$$P = \bigoplus_\lambda |\psi_\lambda\rangle\langle\psi_\lambda|, \qquad \langle d\Gamma_\pm(B)\rangle = \mathrm{Tr}(PB)$$

for channel-diagonal B. Equal classical inputs select equal subsequent operations and equal readouts give equal classical updates. Induction on the operation sequence proves the assertion, including adaptive mean-field feedback. A channel declared to remain empty contributes its vacuum factor and zero one-body density in both constructions. □

For the finite hypercharge-current construction, take $\lambda = (s, a)$ with $a = 1, \ldots, d_s$ and $\mathcal{K}_\lambda = \mathbb{C}^{64} \otimes \mathbb{C}^2$, the prepared site and Spin coordinates. There is one occupied orbital in each of the fifteen channels. Its abelian hopping and phase interventions act within each channel. Replacing the supplied CAR Fock state by the bosonic number state with one particle in each of the same channels preserves

$$P = \bigoplus_s I_{d_s} \otimes |\psi_s\rangle\langle\psi_s|, \qquad 0 \le P \le I, \qquad \operatorname{Tr} P = 15.$$

The projector property in this preparation is therefore not itself a statistics discriminator. The multiplicity index is an orthogonal one-particle label; repeating the same spatial–Spin orbital across different multiplicity indices does not prepare two particles in one complete one-particle mode.

The local Cayley updates use the same one-particle matrices in the two constructions. This comparison second-quantizes each one-particle Cayley unitary; it does not replace it by the Cayley transform of the full many-body Hamiltonian. The midpoint orbitals, and hence the midpoint variational functional and its current, also agree algebraically. Those midpoint averages are not normalized intermediate Fock states. Therefore the charge transfers, electric kicks, and expectation Gauss residuals coincide. The subsequent electric drifts and Wilson plaquette kicks use only the common classical registers, so they preserve this equality before the next matter factor reads the evolved link.

The retained site-charge variance supplies no distinction either. A single particle in channel (s, a) has site probability $p_{s,v} = \|\psi_{s,v}\|^2$, and its site-number projector squares to itself for both statistics. The product over channels gives

$$\operatorname{Var}(Q_v) = \sum_s d_s q_s^2 p_{s,v}(1 - p_{s,v})$$

in both states. The nonzero variance excludes an operator Gauss constraint with the specified classical electric numbers. Its equality does not turn that constraint into a consequence of mean Gauss.

These identical histories concern the executed abelian, channel-preserving restriction. They do not assert equivalence of the full nonabelian or Yukawa theories, of channel-mixing evolutions, or of preparations with two particles in one conserved channel. Nor do they change the validity of the CAR current calculation under its stated assumptions. They show why that current calculation alone does not select those assumptions. An exchange-sensitive comparison must specify additional preparation and readout operations outside this restriction, together with their source realization. A finite comparison of supplied alternatives is distinct from deriving physical spin–statistics from observer consistency.

Proposition B.5 (A two-particle port discriminator)**.** *Let L, R be orthonormal modes of one declared channel, let $a = 3/5$, $b = 4/5$, and supply the one-particle unitary*

$$U = \begin{pmatrix} a & -ib \\ -ib & a \end{pmatrix}.$$

Prepare $\Psi_\pm = (L \otimes R \pm R \otimes L)/\sqrt{2}$ and apply $U \otimes U$. For the coincidence effect

$$F = |LR\rangle\langle LR| + |RL\rangle\langle RL|,$$

the antisymmetric and symmetric preparations have probabilities

$$p_- = 1, \qquad p_+ = (a^2 - b^2)^2 = \frac{49}{625}.$$

A distinguishable-particle preparation $L \otimes R$ gives $p_{\mathrm{dist}} = a^4 + b^4 = 337/625$. All three preparations have the same summed one-particle density I before and after the operation, so this contrast requires a two-particle readout.

Proof. The matrix is unitary because $a^2 + b^2 = 1$, and its determinant is one. Direct expansion gives

$$(U \otimes U)\Psi_- = \Psi_-, \qquad (U \otimes U)\Psi_+ = (a^2 - b^2)\Psi_+ - i\sqrt{2}ab\,(LL + RR).$$

The three vectors Ψ_+, LL, RR are orthonormal, so the bosonic coincidence probability is $(a^2 - b^2)^2$, with probability $2a^2b^2 = 288/625$ at each bunched port. The antisymmetric output has no bunched component. For the distinguishable product input the coincidence amplitudes are a^2 and $-b^2$ in orthogonal labelled outcomes; their squared moduli add to $337/625$. Its bunching probability at each port is $a^2b^2 = 144/625$. Finally the two orthogonal occupied orbitals give summed one-particle density I, and $UIU^\dagger = I$. □

The port labels denote a supplied invariant channel basis. They do not assert that the two physical ports carry identical raw spin-coordinate vectors: a spin-dependent hopping matrix can use different local basis vectors. The rational coefficients agree with a finite hopping factor, but this discriminator supplies its own two-particle preparation and coincidence effect. It inherits no electric flux dressing, operator Gauss constraint, clock identification or laboratory instrument from a one-particle-per-channel history.

Indistinguishability is essential. If the particles carry normalized unobserved internal vectors α, β, unchanged by the port unitary, write $\eta = |\langle \alpha, \beta \rangle|^2$. Expanding the normalized symmetrized or antisymmetrized input and tracing over those vectors gives

$$p_{\mathrm{fermion}} = \frac{337 + 288\eta}{625}, \qquad p_{\mathrm{boson}} = \frac{337 - 288\eta}{625}.$$

The cross term is the product $\langle \alpha, \beta \rangle \langle \beta, \alpha \rangle$; its sign is the exchange sign. Orthogonal hidden states give $\eta = 0$ and erase the contrast. The ideal values above require $\eta = 1$.

There is also a conditional stability bound. Let ρ be either ideal output density and $\widetilde{\rho}$ an actual normalized output density with $\|\widetilde{\rho} - \rho\|_1/2 \le \epsilon$. If an actual effect $0 \le \widetilde{F} \le I$ satisfies $\|\widetilde{F} - F\| \le \delta$, then

$$\left|\mathrm{Tr}(\widetilde{\rho}\widetilde{F}) - \mathrm{Tr}(\rho F)\right| \le \epsilon + \delta.$$

Indeed, split the difference into $\mathrm{Tr}((\widetilde{\rho} - \rho)F)$ and $\mathrm{Tr}(\widetilde{\rho}(\widetilde{F} - F))$. The positive and negative parts of the trace-zero density difference each have trace at most ϵ, so the first term has magnitude at most ϵ; the second is bounded by δ. A preparation trace-distance bound transfers unchanged through an exactly common unitary; imperfect dynamics instead require a bound on the actual output. For a common error budget $\epsilon + \delta < 288/625$, the two ideal-model probability intervals are disjoint because their separation is $576/625$. These are supplied error bounds, not a calibration or a bound on finite sampling error. An observed test additionally needs authenticated preparation, indistinguishability, operation and readout evidence.

B.2 Conditional exclusion from a stable generator and a scalar deformation law

One can select exclusion without assuming it, provided the quantization class and the relation between occupation and dynamics are specified. The following theorem permits a real scalar deformation parameter rather than declaring either canonical commutation or anticommutation relations. Its phase-generator and stability hypotheses are additional physical inputs; a finite spin representation alone supplies neither.

Theorem B.6 (Stability selects exclusion within a scalar deformation class)**.** *Let $\mathcal{K}$ be a finite-dimensional complex one-particle Hilbert space. On a positive Hilbert space $\mathcal{F}$, let $v \mapsto a^\dagger(v)$ be complex-linear, with $a(v)$ its adjoint on a common dense invariant domain $\mathcal{D}$. Assume $\mathcal{D}$ is invariant under these operators and a self-adjoint Hamiltonian H, and is contained in their domains. Supply:*

1. *a common real number q such that, for every unit v,*

$$a(v)a^\dagger(v) - q\, a^\dagger(v)a(v) = I$$

   on $\mathcal{D}$;

2. *the exact occupation/phase law, for every unit v,*

$$N_v = a^\dagger(v)a(v), \qquad [N_v, a^\dagger(v)] = a^\dagger(v);$$

3. *a normalized $\Omega \in \mathcal{D}$ with $a(v)\Omega = 0$ for all v and $H\Omega = E_0\Omega$;*

4. *a unit mode v_- and $\kappa > 0$ such that*

$$[H, a^\dagger(v_-)] = -\kappa a^\dagger(v_-)$$

   on $\mathcal{D}$, and a finite uniform lower bound for H.

Then $q = -1$, and on $\mathcal{D}$,

$$\{a(v), a^\dagger(w)\} = \langle v, w\rangle I, \qquad \{a^\dagger(v), a^\dagger(w)\} = 0, \qquad \{a(v), a(w)\} = 0.$$

Here the inner product is conjugate-linear in its first argument. In particular $(a^\dagger(v))^2 = 0$ for every mode v. The operators extend boundedly to $\mathcal{F}$, where the same relations hold. The reference vacuum Ω is not assumed to be a ground state.

Proof. Fix a unit mode, abbreviate its operators to $a, a^\dagger, N$, and work throughout on the common invariant domain. The deformation law gives

$$Na^\dagger = a^\dagger + q\, a^\dagger N.$$

Comparison with the occupation/phase law yields $(q-1)a^\dagger N = 0$. The vector $|1\rangle = a^\dagger\Omega$ has norm one and $N|1\rangle = |1\rangle$. If $q \neq 1$, it follows that $a^\dagger|1\rangle = 0$. Taking its squared norm using the deformation law gives $0 = 1 + q$. Thus q is either 1 or -1.

If $q = 1$, induction using $aa^\dagger = I + a^\dagger a$ gives

$$|n\rangle = (a^\dagger)^n\Omega, \qquad \| |n\rangle\|^2 = n!, \qquad N|n\rangle = n|n\rangle$$

for every integer $n \geq 0$. Apply this to v_-. The Hamiltonian commutator and the vacuum eigenvalue imply $H|n\rangle = (E_0 - n\kappa)|n\rangle$. All these vectors are nonzero and belong to $\mathcal{D}$, contradicting the lower bound. This is an arbitrary-n argument, not a finite occupation scan. Hence $q = -1$.

For any unit mode, the phase identity then gives $a^\dagger N = 0$, whereas the deformation law gives $Na^\dagger = a^\dagger$. Consequently

$$0 = a^\dagger N a^\dagger = (a^\dagger)^2.$$

Linearity and scaling extend this square-zero identity to every v, including $v = 0$. Expanding it at $v + w$ proves creation anticommutation, and taking adjoints on the dense invariant domain

proves annihilation anticommutation. The diagonal relation is $\{a(v), a^\dagger(v)\} = \|v\|^2 I$ for every v. Polarization at $v + w$ and $v + iw$ gives the mixed relation. Finally, for $\psi \in \mathcal{D}$,

$$\|a(v)\psi\|^2 + \|a^\dagger(v)\psi\|^2 = \|v\|^2\|\psi\|^2.$$

Both operators therefore have bounded extensions, and density extends their identities to all of $\mathcal{F}$. □

Corollary B.7 (Finite-dimensional alternative to the stability hypothesis)**.** *Retain the theorem's common scalar deformation law, linear mode fields, exact occupation/phase law and normalized annihilation vacuum, with $\mathcal{K} \neq \{0\}$. If $\mathcal{F}$ itself is nonzero and finite-dimensional, the same exclusion and CAR conclusions hold without assuming a Hamiltonian or a negative ladder.*

Proof. A dense linear domain in finite dimension is the whole space. The vacuum argument gives $q \in \{1, -1\}$. If $q = 1$, the trace of $aa^\dagger - a^\dagger a = I$ is both zero and $\dim \mathcal{F} > 0$, a contradiction. The square-zero and polarization arguments for $q = -1$ require no Hamiltonian. □

This alternative requires finite dimension of the Hilbert space carrying the full creation and annihilation algebra. A finite one-particle mode space alone is insufficient. A finite observer response space does not by itself identify that operator representation with physical matter. Applying the corollary to observer finite capacity requires precisely this additional identification, as well as the stated deformation and phase laws.

The negative-mode hypothesis has a concrete one-particle instance in the supplied finite matter hopping action. For a unitary spin-link matrix K and $\kappa > 0$, its edge matrix is

$$h_e = \kappa \begin{pmatrix} 0 & K \\ K^\dagger & 0 \end{pmatrix}, \qquad h_e^2 = \kappa^2 I.$$

For any unit spin vector χ, the vector $v_- = (-K\chi, \chi)/\sqrt{2}$ is normalized and satisfies $h_e v_- = -\kappa v_-$. The quantum dynamical law $[H, a^\dagger(v)] = a^\dagger(h_e v)$ would therefore supply the required ladder. This law is a hypothesis linking the one-particle action to its quantization. A one-particle matrix, a measured unitary step, or the existence of a spin lift does not establish that law.

There is a finite nonvacuity witness. In ordinary fermionic Fock space over the edge modes, take the empty vacuum and $H = d\Gamma(h_e)$ with its normal ordering fixed at that vacuum. The negative-energy modes can each be occupied once, and the ground energy is the sum of all negative one-particle eigenvalues. Filling those modes gives the ground state; particles in positive modes and holes in filled negative modes both have excitation energy κ. For the two-spin edge with fifteen internal channels of the supplied matter census, there are thirty negative modes. Its ground energy is -30κ. With

$$(d_s) = (6, 3, 3, 2, 1), \qquad (q_s) = (1, -4, 2, -3, 6), \qquad \sum_s d_s q_s = 0,$$

the filled ground state has total hypercharge zero. Subtracting its constant energy gives a positive excitation Hamiltonian and preserves the field commutator. The reference empty vacuum and the filled ground state are different vectors.

For comparison, canonical bosonic Fock space on the same fixed edge action admits occupation n in one negative mode of every internal channel. These states have total hypercharge zero and energy $-15n\kappa$. Thus even the globally neutral sector has no uniform lower bound. Adding a chemical potential proportional to total hypercharge cannot repair this sequence. This neutrality

statement does not impose local quantum Gauss constraints or nonabelian singlet constraints, and is not a claim about a gauge-invariant physical Hilbert space.

The hypotheses exclude several concrete alternatives. At fixed total particle number the bosonic Hamiltonian is bounded below, so stability across all occupation sectors is essential. Replacing the one-particle law by $h_e + cI$ with $c \geq \kappa$ stabilizes its full bosonic Fock Hamiltonian: its second-quantized change is cN_{total}, not a constant vacuum-energy subtraction. Positive number interactions can also bound the energy, while changing the negative ladder. Neither change is ruled out by the theorem. A positive spectral replacement such as $|h_e|$ likewise changes the supplied dynamics.

Even the phase law and stability together do not force exclusion. On a three-dimensional Hilbert space with basis $|0\rangle, |1\rangle, |2\rangle$, set

$$a^\dagger|0\rangle = |1\rangle, \qquad a^\dagger|1\rangle = \sqrt{2}|2\rangle, \qquad a^\dagger|2\rangle = 0.$$

Then $N = a^\dagger a = \operatorname{diag}(0, 1, 2)$ satisfies $[N, a^\dagger] = a^\dagger$, and $H = -\kappa N$ is bounded below, but $(a^\dagger)^2|0\rangle \neq 0$. This model violates the scalar deformation hypothesis: $aa^\dagger = \operatorname{diag}(1, 2, 0)$, whose middle entry requires $q = 1$ while the top entry then contradicts $aa^\dagger - qN = I$. A hard occupation cutoff or a more general exchange algebra is therefore a genuine alternative outside the stated class.

The theorem is a conditional Pauli-exclusion derivation within a specified quantization class. It is insensitive to spin, so it cannot attach half-integer spin to fermionic statistics or exclude a different quantization class for a different field. The common scalar deformation law, linear field structure, exact occupation/phase law, unrestricted ladder domain and stable physical generator remain supplied. The finite filled-sea construction supplies no continuum vacuum or relativistic locality theorem. Deriving these inputs and selecting the physical matter action from observer consistency remain separate requirements.

B.3 What finite source operations must supply for exclusion

Finite-dimensional response, unitary covariance and a quadratic occupation generator do not alone imply the scalar deformation law used in the stability theorem. A normalized operational interpretation can replace that law, but requires an exact identification between a field operator and an observer operation. The following constructions distinguish these requirements.

Proposition B.8 (Globally covariant finite occupation cutoff)**.** *Let $\mathcal{K} = \mathbb{C}^m$ with $m \geq 1$, and*

$$\mathcal{F}_{\leq 2} = \mathbb{C} \oplus \mathcal{K} \oplus \operatorname{Sym}^2 \mathcal{K}.$$

This positive Hilbert space has dimension $(m+1)(m+2)/2$; the two-mode case has dimension six and the four-mode spin edge has dimension fifteen. Compress the usual bosonic creation and annihilation operators to this total-occupation cutoff, writing $C(v) = P_{\leq 2} b^\dagger(v) P_{\leq 2}$ and $A(v) = C(v)^\dagger$. Then C is linear in v, the normalized empty vacuum is annihilated by all $A(v)$, and for every unit v,

$$N_v = C(v)A(v), \qquad [N_v, C(v)] = C(v).$$

For every genuine unitary U on $\mathcal{K}$,

$$\Gamma_{\leq 2}(U) C(v) \Gamma_{\leq 2}(U)^\dagger = C(Uv).$$

For any Hermitian one-particle h, its restricted second quantization $H = d\Gamma(h)$ obeys $[H, C(v)] = C(hv)$ and is bounded below. Nevertheless, $C(v)^2\Omega \neq 0$ for every nonzero v, and no scalar q satisfies $A(v)C(v) - qC(v)A(v) = I$ even for one fixed unit mode.

Proof. Annihilation lowers total occupation, so $C(v)A(v)$ is the restriction of the ordinary bosonic number operator $d\Gamma(|v\rangle\langle v|)$. It preserves total-occupation sectors and commutes with the cutoff projection. The ordinary one-particle commutator identities therefore survive compression. At total occupation two the creation operator vanishes, so both sides of the claimed raising commutators vanish there as well. The cutoff also commutes with every $\Gamma(U)$ and with $d\Gamma(h)$, proving full unitary covariance and the dynamical identity. The finite spectrum gives

$$H \geq 2\min\{0, \lambda_{\min}(h)\}\, I.$$

The normalized one-mode vectors $|v\rangle$ and $|2v\rangle$ have number eigenvalues one and two, while $\|C(v)^2\Omega\|^2 = 2$. On $|v\rangle$ the proposed scalar law requires $2 - q = 1$, hence $q = 1$. On $|2v\rangle$ it instead requires $-2q = 1$. More generally its exact commutator defect is

$$A(v)C(v) - C(v)A(v) = I - (N_v + I)P_2,$$

where P_2 projects onto total occupation two. This state-dependent boundary term cannot be replaced by a scalar deformation parameter. □

For a supplied hopping matrix with eigenvalues $\pm\kappa$, the cutoff Hamiltonian has minimum -2κ and admits double occupation of its negative mode. Thus its covariance, exact quadratic phase law and stability coexist with failure of exclusion. The construction is a counterexample to this specified finite implication. It is not a model of the complete observer axioms: in particular a projective one-particle transport does not automatically lift to a projective transport on all occupation sectors, since $\Gamma(e^{i\theta}I) = e^{i\theta N_{\text{total}}}$ need not be scalar. No source production of the cutoff field operators is asserted.

Proposition B.9 (Normalized operations and an exact occupation generator)**.** *On a finite-dimensional positive Hilbert space, let C be a creation operator, $A = C^\dagger$, and $N = CA = A^\dagger A$. Assume*

$$0 \leq N \leq I, \qquad [N, C] = C.$$

Then $C^2 = 0$. Consequently, for a complex-linear family $v \mapsto C(v)$ satisfying these conditions for every unit v, one has $\{C(v), C(w)\} = 0$ for all v, w. No scalar deformation law is required.

Proof. Choose an orthonormal eigenbasis of N, with eigenvalues $\lambda_i \in [0, 1]$. The commutator identity gives

$$(\lambda_i - \lambda_j)C_{ij} = C_{ij}.$$

Every nonzero matrix entry therefore maps a zero-eigenvalue vector to a one-eigenvalue vector. A second application vanishes, so $C^2 = 0$. Scaling extends this conclusion from unit vectors to all modes, and expanding $C(v + w)^2 = 0$ gives the anticommutation identity. □

Under the same phase law the converse holds: if $C^2 = 0$, then $CC^\dagger C - C^2C^\dagger = C$ gives $CC^\dagger C = C$. Thus C is a partial isometry and a contraction. Within this phase-law class, contraction is equivalent to exclusion. Identifying an unscaled field operator with a normalized observer operation therefore carries the full selection content; it is not a harmless change of units.

This conclusion alone does not construct ordinary many-fermion Fock space. On $\mathbb{C}\Omega \oplus \mathcal{K}$, the operators $C(v) = |v\rangle\langle\Omega|$ satisfy the contraction and phase hypotheses for every unit v, but allow at most one particle in total. For $\dim \mathcal{K} \geq 2$, their mixed anticommutator is

$$\{C(v)^\dagger, C(w)\} = \langle v, w\rangle|\Omega\rangle\langle\Omega| + |w\rangle\langle v|,$$

which is not generally $\langle v, w\rangle I$. Additional structure is required to recover the full canonical field algebra.

For a linear family on m orthonormal modes, the exclusion test reduces to a finite set of exact operator identities. It suffices to check the contraction and unit phase conditions on every e_i and every $(3e_i + 4e_j)/5$ with $i < j$. The proposition gives $C(e_i)^2 = 0$; expanding the square at the pair vector then gives $\{C(e_i), C(e_j)\} = 0$. Linearity proves exclusion for every mode. To obtain full mixed CAR by the paired-instrument route, it suffices to check its completeness identity at the basis vectors, these real pairs, and the imaginary pairs $(3e_i + 4ie_j)/5$. The two pair identities force the symmetric and antisymmetric cross terms to vanish. These are exact matrix identities, not conclusions from finitely sampled outcome frequencies.

Basis-mode checks alone are insufficient. Set $\sigma_+ = |1\rangle\langle 0|$. Two commuting hard-core bit raisers $C_1 = \sigma_+ \otimes I$, $C_2 = I \otimes \sigma_+$ each satisfy the contraction and phase conditions, but $C = (3C_1 + 4C_2)/5$ has $C^2|00\rangle = (24/25)|11\rangle \neq 0$ and fails the unit phase law. Inserting the fermionic sign on the second raiser changes no occupation-basis transition probability. Coherent superposition tests, or source-derived identities that imply them, must therefore carry the relative-sign information.

The bound $N \leq I$ follows if the unscaled removal operator A is itself a Kraus operator of a trace-nonincreasing observer operation. Equivalently, A is a contraction; its adjoint C is a contraction too. This is an operational sufficient condition for exclusion only after the same operator's quadratic form $A^\dagger A$ is identified with the exact unit-step occupation generator. A normalized probability effect and a particle-number operator are not interchangeable merely because both are positive.

A stronger operational interface supplies the mixed relation as well. If a complete two-outcome instrument has exactly the adjoint-paired Kraus operators $A(v), C(v)$ for each unit v, trace preservation gives

$$C(v)A(v) + A(v)C(v) = I.$$

With the exact phase law and linearity, the preceding proposition and complex polarization recover all canonical anticommutation relations. The adjoint pairing, exhaustive two-outcome list and normalization for every input state are essential hypotheses; they are not consequences of probability normalization alone.

The cutoff example demonstrates the distinction. For a unit mode its operator $N_v + A(v)C(v)$ has eigenvalues at most three. Consequently

$$E_{\text{remove}} = \tfrac{1}{4}N_v, \qquad E_{\text{add}} = \tfrac{1}{4}A(v)C(v), \qquad E_{\text{idle}} = I - \tfrac{1}{4}(N_v + A(v)C(v))$$

are positive and sum to identity. Kraus operators $A(v)/2$, $C(v)/2$ and $\sqrt{E_{\text{idle}}}$ define a normalized instrument which retains the possibility of double occupation. Its removal effect is $N_v/4$, not the occupation generator N_v. In particular the phase commutator with that effect has coefficient $1/4$, not one. The extra idle outcome and the operator normalization cannot be suppressed in transferring an instrument to the exclusion theorem.

Projective readout has a different algebraic role. A Lüders outcome with Kraus operator $P = P^\dagger = P^2$ has $[PP^\dagger, P] = 0$, so a nonzero projector cannot satisfy the unit-step creation law. For the binary coordinate effects, an alternative pair $|1\rangle\langle 0|$, $|0\rangle\langle 1|$ is adjoint-paired, complete and satisfies the single-mode exclusion laws, but swaps the outcome-certain states rather than preserving them. The declared phase instrument and its swap-twisted comparison therefore illustrate different operations, despite carrying the same outcome effects. Their matrix-level semantics do not select a native particle-creation instrument. The distinction is explicit in `Lean/EventAlgebra/LuedersPhaseInstrument.lean` and `Lean/EventAlgebra/SourcePhaseInstrumentOutcomeBridge.lean`.

There is a separate obstruction in the finite history-count interface of `Lean/QFT/SourceHisto`

`ryGNSDynamics.lean`. Its eight three-bit histories carry the repair count

$$Q(x_0, x_1, x_2) = \mathbf{1}_{x_0 \neq x_1} + \mathbf{1}_{x_1 \neq x_2}.$$

The eigenspaces of its diagonal operator have dimensions two, four and two at eigenvalues zero, one and two respectively. No operator A on this full eight-dimensional carrier can obey both $Q = A^\dagger A$ and $[Q, A^\dagger] = A^\dagger$. Indeed the adjoint commutator sends A from the eigenvalue-one space into the eigenvalue-zero space. On the former space $A^\dagger A = Q$ would make A an isometry, which cannot map four dimensions into two. This excludes identifying that particular repair count with one quadratic mode occupation; it excludes neither multiple mode generators nor different source observables or larger carriers.

Counted transitions can supply a weaker positive statement. If a record basis has count labels n_x and an operator T has nonzero entry T_{yx} only when $n_y = n_x + 1$, then $[\mathrm{diag}(n_x), T] = T$ exactly, for arbitrary transition amplitudes. This follows entry by entry. It does not identify $\mathrm{diag}(n_x)$ with $TT^\dagger$: that additional equality constrains transition magnitudes, ranks and degeneracies. The native history packet fixes its diagonal counts and empirical state. Its extension to the complete matrix algebra does not authenticate these occupation-changing operations or their quantum instrument semantics.

Finite covariant response therefore does not force the scalar deformation law, while normalized operations can imply exclusion under an exact quadratic occupation/phase identification. The stated history-count interface supplies neither that identification nor an exhaustive adjoint-paired quantum instrument. These finite results leave open an observer-native construction of the required operations, a different quantization mechanism, and the physical spin–statistics attachment.

Credited prior work

The results of this paper build on the following established work. Each entry names an established result as it is used here and cites its sources.

- A5 representation theory, compact Lie classification, covering groups, exterior unification, and signed graph operators [127, 99, 92, 157].
- Ad-invariant quadratic kinetic forms on reductive Lie algebras and gauge kinetic terms [152].
- Algebraic renormalization and BV/BRST cohomology [144, 95, 94, 124].
- Analytic S-matrix and resonance-pole theory [112].
- Complex-pole renormalization, Nielsen identities, FJ tadpole schemes, and finite-order electroweak calculations [147, 140, 116, 113, 107].
- DHR/DR/Tannaka reconstruction, compact-group representation theory, and Standard Model anomaly algebra [109, 110, 111, 148, 99, 96, 120].
- Finite fermion Fock space and lattice gauge current identities [98, 130].
- Finite lattice connections, Wilson-loop holonomy, vertex recharting, algebraic two-path character phases (an Aharonov–Bohm analogy only), and characters of cyclic groups [155, 91, 145].
- Finite lattice gauge theory and the Standard Model action [155, 138].
- Finite spectral theory, Kraus instruments and truncated symmetric Fock spaces [122, 132, 104, 98].

- Finite-dimensional Hamiltonian ODE enclosures and nonlinear detector Taylor bounds [139, 123].
- Finite-group equivariant current algebras, compact Lie algebras, normed approximation, and exact algebraic representation certificates [115, 129, 97].
- Finite-group heat kernels, character projectors, and lattice-gauge toy models [108, 145, 131].
- Ginsparg-Wilson fermions and determinant-line formulations [118, 136].
- Hamiltonian lattice gauge theory and constrained quantization [130, 124].
- Icosahedral Galois conjugacy and classical topological degree [102, 137].
- Icosahedral graph theory, finite group actions, equiangular tight frames, and universal optimality [100, 134, 101].
- Invariant quadratic forms on compact semisimple Lie algebras and finite Gibbs product laws [129, 93].
- Osterwalder-Schrader reconstruction [141, 142].
- Product gauge algebras and simple-GUT X/Y generators [117, 146].
- Scalar q oscillators, canonical Fock quantization and the finite trace obstruction [119, 153, 98].
- Standard Model fermion bilinears with charge-diagonal structure [143, 133].
- SU(5)-style exterior matter embeddings and Standard Model anomaly algebra [117, 92, 96, 120].
- Super-Tannakian reconstruction, exterior-algebra matter embeddings, binary icosahedral covers, and anomaly algebra [106, 92, 103, 96].
- Symmetric and exterior Fock representations [114, 98].
- Tannaka duality, representation equivalence, and commuting-diagram methods [105, 128, 110].
- Textbook electroweak Lagrangian assembly of gauge kinetic, Higgs potential, gauge mass, and Yukawa sectors [150, 143, 156].
- two-particle bunching and antibunching [125, 135].
- Weinberg and Wilczek-Zee dimension-six baryon-number-violating operator classification and the SMEFT operator basis [151, 154, 121].
- Z6 center-character descent bookkeeping for Standard Model global gauge-group forms [149, 126].

References

[1] L. Bombelli, J. Lee, D. Meyer, and R. D. Sorkin, *Space-time as a causal set*, Phys. Rev. Lett. **59**, 521–524 (1987). https://doi.org/10.1103/PhysRevLett.59.521.

[2] S. Surya, *The causal set approach to quantum gravity*, Living Rev. Relativ. **22**, 5 (2019). https://doi.org/10.1007/s41114-019-0023-1.

[3] L. Bombelli and D. A. Meyer, *The origin of Lorentzian geometry*, Phys. Lett. A **141**, 226–228 (1989). https://doi.org/10.1016/0375-9601(89)90474-X.

[4] G. Brightwell and R. Gregory, *Structure of random discrete spacetime*, Phys. Rev. Lett. **66**, 260–263 (1991). https://doi.org/10.1103/PhysRevLett.66.260.

[5] S. Major, D. Rideout, and S. Surya, *On recovering continuum topology from a causal set*, J. Math. Phys. **48**, 032501 (2007). https://doi.org/10.1063/1.2435599.

[6] S. Major, D. Rideout, and S. Surya, *Stable homology as an indicator of manifoldlikeness in causal set theory*, Class. Quantum Grav. **26**, 175008 (2009). https://doi.org/10.1088/0264-9381/26/17/175008.

[7] L. Glaser and S. Surya, *Towards a definition of locality in a manifoldlike causal set*, Phys. Rev. D **88**, 124026 (2013). https://doi.org/10.1103/PhysRevD.88.124026.

[8] J. Myrheim, *Statistical geometry*, CERN preprint TH-2538 (1978).

[9] D. A. Meyer, *The Dimension of Causal Sets*, PhD thesis, Massachusetts Institute of Technology (1988).

[10] D. D. Reid, *Manifold dimension of a causal set: Tests in conformally flat spacetimes*, Phys. Rev. D **67**, 024034 (2003). https://doi.org/10.1103/PhysRevD.67.024034.

[11] D. M. T. Benincasa and F. Dowker, *The scalar curvature of a causal set*, Phys. Rev. Lett. **104**, 181301 (2010). https://doi.org/10.1103/PhysRevLett.104.181301.

[12] F. Dowker and L. Glaser, *Causal set d'Alembertians for various dimensions*, Class. Quantum Grav. **30**, 195016 (2013). https://doi.org/10.1088/0264-9381/30/19/195016.

[13] L. Bombelli, J. Henson, and R. D. Sorkin, *Discreteness without symmetry breaking: A theorem*, Mod. Phys. Lett. A **24**, 2579–2587 (2009). https://doi.org/10.1142/S0217732309031958.

[14] S. W. Hawking, A. R. King, and P. J. McCarthy, *A new topology for curved space–time which incorporates the causal, differential, and conformal structures*, J. Math. Phys. **17**, 174–181 (1976). https://doi.org/10.1063/1.522874.

[15] D. B. Malament, *The class of continuous timelike curves determines the topology of spacetime*, J. Math. Phys. **18**, 1399–1404 (1977). https://doi.org/10.1063/1.523436.

[16] B. Müller, J. Kim, D. Matscheko, and J. Hill, *Observation-Determined Normal Forms: Stability, Obstructions, and Refinement in Constraint and Rewrite Systems*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2051/extra/observable_normal_forms.pdf.

[17] B. Müller, M. Poneder, A. Osika, K. Xue, P. Nguyen, M. A. Visser, D. Matscheko, and B. de La Fournière, *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency*, 2026. Available at GitHub PDF.

[18] B. Müller, M. Poneder, A. Osika, K. Xue, P. Nguyen, and D. Matscheko, *Deriving Standard Model Gauge Structure from Observer Overlap Consistency*, 2026. Available at GitHub PDF.

[19] B. Müller, K. Xue, J. Kim, K. A. Anirudha, D. Matscheko, and J. Hill, *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2051/paper/reality_as_consensus_protocol.pdf.

[20] B. Müller, A. Osika, K. Xue, B. Cassie, and B. de La Fournière, *Federated Echosahedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in Observer-Patch Holography*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2051/paper/screen_microphysics_and_observer_synchronization.pdf.

[21] B. Müller, M. Poneder, A. Osika, and K. Xue, *Deriving the Particle Zoo from Observer Consistency*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2051/paper/deriving_the_particle_zoo_from_observer_consistency.pdf.

[22] B. Müller and D. Matscheko, *Observer-Patch Holography and the Dark Sector: Modular Charge, the Anomalous Collar Source, and the Deep Galaxy Law*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2051/cosmology/oph_dark_matter_paper.pdf.

[23] B. Müller and J. Hill, *Explaining the Yang–Mills Mass Gap with Observer-Patch Repair Dynamics: A Support-Visible OPH Route to the Clay Problem: A Conditional Identity*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2051/extra/yang_mills_gap_clay_problem.pdf.

[24] B. Müller, *Observer-Patch Holography as a String-Vacuum Selector: Observers, Clocks, Edge Strings, and the Bouchard-Donagi Witness*, 2026. Archived manuscript. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2051/paper/archive/observer_patch_holography_as_string_vacuum_selector.pdf.

[25] B. Müller, M. Poneder, A. Osika, K. Xue, B. Cassie, P. Nguyen, J. Kim, D. Matscheko, J. Hill, W. T. Glynn, M. A. Visser, K. A. Anirudha, and B. de La Fournière, *Observers Who Read Their Universe into Existence: Finite Consensus, an Executable Observer, and Standard Physics*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2051/flagship/from_observer_consensus_to_standard_physics.pdf.

[26] B. Müller, M. Poneder, and J. Kim, *An Executable Observer for Observer Patch Holography: Specification, Certified Components and Acceptance Criteria*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2051/paper/executable_observer_construction.pdf.

[27] B. Müller, *The de Sitter Time-Advance Sign from a Finite Screen with Fixed Capacity*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2051/extra/de_sitter_time_advance_sign_from_fixed_screen_capacity.pdf.

[28] B. Müller, *OPH-FPE: finite simulator and receipt engine for Observer-Patch Holography physics experiments*, 2026. Source available at https://github.com/muellerberndt/oph-physics-sim. Large-run source revision: https://github.com/muellerberndt/oph-physics-sim/tree/ce17921eb7504106fef1ba445e1349b5367aa676. Exact-federation archive: https://github.com/FloatingPragma/observer-patch-holography/tree/r2051/evidence/exact_federation_L6_canonical_20260909. Closure-loop archive: https://github.com/FloatingPragma/observer-patch-holography/tree/r2051/evidence/closure_loop. Active refinement-readout revision: https://github.com/muellerberndt/oph-physics-sim/tree/036b12b608a9ca4c19d69f5e951b3b83a7a51cd8. Finite local-domain revision and data: https://github.com/muellerberndt/oph-physics-sim/tree/550fe77dc67dc8bef2dd8927bcf72ec98cfa3506, https://github.com/muellerberndt/oph-physics

-sim/tree/550fe77dc67dc8bef2dd8927bcf72ec98cfa3506/data/local_domain. Exact repository-local archive and checker: https://github.com/FloatingPragma/observer-patch-holography/tree/r2051/evidence/local_domain. Causal-poset evidence package: https://github.com/FloatingPragma/observer-patch-holography/tree/r2051/evidence/source_net_causal_poset. Closure-loop evidence package: https://github.com/FloatingPragma/observer-patch-holography/tree/r2051/evidence/closure_loop. Evidence index: https://github.com/FloatingPragma/observer-patch-holography/tree/r2051/evidence.

[29] Pragma Research, *Observer-Patch Holography: machine-checked proofs and executable certificates*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography.

[30] Pragma Research, *Frozen Prediction Register and Primitive Twelve-Port Prediction Receipt*, 2026. Immutable source receipt: https://github.com/FloatingPragma/observer-patch-holography/blob/66176656dc1143f9ec50ba1a6e409c403545857f/code/a5_fingerprint/runtime/spin_six_primitive_port_prediction_receipt.json. Append-only custody packet: https://github.com/FloatingPragma/observer-patch-holography/tree/bb73f80d459113ccc82d5a5b5219d1dda78e9a79/evidence/custody/falsification/frozen_targets/fz11_2026-07-31.

[31] DESI Collaboration, "DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints," 2025, arXiv:2503.14738. https://arxiv.org/abs/2503.14738. Official chains and checksum manifest: https://data.desi.lbl.gov/public/papers/y3/bao-cosmo-params/README.html.

[32] M. Filzinger et al., "Improved Limits on the Coupling of Ultralight Bosonic Dark Matter to Photons from Optical Atomic Clock Comparisons," *Physical Review Letters* **130**, 253001 (2023). https://doi.org/10.1103/PhysRevLett.130.253001.

[33] F. Lelli, S. S. McGaugh, and J. M. Schombert, "SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves," *Astronomical Journal* **152**, 157 (2016), arXiv:1606.09251. https://arxiv.org/abs/1606.09251. CDS/VizieR catalogue: https://cdsarc.cds.unistra.fr/ftp/J/AJ/152/157/.

[34] S. S. McGaugh, F. Lelli, and J. M. Schombert, "Radial Acceleration Relation in Rotationally Supported Galaxies," *Physical Review Letters* **117**, 201101 (2016), arXiv:1609.05917. https://arxiv.org/abs/1609.05917.

[35] L. Onsager, "Reciprocal Relations in Irreversible Processes. I," *Physical Review* **37**, 405–426 (1931), https://doi.org/10.1103/PhysRev.37.405.

[36] M. S. Green, "Markoff Random Processes and the Statistical Mechanics of Time-Dependent Phenomena. II," *Journal of Chemical Physics* **22**, 398–413 (1954), https://doi.org/10.1063/1.1740082; R. Kubo, "Statistical-Mechanical Theory of Irreversible Processes. I," *Journal of the Physical Society of Japan* **12**, 570–586 (1957), https://doi.org/10.1143/JPSJ.12.570.

[37] I. Csiszár, "I-Divergence Geometry of Probability Distributions and Minimization Problems," *The Annals of Probability* **3**, 146–158 (1975), https://doi.org/10.1214/aop/1176996454.

[38] J. E. Marsden and M. West, "Discrete Mechanics and Variational Integrators," *Acta Numerica* **10**, 357–514 (2001), https://doi.org/10.1017/S096249290100006X.

[39] I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, I. Martinez-Soler, J. P. Pinheiro, and T. Schwetz, *NuFit-6.0: Updated global analysis of three-flavor neutrino oscillations*, JHEP **12** (2024) 216, arXiv:2410.05380, with the NuFIT 6.1 (2025) profile-table release at https://www.nu-fit.org/?q=node/309.

[40] M. H. A. Newman, "On theories with a combinatorial definition of 'equivalence'," *Ann. of Math. (2)* **43** (1942), 223–243.

[41] E. H. Lieb and D. W. Robinson, "The finite group velocity of quantum spin systems," *Commun. Math. Phys.* **28** (1972), 251–257.

[42] J. J. Bisognano and E. H. Wichmann, "On the duality condition for a Hermitian scalar field," *J. Math. Phys.* **16** (1975), 985–1007.

[43] J. J. Bisognano and E. H. Wichmann, "On the duality condition for quantum fields," *J. Math. Phys.* **17** (1976), 303–321.

[44] R. Brunetti, D. Guido, and R. Longo, "Modular Structure and Duality in Conformal Quantum Field Theory," *Commun. Math. Phys.* **156** (1993), 201–219, arXiv:funct-an/9302008.

[45] H.-W. Wiesbrock, "Half-Sided modular inclusions of von-Neumann-Algebras," *Commun. Math. Phys.* **157** (1993), 83–92.

[46] H. Araki and L. Zsidó, "Extension of the structure theorem of Borchers and its application to half-sided modular inclusions," *Rev. Math. Phys.* **17** (2005), 491–543, arXiv:math/0412061.

[47] D. Guido, R. Longo, and H.-W. Wiesbrock, "Extensions of Conformal Nets and Superselection Structures," *Commun. Math. Phys.* **192** (1998), 217–244, arXiv:hep-th/9703129.

[48] H.-W. Wiesbrock, "Modular Intersections of von-Neumann-Algebras in Quantum Field Theory," *Commun. Math. Phys.* **193** (1998), 269–285.

[49] W. G. Unruh, "Notes on black-hole evaporation," *Phys. Rev. D* **14** (1976), 870–892.

[50] T. Jacobson, "Thermodynamics of spacetime: The Einstein equation of state," *Phys. Rev. Lett.* **75** (1995), 1260–1263, doi:10.1103/PhysRevLett.75.1260, arXiv:gr-qc/9504004.

[51] T. Jacobson, "Entanglement equilibrium and the Einstein equation," *Phys. Rev. Lett.* **116** (2016), 201101, doi:10.1103/PhysRevLett.116.201101, arXiv:1505.04753.

[52] G. W. Gibbons and S. W. Hawking, "Cosmological event horizons, thermodynamics, and particle creation," *Phys. Rev. D* **15** (1977), 2738–2751, doi:10.1103/PhysRevD.15.2738.

[53] Y. Chen, D. Stanford, H. Tang, and Z. Yang, "Negative shocks versus static patch holography," arXiv:2607.14042 [hep-th] (2026). https://arxiv.org/abs/2607.14042

[54] Planck Collaboration, "Planck 2018 results. VI. Cosmological parameters," *Astron. Astrophys.* **641** (2020), A6, doi:10.1051/0004-6361/201833910, arXiv:1807.06209.

[55] R. Bousso, Z. Fisher, J. Koeller, S. Leichenauer, and A. C. Wall, "Proof of the quantum null energy condition," *Phys. Rev. D* **93** (2016), 024017, arXiv:1509.02542.

[56] R. Bousso, Z. Fisher, S. Leichenauer, and A. C. Wall, "Quantum focusing conjecture," *Phys. Rev. D* **93** (2016), 064044, arXiv:1506.02669.

[57] S. Balakrishnan, T. Faulkner, Z. U. Khandker, and H. Wang, "A general proof of the quantum null energy condition," *J. High Energy Phys.* **2019** (2019), 20, arXiv:1706.09432.

[58] E. H. Lieb and M. B. Ruskai, "Proof of the strong subadditivity of quantum-mechanical entropy," *J. Math. Phys.* **14** (1973), 1938–1941.

[59] E. H. Wichmann, "Density matrices arising from incomplete measurements," *J. Math. Phys.* **4** (1963), 884–896.

[60] F. Hiai, M. Ohya, and M. Tsukada, "Sufficiency, KMS condition and relative entropy in von Neumann algebras," *Pacific J. Math.* **96** (1981), 99–109.

[61] M. Takesaki, "Conditional expectations in von Neumann algebras," *J. Funct. Anal.* **9** (1972), 306–321.

[62] D. Petz, "Sufficient subalgebras and the relative entropy of states of a von Neumann algebra," *Commun. Math. Phys.* **105** (1986), 123–131.

[63] D. Petz, "Sufficiency of channels over von Neumann algebras," *Quart. J. Math.* **39** (1988), 97–108.

[64] O. Fawzi and R. Renner, "Quantum conditional mutual information and approximate Markov chains," *Commun. Math. Phys.* **340** (2015), 575–611, arXiv:1410.0664.

[65] P. Hayden, R. Jozsa, D. Petz, and A. Winter, "Structure of states which satisfy strong subadditivity of quantum entropy with equality," *Commun. Math. Phys.* **246** (2004), 359–374.

[66] S. Doplicher and J. E. Roberts, "A new duality theory for compact groups," *Invent. Math.* **98** (1989), 157–218.

[67] S. Doplicher and J. E. Roberts, "Why there is a field algebra with a compact gauge group describing the superselection structure in particle physics," *Commun. Math. Phys.* **131** (1990), 51–107.

[68] T. Tannaka, "Über den Dualitätssatz der nichtkommutativen topologischen Gruppen," *Tohoku Math. J.* **45** (1938), 1–12.

[69] M. G. Krein, "A principle of duality for a bicompact group and a square block algebra," *Dokl. Akad. Nauk SSSR* **69** (1949), 725–728.

[70] H. Georgi and S. L. Glashow, "Unity of all elementary-particle forces," *Phys. Rev. Lett.* **32** (1974), 438–441.

[71] S. L. Glashow, J. Iliopoulos, and L. Maiani, "Weak interactions with lepton-hadron symmetry," *Phys. Rev. D* **2** (1970), 1285–1292.

[72] E. Witten, "An SU(2) anomaly," *Phys. Lett. B* **117** (1982), 324–328.

[73] S. Dimopoulos, S. Raby, and F. Wilczek, "Supersymmetry and the scale of unification," *Phys. Rev. D* **24** (1981), 1681–1683.

[74] U. Amaldi, W. de Boer, and H. Fürstenau, "Comparison of grand unified theories with electroweak and strong coupling constants measured at LEP," *Phys. Lett. B* **260** (1991), 447–455.

[75] D. J. Gross and W. Taylor, "Two-dimensional QCD is a string theory," *Nucl. Phys. B* **400** (1993), 181–208, arXiv:hep-th/9301068.

[76] F. Peter and H. Weyl, "Die Vollständigkeit der primitiven Darstellungen einer geschlossenen kontinuierlichen Gruppe," *Math. Ann.* **97** (1927), 737–755.

[77] A. W. Knapp, *Lie Groups Beyond an Introduction*, second edition, Progress in Mathematics **140**, Birkhäuser, 2002.

[78] A. Bullivant, M. Calçada, Z. Kádár, P. Martin, and J. Faria Martins, "Topological phases from higher gauge symmetry in 3+1D," *Phys. Rev. B* **95** (2017), 155118, arXiv:1606.06639.

[79] S. Chandrasekharan and U.-J. Wiese, "Quantum link models: A discrete approach to gauge theories," *Nucl. Phys. B* **492** (1997), 455–471, arXiv:hep-lat/9609042.

[80] W. Donnelly and A. C. Wall, "Entanglement entropy of electromagnetic edge modes," *Phys. Rev. Lett.* **114** (2015), 111603, arXiv:1412.1895.

[81] F. Pastawski, B. Yoshida, D. Harlow, and J. Preskill, "Holographic quantum error-correcting codes: Toy models for the bulk/boundary correspondence," *JHEP* **06** (2015), 149, arXiv:1503.06237.

[82] M. A. Levin and X.-G. Wen, "String-net condensation: A physical mechanism for topological phases," *Phys. Rev. B* **71** (2005), 045110, arXiv:cond-mat/0404617.

[83] D. Laghi, G. Carullo, J. Veitch, and W. Del Pozzo, "Quantum black hole spectroscopy: probing the quantum nature of the black hole area using LIGO-Virgo ringdown detections," *Class. Quantum Grav.* **38** (2021), 095005, arXiv:2011.03816.

[84] Clay Mathematics Institute, "Yang–Mills & the Mass Gap." Available at https://www.claymath.org/millennium/yang-mills-the-maths-gap/.

[85] A. Jaffe and E. Witten, "Quantum Yang–Mills Theory," official Clay Mathematics Institute problem description. Available at https://www.claymath.org/wp-content/uploads/2022/06/yangmills.pdf.

[86] L. Wu, "Poincaré and transportation inequalities for Gibbs measures under the Dobrushin uniqueness condition," *Annals of Probability* **34** (2006), 1960–1989, arXiv:math/0611635.

[87] K. Osterwalder and R. Schrader, "Axioms for Euclidean Green's functions," *Communications in Mathematical Physics* **31** (1973), 83–112.

[88] K. Osterwalder and R. Schrader, "Axioms for Euclidean Green's functions II," *Communications in Mathematical Physics* **42** (1975), 281–305.

[89] Particle Data Group, "Review of Particle Physics," *Phys. Rev. D* **110** (2024), 030001. Available at https://pdg.lbl.gov/2024/download/db2024.pdf.

[90] Particle Data Group, "Electroweak Model and Constraints on New Physics," Available at https://pdg.lbl.gov/2024/reviews/rpp2024-rev-standard-model.pdf.

[91] Aharonov, Y., and Bohm, D. (1959). "Significance of electromagnetic potentials in the quantum theory." *Physical Review* **115**, 485–491. https://doi.org/10.1103/PhysRev.115.485

[92] Baez, J., and Huerta, J. (2010). "The algebra of grand unified theories." *Bulletin of the American Mathematical Society* **47**, 483–552. https://doi.org/10.1090/S0273-0979-10-01294-2

[93] Barndorff-Nielsen, O. (2014). *Information and Exponential Families: In Statistical Theory*. Wiley (Wiley Series in Probability and Statistics). https://doi.org/10.1002/9781118857281

[94] Barnich, G., Brandt, F., and Henneaux, M. (2000). "Local BRST cohomology in gauge theories." *Physics Reports* **338**, 439–569. https://doi.org/10.1016/S0370-1573(00)00049-1

[95] Becchi, C., Rouet, A., and Stora, R. (1976). "Renormalization of gauge theories." *Annals of Physics* **98**, 287–321. https://doi.org/10.1016/0003-4916(76)90156-1

[96] Bouchiat, C., Iliopoulos, J., and Meyer, P. (1972). "An anomaly-free version of Weinberg's model." *Physics Letters B* **38**, 519–523. https://doi.org/10.1016/0370-2693(72)90532-1

[97] Boyd, S., and Vandenberghe, L. (2004). *Convex Optimization*. Cambridge University Press. https://doi.org/10.1017/CBO9780511804441

[98] Bratteli, O., and Robinson, D. W. (1997). *Operator Algebras and Quantum Statistical Mechanics 2*. Springer. https://doi.org/10.1007/978-3-662-03444-6

[99] Bröcker, T., and tom Dieck, T. (1985). *Representations of Compact Lie Groups*. Springer. https://doi.org/10.1007/978-3-662-12918-0

[100] Brouwer, A. E., Cohen, A. M., and Neumaier, A. (1989). *Distance-Regular Graphs*. Springer, Berlin. https://doi.org/10.1007/978-3-642-74341-2

[101] Cohn, H., and Kumar, A. (2007). "Universally optimal distribution of points on spheres." *Journal of the American Mathematical Society* **20**, 99–148. https://doi.org/10.1090/S0894-0347-06-00546-7

[102] Coxeter, H. S. M. (1973). *Regular Polytopes*. Dover. ISBN 0-486-61480-8.

[103] Coxeter, H. S. M., and Moser, W. O. J. (1980). *Generators and Relations for Discrete Groups*. Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-662-21943-0

[104] Davies, E. B., and Lewis, J. T. (1970). "An operational approach to quantum probability." *Communications in Mathematical Physics* **17**, 239–260. https://doi.org/10.1007/BF01647093

[105] Deligne, P., and Milne, J. S. (1982). "Tannakian Categories." In *Hodge Cycles, Motives, and Shimura Varieties*, 101–228. Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-540-38955-2_4

[106] Deligne, P. (2002). "Catégories tensorielles." *Moscow Mathematical Journal* **2**, 227–248. https://doi.org/10.17323/1609-4514-2002-2-2-227-248

[107] Denner, A. (1993). "Techniques for the calculation of electroweak radiative corrections at the one-loop level and results for W-physics at LEP 200." *Fortschritte der Physik* **41**, 307–420. https://doi.org/10.1002/prop.2190410402

[108] Diaconis, P. (1988). *Group Representations in Probability and Statistics*. Institute of Mathematical Statistics (Lecture Notes–Monograph Series). https://doi.org/10.1214/lnms/1215467407

[109] Doplicher, S., Haag, R., and Roberts, J. E. (1971). "Local observables and particle statistics I." *Communications in Mathematical Physics* **23**, 199–230. https://doi.org/10.1007/BF01877742

[110] Doplicher, S., and Roberts, J. E. (1989). "A new duality theory for compact groups." *Inventiones Mathematicae* **98**, 157–218. https://doi.org/10.1007/BF01388849

[111] Doplicher, S., and Roberts, J. E. (1990). "Why there is a field algebra with a compact gauge group describing the superselection structure in particle physics." *Communications in Mathematical Physics* **131**, 51–107. https://doi.org/10.1007/BF02097680

[112] Eden, R. J., Landshoff, P. V., Olive, D. I., and Polkinghorne, J. C. (1966). *The Analytic S-Matrix*. Cambridge University Press. ISBN 0-521-04869-9.

[113] Fleischer, J., and Jegerlehner, F. (1981). "Radiative corrections to Higgs-boson decays in the Weinberg-Salam model." *Physical Review D* **23**, 2001–2026. https://doi.org/10.1103/PhysRevD.23.2001

[114] Fock, V. (1932). "Konfigurationsraum und zweite Quantelung." *Zeitschrift für Physik* **75**, 622–647. https://doi.org/10.1007/BF01344458

[115] Fulton, W., and Harris, J. (1991). *Representation Theory: A First Course*. Springer, New York (Graduate Texts in Mathematics). https://doi.org/10.1007/978-1-4612-0979-9

[116] Gambino, P., and Grassi, P. A. (2000). "Nielsen identities of the SM and the definition of mass." *Physical Review D* **62**, 076002. https://doi.org/10.1103/PhysRevD.62.076002

[117] Georgi, H., and Glashow, S. L. (1974). "Unity of All Elementary-Particle Forces." *Physical Review Letters* **32**, 438–441. https://doi.org/10.1103/PhysRevLett.32.438

[118] Ginsparg, P. H., and Wilson, K. G. (1982). "A remnant of chiral symmetry on the lattice." *Physical Review D* **25**, 2649–2657. https://doi.org/10.1103/PhysRevD.25.2649

[119] Greenberg, O. W. (1991). "Particles with small violations of Fermi or Bose statistics." *Physical Review D* **43**, 4111–4120. https://doi.org/10.1103/PhysRevD.43.4111

[120] Gross, D. J., and Jackiw, R. (1972). "Effect of Anomalies on Quasi-Renormalizable Theories." *Physical Review D* **6**, 477–493. https://doi.org/10.1103/PhysRevD.6.477

[121] Grzadkowski, B., Iskrzyński, M., Misiak, M., and Rosiek, J. (2010). "Dimension-six terms in the Standard Model Lagrangian." *Journal of High Energy Physics* **2010**, 85. https://doi.org/10.1007/JHEP10(2010)085

[122] Halmos, P. R. (1958). *Finite-Dimensional Vector Spaces*. Van Nostrand. https://doi.org/10.1007/978-1-4612-6387-6

[123] Hartman, P. (2002). *Ordinary Differential Equations*. Society for Industrial and Applied Mathematics (Classics in Applied Mathematics). https://doi.org/10.1137/1.9780898719222

[124] Henneaux, M., and Teitelboim, C. (1992). *Quantization of Gauge Systems*. Princeton University Press. https://doi.org/10.1515/9780691213866

[125] Hong, C. K., Ou, Z. Y., and Mandel, L. (1987). "Measurement of subpicosecond time intervals between two photons by interference." *Physical Review Letters* **59**, 2044–2046. https://doi.org/10.1103/PhysRevLett.59.2044

[126] Hucks, J. (1991). "Global structure of the standard model, anomalies, and charge quantization." *Physical Review D* **43**, 2709–2717. https://doi.org/10.1103/PhysRevD.43.2709

[127] James, G., and Liebeck, M. (2001). *Representations and Characters of Groups*. Cambridge University Press. https://doi.org/10.1017/CBO9780511814532

[128] Joyal, A., and Street, R. (1991). "An introduction to Tannaka duality and quantum groups." In *Category Theory (Como, 1990)*, 413–492. Springer Berlin Heidelberg. https://doi.org/10.1007/BFb0084235

[129] Knapp, A. W. (1996). *Lie Groups Beyond an Introduction*. Birkhäuser, Boston (Progress in Mathematics). https://doi.org/10.1007/978-1-4757-2453-0

[130] Kogut, J., and Susskind, L. (1975). "Hamiltonian formulation of Wilson's lattice gauge theories." *Physical Review D* **11**, 395–408. https://doi.org/10.1103/PhysRevD.11.395

[131] Kogut, J. B. (1979). "An introduction to lattice gauge theory and spin systems." *Reviews of Modern Physics* **51**, 659–713. https://doi.org/10.1103/RevModPhys.51.659

[132] Kraus, K. (1983). *States, Effects, and Operations: Fundamental Notions of Quantum Theory*. Springer. https://doi.org/10.1007/3-540-12732-1

[133] Langacker, P. (2017). *The Standard Model and Beyond*. CRC Press. https://doi.org/10.1201/b22175

[134] Lemmens, P. W. H., and Seidel, J. J. (1973). "Equiangular lines." *Journal of Algebra* **24**, 494–512. https://doi.org/10.1016/0021-8693(73)90123-3

[135] Loudon, R. (1998). "Fermion and boson beam-splitter statistics." *Physical Review A* **58**, 4904–4909. https://doi.org/10.1103/PhysRevA.58.4904

[136] Lüscher, M. (1999). "Abelian chiral gauge theories on the lattice with exact gauge invariance." *Nuclear Physics B* **549**, 295–334. https://doi.org/10.1016/S0550-3213(99)00115-7

[137] Milnor, J. W. (1997). *Topology from the Differentiable Viewpoint*. Princeton University Press (Princeton Landmarks in Mathematics). ISBN 0-691-04833-9.

[138] Montvay, I., and Münster, G. (1994). *Quantum Fields on a Lattice*. Cambridge University Press. https://doi.org/10.1017/CBO9780511470783

[139] Moore, R. E., Kearfott, R. B., and Cloud, M. J. (2009). *Introduction to Interval Analysis*. SIAM. https://doi.org/10.1137/1.9780898717716

[140] Nielsen, N. K. (1975). "On the gauge dependence of spontaneous symmetry breaking in gauge theories." *Nuclear Physics B* **101**, 173–188. https://doi.org/10.1016/0550-3213(75)90301-6

[141] Osterwalder, K., and Schrader, R. (1973). "Axioms for Euclidean Green's functions." *Communications in Mathematical Physics* **31**, 83–112. https://doi.org/10.1007/BF01645738

[142] Osterwalder, K., and Schrader, R. (1975). "Axioms for Euclidean Green's functions II." *Communications in Mathematical Physics* **42**, 281–305. https://doi.org/10.1007/BF01608978

[143] Peskin, M. E., and Schroeder, D. V. (1995). *An Introduction to Quantum Field Theory*. Addison-Wesley. https://doi.org/10.1201/9780429503559

[144] Piguet, O., and Sorella, S. P. (1995). *Algebraic Renormalization: Perturbative Renormalization, Symmetries and Anomalies*. Springer. https://doi.org/10.1007/978-3-540-49192-7

[145] Serre, J. P. (1977). *Linear Representations of Finite Groups*. Springer. https://doi.org/10.1007/978-1-4684-9458-7

[146] Slansky, R. (1981). "Group theory for unified model building." *Physics Reports* **79**, 1–128. https://doi.org/10.1016/0370-1573(81)90092-2

[147] Stuart, R. G. (1991). "Gauge invariance, analyticity and physical observables at the Z0 resonance." *Physics Letters B* **262**, 113–119. https://doi.org/10.1016/0370-2693(91)90653-8

[148] Tannaka, T. (1939). "Über den Dualitätssatz der nichtkommutativen topologischen Gruppen." *Tohoku Mathematical Journal, First Series* **45**, 1–12. https://www.jstage.jst.go.jp/article/tmj1911/45/0/45_0_1/_article

[149] Tong, D. (2017). "Line operators in the Standard Model." *Journal of High Energy Physics* **2017**, 104. https://doi.org/10.1007/JHEP07(2017)104

[150] Weinberg, S. (1967). "A model of leptons." *Physical Review Letters* **19**, 1264–1266. https://doi.org/10.1103/PhysRevLett.19.1264

[151] Weinberg, S. (1979). "Baryon- and Lepton-Nonconserving Processes." *Physical Review Letters* **43**, 1566–1570. https://doi.org/10.1103/PhysRevLett.43.1566

[152] Weinberg, S. (1996). *The Quantum Theory of Fields, Volume II*. Cambridge University Press. https://doi.org/10.1017/CBO9781139644174

[153] Wielandt, H. (1949). "Über die Unbeschränktheit der Operatoren der Quantenmechanik." *Mathematische Annalen* **121**, 21. https://doi.org/10.1007/BF01329611

[154] Wilczek, F., and Zee, A. (1979). "Operator Analysis of Nucleon Decay." *Physical Review Letters* **43**, 1571–1573. https://doi.org/10.1103/PhysRevLett.43.1571

[155] Wilson, K. G. (1974). "Confinement of quarks." *Physical Review D* **10**, 2445–2459. https://doi.org/10.1103/PhysRevD.10.2445

[156] Yukawa, H. (1935). "On the interaction of elementary particles. I." *Proceedings of the Physico-Mathematical Society of Japan. 3rd Series* **17**, 48–57. https://doi.org/10.11429/ppmsj1919.17.0_48

[157] Zaslavsky, T. (1982). "Signed graphs." *Discrete Applied Mathematics* **4**, 47–74. https://doi.org/10.1016/0166-218X(82)90033-6

AI Assistance Disclosure

This research project used research-grade commercial models, including Anthropic's Fable and OpenAI's GPT-6 Astra, for research support, software development, editing, and synthesis. The authors are responsible for the paper's claims, methods, and final text.